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Ambulance Service Clinical Procedure Manual

CPM 2024
V4.0

INTRODUCTION

CLINICAL PRIVILEGING LEVELS

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1. CLINICAL SKILLS PRIVILEGING

VERSION CONTROL

VERSION HISTORY			
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APPROVALS			
Date	**Document Version**	**Approver Name & Title**	**Approver Signature**
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I. Introduction

This Clinical Procedure Manual is an essential reference tool for Ambulance Paramedics (AP) and Critical Care Paramedics (CCP) at Hamad Medical Corporation Ambulance Service, Qatar. It is designed to provide a comprehensive guide to the skills within the practitioner's scope of practice.

The manual aims to enhance and support the existing knowledge of practitioners by offering detailed descriptions of permissible procedures according to their qualifications and privilege levels. Practitioners are advised to regularly review this manual and engage in continued practice to ensure skill retention and proficiency.

Clinical practitioners must perform **only** those procedures approved by the Ambulance Service and the Department of Healthcare Professions (DHP) that correspond to their specific credentialing level:

SYMBOL	PRIVILEGING LEVEL
I	**LEVEL 1 PRIVILEGES – First Responder (Basic Life Support)**
II	**LEVEL 2 PRIVILEGES – reduced scope Ambulance Paramedic**
III	**LEVEL 3 PRIVILEGES – full scope Ambulance Paramedic**
IV	**LEVEL 4 PRIVILEGES – full scope Critical Care Paramedic**

II. Using the Clinical Procedure Manual

Initially, the manual contains tables summarizing all clinical procedures associated with each clinical privilege level within the Ambulance Service.

The table has a Privileging Level approved icon illustrated as follows I II III IV indicating the appropriate level at which the procedure may be performed or assist to perform.

Every clinical procedure is outlined in a table that describes the clinical privilege to perform or assist to perform the procedure, the indication/s, contraindication/s. adverse effect/s, standard precautions and finally the procedure to complete the clinical skill:

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Indications

The indication is the clinical condition that leads to the recommendation or necessity of the procedure, intervention, medication or assessment

Contraindications

The contraindication is the clinical condition (a situation or factor) that prevents the performance of the procedure, intervention, medication or assessment due to the potential harm that it could cause the patient.

Precautions

Precautions refer to steps or measures that should be taken to minimize the risk of adverse events or complications when performing a particular procedure or administrating a specific medication.

Adverse Effects

An adverse event is an undesired occurrence that results from performing a particular procedure or administrating a specific medication.

Standard precautions

Wear the respective PPE as directed by the following PPE icons key

Procedure

The sequence of steps to be followed to perform or direct an intervention on a patient with the object to assess or treat.

Practitioners must ensure that the procedure aligns with their designated Privileging Level before proceeding. In cases of uncertainty in carrying out a clinical procedure, practitioners should refer to these tables for guidance.

III. Standard steps in carrying out a procedure

Standard steps are specific critical elements identify the minimal safe competencies that are inherent in the safe performance of a procedure. These standard steps are applicable to all procedures outlined in the manual and must be performed consistently with each patient for patient and crew safety.

1. Ensuring Scene Safety:

- Assess the environment to confirm it is secure for both the patient and the paramedics.
- Identify and evaluate any immediate dangers or hazards present.
- Verify that there are no threats to safety before proceeding with the procedure.

2. Applying Standard precautions

- Adhere strictly to infection control guidelines and protocols. (see the latest infection control guidelines as per SOP)
- Each skill is preceded by infection control and personal protective equipment (PPE) icons
- Wear the respective PPE as directed by the following PPE icons key

Wash Hands	**Examination Gloves**	**Surgical face Mask**	**N95 Mask**	**Goggles**
				
Face Shield	**Safety Vest**	**Helmets**	**Earmuffs**	**Safety Gown**
				

3. **Assessing the need for the procedure**
    - Conduct a thorough assessment of the patient's condition to determine the necessity of the procedure.
    - Evaluate clinical signs and symptoms to justify the intervention.
    - Confirm that the procedure is indicated and appropriate for the specific patient at the right time.
4. **Obtaining patient/family consent**
    - Provide a clear explanation of the procedure to the patient and/or their family members.
    - Discuss the potential risks, benefits, and alternatives associated with the procedure.
    - Obtain verbal consent from the patient or their legal representative prior to proceeding.
5. **Ensuring patient comfort**
    - Position the patient in a manner that is both comfortable and conducive to the procedure.
    - Offer reassurance and address any concerns or anxieties the patient may express.
6. **Explaining the procedure**
    - Describe each step of the procedure using clear and simple language.
    - Inform the patient about what to expect during and after the procedure to ensure they are well-informed.

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7. **Documenting the procedure in the electronic patient care report (EPCR)**
    - Accurately record all relevant details of the procedure in the EPCR.
    - Include essential information such as the rationale for the procedure, assessment findings, steps undertaken, and any observations made.
8. **Evaluating post-procedure condition**
    - Monitor the patient's condition following the procedure to identify any immediate adverse reactions or complications.
    - Provide appropriate follow-up care instructions and ensure that the patient is stable before concluding the care process.
9. **Handover procedure**
    - Provide a comprehensive and concise clinical handover to receiving practitioner on-scene, via tetra radio or at the receiving facility using IMIST-AMBO.

IV. International patient safety goals

Ensuring patient safety is an ongoing concern for all HMCAS paramedics as it is a fundamental component to the delivery of safe high-quality healthcare.

Identify patients correctly:

- Ensure patients are accurately identified using either the patients' health card, Qatar ID or Passport.
- If the patient's identity cannot be confirmed, record the patient as unknown.
- Be sure to record the HC number used to register the patient at hospital in the ePCR

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1. **Improve Effective Communication**

- Use the IMIST-AMBO to provide a structured handover with all the relevant information.
- This applicable to handovers between Ambulance Service Staff and handovers conducted at healthcare facilities.

2. **Improve the safety of High-Alert Medications**

- Ensure all medications are labeled where appropriate.
- Double check all high alert medications with your partner prior to administration to confirm you are giving the right medication to the right patient.

3. **Reduce the Risk of Healthcare-Associated Infections**

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- Ensure you perform regular hand hygiene.
- Perform social cleaning of all surfaces, stretcher and equipment used that came into contact with a patient after every encounter.
- Wear PPE if there is an increased risk of infection or if the patient is immunocompromised.

4. Reduce the Risk of Patient Harm Resulting from Falls

- All patients are to be treated as a potential falls risk.
- Ensure patients are securely fastened to the stretcher or patient movement device using straps.
- Adhere to the falls prevention SOP for guidance.

V. Clinical Privileging Levels

PROCEDURE	Level 1	Level 2	Level 3	Level 4
Patient Assessment				
Recording on vital signs (blood pressure, pulse, SpO_2 and respiratory rate) plus use of non-invasive monitoring	yes	yes	yes	yes
Management of invasive monitoring/pressure transducers during transfer	no	no	assist	yes
Capillary blood glucose	yes	yes	yes	yes
AVPU assessment	yes	yes	yes	yes
Use of Capnography	yes	yes	yes	yes
Glasgow Coma Scale	no	yes	yes	Yes
Recording/obtaining a 4 lead ECG	assist	yes	yes	yes
ECG rhythm recognition	no	Yes (VF/VT only)	yes	yes
Recording/obtaining a 12 lead ECG	assist	yes	yes	yes
Interpretation of 12 lead ECG	no	no	conditional	yes
Major incident triage (Sieve)	assist	yes	yes	yes
CBRN triage	no	yes	yes	yes
Major incident triage (Sort)	assist	yes	yes	yes
Perform physical assessment on medical patients	assist	yes	yes	yes
Perform physical assessment on obstetric patients	assist	yes	yes	yes
Perform physical assessment on trauma patients	assist	yes	yes	yes
Limb immobilization				
Basic splinting (e.g. vacuum splints) and bandaging	yes	yes	yes	yes
Traction splint	assist	yes	yes	yes

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Pelvic binder	assist	yes	yes	Yes
Spinal immobilization				
Manual in-line stabilization	assist	yes	yes	yes
C spine collar application	yes	yes	yes	yes
Log roll	assist	yes	yes	yes
Use of scoop and/or spinal board and/or KED and/or vacuum mattress/ Combicarrier	yes	yes	yes	Yes
Patient extrication (controlled)	assist	yes	yes	Yes
Patient extrication (rapid)	assist	yes	yes	yes
Cervical spine clearance as outlined in CPG's	no	yes	yes	yes
Airway management				
Basic airway management (positioning, bag-valve-mask ventilation)	yes	yes	yes	yes
Oral airway insertion (adult)	yes	yes	yes	yes
Oral airway insertion (paediatrics)	no	yes	yes	yes
Oral suctioning	yes	yes	yes	yes
Nasal airway (paediatric)	no	no	no	yes
Nasal airway (adult)	no	yes	yes	yes
Nasal suctioning (adult, child or newborn)	no	yes	yes	yes
Suctioning via a supraglottic airway	no	yes	yes	yes
Suction via endotracheal or tracheostomy	no	yes	yes	yes
Nasal gastric tube insertion	no	no	no	yes
Supra glottic airway placement (adult)	yes	yes	yes	yes
Supra glottic airway placement (over 1 year paediatric)	no	yes	yes	yes
Supra glottic airway placement (under 1 year paediatric)	no	no	no	yes
Use of Magill forceps	no	no	no	yes
Rapid Sequence Intubation (over the age of 8)	no	no	no	yes
Rapid Sequence Intubation (between 2 and 8)	no	no	no	see comments

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Delayed Sequence Intubation/sedation inserted supra glottic airway	no	no	no	yes
Intubation (to include video laryngoscope) over the age of eight years (oral or nasal)	no	no	no	yes
Intubation (to include video laryngoscope) under the age of eight years but greater than 2 years (oral or nasal)	no	no	no	see comments
Intubation under the age of 2 years	no	no	no	see comments
Surgical and/or needle cricothyroidotomy	no	no	no	yes
Intubation via a supraglottic airway (adult or child)	no	no	no	yes
Needle decompression over the age of 14	no	Under CCP supervision	Under CCP supervision	yes
	Ambulance paramedics undertaking the roles and duties of emergency ambulance service Senior Operations manager, operations manager, operations officer, DELTA, Critical Care Assistant and special operations paramedic may (once training has been completed) undertake needle decompression as an independent skill			
Needle decompression under the age of 14 but over the age of four	no	no	Under CCP supervision	yes
	Ambulance paramedics undertaking the roles and duties of emergency ambulance service Senior Operations manager, operations manager, operations officer, DELTA, Critical Care Assistant and special operations paramedic may (once training has been completed) undertake needle decompression as an independent skill			
Needle decompression under the age of four	This skill can only be completed by a Critical Care Paramedic in a child under the age of 4 years			
Inter costal cut down during traumatic cardiac arrest and/or ventilated patient with tension pneumothorax (adult over the age of 14)	no	no	no	see comments
Administration of oxygen during cardiac arrest	yes	yes	yes	yes
Administration of emergency oxygen therapy	yes	yes	yes	yes
Escharotomy	no	no	no	Yes: see Comments

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Vascular access				
Adult intravenous access	no	No: See comments	yes	yes
Paediatric intravenous access (>1 yr)	no	no	yes	yes
Paediatric intravenous access (<1 yr)	no	no	Under CCP supervision	yes
Intra osseous access	no	no	no	yes
External jugular access	no	no	no	yes
Umbilical cannulation	no	no	no	yes
Scalp vein cannulation	no	no	no	yes
Administration of fluids or medication via a nasal gastric tube	no	assist	assist	yes
Ventilation techniques				
CPAP	no	no	no	yes
Controlled volume ventilation (mechanical) with PEEP	no	no	no	yes
Synchronized intermittent mandatory ventilation	no	no	no	yes
BiPAP	no	no	no	yes
Pressure support ventilation	no	no	no	yes
Continuation of hospital/community prescribed ventilation modes e.g. home ventilation or during inter facility transfer	no	yes	yes	yes
Drug route administration				
Drug administration	See list	See list	As per CPG's	As per CPG's
Sub cutaneous drug administration	no	Yes	yes	Yes
Intramuscular drug administration	Adrenaline for anaphylaxis only	yes	yes	yes
Via Intravenous route	no	Under direction of a CCP or Doctor	yes	yes

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Via Intra osseous route	no	Under direction of a CCP or Doctor	Under direction of a CCP or Doctor	yes
Inhaled medication via metered dose inhaler (excluding nebulizer)	yes	yes	yes	yes
Nebulised therapy	no	yes	yes	yes
Oral, sub lingual, sub buccal and intra nasal (within the scope of CPG's)	Oral Paracetamol/ Ibuprofen	yes	yes	yes
Rectal	no	yes	yes	yes
Use of infusion devices such as (but not limited to) syringe drivers and infusion pumps	no	no	See comments	yes
Continuation of previously prescribed medication (outside of normal scope) during transportation to hospital	no	See comments	See comments	yes
Management of intravenous infusion during transport to hospital	no	yes	yes	yes
Cardiac, resuscitation and specialist care				
Basic life support (to include choking)	yes	yes	yes	yes
Termination of CPR as per SOP's	no	yes	yes	yes
Withholding CPR as per SOP's	no	yes	yes	yes
Advanced life support	no	assist	yes	yes
Advisory External Defibrillator (AED)	yes	yes	yes	yes
Manual defibrillation	no	yes	yes	yes
Electrical synchronized cardioversion	no	no	no	yes
External pacing	no	no	no	yes
Percussion pacing	no	no	no	yes
Vagal manoeuvres (over the age of 8)	no	Valsalva maneuver only	Valsalva maneuver only	Yes
Vagal manoeuvres (under the age of 8)	no	no	no	yes
Wound and pressure dressing	yes	yes	yes	Yes

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Specialist trauma and burns management				
Haemostatic dressings and tourniquets	yes	yes	yes	yes
Use of burn dressings	yes	yes	yes	yes
Specialist trauma and burns management				
Haemostatic dressings and tourniquets	yes	yes	yes	yes
Use of burn dressings	yes	yes	yes	yes
Maternal and paediatric specialist skills				
Emergency childbirth	assist	yes	yes	yes
Emergency complex deliveries	assist	assist	yes	yes
Neonatal resuscitation (basic)		Follows basic paediatric CPR	yes	
Neonatal advanced	no	assist	assist	yes
Paediatric basic resuscitation	yes	yes	yes	yes
Paediatric advanced resuscitation	no	no	yes	Yes
Diagnostic				
Use of ultrasound (FAST and/or focused in cardiac arrest)	no	no	no	See comments
Arterial gas analysis	no	no	no	yes
Arterial gas sampling	no	no	no	See comments
Care of patient with central line in situ	no	no	assist	yes
Care of patient with transvenous pacing wire in situ	no	no	assist	yes
Nonspecific				
Use of an incubator	no	assist	assist	See comments
Discharge patient from scene post assessment	no	no	See comments	See comments

MEDICATION	Level 1	Level 2	Level 3	Level 4
Adenosine				✓
Adrenaline NEB			✓	✓
Adrenaline IM		✓	✓	✓
Adrenaline IV			✓	✓
Adrenaline IO				✓
Adrenaline INF				✓
Amiodarone IV			✓	✓
Amiodarone IO				✓
Amiodarone INF				✓
Aspirin	✓	✓	✓	✓
Atropine				✓
Budesonide				✓
Calcium Chloride				✓
Clopidogrel		✓	✓	✓
Dexamethasone				✓
Dextrose PO	✓	✓	✓	✓
Dextrose IV			✓	✓
Dextrose IO				✓
Diclofenac IM			✓	✓
Diphenhydramine				✓
Fentanyl				✓
Furosemide				✓
Glucagon IM		✓	✓	✓
Glucagon IV				✓

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Glycerol Trinitrate SL	✓	✓	✓	✓
Glycerol Trinitrate INF				✓
Hydrocortisone				✓
Ibuprofen	✓	✓	✓	✓
Ipratropium Bromide		✓	✓	✓
Ketamine				✓
Ketorolac IM		✓	✓	✓
Ketorolac IV			✓	✓
Magnesium Sulphate				✓
Methoxyflurane	✓	✓	✓	✓
Metoclopramide				✓
Midazolam				✓

*****Comments related to HMCAS Clinical Privileging**

1. **Interpretation of 12 lead ECG by ambulance paramedics**. This is not a core ambulance paramedic (level 2-3) skill as HMCAS uses diagnostic 12-lead analysis on its prehospital defibrillators. Ambulance paramedics, however, are encouraged to develop ECG analysis skills.
2. **Rapid Sequence Intubation (between 2 and 8).** Critical Care Paramedics are not routinely privileged to perform RSI on children under the age of eight. It is, however, acknowledged that accurate assessment of a child's age is subjective, therefore if this occurs by accident (particularly in the age range of 6 to 8) this is not deemed to be an adverse incident. However, CCP's will not routinely be issued with the required range of endotracheal tubes to facilitate paediatric intubation. The age limit for intubation is subject to continuous review to reflect the needs of the population of Qatar and as such individual CCP's may be privileged to perform paediatric RSI. This will be an additional privilege which will require an amended privileging letter.
3. **Needle decompression** Is limited to CCP, named Ambulance Paramedics (such as Critical Care Assistants and Delta supervisors following additional training) or an AP under CCP supervision. Specific AP's will receive additional privileges to perform needle decompression as an independent skill based on a risk: benefit assessment.

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4. **Finger thoracostomy during traumatic cardiac arrest with tension pneumothorax (adult over the age of 14):** all critical care paramedics (holding CCP privileges) are privileged to perform this skill **BUT** this privilege is only enacted once they have completed HMCAS approved training. This training may be undertaken within Qatar or outside Qatar and appropriate evidence must be placed within their HMCAS training file
5. **Adult intravenous access for paramedics holding privileges of a basic paramedic:** qualified ambulance paramedics, who hold level 1 or 2 privileges will not **NORMALLY** be entitled to perform intravenous cannulation. However, those with appropriate training may perform intravenous cannulation under the supervision of a HMC doctor of CCP.
6. **Continuation of previously prescribed medication by ambulance paramedics which is outside of their normal scope during transportation to hospital:** following discussion with a CCP, either at the patient's bedside, based within the clinical command desk or via an on- call system, an ambulance paramedic may continue to transport a patient with a hospital instigated infusion.
7. **Use of infusion devices such as (but not limited to) syringe drivers and infusion pumps:** ambulance paramedics who have received training may use an infusion device, from hospital to hospital, to facilitate interfacility transfer. This will be an additional privilege which will require an amended privileging letter. CCA trained staff may setup infusion devices under CCP supervision
8. **Arterial gas sampling: Critical care paramedics:** Critical Care Paramedics, who have received training, may perform an arterial stab and/or arterial blood sampling from an inside arterial to facilitate decision-making. This will be an additional privilege which will require an amended privileging letter. Critical Care Paramedics may perform capillary blood gas sampling based on individual privileging.
9. **Discharge patient from scene post assessment:** all critical care paramedics are empowered to assess, review and discharge patients from scene in accordance with agreed Clinical Practice Guidelines (note as of March 2019 such guidelines have not been produced).

VI. List Of Abbreviations

AAA	Abdominal Aortic Aneurysm
ACS	Acute Coronary Syndrome
AF	Atrial Fibrillation
AIDS	Acquired Immune Deficiency Syndrome
ARDS	Adult Respiratory Distress Syndrome
AVPU	Alert, Verbal, Painful, Unresponsive
BIPAP	BI-Level Positive Airway Pressure
BP	Blood Pressure
Bpm	Beats Per Minute
BSA	Body Surface Area
BVM	Bag Valve Mask Resuscitator (Ambu
c/o	Complains of
COPD	Chronic Obstructive Pulmonary Disease
CPAP	Continuous Positive Airway Pressure
CPR	Cardiopulmonary Resuscitation
CRF	Chronic Renal Failure
CSF	Cerebrospinal Fluid
C-spine	Cervical Spine
CVA	Cerebrovascular Accident
CVS	Cardiovascular System
D5W 5%	5% Dextrose in Water
DDx	Differential Diagnosis
DKA	Diabetic Ketoacidosis
Dx	Diagnosis
ECG	Electrocardiogram(s)
ETCO2	End Tidal Carbon Dioxide
ETT	Endo Tracheal Tube(s)
FiO2	FiO2 Fraction of Inspired Oxygen
GCS	GCS Glasgow Coma Scale
gm	Gram
Hbd	Hemoglobin
HIV	Human Immunodeficiency Virus
h/o	History of
HR	Heart Rate
HTN	Hypertension
hx	History
ICP	Intracranial Pressure
IDDM	Insulin Dependent Diabetes Mellitus
I:E Ratio	Inspiratory/Expiratory Ratio

IHD	Ischaemic Heart Disease
IM	Intramuscular
IMV	Intermittent Mandatory Ventilation
Inj	Injection
IPPV	Intermittent Positive Pressure Ventilation
IV	Intravenous
JVP	Jugular Venous Pressure
kg.	Kilogram
KVO	Keep Vein Open
LBB	Long back board
LBBB	Left Bundle Branch Block
LBW	Low Birth Weight
l/min	Liters per Minute
LMA	Laryngeal Mask Airway
LMP	Last Menstrual Period
LOC	Level of Consciousness
LVF	Left Ventricular Failure
MAP	Mean Arterial Pressure
MAS	MAS Meconium Aspiration Syndrome
max	Maximum
MDI	Metered-Dose Inhaler
mg	Milligram
MI	Myocardial Infarction
min	Minute
ml	Milliliter
mmol	Millimoles
mV	Millivolt
MAP	Mean Arterial Pressure
MAS	MAS Meconium Aspiration Syndrome
max	Maximum
MDI	Metered-Dose Inhaler
mg	Milligram
MI	Myocardial Infarction
min	Minute
ml	Milliliter
mmol	Millimoles
mV	Millivolt
N & V	Nausea & Vomiting
NA	Not Applicable
NaCl	Sodium Chloride
NAD	No Abnormality Detected
NGT	Nasogastric Tube

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NIDDM	Non-insulin Dependent Diabetes Mellitus
NSR	Normal Sinus Rhythm
NRBM	Non-Rebreather Mask
NSAID	Nonsteroidal Anti-Inflammatory Drug
NSTEMI	Non-ST Elevation Myocardial Infarction
O2	Oxygen
O2 Sat	Oxygen Saturation
O/E	On Examination
PAC	Premature Atrial Contraction
PAF	Paroxysmal Atrial Fibrillation
PCI	Percutaneous Coronary Intervention
PE	Pulmonary Embolism
PEEP	Positive End Expiratory Pressure
PERRL	Pupils Equal, Round, Reactive to Light
pH	Hydrogen Ion Concentration
PIH	Pregnancy Induced Hypertension
PIP	Peak Inspiratory Pressure
PND	Paroxysmal Nocturnal Dyspnea
po	per oral, by mouth
PPH	Post Partum Hemorrhage
prn	As Often as Necessary
PROM	Premature Rupture of Membranes
pt	Patient
PTB	Pulmonary Tuberculosis
PTL	Preterm Labor
PVC	PVC Premature Ventricular Contraction
RBBB	Right Bundle Branch Block
RBS	Random Blood Sugar
RHF	Right Heart Failure
RR	Respiratory Rate
RSI	Rapid Sequence Induction
RTA	Road Traffic Accident
sec	second
SIDS	Sudden Infant Death Syndrome
SIMV	Synchronized Intermittent Mandatory Ventilation

STEMI	ST-Elevation Myocardial Infarction
SVD	Spontaneous Vaginal Delivery
SVT	Supraventricular Tachycardia
T	Temperature
TV	Tidal Volume
Tx	Treatment
URTI	Upper Respiratory Tract Infection
UTI	Urinary Tract Infection
VF	Ventricular Fibrillation
vol	Volume
Vtach	Ventricular Tachycardia
wk	Week
wt	Weight
x/7	number of days
x/12	number of months
x/52	number of weeks
yrs	Years
>	Greater Than
≥	Greater Than or Equal To
<	Less Than
≤	Less Than or Equal To

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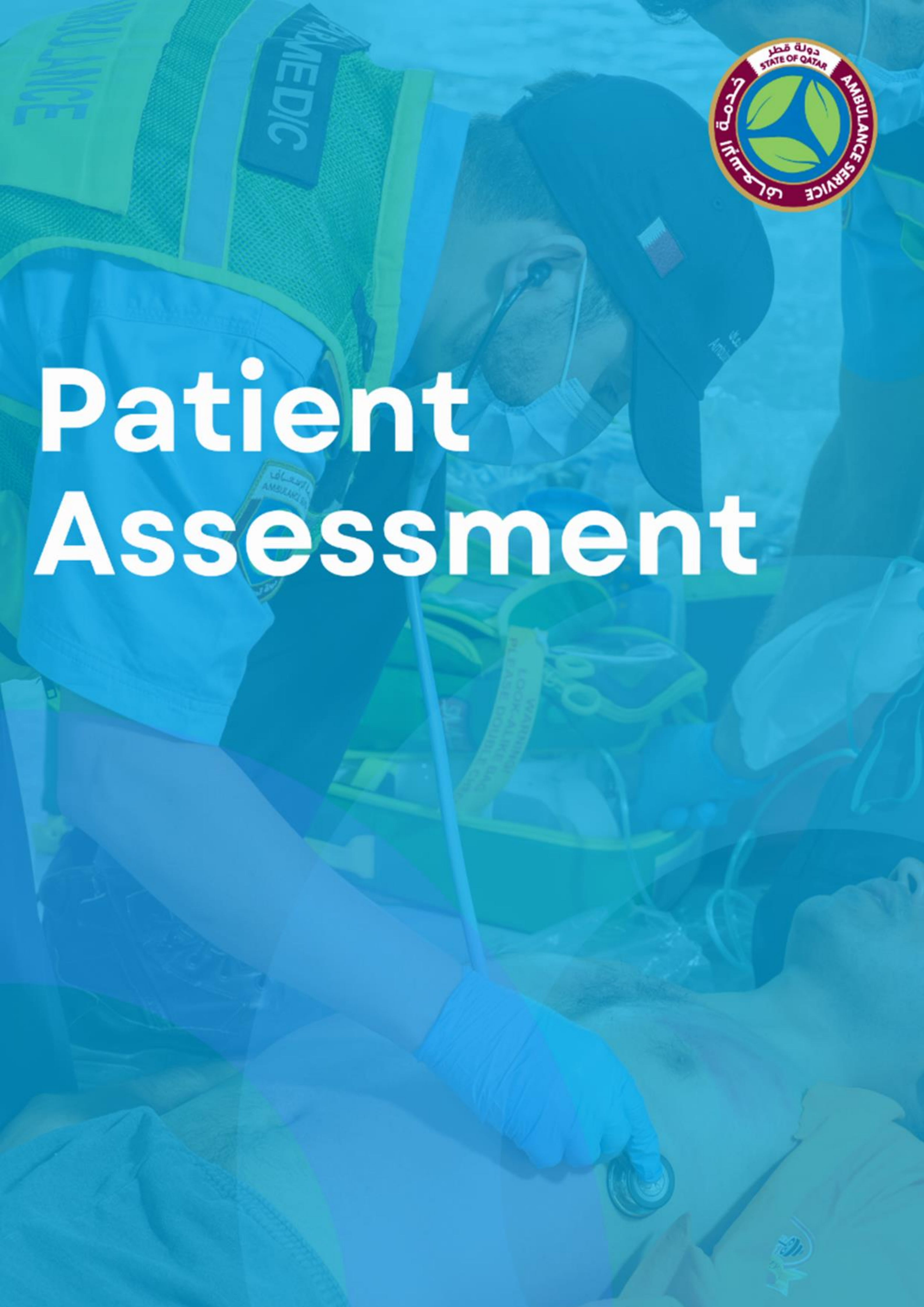
Patient Assessment



1.1 Qatar Early Warning Score Calculation

Indications

1. QEWS should be performed on all adult medical patients.

Contraindications

1. Trauma patients
2. Paediatric patients
3. OB patients
4. Non-emergent scheduled transfers (PTS), unless the patient is observed to be unwell, develops a new complaint during transport or if the patient/ family requests.

Adverse Effects

1. Nil in this setting.

Standard precautions

Procedure

1. Place the patient in their most comfortable position.
2. The following vital signs are important for calculating the QEWS; **Respiratory Rate, Heart Rate, Systolic Blood Pressure, Temperature, SpO_2 on room air and the patients AVPU**.
3. Once each of the vital signs are recorded, calculate the QEWS using the listed table.
4. Allocate a score for each vital sign, then add up the total score.
5. A score of **4 or more is a marker of an increased risk to the patient**, considering patient condition, request CCE consult.

HMCAS QEWS TABLE

QEWS	3	2	1	0	1	2	3
RESPIRATORY RATE	< 8			9–20	21–25	26–35	> 36
HEART RATE	< 31	31– 40	41– 50	51–100	101–110	111–129	> 129
SYSTOLIC BP	< 70	71– 89	90 –109	110–179	180–199	> 200	
TEMPERATURE		C(<36ºC)		N(36º–38ºC)		H(> 38ºC)	
SpO_2 on Air	< 90%	90 – 92%	93 – 94%	≥ 95%			
AVPU	U	P	V	A			
A score of 4 or more is a marker of an increased risk to the patient.							TOTAL =

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1.2 Respiratory Rate Assessment

Indications

1. Respiratory rate assessment must be conducted on **ALL** patients, except for patients presenting with **NO** breathing or agonal breathing.

Contraindications

1. Non-emergent scheduled transfers (PTS), unless the patient is observed to be unwell, develops a new complaint during transport or if the patient/ family requests.

Adverse Effects

1. Expose the patient's chest with discretion.

Standard precautions

1. Wear an N95 mask if the patient exhibits respiratory symptoms.

Procedure

1. Place the patient in their most comfortable position.
2. Expose the patient's chest (with discretion) and inspect for abnormality.
3. Observe the chest wall for rise and fall.
4. Count the number of times the chest wall rises and falls in 15 seconds.
5. Then multiply this number by 4 to determine the number of times the patient's chest rises and falls in 60 seconds (1 minute).
6. Example, if the chest rises and falls 6 times in 15 seconds, then the respiratory rate will be:

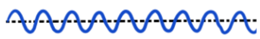
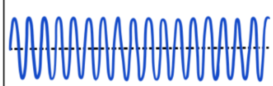
   6 x 4 = 24 breaths per minute

7. Also note the regularity of the patients breathing. Assess if the patient's breathing is regular or irregular.
8. Assess the depth of patient's breathing. Note if the patient's breathing is shallow or deep.
9. RR should be assessed every 15 minutes for stable patients and every 3 minutes for unstable patients.

Breathing Patterns

Eupnea 

Normal breathing rate and pattern

Condition	Description	Causes	Graphic Presentation
Bradypnea	Decreased respiratory rate	Drugs, metabolic disorder, head injury, stroke	
Tachypnea	Increased respiratory rate	Fever, anxiety, shock	
Cheyne-Stokes	Gradual increases and decreases in respirations with periods of apnea.	Increasing intracranial pressure, brain stem injury	
Biot	Rapid, deep respirations (gasps) with short pauses between sets.	CVA or trauma involving the medulla or lower pons, opiate effect	
Kussmaul	Rapid and deep respirations	Severe metabolic acidosis	

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1.3 Lung Sound Assessment

Indications

1. Lung sound assessment must be conducted on **ALL** patients, except for patients presenting with No breathing or agonal breathing.

Contraindications

1. SAS (Scheduled Ambulance Service) patient during routine transportation if not expected as per CPG 16.1

Adverse Effects

1. Expose the patient’s chest with discretion.

Standard precautions

1. Wear an N95 mask if the patient exhibits respiratory symptoms.

Procedure

1. Expose the patient’s chest (with discretion) and inspect for abnormality.
2. Observe the chest wall for rise and fall.
3. Direct the patient to inhale deeply through an open mouth and then exhale.
4. Place the stethoscope diaphragm directly on the patient's skin.
5. Ensure that the stethoscope is held still and that the conductive tubing is kept clear of contact with any surface to avoid extraneous noise.
6. Listen to both sides of the chest in a methodical manner. Listen in a minimum of 4 fields anterior (front) and posterior (back).
7. **Anterior:**
   - 3 fingers breadth below the clavicle at the mid-clavicular line, on right and left sides.
   - Just below the nipple line (6th intercostal space) mid-axillary, on right and left sides.
8. **Posterior:**
   - Upper back, 3 fingers breadth below the shoulder (not over the scapular bone), on right and left sides.
   - 3 fingers breadth below the scapular bone, on right and left sides.
9. Compare lung sounds between the right and left lung fields.

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1.4 Pulse Rate Assessment

Indications

1. Pulse rate assessment must be conducted on **ALL** patients, except for patients in suspected adult medical cardiac arrest.

Contraindications

1. SAS patient during routine transportation if not excepted as per CPG 16.1

Adverse Effects

1. Palpating both carotid pulses (central) at the same time may cause the patient to present with bradycardia.

Standard precautions

Procedure

1. Locate the appropriate point to measure the pulse rate, as depicted in the diagram below.
2. Count the number of times the pulse wave is felt in 15 seconds over the specifically identified pulse check point.
3. Then multiply this number by 4 to determine the patient's pulse rate in 60 seconds (1 minute).
4. Example, if the pulse wave is felt 20 times in 15 seconds, then the patient's pulse rate will be:

   20 x 4 = 80 beats per minute
5. Also note the regularity of the patient's pulse rate. Assess if the patient's pulse rate is regular or irregular.
6. Assess the character of patient's pulse rate. Note if the patient's pulse rate is weak or strong.

Normal vital signs for adults

NORMAL VITAL SIGNS FOR ADULTS	
VITAL SIGN	NORMAL RANGE
Blood pressure	SBP 110 – 140 mmHg/ DBP 60 – 90 mmHg
Heart rate	51 – 100 beats/min
Respiratory rate	12 – 20 breaths/min
Temperature	36 – 37.6°C
RBS	4 – 6.7 mmol/L

Normal vital signs for paediatrics

NORMAL VITAL SIGNS FOR PAEDIATRICS							
Age Group	Weight	RR (breaths/min)	HR (beats/min)	SBP (mmHg)	DBP (mmHg)	Temp (°C)	RBS (mmol/L)
Preterm	2kg	40-70	110-170	55-75	35-45	36.5	2.6-6
Term – 2 months	3 – 4kg	35-55	110-160	65-85	45-55	36.5	2.6-6
3 – 7 months	5 – 7kg	35-45	110-160	70-90	50-65	36.5	4-6
8 – 12 months	8 – 11kg	22-38	90-160	80-100	55-65	36.5	4-6
1 – 3 years	12 – 15kg	22-30	80-150	90-105	55-70	36.5	4-6
4 – 5 years	16 – 21kg	20-24	70-120	95-110	60-75	36.5	4-6
6 – 7 years	22 – 28kg	20-24	70-120	95-110	60-75	36.5	4-6
8 – 9 years	29 – 34kg	16-22	60-110	100-120	60-75	36.5	4-6
10 – 11 years	35 – 40kg	16-22	60-110	100-120	60-75	36.5	4-6
12 – 13 years	41 – 45kg	12-20	60-100	110-135	65-85	36.5	4-6
≥ 14 years	46 – 50kg	12-20	60-100	110-135	65-85	36.5	4-6

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1.5 Random Blood Sugar Assessment

Indications

1. Seizures
2. Sick paediatric patients
3. Impaired consciousness
4. Post collapse
5. Abnormal behaviour
6. Any patient who is suspected of being hypoglycaemic (stroke, ROSC, history of fasting)
7. History of diabetes mellitus

NOTE: A patient with impaired consciousness must have RBS measured, whenever practical.

Contraindications

1. There are no actual contraindications to checking and recording an RBS. But it must be remembered that this procedure is invasive and so judgement must be used as to the appropriateness of performing the procedure.
2. Non-emergent scheduled transfers (PTS), unless the patient is observed to be unwell, develops a new complaint during transport or if the patient/ family requests.

Precautions

1. Ensure glucometer is calibrated as per manufacturer's recommendation.
2. Alcohol can affect the RBS result.
3. Use of a band aid represents a readily available near-sterile option for drying the site.

Standard precautions

Procedure

1. Check and prepare the Glucometer, lancets, Glucostrips, swab and band aid.
2. Switch on Glucometer and confirm the code matches the Glucostrips code.
3. Disinfect the area using 70% alcohol wipes.
4. Prepare lancing device, ensuring it is a fully disposable lancet.
5. Without touching the 'blood target area', insert the sensor electrode into the test port of the glucometer, ensuring it turns on.
6. Lance the side of the finger with the lancet and obtain a hanging drop of blood.
7. Move the glucometer to the lanced finger and apply a drop of blood to the target area of the sensor strip.
8. The test will start automatically, and the RBS reading will appear on the screen.
9. Discard the lancet and sensor electrode into the sharps container.
10. Cover the wound with a band aid.

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11. For normal RBS, only test and record on the ePCR once.
12. For hypoglycaemic patients, recheck the RBS 5 minutes after correction. Repeat RBS checks if the patient's condition changes.
13. Clean the glucometer using HMC-approved disinfectant wipes.

CONVERSION TABLE	
mmol/L	**mg/dL**
1	18
2	36
3	54
4	72
5	90
6	108
7	126
8	144
9	162
10	180
11	198
12	216
13	234
14	252
15	270
16	288
17	306
18	324
19	342
20	360

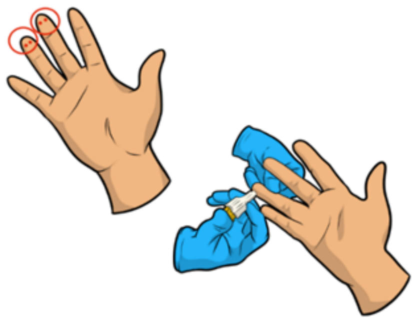
Hints

1. Place the patients hand below the level of the heart to allow for gravitational movement of blood to the tip of the fingers to improve success rates of RBS checking and recording.

2. Milk the finger to improve blood flow to the tip to improve success rates of RBS checking and recording.

Sites for RBS Assessment

The middle or ring finger is typically preferred in *adults* and *children*. The thumb has a pulse and is likely to bleed excessively. The index finger can be calloused or sensitive and the little finger does not have enough tissue to prevent hitting the bone with the lancet. The lancet puncture is done to the left or right of the midline of the palmar surface of the fingertip, staying away from the fingernail.



Sites for RBS Assessment <1year old

This puncture site is used for *newborns* and *infants*, less than one year old, because their fingers are too tiny for finger-stick blood collection. The puncture is done on the farthest lateral or medial aspect of the plantar surface of the heel, not on the bottom. Punctures done on the plantar surface can potentially damage cartilage or bone.

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1.6 Blood Pressure Measurement: Manual (adult)

Indications

1. A manual BP should be recorded when a non-inflatable BP (NIBP) is unrecordable.
2. A manual BP should also be recorded when the NIBP reading is outside the normal BP values.

Contraindications

1. SAS patient during routine transportation if not excepted as per CPG 16.1

Precautions

1. A blood pressure is not to be taken on the arm of a patient that has an AV fistula present.
2. Do not take a blood pressure on an injured arm.

Standard precautions

3. 

Procedure

1. Make the patient comfortable and expose the patient's arm.
2. Apply the appropriately sized BP cuff to the exposed arm at 2 finger width above the antecubital area.
3. Palpate the brachial pulse and place the ARTERY marker point of the BP cuff directly over the brachial pulse.
4. Hold the patient's arm at the level of the heart. The patient's elbow must not be lower than the lowest rib cage and must not be raised higher than the shoulder.
5. Locate and palpate the radial pulse of the same arm. Inflate the BP cuff slowly and at the point the radial pulse disappears, this becomes the palpable BP. Then deflate the cuff completely.
6. Place the diaphragm of the stethoscope over the brachial artery and inflate the BP cuff to 20 mmHg above the palpable blood pressure.
7. Deflate the cuff at a rate of 2-3 mmHg per second. At the point the Korotkoff sound is heard is the systolic blood pressure. At the point the heart sounds disappear is the diastolic blood pressure.
8. Deflate the cuff completely and remove, redress the patient (if required) and record the blood pressure on the ePCR.
9. BP should be assessed every 15 minutes for stable patients and every 3 minutes for unstable patients.
10. Hypotensive patients should be treated according to the CPG.
11. Clean the BP apparatus using HMC-approved disinfectant wipes.

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1.7 Non-invasive blood pressure measurement Using the LIFEPAK15

Indications

1. **All** patients being assessed by Ambulance Paramedics and Critical Care Paramedics.

Contraindications

1. Non-emergent scheduled transfers (PTS), unless the patient is observed to be unwell, develops a new complaint during transport or if the patient/ family requests.

Precautions

1. A blood pressure is not to be taken on the arm of a patient that has an AV fistula present.
2. Do not take a blood pressure on an injured arm.
4. Machine BP measurements must be correlated with patients' condition.

Adverse Effects

1. Prolonged inflation/ deflation of the NIBP may cause bruising and skin irritation from compression.
2. Prolonged periods of frequent NIBP measurements may also cause pain and patient discomfort, limb swelling, phlebitis compartment syndrome, peripheral neuropathy, thrombophlebitis, venous stasis, ecchymosis, and petechiae.

Standard precautions

Procedure

1. Make the patient comfortable and expose the arm (With discretion and consent from the patient).
2. Turn ON the LIFEPAK 15 and confirm that the NIBP cable is connected to the machine.
3. Apply the appropriately sized BP cuff to the arm at 2 finger width above the antecubital area.
4. Inform the patient that the BP cuff will inflate automatically based on the preset time interval.
5. Inform the patient to remain still as movement to the arm may cause erroneous readings.
6. Activate BP recording by pushing the **NIBP** button situated on the right side of the screen. The BP will record automatically.
7. In an emergency setting, BP must be recorded every 15 minutes in a stable patient and every 3 minutes in an unstable patient. Remember to change the time interval of recording BP on the LIFEPAK 15
8. Clean the LIFEPAk 15 and its attachments using HMC-approved disinfectant wipes.

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1.8 Temperature assessment using Genius 2 tympanic thermometer

Indications
1. To obtain a baseline temperature to enable comparisons to be made with future recordings. 2. To enable close observation in resolving hypothermia/ hyperthermia. 3. To observe and monitor patients for changes indicating an infection.
Contraindications
1. Trauma, ear drainage or obstructions to the affected ear. Consider the alternate ear. 2. Non-emergent scheduled transfers (PTS), unless the patient is observed to be unwell, develops a new complaint during transport or if the patient/ family requests.
Adverse Effects
1. Recording a tympanic temperature may cause discomfort to certain patients.
Standard precautions

Procedure
1. Visually inspect the patients' external ear and ear canal for trauma, ear drainage or obstructions. 2. Remove the Genius 2 Tympanic Thermometer from the storage base. 3. Inspect the probe lens. If any debris is present, clean the probe tip. 4. Press the scan button to verify functionality and mode selection on the LCD screen. 5. Install a probe cover by firmly inserting the probe tip into a probe cover. The thermometer will then perform a system reset. 6. Inspect the probe cover to make sure it is fully seated (no space between cover and tip base) and no holes, tears, or wrinkles are present in the plastic film. 7. Place the probe in the ear canal and seal the opening with the probe tip. Ensure that the probe shaft is aligned with the ear canal. 8. Once positioned lightly in the ear canal press and release the scan button. Wait for the triple beep before removing the thermometer. 9. Remove the probe from the ear as soon as the triple beep is heard and read. 10. The patient temperature and the probe eject icons will be displayed. 11. Press the eject button to eject the probe cover and dispose of into a suitable waste receptacle. 12. Always return the thermometer to base for storage. 13. Use the patient's temperature reading to determine the patient's QEWS score.

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14.	Clean the tympanic thermometer using HMC-approved disinfectant wipes.

Hints

1. Conversion of Fahrenheit (˚F) to Centigrade (˚C);

   ˚C = (˚F-32) X 5/9

2. Always install a new probe cover prior to taking a temperature. The probe cover membrane should be smooth with no holes, tears, or wrinkles.
3. Temperature measurement range of the Genius 2 Thermometer is 33.0°C to 42.0°C (91.4°F to 107.6°F).
4. Disposable single use probe covers aid in the prevention of a cross contamination of infectious diseases.
5. Ensure that the probe tip seals the ear canal prior to taking a temperature. Failing to seal the ear canal will result in a loss of accuracy.
6. Patients with removable hearing aids should remove the device at least 10 minutes prior to ear temperature assessment. Implanted devices generally do not affect ear temperature.
7. Always wait at least two minutes before taking another measurement in the same ear.
8. Used probe covers must be treated as infectious biological waste and disposed of in accordance with current medical practices.
9. After a temperature has been acquired, the thermometer will enter "off" mode after approximately 10 seconds. The temperature can be recalled by pressing and releasing the scan button or by pressing and holding the °C/°F button.

Genius 2 Tympanic Thermometer

CONVERSION TABLE	
Fahrenheit (F̊)	**Centigrade (C̊)**
96.8	36.0
97.7	36.5
98.6	37.0
99.5	37.5
100.4	38.0
101.0	38.5
102.2	39.0
103.1	39.5
104.0	40.0

1.9 Glasgow Coma Score Assessment

Indications

1. All patients as continuous monitoring and trending.

Contraindications

1. No contra-indication in the intended setting.

Adverse Effects

1. The applying of painful stimuli may cause the patient discomfort and bruising to the skin.

Standard precautions

Procedure

1. Score the patient based on the 3 criteria: their best eye response (4/15), best verbal response (5/15) and best motor response (6/15). The three individual scores are then added to achieve a total score out of 15. The lowest GCS score recordable is 3/15.
2. **To assess the best eye response (4/15):**
    a. **Spontaneous (4)**: Observe the patient's eyes. A patient that has eyes that are opening spontaneously receives a score of 4.
    b. **Voice (3):** Supply vocal stimulus by asking the patient loudly and clearly to open their eyes. If the patient responds by opening their eyes, they receive a score of 3.
    c. **Pain (2):** Tap on the shoulder. You can also push down on the patient's fingernail bed. If the patient opens their eyes they receive a score of 2.

   None (1): If there is no response to pain the patient receives a score of 1.
3. **To assess the Best verbal response (5/15):**
    a. **Orientated (5):** Ascertain whether the patient is orientated to time and place. Patients that respond appropriately receive a score of 5. Ask the patient questions which you know the answer to, such as; 'What day is it today'? and 'Do you know where you are at the moment?'
4. **Confused (4):** If the patient appears slightly confused and/ or disorientated during conversation they receive a score of 4.
5. **Inappropriate speech (3):** If the patient has random or muddled speech without exchange of information during conversation, they receive a score of 3.
6. **Incomprehensible (2):** If the patient is making sounds but is unable to formulate words, they receive a score of 2.
7. **None (1):** A patient that is unable to produce sounds receives a score of

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This does not refer to aphasia due to any cause, such as airway obstruction or laryngeal injury.

8. **To assess the best motor response (6/15):**
    a. **Obeys Commands (6):** A patient who responds to you and does what you ask receives a score of 6. In order to assess this, shake the person's hand upon arrival or ask them 'can I hold your wrist to take your pulse'?
    b. **Localizing to pain (5):** Elicit a pain response through the techniques previously mentioned. If the patient purposefully attempts to remove the stimulus, they receive a score of 5. E.g. the patient pushes your hand away if you elicit nail bed pressure.
    c. **Withdraws to pain (4):** Elicit a pain response through techniques previously mentioned. If the patient pulls away from the stimulus, they receive a score of 4.
    d. **Abnormal Flexion (Decorticate) (3)**: Elicit a pain response through techniques previously mentioned. If the patient's arms move toward their chest, their fingers and wrists flex on their chest and they point at their toes, then they are said to have decorticate posturing and receive a score of 3. This posture indicates head injury, and a patient may be present in this position prior to any painful stimuli.
    e. **Abnormal Extension (Decerebrate) 2**: Elicit a pain response through techniques previously mentioned. If the patient's arms and legs extend, their wrists rotate away from their body and they point at their toes, then they are said to have decerebrate posturing and receive a score of 2. This posture also indicates head injury, and a patient may be present in this position before any painful stimuli.
    f. **No Response (1):** A patient without a motor response receives a score of 1.
9. Note the GCS prior to the administration of analgesic, sedative and paralytic agents.

Glasgow Coma Score Assessment

Behavior	Response: Infant (0-12months)	Response: Child/Adult	Score
E BEST EYE RESPONSE	Eyes opens spontaneously	Eyes opens spontaneously	4
	Eyes opens to verbal command	Eyes opens to verbal command	3
	Eyes opens to pain	Eyes opens to pain	2
	Eyes does not open	Eyes does not open	1
V BEST VERBAL RESPONSE	Coos, babbles, follows objects	Oriented to time, place, peson	5
	Irritable, cries	Confused	4
	Cries to pain	Inappropriate words	3
	Moans and grunts	Incomprehensible souds	2
	No verbal response	No verbal response	1
M BEST MOTOR RESPONSE	Moves spontaneously and purposefully	Obeys command	6
	Localises to pain and withdraws to touch	Localises to pain	5
	Withdraws from pain	Withdraws from pain	4
	Flexion in response to pain	Flexion in response to pain	3
	Extension in response to pain	Extension in response to pain	2
	No motor response	No motor response	1

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1.10 Level of consciousness: AVPU scale

Indications

1. All patients.

Contraindications

1. Nil contra-indications in the intended setting.

Adverse Effects

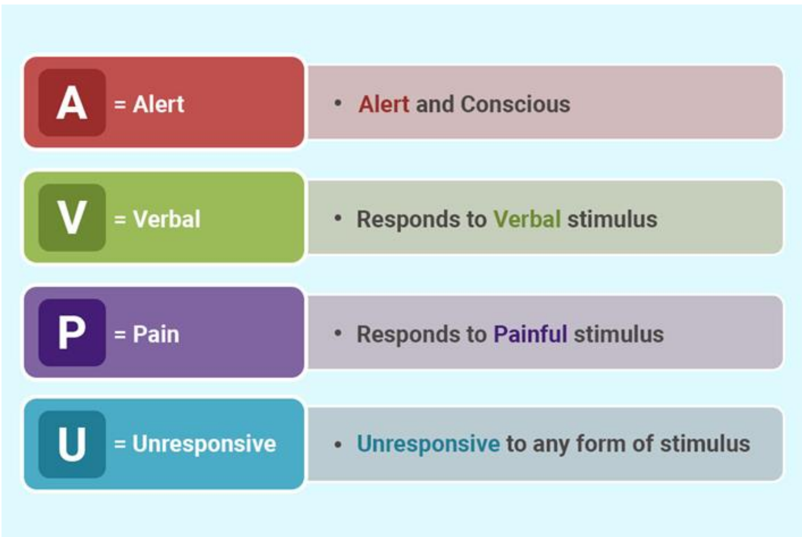
1. Applying painful stimulus to a patient may cause discomfort and bruising to the patient's skin.

Standard precautions

2.

Procedure

1. **Alert:** If the patient is alert, then it signifies orientation to person, place, time, and event. Ask your patient simple open-ended questions that cannot be answered with yes or no to determine the LOC. For example, "Where are you right now?" and "What time is it?" Do not ask your patient, "Do you know where you are right now?"
2. **Verbal:** Patient appears drowsy, confused or unresponsive, but responds when spoken to or to a loud shout.
3. **Painful:** Appears unresponsive, but responds with movement or groaning when painful stimulus is applied (Use shoulder shaking and tapping to apply painful stimulus)
4. **Unresponsive:** the patient does not show any eye, voice or motor response to voice or pain.



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1.11 Pain Assessment

Indications

1. All patients experiencing pain.

Contraindications

1. Nil contra-indications in the setting of pain assessment.

Adverse Effects

1. Nil.

Standard precautions

2.

Procedure

1. Remember that the assessment of pain is dependent on age, verbal, and cognitive ability of the patient. A commonly accepted mnemonic used for the assessment of pain is **OPQRST**:

   a. **Onset:** What was the patient doing when the pain started (active, inactive, stressed), and was the onset sudden, gradual, or part of an ongoing chronic problem?

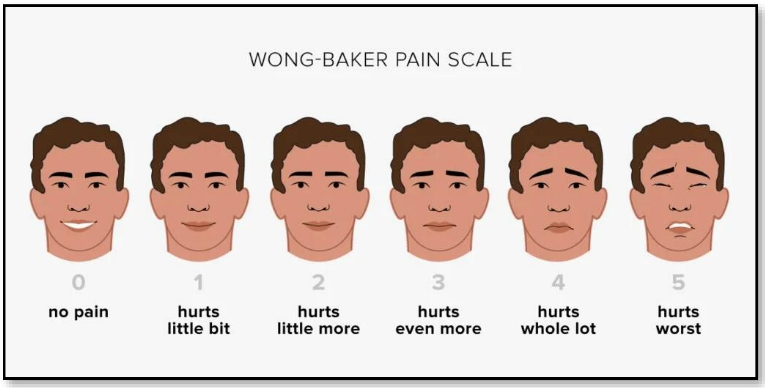
   b. **Position/Palliation**: Where is the pain? Does anything make the pain better or worse?

   c. **Quality**: Describe the pain. For example, is it dull, sharp, or crushing?

   d. **Region/Radiation:** Does the pain radiate or move anywhere?

   e. **Severity**: How severe is the pain. To determine the severity score of the pain, use the Wong-Baker Pain Rating Scale (0 denoting no pain through to 10 denoting the worst pain imaginable).

   f. **Timing**: When did the pain start and does it come and go?



Morales-Brown, L. (2022, December 23). *What to know about the Wong-Baker pain scale*. https://www.medicalnewstoday.com/articles/wong-baker-pain-scale

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1.12 BEFAST assessment for stroke

Indications

1. Any patient presenting with a sudden loss of motor tone or numbness in one side of the face or body.
2. Any patient presenting with a sudden onset of confusion, speech difficulties, loss of vision in one or both eyes.
3. Any patient presenting with a sudden onset of severe headache and collapse

Contraindications

1. Nil contra-indications in the setting of an assessment for stroke.

Adverse Effects

1. Nil in the setting of the assessment of a stroke.

Standard precautions

Procedure

1. Make the patient comfortable and assess for sudden loss of **B**alance, sudden **E**ye abnormality, **F**acial drooping, **A**rm weakness, **S**peech difficulties and **T**ime of onset of symptoms.
2. **B - Balance**: Watch for a sudden loss of balance. Is the patient leaning to one side or staggering when walking?
3. **E-Eyes:** is there a sudden loss of vision in one or both eyes? Double vision that does not go away when you blink your eyes? No side vision or vision above midline?
4. **F - Face:** Ask the patient to smile or stick out their tongue. Is the smile uneven, does the tongue deviate to one side or does one side of the face droop?
5. **A - Arms:** Ask the person to raise both arms. Does one arm drift downward? Is there sudden loss of coordination, numbness, weakness of that arm? Does the arm or leg feel numb?
6. **S - Speech:** Sudden difficulty in speaking or understanding. Can the person repeat a simple phrase? Does the speech sound slurred/strange/garbled?
7. **T-Time:** Time last seen normal or of symptom onset.
8. If the BEFAST assessment is positive, transport the patient P1 to HGH with pre-notification via CCE.
9. Patients with symptoms onset of less than **15 hours** or who wake up from their sleep with symptoms of stroke must also be transported P1 to HGH with pre- notification via CCE.
10. Patients with symptoms of stroke greater than 15 hours, transport P2 to HGH.

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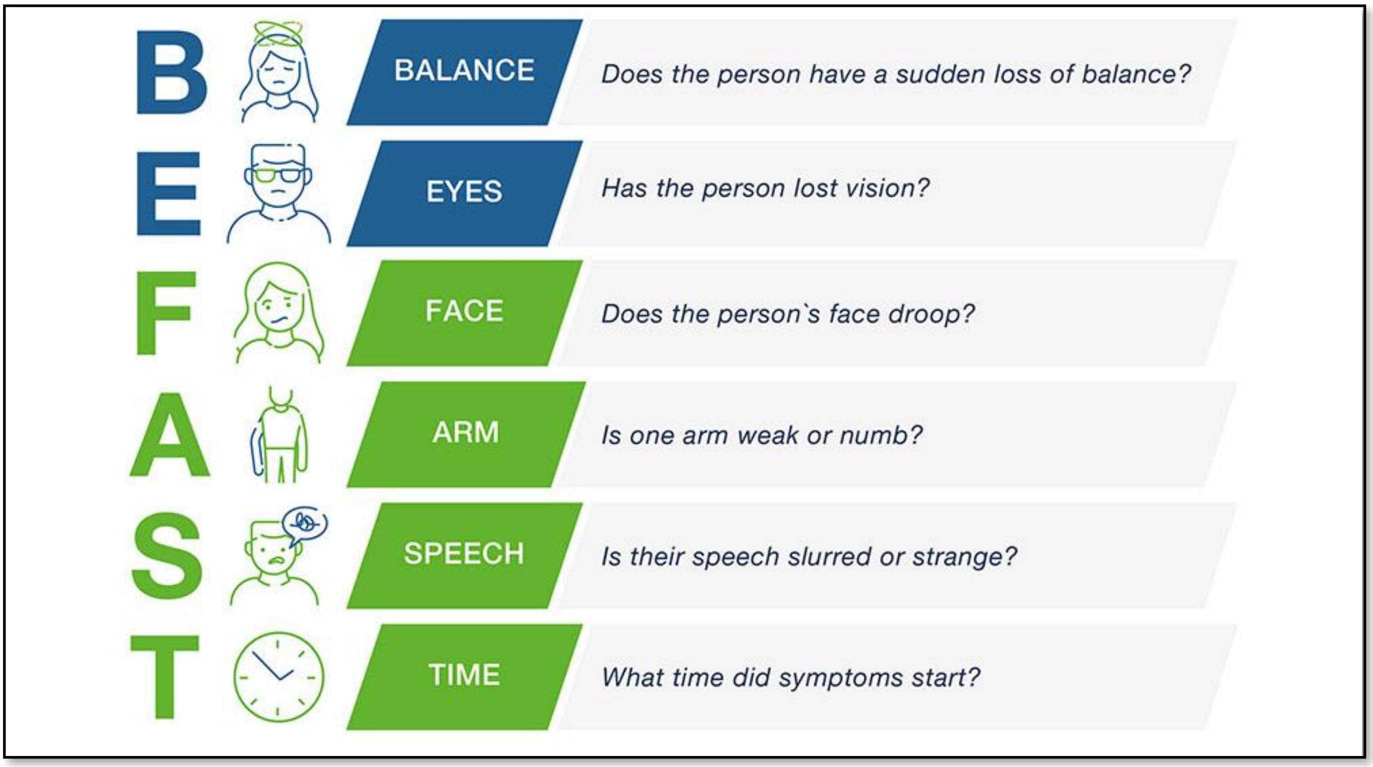
BEFAST Assessment for Stroke



Warning Signs and Risks. (n.d.). Trinity Health System.

https://www.holycrosshealth.org/services/neurosciences/neurosurgery/stroke/warning-signs-and-risks

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1.13 Pulse oximetry using the LIFEPAK15

Indications

1. To determine a patient's oxygen saturation.
2. To monitor Pleth waveform and pulse bar strength

Contraindications

1. Non-emergent scheduled transfers (PTS), unless the patient is seen to be unwell, develops a new complaint during transport or if the patient/ family requests.

Precautions

1. The reliability of SpO_2 readings depends on the correct sensor size and placement and adequate blood flow through the sensor site.
2. Inaccurate pulse oximetry readings may occur when the following factors are present:
   - Excessive patient movement.
   - The sensor is exposed to excessive ambient light.
   - Dirt or nail polish under the sensor site.
   - Carbon monoxide poisoning.
   - Poor blood flow to the area, e.g., cold extremities.
3. The SpO_2 of arterial blood is usually 95- 100%.
   Pulse oximetry is not a complete measure of respiratory or circulatory sufficiency.

Standard precautions

Procedure

1. Turn the LIFEPAK 15 ON and attach the pulse oximeter cable to the device.
2. Select a clean and the non-dominant hand and finger of the patient. If needed clean dirt of the finger with wipes and swabs. Henna and nail polish will affect on the SpO_2 reading.
3. Use the ear lobe as an alternative spo2 reading point if it cannot be obtained from the finger.
4. Do not apply the SpO_2 probe and NIBP cuff on the same limb. Using the same limb will result in lower SpO_2 readings when a NIBP is being taken.
5. Attach the proper pulse oximeter sensor to the patients' selected finger (adult/ paediatric).
6. Make sure the sensors are secure and properly aligned.
7. Observe the pulse wave for amplitude; this indicates relative signal strength.
8. Clean the LIFEPAK 15 and its attachments using HMC-approved disinfectant wipes.

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1.14 Capnography using the LIFEPAK15

Indications
1. The oral/ nasal $ETCO_2$ cannula is used to monitor the respiratory status of non-intubated critically ill or injured patient and patient presenting with respiratory compromise. • Pulmonary oedema • Asthma and COPD • Measurement of hypoventilation and hyperventilation. 2. Patients who receive sedation/ analgesia 3. In an intubated patient $ETCO_2$ is used to: • Verify ETT placement, • Monitor and detection of ETT dislodgement, • Detection of loss of circulatory function, • Determination of adequate CPR compressions, • Confirmation of return of spontaneous circulation.
Contraindications
1. There are no contra-indications to the use of $ETCO_2$
Adverse Effects
1. Nil
Standard precautions

Procedure
1. Turn the LifePak 15 ON and attach the orange colored $EtCO_2$ connector of the appropriately sized Microstream® to the device first, prior to connecting to patient. This allows for calibration of the $EtCO_2$ readings. 2. The adult/ paediatric CO_2 sampling line and airway adaptor should be connected directly to the ETT or LTA in an intubated patient. 3. The adult/ paediatric oral/ nasal sampling set with O_2 tubing should be connected externally to the patient's nostrils. 4. Nasal cannula O_2 can be provided using the adult/ paediatric oral/ nasal sampling set with O_2 tubing at a flow rate of $\leq$ 5 LPM. The O_2 flowrate may be increased to 15 LPM for pre-oxygenation and apnoeic period of intubation. 5. Practitioners must monitor capnography numeric reading (amount) of the $ETCO_2$ and a graphic display (waveform) of $ETCO_2$ throughout the respiratory cycle. 6. Normal airway $ETCO_2$ = 35-45 mmHg. 7. Clean the LIFEPAK 15 and its attachments using HMC-approved disinfectant wipes.

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1.15 Adult Medical Patient Assessment

Indications
1. All adult medical patients
Contraindications
1. There are no contra-indications.
Precautions
1. Expose the patient with discretion
Standard precautions
 1. Wear an N95 mask if the patient exhibits respiratory symptoms. 2. A face shield and gown are required when treating patients with a travel history from high-risk countries. (emerging and reemerging infectious diseases)
Procedure
1. Gather general impression: a. Position, gender, age b. Any life-threatening condition. c. If fever, cough and travel history to high-risk countries, consider risk of infectious diseases. 2. Conduct AVPU examination and look for signs of life within 10 seconds: if unconscious patient with no or no normal (agonal) breathing, refer to HMCAS CPG 2.1. 3. Assess and Manage life threatening conditions (airway, breathing, and circulation) 4. Rest and reassure patient by keeping eye contact while assessing radial pulse. Place the patient in their most. 5. Perform or delegate vital signs survey, RBS, Perfusion Status Assessment, Respiratory Status Assessment (including auscultation) and pain score. 6. Gather a brief history and perform a brief physical exam guided by the chief complaint. 7. Calculate QEWS score and report situation to NCC within 2 minutes and specify to NCC if help is required (CCP, Delta, LifeFlight, Police, Civil Defence etc.). 8. Conduct a specific history taking for every highlighted medical emergency including OPQRST. 9. Assess the relevant parts of the various systems based on the patient's chief complaint. 10. Conduct a specific assessment for every highlighted medical emergency including

RBS, BEFAST, Auscultation and 12-Lead ECG

11. Manage the patient according to the appropriate CPG.
12. Consider transport and/ or rendezvous point with CCP.
13. Initiate transport to the appropriate facility with prenotification through CCE
14. Follow transport pathways criteria if indicated (Heart hospital, Stroke pathway, HGH resuscitation room)
15. Reassess the patient's condition and interventions.
16. Keep continuous observational and vital signs monitoring until handover at the receiving facility.
17. Apply defib pads and Consider ETCO2 monitoring for critically ill patient.

1.16 Adult Trauma Patient Assessment

Indications

1. All adult trauma patients

Contraindications

1. Nil in this setting.

Precautions

1. Expose the patient with discretion.

Standard precautions

1. A face shield and gown are required if the patient has catastrophic bleeding.

Procedure

1. Gather general impression:
   a. Mechanism of injury
   b. Position, gender, age
   c. Any life-threatening condition (immediately control catastrophic haemorrhage if present).
   d. Need for c-spine immobilisation (maintain c-spine control simultaneously with controlling haemorrhage).
2. Conduct AVPU examination and rapid assessment of clinical status (airway, breathing and circulation).
3. Perform Life Saving On-Scene Intervention, if applicable, according to CC-LOC-ABC approach:
   a. Airway: Jaw thrust, suction or OPA if indicated.
   b. Breathing: O_2 therapy or assist ventilation.
   c. Circulation: CPR if unconscious and no/ agonal breathing and no carotid pulse.
4. Perform vital signs survey, GCS, pain scale and RBS.
5. Conduct rapid trauma survey looking for other significant injuries. Focus on the head, neck, chest, abdomen, pelvis and femurs to be completed in 30 seconds.
6. Inspect and palpate the head and face (i.e. look for skull depression).
7. Inspect and palpate the neck (i.e. look for JVD, swelling, or tracheal deviation).
8. Apply appropriate size of C-collar if indicated.
9. Inspect, palpate, and auscultate the chest (percussion if diminished or absent breath sounds):
   a. If tension pneumothorax, immediately perform chest decompression.
   b. If sucking chest wound, immediately apply chest seal.

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c. If flail chest, immediately treat to achieve chest stabilisation.
d. If impaled object, immediately stabilise it.

10. Inspect and palpate the abdomen (i.e. look for distension, tenderness)
11. Inspect the pelvis for obvious injuries (don't rock the pelvis): If pain and/or signs of pelvic injuries, unexplained hypotension or based on extreme mechanism of injury then apply the pelvic binder.
12. Inspect and palpate the lower extremities i.e. look for long bone injuries and assess PMS.
13. Inspect and palpate the upper extremities i.e. look for long bone injuries and assess PMS.
14. Logroll the patient (if indicated) without excessive movement of the c-spine and back.
15. Inspect and palpate the back (look for signs of spinal cord injury).
16. Triage decision for patient with major trauma (Actual, Emergent or Potential patient).
17. Report to NCC and request additional resources.
18. Minimize on-scene time.
19. Consider transport and/or rendezvous point with CCP.
20. Ensure appropriate receiving hospital and early notification according to Trauma Bypass Criteria (through CCE).
21. Commence pain management as soon as practicable.
22. Gather a SAMPLE history.
23. Perform secondary survey (pertinent physical exam).
24. Manage the patient according to the relevant HMCAS CPG.
25. Reassess the patient's condition and interventions regularly.
26. Keep continuous observational and vital signs monitoring until handover at the receiving facility.
27. Apply defib pads and Consider ETCO2 monitoring for critically ill patient.

HEAD-TO-TOE-EXAM

HEAD-TO-TOE-EXAM	
HEAD	
General	*Inspect for:* lacerations, deformity, facial muscle tone, asymmetry. *Palpate for:* crepitus, bony tenderness, subcutaneous emphysema.
Eyes	*Inspect:* pupils, Raccoon eyes
Ears	*Inspect:* Blood or CSF, Battle's sign
Nose	*Inspect:* deformity, epistaxis
Mouth	*Inspect:* loose teeth, bite malocclusion, airway/tongue swelling
Voice	*Inspect:* hoarseness
NECK	
Inspect	Deformity, lacerations, raised JVP
Palpate	Tracheal position, bony tenderness, carotid pulse, subcutaneous emphysema, lymphadenopathy
CHEST	
Inspect	Expansion, paradoxical movement, accessory muscle use, lacerations, deformity
Palpate	Tenderness, subcutaneous emphysema, bony crepitus
Auscultate	Air entry, breath sounds
ABDOMEN	
Inspect	Laceration, bruising, distention
Palpate	Tenderness, guarding, rigidity, rebound tenderness
PELVIS	
Inspect	Lacerations, bruising, deformity, priapism (spinal trauma)
BACK	
Inspect	Laceration, bruising, deformity
Palpate	Bony tenderness, evidence of a bony step, subcutaneous emphysema
UPPER & LOWER LIMBS	
Inspect	Laceration, bruising, deformity, shortening or rotation
Palpate	Bony tenderness, crepitus, PMS

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1.17 Mental Status Examination for ABD

I II III IV

Indications

1. Patients presenting with Acute Behavioural Disturbance (ABD) can pose a significant risk to themselves and others.
2. The mnemonic ASEPTIC can be used to remember the components of the Mental Status Examination.

   A - Appearance/Behaviour. **S** – Speech. **E** - Emotion (Mood and Affect). **P** - Perception (Auditory/Visual Hallucinations). **T** - Thought Content (Suicidal/Homicidal Ideation) and Process. **I** - Insight and Judgement. **C** - Cognition

Contraindications

1. Nil in this setting

Adverse effects

1. Unpredictable behavior including agitation and aggression

Standard precautions

2.

Procedure

1. Assess the patient's **Appearance** by observing the following indicators:

   Appearance: Note the posture, clothes, grooming, and cleanliness and make note of any evidence of self-harm (cuts on wrists/legs)
   Movement and Gait: Look for any unusual psychomotor agitation or retardation, tremor.
   Behaviors: Note any mannerisms, gestures, expression, eye contact, ability to follow commands
   Attitude: Note if the patient is cooperative, hostile, suspicious, easily distracted, focused, or defensive
   Level of Consciousness: Is the patient vigilant, alert, drowsy, lethargic, asleep, confused, fluctuating.
   Rapport: Does the patient trust you and do you have a good connection/relationship?

2. Observe the patient's **Speech** looking for:

   Quantity of speech: Talkative, spontaneous, expansive, poverty of speech (i.e. - very little is said)
   Rate of speech: Fast, slow, normal, pressured.
   Volume (tone) of speech: oud, soft, monotone, weak, strong
   Fluency and rhythm of speech: Slurred, clear, hesitant, with good articulation,
   Response latency: How long does it take the patient to respond?

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3. Ask the patient to describe their **Mood/Emotion** subjectively and note the appropriateness of the patient's affect to the current situation. **Quality of affect:** sad, angry, hostile, indifferent, euthymic, dysphoric, detached, elated, euphoric, anxious, animated, irritable.
4. Assess the patient’s Perceptions by inquiring for: **Hallucinations:** **Auditory:** Are you hearing any things that others can’t hear? Visual: Are you seeing any things that others can’t hear? Olfactory: Any unusual smells that you notice, e.g. burning smells? (classically seen in temporal lobe epilepsy) **Illusions:** Distortions of real images or sensations **Depersonalization:** Patient feels that they are not real **Derealisation:** Patient feels that the world is not real
5. Assess the patient’s **Thought** content by observing: **Disturbances in content of thought:** Delusions, Preoccupations (which may involve the patient's illness), Obsessions, Compulsions, Phobias, Plans and Intentions Does the patient have thoughts of doing self-harm? Is there a plan? *Note homicidal ideation, intent or planed*
6. Assess the patient’s **Insight** by observing if: **Good insight:** Patient understands they are ill and need treatment (like being in action phase)) **Partial insight:** may indicate that the patient acknowledges a problem, but is not willing to seek appropriate help or treatment (like being contemplative) **Poor insight:** means that the patient does not see that they are ill nor does the patient need any help or treatment (like being pre-contemplative) Is the patient able to use facts and make reasonable decisions? May be good, fair, impaired.
7. Asses the patient’s **Cognition** by observing: The Orientation for time, person and place Registration - ability to repeat new information such as a name and address Concentration and attention General knowledge - recent news items etc.

Airway Management

2.1 Finger Sweep Manoeuvre

I II III IV

Indications

1. Upper airway obstruction in an unconscious patient.

Contraindications

1. Conscious patients.
2. **Do not perform** a **blind finger sweep** because this could push an object further into the airway.

Adverse Effects

1. Might stimulate a gag reflex and cause the patient to vomit and aspirate.

Standard precautions

2.

Procedure

1. Identify the need to perform a finger sweep manoeuvre. A finger sweep manoeuvre is used to clear a visible mechanical obstruction from the upper airway of an unconscious patient.
2. Open the patient's mouth by grasping the lower jaw and tongue between the thumb and fingers of one hand.
3. Visually inspect the mouth for a mechanical obstruction.
4. Insert your index finger of your other hand into the patients' mouth. Insert little finger for infant and pediatric patients.
5. Attempt to sweep the patients' mouth for foreign objects.
6. Remove the foreign objects from the patients' mouth using your fingers.
7. Suction the oropharynx, if required.
8. Visually inspect the mouth again before commencing oxygen therapy.

2.2 Head-tilt-chin lift manoeuvre

Indications

1. Unconscious **MEDICAL** patient with or without breathing.
2. Medical MCI scene, patient not breathing.
3. To assist ventilation with a BVM.

Contraindications

1. Unconscious patients with suspected cervical spine injuries.

Adverse Effects

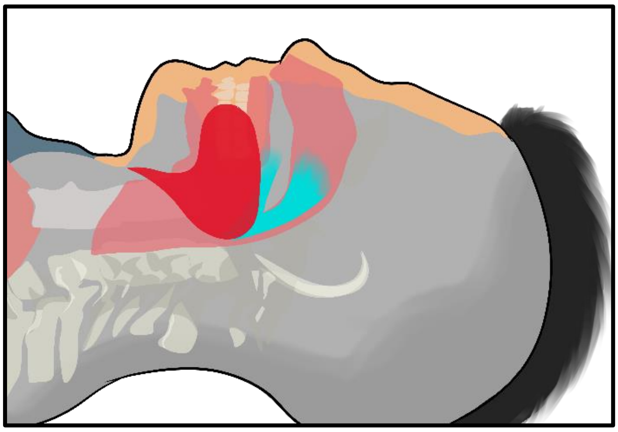
1. Can aggravate a cervical spinal injury
2. Aspiration

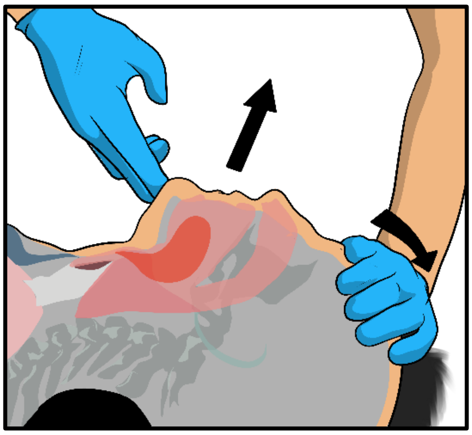
Standard precautions

Procedure

1. Exclude the incidence of cervical spine injury based on the history and mechanism of injury.
2. Place the patient in the supine position.
3. Clear the oropharynx of any foreign objects using the finger sweep manoeuvre. Suction the oropharynx, if required.
4. Logroll and finger sweep/ suction, if required.
5. Place the palm of one hand on the patient’s forehead.
6. Place the fingers of the other hand under the bony part of the lower jaw, near the chin.
7. While applying downward pressure on the forehead, simultaneously lift the chin and tilt the head backwards.
8. Measure and insert an oropharyngeal airway, if required.
9. Provide supplemental oxygen as required (NRB/ BVM).
10. Reassess the patient’s airway and breathing regularly.
11. AP staff to request CCP assistance for airway protection, if required.

Head-tilt-chin lift Maneuver





- Head tilt stretches anterior neck muscles, lifts tongue away from posterior pharyngeal wall and epiglottis away from laryngeal inlet.
- Chin lift stretches structures more and pulls mandible and tongue forward.

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2.3 Jaw-thrust manoeuvre

Indications

1. Unconscious TRAUMA patient with or without breathing with suspected cervical spine injury.
2. Trauma MCI scene, with unconscious, not breathing patient.
3. To assist ventilation in a patient who has sustained traumatic injuries.
4. During the apnoeic phase of RSI, prior to ETT placement.

Contraindications

1. Fracture or dislocated jaw
2. A conscious patient

Adverse Effects

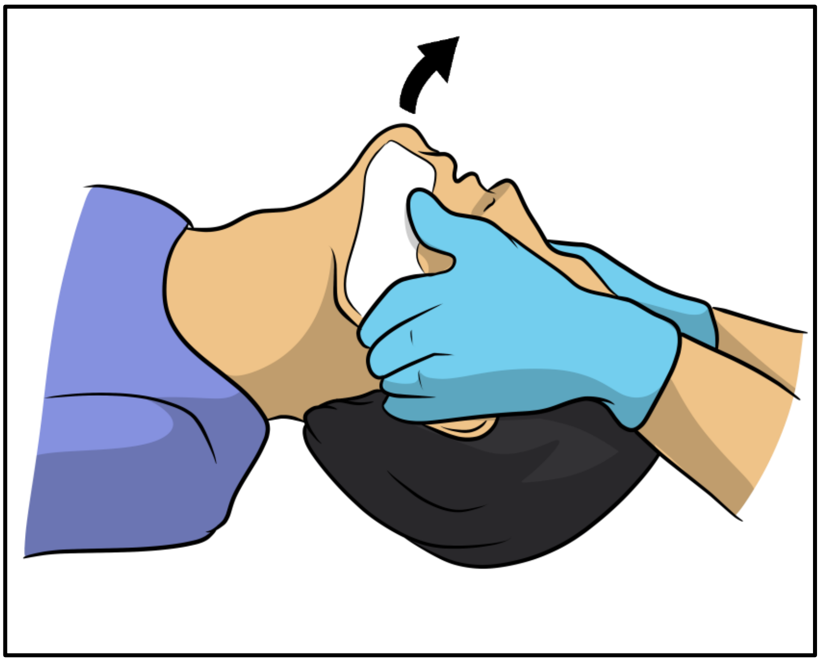
1. Posterior mandibular bruising

Standard precautions

Procedure

1. Place the patient in the supine position.
2. Position yourself at the top end of the patient's head.
3. Clear the oropharynx of any foreign objects using the finger sweep manoeuvre.
4. Suction the oropharynx, if required.
5. Maintain in-line support of the patient's head and neck.
6. Carefully place the heel of your hands on the corresponding zygoma of the patient. That is, the heel of your left hand on the patients' left zygoma and the heel of your right hand on the patients right zygoma. Consider modified jaw thrust manoeuvre if there is any Zygoma fracture.
7. Place your thumbs just inferior to the eyes and lateral to the nose.
8. Gently grip the angle of the jaw, just below the ears, with your fingers.
9. Gently pull the jaw upwards, using the zygomas for leverage and without displacing the cervical spine.
10. Measure and insert an oropharyngeal airway, if required.
11. Provide supplemental oxygen, as required.
12. This technique may be used in conjunction with the two-person technique for BVM ventilation.

Jaw-thrust Maneuver



2.4 Oral Suctioning

Indications

1. Liquid obstruction by vomitus, blood or secretions of the patients' oropharynx and hypopharynx.

Contraindications

1. Trismus

Adverse Effects

1. Prolonged oral suctioning can result in the patient experiencing hypoxia.
2. Activation of the gag reflex can stimulate further vomiting and possible aspiration.
3. May cause an increase in the intracranial pressure.
4. Unsterile techniques may increase the incidence of infection.

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms, such as coughing.

Procedure

1. Do not manipulate the head or neck if suspected cervical spinal injuries.
2. Logroll the patient (tilt the combi carrier for trauma patient) to facilitate gravitational drainage of the liquid obstruction, if required.
3. If a patient has trismus, logroll the patient (tilt the combi carrier for trauma patient) to allow for gravitational drainage of the liquid obstruction.
4. Prepare the suction unit, A5 pre-pack, container of sterile water to clear the catheter and LifePak 15 monitoring of HR, BP and pulse oximetry.
5. Turn on the suction unit and adjust the pressure to between 120-150mm Hg (Adults) and 80-100mm Hg (Paediatrics).
6. Select a rigid suction catheter (Yankauer) and attach it to the suctioning tubing.
7. Measure the distance from the corner of the mouth to the earlobe. Mark this length on the catheter with your index finger.
8. Insert the catheter until the measured length reaches the patient's lips and occlude the vacuum port on the catheter to suction.

   Oral suctioning should be performed for a maximum of 15 seconds before supplemental oxygen is continued.

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2.5 Stoma and tracheostomy tube suctioning

II III IV

Indications

1. Liquid obstruction by vomitus, blood, or secretions of the patients' stoma/ tracheostomy tube.
2. Respiratory distress associated with secretions in the trachea of the patient's stoma/ tracheostomy.

Contraindications

1. Nil in this setting

Adverse Effects

1. Prolonged suctioning can result in the patient experiencing hypoxia.
2. May stimulate further vomiting and possible aspiration.
3. May cause an increase in the intracranial pressure.
4. Unsterile techniques may increase the incidence of infection.

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms, such as coughing.

Procedure

1. Get consent from the patient/ family. Involve the family members/ caregiver in the patient's care as they may possess specific skills and knowledge regarding the tube.
2. Place the patient in their most comfortable position, if conscious.

   If the patient is unconscious, place the patient in the lateral position to facilitate gravitational drainage of the liquid obstruction.
3. Remove all tracheostomy accessories if present (speaking valve, humidifier, etc.)
4. Pre-oxygenate the patient with a NRB at 10-15 LPM over the stoma/ tracheostomy tube (and face if more O2 source available)
5. Prepare the suction unit, pre-packs (A5 and A6/ A7), container of sterile water to clear the catheter and LIFEPAK15 monitoring of ECG, HR, BP, ETCO2 and SPO2
6. Turn on the suction unit and adjust the pressure to between 120-150mm Hg (Adults) and 80-100mm Hg (Paediatrics).
7. Select a soft suction catheter and attach it to the suctioning tubing. Catheter diameter is determined by the size of the stoma/ tracheostomy tube in place (tracheostomy tube size multiplied by 2).
8. The approximate length of insertion of the suction tube is determined by multiplying the patients' small finger by 2
9. Insert the catheter gently into the stoma/ tracheostomy tube. Apply suction while

gently withdrawing the catheter in a rotating motion for not more than 10 seconds (from the time of removing O2)

10. Start oxygenation immediately with a NRB at 10-15 LPM over the stoma/ tracheostomy tube.
11. Immerse the suction catheter in sterile water if intermittently used.
12. Do not perform suction more than 3 consecutive times. If needed, allow more time for oxygenation and change catheter.

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2.6 Endotracheal tube suctioning

Indications

1. Liquid obstruction by vomitus, blood or secretions of the ETT.

Contraindications

1. Nil in this setting

Adverse Effects

1. Prolonged ETT suctioning can result in the patient experiencing hypoxia.
2. May cause an increase in intracranial pressure.
3. May induce bradycardia.
4. Unsterile techniques may increase the incidence of infection.

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms, such as coughing.

Procedure

1. Be cautious not to manipulate the head or neck in a patient with suspected cervical spinal injuries.
2. Prepare the suction unit, C3/ C4, (A5 pre-pack, if required) by AP staff, container of sterile water to clear the catheter.
3. Ensure that the patient's BP, ECG, SpO_2, $EtCO_2$, HR and RR are being monitored.
4. Turn on the suction unit and adjust the pressure to between 120-150mm Hg (Adult) and 80-100mm Hg (Paediatrics).
5. Connect the (KimVent) Turbo-cleaning closed suction system (14F for adults) to the ETT and attach it to the suctioning tubing.
6. Do not remove the oxygen delivery device and $ETCO_2$ from the tip of the ETT.
7. Stabilise the KimVent and ETT with one hand and with the other hand insert the catheter into the ETT.
8. Insert the catheter until the numbers on the Kimvent line up with the numbers on the ETT. This will ensure that the catheter tip is within 1 cm to the end of the ETT.
9. Depress and hold the thumb control valve, then gently withdraw the catheter.
10. Stop withdrawal when black marking ring is visible inside sleeve.
11. Release thumb control valve.
12. Repeat as necessary.

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Endotracheal tube suctioning using the KimVent closed circuit system

2.7 Laryngeal tube suctioning

Indications

1. Liquid obstruction by vomitus, blood or secretions of the LTA.

Contraindications

1. Nil in this setting

Adverse Effects

1. Prolonged LTA suctioning can result in the patient experiencing hypoxia.
2. May stimulate further vomiting and possible aspiration.
3. May cause an increase in the intracranial pressure.
4. Unsterile techniques may increase the incidence of infection.

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms, such as coughing.

Procedure

1. Be cautious not to manipulate the head or neck in a patient with suspected cervical spinal injuries.
2. Prepare the suction unit, C3/ C4, A5 pre- pack (AP staff), container of sterile water to clear the catheter.
3. Ensure that the patient's BP, ECG, SpO_2, $EtCO_2$, HR and RR are being monitored.
4. Turn on the suction unit and adjust the pressure to between 120-150mm Hg (Adult) and 80-100mm Hg (Paediatrics).
5. Connect the (KimVent) Turbo-cleaning closed suction system (14F - Adult) to the LTA and attach it to the suctioning tubing.
6. Do not remove the oxygen delivery device and $EtCO_2$ from the tip of the LTA.
7. Stabilise the KimVent and LTA with one hand and with the other hand insert the catheter into the LTA.
8. Insert the catheter until the catheter can no longer advance.
9. Depress and hold the thumb control valve, then gently withdraw the catheter.
10. Release thumb control valve.
11. Repeat as necessary.
12. The LTA drain tube can also be suctioned to empty abdominal contents. Use a soft tip. (16F – Adults) catheter connected directly to the suction drain tube.

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2.8 Foreign body airway obstruction in adults

Indications

1. Foreign body airway obstructions in adults

Contraindications

1. Nil in this setting.

Adverse Effects

1. The back slaps and abdominal thrusts may cause the patient some discomfort and pain.

Standard precautions

1. A face shield is required if there is a risk of coughing or vomiting.

Procedure

1. If the patient is conscious and able to talk, then encourage the patient to cough and monitor for complete obstruction.
2. Stand behind the patient and support him/ her.
3. Notify NCC of a complete airway obstruction. AP staff to immediately request CCP assistance.
4. If the patient is conscious but is unable to talk or cough, then apply up to five back slaps.
5. In the presence of a complete airway obstruction, APs should initiate early P1 transport and meet a CCP en route.
6. Check the mouth to see if the foreign body has been released.
7. If no relief, then give up to five abdominal thrusts. If the patient is obese, perform five chest thrusts.
8. Continue alternating 5 back slaps with 5 abdominal thrusts, checking to see if the foreign body has been cleared between cycles.
9. If the foreign body clears, provide supportive care and perform a complete assessment. Encourage the patient to be transported by ambulance to hospital for evaluation.
10. If the foreign body was not released and the patient becomes unconscious, or if the patient is found unconscious with a reliable witness confirming that the patient was choking before the event.
11. Support or move the patient carefully to the ground.
12. Start CPR with a compression ventilation ratio of 30:2
13. If BVM ventilation not successful, reposition airway, look for the foreign body if can

be visualized, perform a finger sweep, and remove it and then attempt ventilations with a BVM. Do not perform a blind finger sweep manoeuvre.

14. If no foreign body can be visualized, perform 30 manual chest compressions.
15. If BVM ventilation is successful (visible chest rise), continue rescue breaths and assess the need to initiate standard cardiac arrest care (no return of spontaneous breathing).
16. After completing 30 manual chest compressions inspect the mouth and if the foreign body can be seen, perform a finger sweep to remove.
17. Attempt 2 BVM ventilations. If ventilations are successful (visible chest rise) then continue rescue breaths and assess the need to initiate standard cardiac arrest care.
18. DO NOT insert a LTA in the setting of FBAO.
19. If the FBAO remains, continue the cycle of 30 manual chest compressions, visual inspection of the mouth and 2 attempted rescue breaths until the airway is cleared.
20. Following four cycles of 30 compressions and 2 attempted ventilations (approximately 2 minutes)
21. Following removal of the FBAO if spontaneous breathing does not return, initiate standard cardiac arrest care. If spontaneous breathing does return, support ventilations as needed, move the patient to the recovery position and consider the use of suction to clear residual objects in the airway as needed.
22. Provide oxygen support if SpO2 < 94%.

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2.9 Use of Magill's forceps

Indications

1. Removal of foreign bodies from the retropharyngeal area

Contraindications

1. Nil in this setting.

Adverse Effects

1. Airway trauma from laryngoscopy and Magill's forceps use.
2. May precipitate vomiting/ aspiration in patient with intact gag reflex

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms and there is a risk of vomiting.

Procedure

1. Notify NCC of a complete airway obstruction.
2. Prepare pre-packs Airway Roll, Adult Airway Roll/ Paediatric Airway Roll and the Magill's forceps.
3. Apply up to five back slaps whilst preparing the equipment.
4. Check if the foreign body has been released.
5. Give up to five abdominal thrusts. If the patient is obese, perform five chest thrusts.
6. Continue alternating 5 back slaps with 5 abdominal thrusts, checking to see if the foreign body has been cleared between cycles.
7. Position the patient supine.
8. Open the airway and check the patient's mouth for any foreign body that may have dislodged.
9. Under direct laryngoscopy, using the Magill's forceps to remove the foreign body from the retropharyngeal area.
10. If the foreign body clears, provide supportive care and perform a complete assessment. Encourage the patient to be transported by ambulance to hospital for evaluation.

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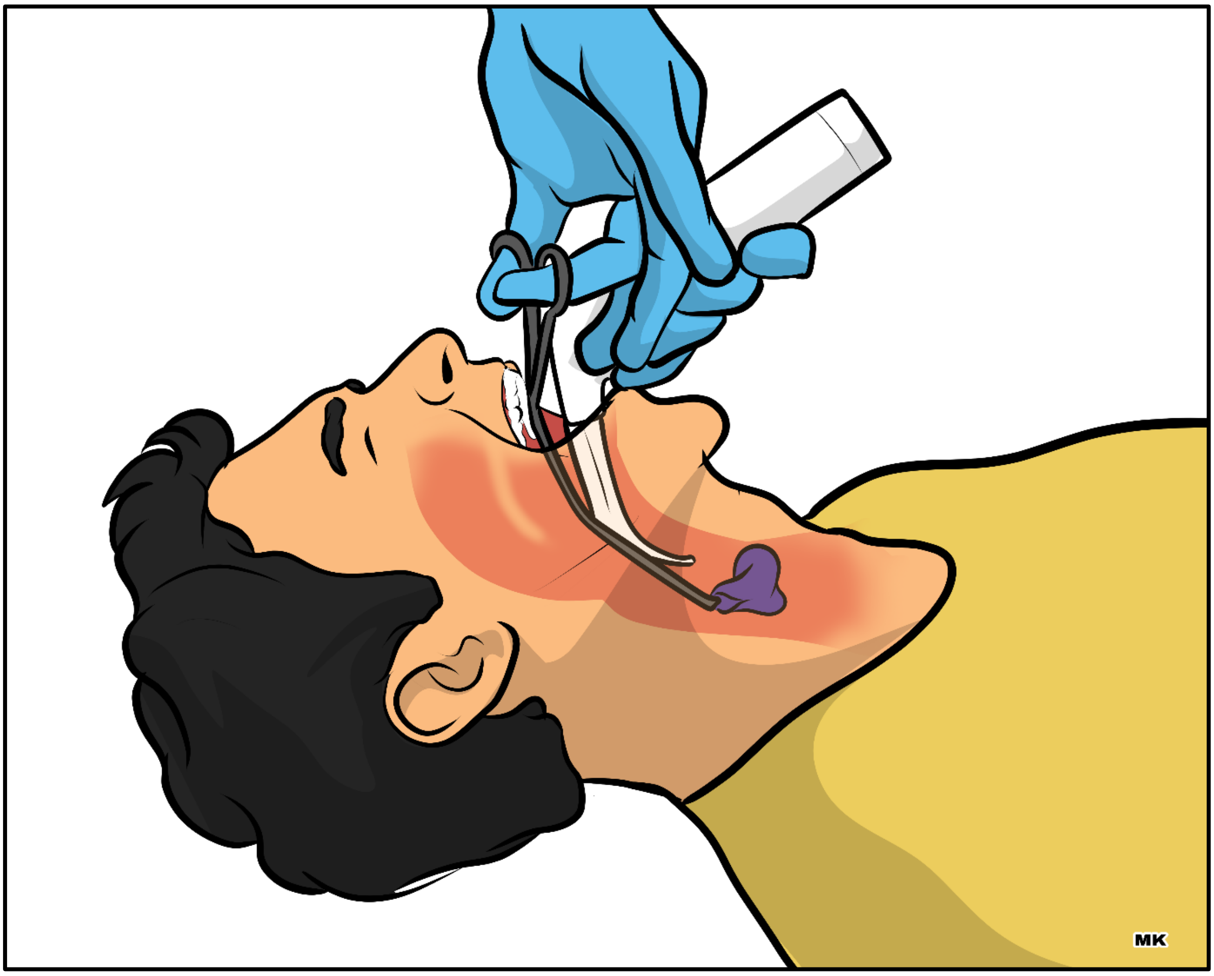
Use of Magill's Forceps with laryngoscopy to remove foreign bodies



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2.10 Oropharyngeal airway insertion

Indications

1. Failed insertion of LTA during adult cardiac arrest (AP).
2. Maintain airway patency for unconscious patients with no gag reflex and to improve difficult BVM ventilation.
3. Maintain airway patency during re- oxygenation phase of intubation

Contraindications

1. Awake patient
2. Unable to open the patient’s mouth

Adverse Effects

1. Airway trauma from OPA placement.
2. Intolerance of OPA requiring removal.
3. Insertion of the OPA can precipitate vomiting and/ aspiration in patient with intact gag reflex.
4. Incorrect size or placement can potentially exacerbate airway obstruction.
5. Improper technique of insertion may cause complete airway obstruction from the tongue.

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms and there is a risk of vomiting.

Procedure

1. Prepare pre-pack Airway Roll (AP)/ Adult/ Paediatric Airway Roll (CCP), A5 and suction unit.
2. Ensure that the airway has been opened using head tilt-chin lift (medical patient) or jaw thrust (trauma patient) manoeuvres.
3. Select an appropriately sized OPA by measuring the OPA from the centre of the patient’s mouth to the angle of their jaw/ bottom of the earlobe.
4. Open the mouth using the cross-finger manoeuvre or tongue-jaw lift manoeuvre.
5. Insert the OPA upside down into the mouth without pushing the tongue back.
6. As the OPA is inserted, rotate it 180 degrees as it reaches the soft palate.
7. Advance the OPA until the flange rests on the patient’s teeth/ lips.
8. Check for adequate ventilation (chest rise) after insertion of the OPA.
9. Reassess the patient’s airway repeatedly.

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2.11 Naso-pharygeal airway insertion

Indications

1. Airway adjunct for use in patients with potential or actual airway obstruction, particularly in circumstances where an oropharyngeal airway is inappropriate (e.g. patients with trismus or an intact gag reflex).

Contraindications

1. Suspected base of skull fractures (blood and/ cerebrospinal fluid exiting the nose).
2. Nasal trauma.
3. Nasal polyps

Adverse Effects

1. Airway trauma, particularly epistaxis.
2. Incorrect size or placement will compromise effectiveness.
3. Exacerbate injury in base of skull fracture, with NPA potentially displacing into the cranial vault.
4. Can still stimulate a gag reflex in sensitive patients, precipitating vomiting or aspiration.

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms and there is a risk of vomiting.

Procedure

1. Prepare Airway Roll (AP), Adult Airway Roll/ Paediatric Airway Roll (CCP), A5 and suction unit.
2. Select an appropriately sized NPA with the correct size reaching from the tip of the patient’s nose to their ear lobe. Also compare the diameter of the NPA to the diameter of the patient’s nostril or compare the diameter of the NPA to the diameter of the patient’s little finger.
3. Ensure that the safety pin has been pinned across the tube immediately inferior to the flange.
4. Lubricate the tube.
5. Insert the bevelled end of the tube into the right nostril and advance with a twisting motion.
6. If resistance is encountered, withdraw the NPA and attempt to insert the NPA into the left nostril.
7. Check for adequate ventilation after insertion of the NPA.
8. Repeatedly reassess the patient’s airway

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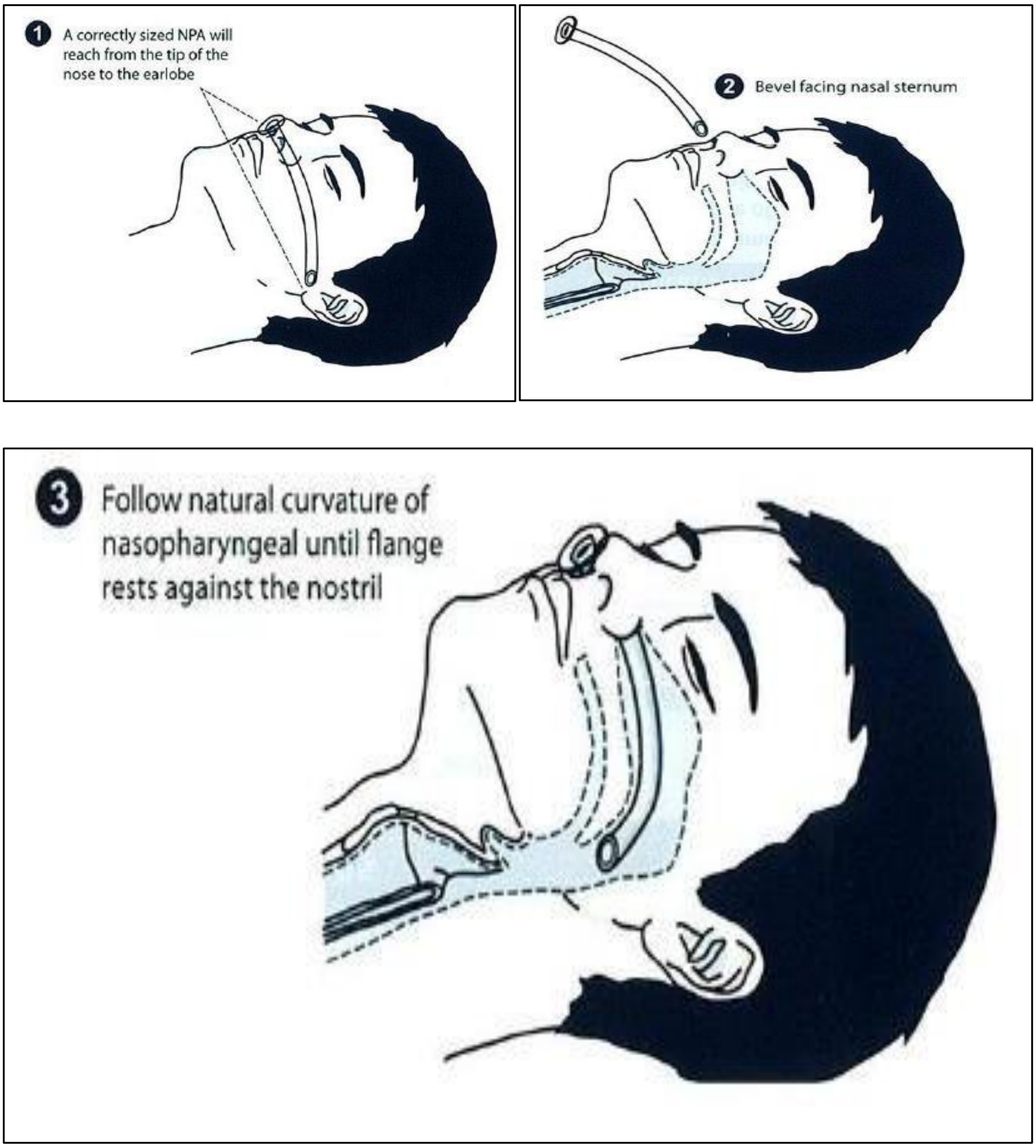
Naso-pharyngeal airway insertion



2.12 Cricoid pressure manoeuvre

Indications

1. Cricoid pressure manoeuvre for ETT placement.

Contraindications

1. Neck trauma

Adverse Effects

1. Airway trauma.

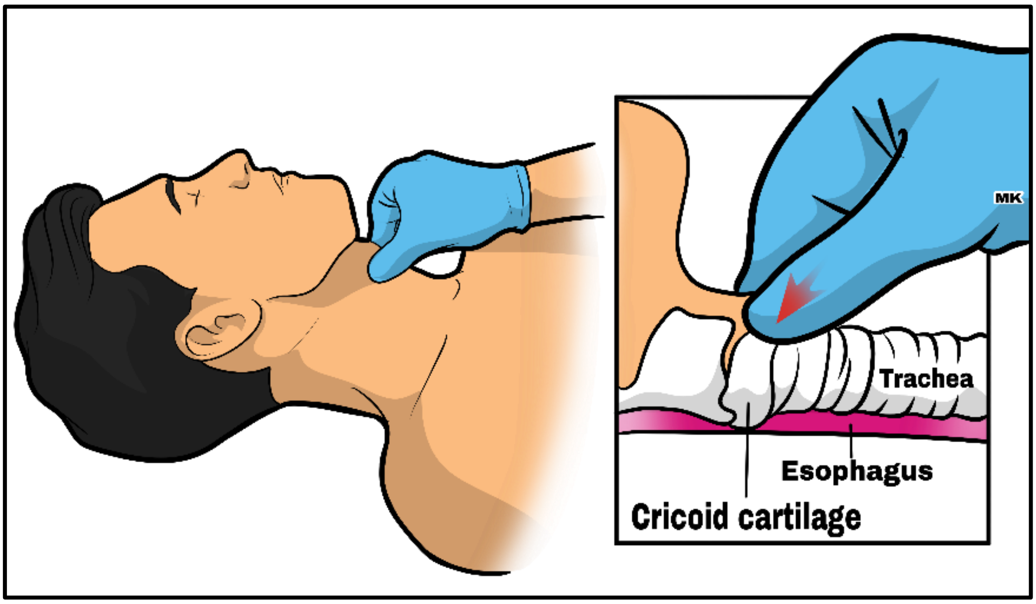
Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms and there is a risk of vomiting.

Procedure

1. Prepare for intubation using pre-pack Adult/ Peadiatric Airway Roll, C6, oxygen, Frova, Oxylog 3000 plus and RSI medication.
2. Follow Procedure 2.16 for equipment preparation, patient positioning, medication administration, ETT placement and post-intubation management.
3. Locate the cricoid cartilage, larynx, trachea, and the hyoid bone by palpation so that pressure is applied to the right structure.
4. Hold the cricoid cartilage between the thumb and the middle finger.
5. Place the index finger on the cricoid
6. cartilage.
7. With the patient in the supine position, push the cricoid cartilage backward against the spine. Push with about 9 pounds of pressure (40 Newtons).
8. Continue with ETT placement and post- intubation management.
9. Release cricoid pressure after placement of the ETT.
10. Manage the patient according to the appropriate CPG.
11. Reassess the patient’s condition and interventions.
12. Ensure notification of the receiving facility as indicated (through CTL).
13. Transport the patient to the appropriate facility

Cricoid pressure manoeuvre



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2.13 BURP manoeuvre

Indications

1. BURP manoeuvre for ETT placement, as part of the failed intubation drill.

Contraindications

1. Neck trauma

Adverse Effects

1. Airway trauma.

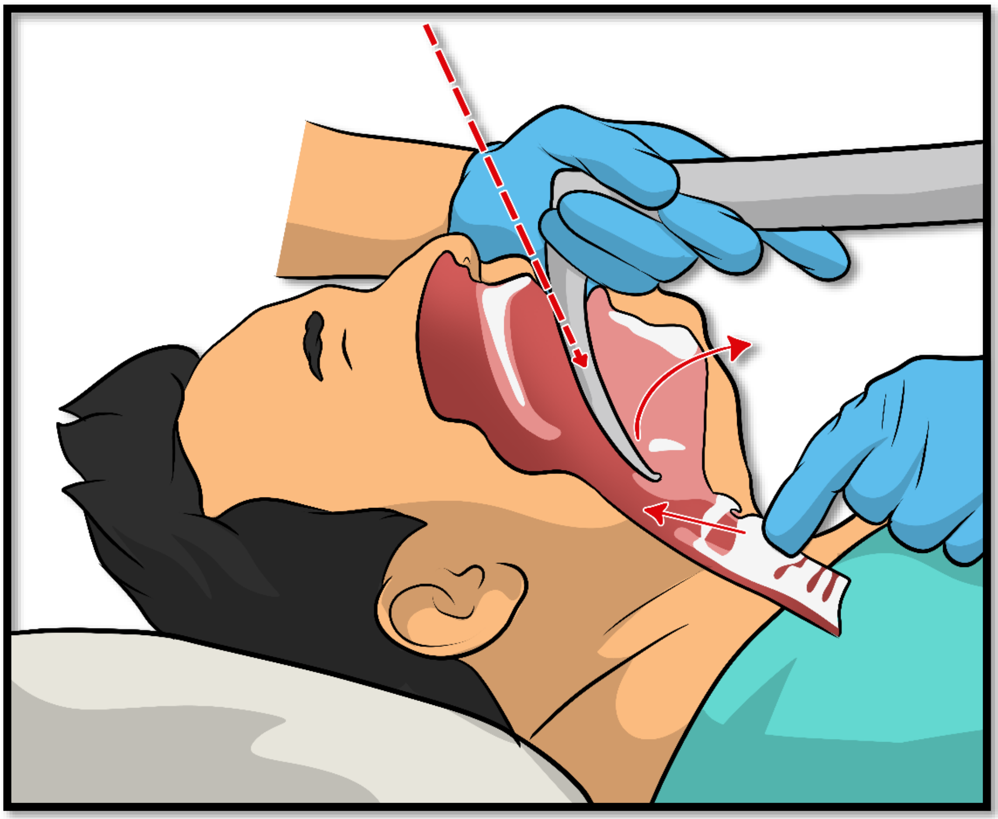
Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms and there is a risk of vomiting.

Procedure

1. Prepare for intubation using pre-pack, Adult/ Pediatric Airway Roll, C6, oxygen, Frova, Oxylog 3000 plus and medication.
2. Follow Procedure 2.16 for equipment preparation, patient positioning, medication administration, ETT placement and post-intubation management.
3. Locate the cricoid cartilage, larynx, trachea, and the hyoid bone by palpation so that pressure is applied to the right structure.
4. Hold the thyroid cartilage between the thumb and the middle finger.
5. Place the index finger on the thyroid.
6. Push the thyroid cartilage by providing BACKWARDS, UPWARDS, RIGHTWARDS PRESSURE. This will displace the larynx superiorly, posteriorly and rightward laterally to improve visualization of the cords.
7. Continue with ETT placement and post- intubation management.
8. Manage the patient according to the appropriate CPG. Reassess the patient's condition and interventions.
9. Ensure notification of the receiving facility as indicated (through CCE).
10. Transport the patient to the appropriate facility.

BURP manoeuvre



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2.14 Adult orotracheal intubation using the FROVA

Indications

1. Failure of airway maintenance or protection.
2. Hypercapnic or Hypoxemic ventilatory failure.
3. The expected clinical course of the patient will require intubation.
4. Use of the Frova device is indicated in all intubations in which the Glidescope is not utilized.

Contraindications

1. Airway which can be safely and adequately managed by less invasive means e.g., CPAP/BIPAP.
2. Procedure should not be performed prior to evaluating for, and attempting to treat, any reversible causes e.g., hypoglycaemia, seizure, overdose, etc.
3. There are no known contraindications to the use of the Frova intubating introducer for intubation.

Adverse Effects

1. Failure to intubate
2. Unrecognized esophageal intubation
3. Right main bronchus intubation
4. Aspiration
5. Hypoxia
6. Hypotension
7. Laryngospasm/Bronchospasm
8. Oropharyngeal trauma
9. Malposition
10. Vagal stimulation
11. Equipment failure
12. Raised ICP

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms and there is a risk of vomiting.

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Procedure
1. Prepare yourself mentally. Identify risks for potential difficult airway using the mnemonic LEON (**L**ook externally, **E**valuate, **O**bstructions and **N**eck mobility). 2. Position the patient appropriately. a. If a medical patient is lying on the floor/ bed, raise the head 10 cm and extend in the sniffing position to line the external auditory meatus with the sternal notch. The face will then be parallel to the ceiling. b. If a medical patient is lying on the ambulance stretcher, elevate the head section of the stretch by 20 degrees. This will line the external auditory meatus with the sternal notch. c. If trauma patient and is secured on the scoop stretcher (SMR), elevate the head section of the scoop stretcher by 30 degrees (reverse Trendelenburg position). This will line the external auditory meatus with the sternal notch. d. Obese patients should be placed in the **RAMP** position. This will line the external auditory meatus with the sternal notch. 3. Pre-oxygenate the patient appropriately (refer to Procedure 3.6). 4. Place the nasal $ETCO_2$ and connect to O_2 supply. Start O_2 delivery at 6 LMP (or as much as can be tolerated by the patient) aiming for 15 LPM. 5. Prepare your team by giving them clear and concise commands. 6. Establish 2 IV access and infuse Normal Saline, as required. (Up to 1 liter) 7. Ensure that the patient's BP, ECG, SpO_2, HR, $ETCO_2$ and RR are being monitored. Attach the defibrillation pads to the patient. 8. Change BP time interval on the Lifepak 15 to 2/ 3 minutes. Vitals signs should be monitored pre-RSI, per-RSI and post-RSI. 9. Prepare for standard intubation using the Adult Airway Roll, C6, RSI medication administration pack and medication. 10. Prepare an appropriately sized OPA and oxygen with adult BVM. 11. Tested suction unit with Yankeur suction catheter. 12. Prepare laryngoscope with appropriate blade and size. Prepare the spare laryngoscope handle as backup. 13. Select ETT with 3 size (1 correct size, 1 size big and 1 size small) options and lubricated Frova. 14. Prefill a 10 ml syringe with the desired amount of air to inflate the ETT cuff. 15. Prepare back up equipment, LTA for failed intubation (Safeguard) and Quick Trach II for failed ventilation (Rescue). 16. Prepare the Glidescope/ McGrath video laryngoscope, if available. 17. Prepare and place the tube holder in position under the patient's neck. 18. Preload an appropriately sized, lubricated ETT onto the Adult Frova device (14.0

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French).

19. Optimize haemodynamics
20. Calculate Shock Index
21. Prepare pharmacology: Fentanyl, Ketamine, Rocuronium, Push dose pressor (Phenylephrine/Adrenaline) and Adrenaline IM dose for anaphylaxis.
22. Prepare for mechanical ventilation.
23. Proceed with RSI Checklist.
24. Final briefing to team on roles and responsibilities during procedure.
25. Paralyze and induce the patient:
26. Administer:

a. **Fentanyl 1mcg/kg** based on hemodynamic status (Fentanyl to be diluted to 10ml). Administer Fentanyl 5 minutes prior to induction and paralysis. Administer Fentanyl slowly over 3 minutes.

b. **Ketamine 1-2mg/kg** based on hemodynamic status

c. **Rocuronium 100-150mg (1.5mg/kg)** for adults and **1.5mg/kg** for paediatrics.

27. During the apneic phase, jaw thrust is performed to maintain pharyngeal patency and increase apneic O_2 to 15 LPM.
28. Wait 45-60 seconds. Then check for jaw flaccidity before attempting laryngoscopy.
29. Change $ETCO_2$ probe on Lifepak 15 from nasal (Smart CapnoLine) to the Filter-Line Set during the apneic phase.
30. Place the patient's head in desired position (Align the laryngeal, pharyngeal and oral airway axes) and place laryngoscope in patient's mouth. Perform BURP /External Laryngeal manipulation as required.
31. Using laryngoscopy, introduce the tip of the Frova beyond the epiglottis and advance it in a straight line toward the glottis.
32. Advance the Frova into the trachea. If resistance is encountered, do not force the Frova; gently rotate and advance.
33. Inform CCA to hold Frova in place and follow your commands.
34. Note: During the introduction, you may feel tracheal rings. The Frova should not be advanced beyond the carina.
35. Maintain position of the Frova; advance the preloaded ETT into the trachea to the appropriate distance.
36. Maintain the position of the ETT, remove the Frova and laryngoscope.

37. Inflate ETT cuff inlet port with an appropriate amount of air.

38. Place BVM on ETT with ETCO2 monitoring, ventilate while listening over epigastrium, then left lung and then right lung to ensure proper placement.
39. Confirm ETT placement with ETCO2 waveform on monitor.

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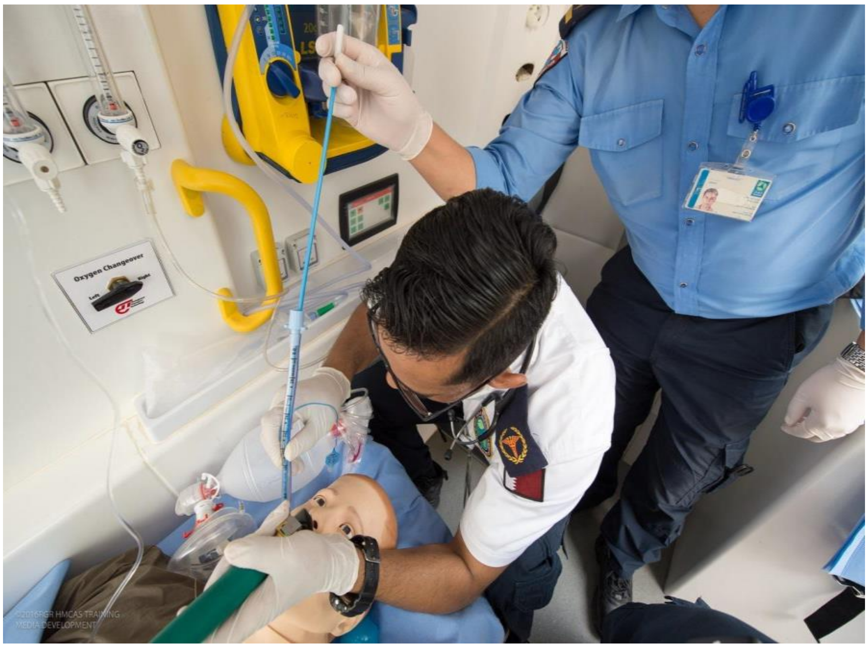
40. If ETT cannot be confirmed through ETCO2 monitoring or confirmation of lung sounds, the ETT should be removed, and an alternate airway device should be utilized (Follow the Failed Intubation Drill).
41. After ETT is confirmed to be in the trachea, secure the ETT with standard commercial tube holder.
42. Ensure that the ETT cuff is adequately inflated.
43. Place the patient on the mechanical ventilator.
44. Continue to ventilate patient while monitoring vital signs.
45. Plan for post intubation hypotension and desaturation. Remember to prepare a push dose pressor. Follow the post intubation checklist.
46. Post intubation management, maintain sedation and paralysis
    a. **Fentanyl 0.5 mcg/kg**, repeated every 15-20 minutes, as necessary OR Infusion of 1-10mcg/kg/hr.
    b. **Ketamine 0.25 - 0.5mg/ kg** every 15-20 minutes, as necessary OR infusion of 0.01-0.05mg/kg/min.
    c. **Rocuronium 0.25mg/kg** after 30-45 minutes of initial dose.
47. Fill in appropriate ePCR documentation requirements.
48. Note that the Frova may be utilized as a one-person or two-person technique. The two-person technique is advocated to minimize risks of failure to appropriately place the ETT.

Adult orotracheal intubation using the Frova



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2.15 Adult orotracheal intubation using the Glidescope

Indications

Clinician identifies a suspected difficult airway in the presence of one of the following:

1. Failure of airway maintenance or protection.
2. Hypercapnic or Hypoxemic ventilatory failure.
3. The expected clinical course of the patient will require intubation.

Contraindications

1. Airway which can be safely and adequately managed by less invasive means e.g., CPAP/BIPAP.
2. Procedure should not be performed prior to evaluating for, and attempting to treat, any reversible causes e.g., hypoglycaemia, seizure, overdose, etc.
3. There are no known contraindications to the use of the Frova intubating introducer for intubation.

Adverse Effects

1. Failure to intubate
2. Unrecognized esophageal intubation
3. Right main bronchus intubation
4. Aspiration
5. Hypoxia
6. Hypotension
7. Laryngospasm/Bronchospasm
8. Oropharyngeal trauma
9. Malposition
10. Vagal stimulation
11. Equipment failure
12. Raised ICP

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms and there is a risk of vomiting.

Procedure

1. Prepare yourself mentally. Identify risks for potential difficult airway using the mnemonic LEON (**L**ook externally, **E**valuate, **O**bstructions and **N**eck mobility).

2. Position the patient appropriately.
   a. If a medical patient is lying on the floor/ bed, raise the head 10 cm and extend in the sniffing position to line the external auditory meatus with the sternal notch. The face will then be parallel to the ceiling.
   b. If a medical patient is lying on the ambulance stretcher, elevate the head section of the stretch by 20 degrees. This will line the external auditory meatus with the sternal notch.
   c. If trauma patient and is secured on the scoop stretcher (SMR), elevate the head section of the scoop stretcher by 30 degrees (reverse Trendelenburg position). This will line the external auditory meatus with the sternal notch.
   d. Obese patients should be placed in the **RAMP** position. This will line the external auditory meatus with the sternal notch.
3. Pre-oxygenate the patient appropriately (refer to Procedure 3.6).
4. Place the nasal $ETCO_2$ and connect to O_2 supply. Start O_2 delivery at 6 LMP (or as much as can be tolerated by the patient) aiming for 15 LPM.
5. Prepare your team by giving them clear and concise commands.
6. Establish 2 IV access and infuse Normal Saline, as required. (Up to 1 liter)
7. Ensure that the patient's BP, ECG, SpO_2, HR, $ETCO_2$ and RR are being monitored. Attach the defibrillation pads to the patient.
8. Change BP time interval on the Lifepak 15 to 2/ 3 minutes. Vitals signs should be monitored pre-RSI, per-RSI and post-RSI.
9. Prepare for standard intubation using the Adult Airway Roll, C6, RSI medication administration pack and medication.
10. Prepare an appropriately sized OPA and oxygen with adult BVM.
11. Tested suction unit with Yankeur suction catheter.
12. Prepare laryngoscope with appropriate blade and size. Prepare the spare laryngoscope handle as backup.
13. Select ETT with 3 size (1 correct size, 1 size big and 1 size small) options and lubricated Frova.
14. Prefill a 10 ml syringe with the desired amount of air to inflate the ETT cuff.
15. Prepare back up equipment, LTA for failed intubation (Safeguard) and Quick Trach II for failed ventilation (Rescue).
16. Prepare the Glidescope/ McGrath video laryngoscope, if available.
17. Insert the GlideRite rigid stylet into the ETT.
18. Prepare and place the tube holder in position under the patient's neck.
19. Preload an appropriately sized, lubricated ETT onto the Adult Frova device (14.0 French).
20. Optimize haemodynamics

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21. Calculate Shock Index
22. Prepare pharmacology: Fentanyl, Ketamine, Rocuronium, Push dose pressor (Phenylephrine/Adrenaline) and Adrenaline IM dose for anaphylaxis.
23. Prepare for mechanical ventilation.
24. Proceed with RSI Checklist.
25. Final briefing to team on roles and responsibilities during procedure.
26. Paralyze and induce the patient:
27. Administer:

a. **Fentanyl 1mcg/kg** based on hemodynamic status (Fentanyl to be diluted to 10ml). Administer Fentanyl 5 minutes prior to induction and paralysis. Administer Fentanyl slowly for over 3 minutes.

b. **Ketamine 1-2mg/kg** based on hemodynamic status

c. **Rocuronium 100-150mg (1.5mg/kg)** for adults and **1.5mg/kg** for paediatrics.

28. During the apneic phase, jaw thrust is performed to maintain pharyngeal patency and increase apneic O_2 to 15 LPM.
29. Wait 45-60 seconds. Then check for jaw flaccidity before attempting laryngoscopy.
30. Change $ETCO_2$ probe on Lifepak 15 from nasal (Smart CapnoLine) to the Filter-Line Set during the apneic phase.

31. Place the patient's head in desired position and place Glidescope midline in patient's mouth with your left hand.

32. With the laryngoscope inserted, look to the monitor to identify the epiglottis, then manipulate the scope to obtain the best glottic view.

33. Look into the patients' mouth and introduce the ETT.

34. Look at the monitor and place ETT into trachea until the black line on the ETT passes the vocal cords.

35. Flick the GlideRite® Rigid Stylet approximately 5 cm to ensure ease of passing of the ETT.
36. Have your partner remove the rigid stylet.
37. Inflate ETT cuff inlet port with an appropriate amount of air.
38. Place BVM on ETT with ETCO2 monitoring, ventilate while listening over epigastrium, then left lung and then right lung to ensure proper placement.
39. Confirm ETT placement with ETCO2 waveform on monitor.
40. If ETT cannot be confirmed through ETCO2 monitoring or confirmation of lung sounds, the ETT should be removed, and an alternate airway device should be utilized (Follow the Failed Intubation Drill).

41. After ETT is confirmed to be in the trachea, secure the ETT with standard commercial tube holder.
42. Ensure that the ETT cuff is adequately inflated.
43. Place the patient on the mechanical ventilator.
44. Continue to ventilate patient while monitoring vital signs.
45. Plan for post intubation hypotension and desaturation. Remember to prepare push dose pressor. Follow the post intubation checklist.
46. Post intubation management, maintain sedation and paralysis
    a. **Fentanyl 0.5 mcg/kg**, repeated every 15-20 minutes, as necessary OR Infusion of 1-10mcg/kg/hr.
    b. **Ketamine 0.25 - 0.5mg/ kg** every 15-20 minutes, as necessary OR infusion of 0.01-0.05mg/kg/min.
    c. **Rocuronium 0.25mg/kg** after 30-45 minutes of initial dose.
47. Fill in appropriate ePCR documentation requirements.

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2.16 Adult orotracheal intubation using the McGrath

Indications

1. Clinician identifies a suspected difficult airway in the presence of one of the following:
   a. Failure of airway maintenance or protection
   b. Hypercapnic or Hypoxemic ventilatory failure.
   c. The expected clinical course of the patient will require intubation.

Contraindications

1. Airway which can be safely and adequately managed by less invasive means e.g., CPAP/BIPAP.
2. Procedure should not be performed prior to evaluating for, and attempting to treat, any reversible causes e.g., hypoglycaemia, seizure, overdose, etc.
3. There are no known contraindications to the use of the Frova intubating introducer for intubation.

Adverse Effects

1. Failure to intubate
2. Unrecognized esophageal intubation
3. Right main bronchus intubation
4. Aspiration
5. Hypoxia
6. Hypotension
7. Laryngospasm/Bronchospasm
8. Oropharyngeal trauma
9. Malposition
10. Vagal stimulation
11. Equipment failure
12. Raised ICP

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms and there is a risk of vomiting.

Procedure

1. Prepare yourself mentally. Identify risks for potential difficult airway using the

mnemonic LEON (**L**ook externally, **E**valuate, **O**bstructions and **N**eck mobility).

2. Position the patient appropriately.
3. If a medical patient is lying on the floor/ bed, raise the head 10 cm and extend in the sniffing position to line the external auditory meatus with the sternal notch. The face will then be parallel to the ceiling.
4. If a medical patient is lying on the ambulance stretcher, elevate the head section of the stretch by 20 degrees. This will line the external auditory meatus with the sternal notch.
5. If trauma patient and is secured on the scoop stretcher (SMR), elevate the head section of the scoop stretcher by 30 degrees (reverse Trendelenburg position). This will line the external auditory meatus with the sternal notch.
6. Obese patients should be placed in the **RAMP** position. This will line the external auditory meatus with the sternal notch.
7. Pre-oxygenate the patient appropriately (refer to Procedure 3.6).
8. Place the nasal $ETCO_2$ and connect to O_2 supply. Start O_2 delivery at 6 LMP (or as much as can be tolerated by the patient) aiming for 15 LPM.
9. Prepare your team by giving them clear and concise commands.
10. Establish 2 IV access and infuse Normal Saline, as required. (Up to 1 liter)
11. Ensure that the patient's BP, ECG, SpO_2, HR, $ETCO_2$ and RR are being monitored. Attach the defibrillation pads to the patient.
12. Change BP time interval on the Lifepak 15 to 2/ 3 minutes. Vitals signs should be monitored pre-RSI, per-RSI and post-RSI.
13. Prepare for standard intubation using the Adult Airway Roll, C6, RSI medication administration pack and medication.
14. Prepare an appropriately sized OPA and oxygen with adult BVM.
15. Tested suction unit with Yankeur suction catheter.
16. Prepare laryngoscope with appropriate blade and size. Prepare the spare laryngoscope handle as backup.
17. Select ETT with 3 size (1 correct size, 1 size big and 1 size small) options and lubricated Frova.
18. Prefill a 10 ml syringe with the desired amount of air to inflate the ETT cuff.
19. Prepare back up equipment, LTA for failed intubation (Safeguard) and Quick Trach II for failed ventilation (Rescue).
20. Prepare the Glidescope/ McGrath video laryngoscope, if available.
21. Prepare and place the tube holder in position under the patient's neck.
22. Preload an appropriately sized, lubricated ETT onto the Adult Frova device (14.0 French).
23. Optimize haemodynamics

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24. Calculate Shock Index
25. Prepare pharmacology: Fentanyl, Ketamine, Rocuronium, Push dose pressor (Phenylephrine/Adrenaline) and Adrenaline IM dose for anaphylaxis.
26. Prepare for mechanical ventilation.
27. Proceed with RSI Checklist.
28. Final briefing to team on roles and responsibilities during procedure.
29. Paralyze and induce the patient:
30. Administer:

a. **Fentanyl 1mcg/kg** based on hemodynamic status (Fentanyl to be diluted to 10ml). Administer Fentanyl 5 minutes prior to induction and paralysis. Administer Fentanyl slowly for over 3 minutes.
b. **Ketamine 1-2mg/kg based** on hemodynamic status
c. **Rocuronium 100-150mg (1.5mg/kg)** for adults and **1.5mg/kg** for paediatrics.

31. During the apneic phase, jaw thrust is performed to maintain pharyngeal patency and increase apneic O_2 to 15 LPM.
32. Wait 45-60 seconds. Then check for jaw flaccidity before attempting laryngoscopy.
33. Change $ETCO_2$ probe on Lifepak 15 from nasal (Smart CapnoLine) to the Filter-Line Set during the apneic phase.
34. Place the patient's head in desired position (Align the laryngeal, pharyngeal and oral airway axes) and place the McGrath in the right side of the patient's mouth with your left hand.
35. Move the device to the central position of the patient's mouth while sweeping the tongue to the left.
36. Advance the tip of the McGrath blade into the vallecula.
37. Visualize the epiglottis on the screen. Then lift the anatomy forward and upwards to expose the glottic aperture. When the device is in the optimal position, the glottis should be viewed in the central upper section of the screen.
38. Introduce the tip of the Frova beyond the epiglottis and advance it in a straight line toward the glottis.
39. Advance the Frova into the trachea. If resistance is encountered, do not force the Frova; gently rotate and advance.
40. Maintain position of the Frova; advance the preloaded ETT into the trachea to the appropriate distance.
41. Maintain the position of the ETT, remove the Frova and laryngoscope.
42. Inflate ETT cuff inlet port using a 10ml syringe with air to an appropriate amount.
43. Place BVM on ETT with ETCO2 monitoring, ventilate while listening over epigastrium, then left lung and then right lung to ensure proper placement.
44. Confirm ETT placement with ETCO2 waveform on monitor.
45. If ETT cannot be confirmed through ETCO2 monitoring or confirmation of lung

sounds, the ETT should be removed, and an alternate airway device should be utilized (Follow the Failed Intubation Drill).

46. After ETT is confirmed to be in the trachea, secure the ETT with standard commercial tube holder.
47. Ensure that the ETT cuff is adequately inflated.
48. Place the patient on the mechanical ventilator.
49. Continue to ventilate patient while monitoring vital signs.
50. Plan for post intubation hypotension and desaturation.
51. Follow the post intubation checklist.
52. Post intubation management, maintain sedation and paralysis.
    a. **Fentanyl 0.5 mcg/kg**, repeated every 15-20 minutes, as necessary OR Infusion of 1-10mcg/kg/hr.
    b. **Ketamine 0.25 - 0.5mg/ kg** every 15-20 minutes, as necessary OR infusion of 0.01-0.05mg/kg/min.
    c. **Rocuronium 0.25mg/kg** after 30-45 minutes of initial dose.
53. Fill in appropriate ePCR documentation requirements.

2.17 Adult orotracheal intubation using the RSI

Indications

1. Loss of airway protection/patency
2. Ventilation/oxygenation failure
3. Neuroprotection of the TBI and CVA patient
4. Anticipated clinical course requiring anesthesia
5. Humanitarian reasons (e.g. major polytrauma and significant BSA burns patients with a relatively high level of consciousness.

Contraindications

1. Hypersensitivity to any of the RSI medication.
2. Procedure should not be performed prior to evaluating for, and attempting to treat, any reversible causes e.g., hypoglycaemia, seizure, overdose, etc.

Adverse Effects

1. Failure to intubate
2. Unrecognized esophageal intubation
3. Right main bronchus intubation
4. Aspiration
5. Hypoxia
6. Hypotension
7. Laryngospasm/Bronchospasm
8. Oropharyngeal trauma
9. Malposition
10. Vagal stimulation
11. Equipment failure
12. Raised ICP

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms and there is a risk of vomiting.

Procedure

1. Prepare yourself mentally. Identify risks for potential difficult airway using the mnemonic LEON (**L**ook externally, **E**valuate, **O**bstructions and **N**eck mobility).
2. Position the patient appropriately.

a. If a medical patient is lying on the floor/ bed, raise the head 10 cm and extend in the sniffing position to line the external auditory meatus with the sternal notch. The face will then be parallel to the ceiling.
b. If a medical patient is lying on the ambulance stretcher, elevate the head section of the stretch by 20 degrees. This will line the external auditory meatus with the sternal notch.
c. If trauma patient and is secured on the scoop stretcher (SMR), elevate the head section of the scoop stretcher by 30 degrees (reverse Trendelenburg position). This will line the external auditory meatus with the sternal notch.
d. Obese patients should be placed in the **RAMP** position. This will line the external auditory meatus with the sternal notch.

3. Pre-oxygenate the patient appropriately (refer to Procedure 3.6).
4. Place the nasal $ETCO_2$ and connect to O_2 supply. Start O_2 delivery at 6 LMP (or as much as can be tolerated by the patient) aiming for 15 LPM.
5. Prepare your team by giving them clear and concise commands.
6. Establish 2 IV access and infuse Normal Saline, as required. (Up to 1 liter)
7. Ensure that the patient's BP, ECG, SpO_2, HR, $ETCO_2$ and RR are being monitored. Attach the defibrillation pads to the patient.
8. Change BP time interval on the Lifepak 15 to 2/ 3 minutes. Vitals signs should be monitored pre-RSI, per-RSI and post-RSI.
9. Prepare for standard intubation using the Adult Airway Roll, C6, RSI medication administration pack and medication.
10. Prepare an appropriately sized OPA and oxygen with adult BVM.
11. Tested suction unit with Yankeur suction catheter.
12. Prepare laryngoscope with appropriate blade and size. Prepare the spare laryngoscope handle as backup.
13. Select ETT with 3 size (1 correct size, 1 size big and 1 size small) options and lubricated Frova.
14. Prefill a 10 ml syringe with the desired amount of air to inflate the ETT cuff.
15. Prepare back up equipment, LTA for failed intubation (Safeguard) and Quick Trach II for failed ventilation (Rescue).
16. Prepare the Glidescope/ McGrath video laryngoscope, if available.
17. Prepare and place the tube holder in position under the patient's neck.
18. Preload an appropriately sized, lubricated ETT onto the Adult Frova device (14.0 French).
19. Optimize haemodynamics
20. Calculate Shock Index
21. Prepare pharmacology: Fentanyl, Ketamine, Rocuronium, Push dose pressor

(Phenylephrine/Adrenaline) and Adrenaline IM dose for anaphylaxis.

22. Prepare for mechanical ventilation.
23. Proceed with RSI Checklist.
24. Final briefing to team on roles and responsibilities during procedure.
25. Paralyze and induce the patient:
26. Administer:

a. **Fentanyl 1mcg/kg** based on hemodynamic status (Fentanyl to be diluted to 10ml). Administer Fentanyl 5 minutes prior to induction and paralysis. Administer Fentanyl slowly for over 3 minutes.

b. **Ketamine 1-2mg/kg** based on hemodynamic status

c. **Rocuronium 100-150mg (1.5mg/kg)** for adults and **1.5mg/kg** for paediatrics.

27. During the apneic phase, jaw thrust is performed to maintain pharyngeal patency and increase apneic O_2 to 15 LPM.
28. Wait 45-60 seconds. Then check for jaw flaccidity before attempting laryngoscopy.
29. Change $ETCO_2$ probe on Lifepak 15 from nasal (Smart CapnoLine) to the Filter-Line Set during the apneic phase.
30. Adjust step 30-37 based on device and technique use for laryngoscopy. Place the patient's head in desired position (Align the laryngeal, pharyngeal and oral airway axes) and place the McGrath in the right side of the patient's mouth with your left hand.
31. Move the device to the central position of the patient's mouth while sweeping the tongue to the left.
32. Advance the tip of the McGrath blade into the vallecula.
33. Visualize the epiglottis on the screen. Then lift the anatomy forward and upwards to expose the glottic aperture. When the device is in the optimal position, the glottis should be viewed in the central upper section of the screen.
34. Introduce the tip of the Frova beyond the epiglottis and advance it in a straight line toward the glottis.
35. Advance the Frova into the trachea. If resistance is encountered, do not force the Frova; gently rotate and advance.
36. Maintain position of the Frova; advance the preloaded ETT into the trachea to the appropriate distance.
37. Maintain the position of the ETT, remove the Frova and laryngoscope.
38. Inflate ETT cuff inlet port using a 10ml syringe with air to an appropriate amount.
39. Place BVM on ETT with ETCO2 monitoring, ventilate while listening over epigastrium, then left lung and then right lung to ensure proper placement.
40. Confirm ETT placement with ETCO2 waveform on monitor.
41. If ETT cannot be confirmed through ETCO2 monitoring or confirmation of lung sounds, the ETT should be removed, and an alternate airway device should be

utilized (Follow the Failed Intubation Drill).

42. After ETT is confirmed to be in the trachea, secure the ETT with standard commercial tube holder.
43. Ensure that the ETT cuff is adequately inflated.
44. Place the patient on the mechanical ventilator.
45. Continue to ventilate patient while monitoring vital signs.
46. Plan for post intubation hypotension and desaturation.
47. Follow the post intubation checklist.
48. Post intubation management, maintain sedation and paralysis
    a. **Fentanyl 0.5 mcg/kg**, repeated every 15-20 minutes, as necessary OR Infusion of 1-10mcg/kg/hr.
    b. **Ketamine 0.25 - 0.5mg/ kg** every 15-20 minutes, as necessary OR infusion of 0.01-0.05mg/kg/min.
    c. **Rocuronium 0.25mg/kg** after 30-45 minutes of initial dose.
49. Fill in appropriate ePCR documentation requirements.

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2.18 Adult Orotracheal intubation using Blind Bougie Technique

Indications

1. Included in the failed intubation drill as an option for a grade III Cormack- Lehane laryngoscopic view.

Contraindications

1. In a “can't intubate, can’t oxygenate” (CICO) situation, where there is no time to evaluate and execute various options and the airway must be secured immediately.

Adverse Effects

1. Failure to intubate
2. Unrecognized esophageal intubation
3. Right main bronchus intubation
4. Aspiration
5. Hypoxia
6. Hypotension
7. Laryngospasm/Bronchospasm
8. Oropharyngeal trauma
9. Malposition
10. Vagal stimulation
11. Equipment failure
12. Raised ICP

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms and there is a risk of vomiting.

Procedure

1. Prepare yourself mentally. Identify risks for potential difficult airway using the mnemonic LEON (**L**ook externally, **E**valuate, **O**bstructions and **N**eck mobility).
2. Position the patient appropriately.
   a. If a medical patient is lying on the floor/ bed, raise the head 10cm and extend in the sniffing position to line the external auditory meatus with the sternal notch. The face will then be parallel to the ceiling.
   b. If a medical patient is lying on the ambulance stretcher, elevate the head section of the stretch by 20 degrees. This will line the external auditory meatus with the sternal notch.

c. If trauma patient and is secured on the scoop stretcher (SMR), elevate the head section of the scoop stretcher by 30 degrees (reverse Trendelenburg position). This will line the external auditory meatus with the sternal notch.
d. Obese patients should be placed in the **RAMP** position. This will line the external auditory meatus with the sternal notch.

3. Pre-oxygenate the patient appropriately (refer to Procedure 3.6).
4. Place the nasal $ETCO_2$ and connect to O_2 supply. Start O_2 delivery at 6 LMP (or as much as can be tolerated by the patient) aiming for 15 LPM.
5. Prepare your team by giving them clear and concise commands.
6. Establish 2 IV access and infuse Normal Saline, as required. (Up to 1 liter)
7. Ensure that the patient's BP, ECG, SpO_2, HR, $ETCO_2$ and RR are being monitored. Attach the defibrillation pads to the patient.
8. Change BP time interval on the Lifepak 15 to 2/ 3 minutes. Vitals signs should be monitored pre-RSI, per-RSI and post-RSI.
9. Prepare for standard intubation using the Adult Airway Roll, C6, RSI medication administration pack and medication.
10. Prepare an appropriately sized OPA and oxygen with adult BVM.
11. Tested suction unit with Yankeur suction catheter.
12. Prepare laryngoscope with appropriate blade and size. Prepare the spare laryngoscope handle as backup.
13. Select ETT with 3 size (1 correct size, 1 size big and 1 size small) options and lubricated Frova.
14. Prefill a 10 ml syringe with the desired amount of air to inflate the ETT cuff.
15. Prepare back up equipment, LTA for failed intubation (Safeguard) and Quick Trach II for failed ventilation (Rescue).
16. Prepare the Glidescope/ McGrath video laryngoscope, if available.
17. Prepare and place the tube holder in position under the patient's neck.
18. Preload an appropriately sized, lubricated ETT onto the Adult Frova device (14.0 French).
19. Optimize haemodynamics
20. Calculate Shock Index
21. Prepare pharmacology: Fentanyl, Ketamine, Rocuronium, Push dose pressor (Phenylephrine/Adrenaline) and Adrenaline IM dose for anaphylaxis.
22. Prepare for mechanical ventilation.
23. Proceed with RSI Checklist.
24. Final briefing to team on roles and responsibilities during procedure.
25. Paralyze and induce the patient:
26. Administer:

a. **Fentanyl 1mcg/kg** based on hemodynamic status (Fentanyl to be diluted to 10ml).

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Administer Fentanyl 5 minutes prior to induction and paralysis. Administer Fentanyl slowly for over 3 minutes.

b. **Ketamine 1-2mg/kg** based on hemodynamic status

c. **Rocuronium 100-150mg (1.5mg/kg)** for adults and **1.5mg/kg** for paediatrics.

27. During the apneic phase, jaw thrust is performed to maintain pharyngeal patency and increase apneic O_2 to 15 LPM.
28. Wait 45-60 seconds. Then check for jaw flaccidity before attempting laryngoscopy.
29. The first attempt at ETT placement was unsuccessful and you have a Cormack-Lehane Grade III view.
30. Grasp the Frova/ bougie in the right hand at the 20 to 25cm mark.
31. Pass it alongside the laryngoscope with the 30-degree angled tip (caudad) anteriorly.
32. The tip is advanced underneath the epiglottis and past the vocal cords blindly.
33. The tip lifts the floppy epiglottis and then is advanced blindly past the vocal cords. You may feel the cricoid rings located anteriorly.
34. Follow advancement into the trachea to a depth of 22 to 26cm in average adult.
35. Tip may be gently advanced further (28-36 cm) to contact carina/main stem bronchus when a hangup occurs.
36. Maintain tongue displacement with laryngoscopy/ hand grasp.
37. Pass the ETT to 3 X internal diameter (ID) (e.g., 7.0 ID = 21cm at the level of the teeth, but do not force the advancement (an assistant should grasp the proximal end of the bougie to stabilize it).
38. Inflate ETT cuff inlet port with the appropriate amount air.
39. Change $ETCO_2$ probe on Lifepak 15 from nasal (Smart CapnoLine) to the Filter-Line Set during the apneic phase.
40. Place BVM on ETT with ETCO2 monitoring, ventilate while listening over epigastrium, then left lung and then right lung to ensure proper placement.
41. Confirm ETT placement with at least 5 ETCO2 waveforms on monitor.
42. After ETT is confirmed to be in the trachea, secure the ETT with standard commercial tube holder.
43. Ensure that the ETT cuff is adequately inflated.
44. Place the patient on the mechanical ventilator.
45. Continue to ventilate patient while monitoring vital signs.
46. Plan for post intubation hypotension and desaturation. Remember to prepare push dose pressor.
47. Follow the post intubation checklist.
48. Post intubation management, maintain sedation and paralysis.
    a. **Fentanyl 0.5 mcg/kg**, repeated every 15-20 minutes, as necessary OR Infusion of 1-10mcg/kg/hr.
    b. **Ketamine 0.25 - 0.5mg/ kg** every 15-20 minutes, as necessary OR infusion of 0.01-0.05mg/kg/min.
    c. **Rocuronium 0.25mg/kg** after 30-45 minutes of initial dose.
49. Fill in appropriate ePCR documentation requirements.

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2.19 Failed Airway Drill

Indications

1. Failure to successfully place an ETT in the first attempt of laryngoscopy

Contraindications

1. Hypersensitivity to any of the RSI medication.
2. Patient in whom cricothyrotomy would be impossible.
3. Patient in whom intubation would be impossible (e.g., complete upper airway obstruction).
4. Pediatric patients under 8 years.
5. Procedure should not be performed prior to evaluating for, and attempting to treat, any reversible causes e.g., hypoglycaemia, seizure, overdose, etc.

Adverse Effects

1. Failure to intubate
2. Unrecognized esophageal intubation
3. Right main bronchus intubation
4. Aspiration
5. Hypoxia
6. Hypotension
7. Laryngospasm/Bronchospasm
8. Oropharyngeal trauma
9. Malposition
10. Vagal stimulation
11. Equipment failure
12. Raised ICP

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms and there is a risk of vomiting.

Procedure

1. Prepare yourself mentally. Identify risks for potential difficult airway using the mnemonic LEON (**L**ook externally, **E**valuate, **O**bstructions and **N**eck mobility).
2. Position the patient appropriately.

a. If a medical patient is lying on the floor/ bed, raise the head 10 cm and extend in the sniffing position to line the external auditory meatus with the sternal notch. The face will then be parallel to the ceiling.

b. If a medical patient is lying on the ambulance stretcher, elevate the head section of the stretch by 20 degrees. This will line the external auditory meatus with the sternal notch.
c. If trauma patient and is secured on the scoop stretcher (SMR), elevate the head section of the scoop stretcher by 30 degrees (reverse Trendelenburg position). This will line the external auditory meatus with the sternal notch.
d. Obese patients should be placed in the **RAMP** position. This will line the external auditory meatus with the sternal notch.

3. Pre-oxygenate the patient appropriately (refer to Procedure 3.6).
4. Place the nasal $ETCO_2$ and connect to O_2 supply. Start O_2 delivery at 6 LMP (or as much as can be tolerated by the patient) aiming for 15 LPM.
5. Prepare your team by giving them clear and concise commands.
6. Establish 2 IV access and infuse Normal Saline, as required. (Up to 1 liter)
7. Ensure that the patient's BP, ECG, SpO_2, HR, $ETCO_2$ and RR are being monitored. Attach the defibrillation pads to the patient.
8. Change BP time interval on the Lifepak 15 to 2/ 3 minutes. Vitals signs should be monitored pre-RSI, per-RSI and post-RSI.
9. Prepare for standard intubation using the Adult Airway Roll, C6, RSI medication administration pack and medication.
10. Prepare an appropriately sized OPA and oxygen with adult BVM.
11. Tested suction unit with Yankeur suction catheter.
12. Prepare laryngoscope with appropriate blade and size. Prepare the spare laryngoscope handle as backup.
13. Select ETT with 3 size (1 correct size, 1 size big and 1 size small) options and lubricated Frova.
14. Prefill a 10ml syringe with the desired amount of air to inflate the ETT cuff.
15. Prepare back up equipment, LTA for failed intubation (Safeguard) and Quick Trach II for failed ventilation (Rescue).
16. Prepare the Glidescope/ McGrath video laryngoscope, if available.
17. Prepare and place the tube holder in position under the patient's neck.
18. Preload an appropriately sized, lubricated ETT onto the Adult Frova device (14.0 French).
19. Optimize haemodynamics
20. Calculate Shock Index
21. Prepare pharmacology: Fentanyl, Ketamine, Rocuronium, Push dose pressor (Phenylephrine/Adrenaline) and Adrenaline IM dose for anaphylaxis.
22. Prepare for mechanical ventilation.
23. Proceed with RSI Checklist.

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24. Final briefing to team on roles and responsibilities during procedure.
25. Paralyze and induce the patient:
26. Administer:
    a. **Fentanyl 1mcg/kg** based on hemodynamic status (Fentanyl to be diluted to 10ml). Administer Fentanyl 5 minutes prior to induction and paralysis. Administer Fentanyl slowly for over 3 minutes.
    b. **Ketamine 1-2mg/kg** based on hemodynamic status
    c. **Rocuronium 100-150mg (1.5mg/kg)** for adults and **1.5mg/kg** for paediatrics.
27. During the apneic phase, jaw thrust is performed to maintain pharyngeal patency and increase apneic O_2 to 15 LPM.
28. Wait 45-60 seconds. Then check for jaw flaccidity before attempting laryngoscopy.
29. Change $ETCO_2$ probe on Lifepak 15 from nasal (Smart CapnoLine) to the Filter-Line Set during the apneic phase.
30. Place the patient's head in desired position (Align the laryngeal, pharyngeal and oral airway axes) and place laryngoscope in the patient's mouth.
31. **FIRST LARYNGOSCPIC** attempt and you fail to visualize the glottic aperture. Then within 30 seconds check the head position and undertake the BURP maneuver. If the laryngoscopic grade improves then place the ETT and
32. If the laryngoscopic grade does not improve then remove the laryngoscope, insert an OPA and reoxygenation - BVM.
33. Once the patient is reoxygenated, undertake the **SECOND LARYNGOSCOPIC** attempt. This time use a longer blade and if the Laryngoscopic grade improves then place the ETT and continue with post intubation management.
34. If the Laryngoscopic grade does not improve, then perform the blind bougie technique, as per Procedure 1.18.
35. If the blind bougie technique is unsuccessful, then undertake LTA/ iLTA insertion attempt (Safeguard), as per Procedure 2.21
36. Consider a **THIRD LARYNGOSCOPIC** attempt if saturation improved to > 95% with LTA.
37. Failed LTA placement OR unable to achieve saturation > 80% with LTA or BVM then undertake an emergency cricothyrotomy (Rescue), as per Procedure 2.24.
38. After successful placement of the ETT at any step above, then inflate ETT cuff inlet port with the appropriate amount air.
39. Place BVM on ETT with ETCO2 monitoring, ventilate while listening over epigastrium, then left lung and then right lung to ensure proper placement.
40. Confirm ETT placement with ETCO2 waveform on monitor.
41. If ETT cannot be confirmed through ETCO2 monitoring or confirmation of lung sounds, the ETT should be removed, and an alternate airway device should be utilized (Follow the Failed Intubation Drill).

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42. After ETT is confirmed to be in the trachea, secure the ETT with standard commercial tube holder.
43. Ensure that the ETT cuff is adequately inflated.
44. Place the patient on the mechanical ventilator.
45. Continue to ventilate patient while monitoring vital signs.
46. Plan for post intubation hypotension and desaturation.
47. Follow the post intubation checklist.
48. Post intubation management, maintain sedation and paralysis
    a. **Fentanyl 0.5 mcg/kg**, repeated every 15-20 minutes, as necessary OR Infusion of 1-10mcg/kg/hr.
    b. **Ketamine 0.25 - 0.5mg/ kg** every 15-20 minutes, as necessary OR infusion of 0.01-0.05mg/kg/min.
    c. **Rocuronium 0.25mg/kg** after 30-45 minutes of initial dose.
49. Fill in appropriate ePCR documentation requirements.

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2.20 Adult orotracheal intubation using Delayed Sequence Induction

IV

Indications

1. Patients who are semi-conscious and agitated with or without a partially occluded airway and or hypoxia from gas exchange failure.
2. Patient who is agitated or is otherwise intolerant of preoxygenation via nasal cannula, non-rebreather mask, bag- valve-mask, and/or CPAP.

Contraindications

1. Hypersensitivity to any of the RSI medication.
2. Patient in whom cricothyrotomy would be impossible.
3. Patient in whom intubation would be impossible (e.g., complete upper airway obstruction).
4. Pediatric patients under 8 years.
5. Procedure should not be performed prior to evaluating for, and attempting to treat, any reversible causes e.g., hypoglycaemia, seizure, overdose, etc.

Adverse Effects

1. Failure to intubate
2. Unrecognized esophageal intubation
3. Right main bronchus intubation
4. Aspiration
5. Hypoxia
6. Hypotension
7. Laryngospasm/Bronchospasm
8. Oropharyngeal trauma
9. Malposition
10. Vagal stimulation
11. Equipment failure
13. Raised ICP

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms and there is a risk of vomiting.

Procedure

1. Administer **Ketamine 25mg IV/ IO slow IV push over 15-30 seconds to maintain spontaneous breathing.** Repeat Ketamine 25mg slow IV/ IO push over 15-30 seconds

to achieve the desired effect. **OR** 100 mcg Fentanyl as a slow IV/ IO push over 15-30 seconds to achieve the desired effect.

2. Provide two-handed basic airway support and assisted ventilation with BVM, if required.
3. Prepare yourself mentally. Identify risks for potential difficult airway using the mnemonic LEON (**L**ook externally, **E**valuate, **O**bstructions and **N**eck mobility).
4. Position the patient appropriately.
5. If a medical patient is lying on the floor/ bed, raise the head 10cm and extend in the sniffing position to line the external auditory meatus with the sternal notch. The face will then be parallel to the ceiling.
6. If a medical patient is lying on the ambulance stretcher, elevate the head section of the stretch by 20 degrees. This will line the external auditory meatus with the sternal notch.
7. If trauma patient and is secured on the scoop stretcher (SMR), elevate the head section of the scoop stretcher by 30 degrees (reverse Trendelenburg position). This will line the external auditory meatus with the sternal notch.
8. Obese patients should be placed in the **RAMP** position. This will line the external auditory meatus with the sternal notch.
9. Pre-oxygenate the patient appropriately (refer to Procedure 3.6).
10. Place the nasal $ETCO_2$ and connect to O_2 supply. Start O_2 delivery at 6 LMP (or as much as can be tolerated by the patient) aiming for 15 LPM.
11. Prepare your team by giving them clear and concise commands.
12. Establish 2 IV access and infuse Normal Saline, as required. (Up to 1 liter)
13. Ensure that the patient's BP, ECG, SpO_2, HR, $ETCO_2$ and RR are being monitored. Attach the defibrillation pads to the patient.
14. Change BP time interval on the Lifepak 15 to 2/ 3 minutes. Vitals signs should be monitored pre-RSI, per-RSI, and post-RSI.
15. Prepare for standard intubation using the Adult Airway Roll, C6, RSI medication administration pack and medication.
16. Prepare an appropriately sized OPA and oxygen with adult BVM.
17. Tested suction unit with Yankeur suction catheter.
18. Prepare laryngoscope with appropriate blade and size. Prepare the spare laryngoscope handle as backup.
19. Select ETT with 3 size (1 correct size, 1 size big and 1 size small) options and lubricated Frova.
20. Prefill a 10ml syringe with the desired amount of air to inflate the ETT cuff.
21. Prepare back up equipment, LTA for failed intubation (Safeguard) and Quick Trach II for failed ventilation (Rescue).

22. Prepare the Glidescope/ McGrath video laryngoscope, if available.
23. Prepare and place the tube holder in position under the patient's neck.
24. Preload an appropriately sized, lubricated ETT onto the Adult Frova device (14.0 French).
25. Optimize haemodynamics
26. Calculate Shock Index
27. Prepare pharmacology: Fentanyl, Ketamine, Rocuronium, Push dose pressor (Phenylephrine/Adrenaline) and Adrenaline IM dose for anaphylaxis.
28. Prepare for mechanical ventilation.
29. Proceed with RSI Checklist.
30. Final briefing to team on roles and responsibilities during procedure.
31. Paralyze and induce the patient:
32. Administer:
    d. **Fentanyl 1mcg/kg** based on hemodynamic status (Fentanyl to be diluted to 10ml). Administer Fentanyl 5 minutes prior to induction and paralysis. Administer Fentanyl slowly for over 3 minutes.
    e. **Ketamine 1-2mg/kg** based on hemodynamic status
    f. **Rocuronium 100-150mg (1.5mg/kg)** for adults and **1.5mg/kg** for paediatrics.
33. During the apneic phase, jaw thrust is performed to maintain pharyngeal patency and increase apneic O_2 to 15 LPM.
34. Wait 45-60 seconds. Then check for jaw flaccidity before attempting laryngoscopy.
35. Change $ETCO_2$ probe on Lifepak 15 from nasal (Smart CapnoLine) to the Filter-Line Set during the apneic phase.
36. Adjust step 36-43 based on device and technique use for laryngoscopy. Place the patient’s head in desired position (Align the laryngeal, pharyngeal and oral airway axes) and place the McGrath in the right side of the patient’s mouth with your left hand.
37. Move the device to the central position of the patient’s mouth while sweeping the tongue to the left.
38. Advance the tip of the McGrath blade into the vallecula.
39. Visualize the epiglottis on the screen. Then lift the anatomy forward and upwards to expose the glottic aperture. When the device is in the optimal position, the glottis should be viewed in the central upper section of the screen.
40. Introduce the tip of the Frova beyond the epiglottis and advance it in a straight line toward the glottis.
41. Advance the Frova into the trachea. If resistance is encountered, do not force the Frova; gently rotate and advance.
42. Maintain position of the Frova; advance the preloaded ETT into the trachea to the

appropriate distance.

43. Maintain the position of the ETT, remove the Frova and laryngoscope.
44. Inflate ETT cuff inlet port using a 10ml syringe with air to an appropriate amount.
45. Place BVM on ETT with ETCO2 monitoring, ventilate while listening over epigastrium, then left lung and then right lung to ensure proper placement.
46. Confirm ETT placement with ETCO2 waveform on monitor.
47. If ETT cannot be confirmed through ETCO2 monitoring or confirmation of lung sounds, the ETT should be removed, and an alternate airway device should be utilized (Follow the Failed Intubation Drill).
48. After ETT is confirmed to be in the trachea, secure the ETT with standard commercial tube holder.
49. Ensure that the ETT cuff is adequately inflated.
50. Place the patient on the mechanical ventilator.
51. Continue to ventilate patient while monitoring vital signs.
52. Plan for post intubation hypotension and desaturation.
53. Follow the post intubation checklist.
54. Post intubation management, maintain sedation and paralysis
    a. **Fentanyl 0.5 mcg/kg**, repeated every 15-20 minutes, as necessary OR Infusion of 1-10mcg/kg/hr.
    b. **Ketamine 0.25 - 0.5mg/ kg** every 15-20 minutes, as necessary OR infusion of 0.01-0.05mg/kg/min.
    c. **Rocuronium 0.25mg/kg** after 30-45 minutes of initial dose.
55. Fill in appropriate ePCR documentation requirements.

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2.21 Laryngeal Tube airway insertion

Indications

1. Cardiac arrest.
2. Failed endotracheal intubation (CCP Only).

Contraindications

1. Awake patient.
2. Unable to open the patient's mouth.
3. Upper airway swelling/ FBAO.

Adverse Effects

1. Insertion of the LTA might trigger the gag reflex and cause vomiting and possible aspiration of gastric contents.
2. Local irritation.
3. Upper airway trauma.
4. Over-inflation of the cuff.

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms and there is a risk of vomiting.

Procedure

1. Prepare pre-packs Airway Roll (AP), Adult/ Paediatric Airway Roll (CCP), A5, oxygen source and suction.
2. Choose the correct size LTA.
3. Test the LTA cuff by inflating and then deflating it with the appropriate amount of air, based on the size then connect the HME filter.
4. Apply lubricant to the bevelled distal tip and posterior aspect of the LTA.
5. Place the patient's head in the sniffing position (neutral if suspicion of C-spine injury).
6. With your non-dominant hand perform a tongue-jaw lift. With your dominant hand hold the LTA at the connector end.
7. Insert the LTA into the midline of the patient's mouth, gently sliding it against the hard palate.
8. Insert the LTA until the indicator line on the device is aligned with the patient's teeth or gums.
9. Inflate the LTA cuff with the required volume of air according to the LTA size (Colour coded syringe).

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10. Attach the side stream $ETCO_2$ to the Lifepak15 and note the reading.
11. Attach the BVM/ ventilator tubing to the $ETCO_2$ connector and ventilate the patient as required.
12. Secure the LTA with Thomas tube holder.

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2.22 Intubating laryngeal tube airway insertion (iLTS- D)

I II III IV

Indications

1. Cardiac arrest.
2. Failed endotracheal intubation (CCP Only).

Contraindications

1. Awake patient.
2. Unable to open the patient's mouth.
3. Upper airway swelling/ FBAO.

Adverse Effects

1. Insertion of the iLTS-D might trigger the gag reflex and cause vomiting and possible aspiration of gastric contents.
2. Local irritation.
3. Upper airway trauma.
4. Over-inflation of the cuff.

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms and there is a risk of vomiting.

Procedure

1. Prepare Airway Roll (AP), Adult/ Paediatric Airway Roll (CCP), A5, oxygen source and suction.
2. Choose the correct size iLTS-D according to patient height. Size 2.5/3 (125-155 cm tall). Size 4/5 (Size 4: 155 cm – 180cm tall Size 5: >180cm tall).
3. Test the iLTS-D cuff by inflating and then deflating it with the appropriate amount of air, based on the size then connect the HME filter
4. Apply lubricant to the bevelled distal tip and posterior aspect of the iLTS-D.
5. Place the patient's head in the sniffing position (neutral if suspicion of C-spine injury).
6. With your non-dominant hand perform a tongue-jaw lift. With your dominant hand hold the iLTS-D at the connector end.
7. Insert the iLTS-D into the midline of the patient's mouth, gently sliding it against the hard palate.
8. Insert the iLTS-D until the indicator line on the device (choose the appropriate line based on size) is aligned with the patient's teeth or gums.
9. Inflate the iLTS-D cuff with the required volume of air according to the iLTS-D size (Colour coded syringe).

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10. Attach the side stream $ETCO_2$ and note the reading.
11. Attach the BVM/ ventilator tubing to the $ETCO_2$ connector and ventilate the patient as required.
12. Secure the iLTS-D with a commercial tube holder.

2.23 Armored ETT insertion using the iLTS- D

Indications

1. Cardiac arrest.
2. Failed endotracheal intubation (CCP Only).

Contraindications

1. Awake patient.
2. Unable to open the patient's mouth.
3. Upper airway swelling/ FBAO.

Adverse Effects

1. Insertion of the iLTS-D might trigger the gag reflex and cause vomiting and aspiration of gastric contents.
2. Local irritation.
3. Upper airway trauma.
4. Over-inflation of the cuff.

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms and there is a risk of vomiting.

Procedure

1. Prepare Adult/ Paediatric Airway Roll (CCP), A5, oxygen source, Frova and suction.
2. Choose the correct size iLTS-D. Size 2.5/3 (two sizes in one device): fits patients that are 125-155 cm tall. Size 4/5 (two sizes in one device): fits patients that are ≥ 155 cm tall.
3. Test the iLTS-D cuff by inflating and then deflating it with the appropriate amount of air, based on the size. Test the cuff of the armored ETT and deflate fully.
4. Apply lubricant to the beveled distal tip and posterior aspect of the iLTS-D.
5. Place the patient's head in the sniffing position (neutral if suspicion of C-spine injury).
6. With your non-dominant hand perform a tongue-jaw lift. With your dominant hand hold the iLTS-D at the connector end.
7. Insert the iLTS-D into the midline of the patient's mouth, gently sliding it against the hard palate.
8. Insert the iLTS-D until the indicator line on the device (choose the appropriate line based on size) is aligned with the patient's teeth or gums.
9. Inflate the iLTS-D cuff with the required volume of air according to the iLTS-D size (Colour coded syringe).

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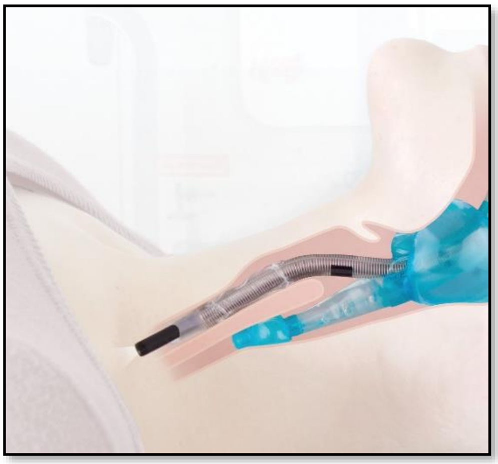
10. Attach the side stream $ETCO_2$ and note the reading.
11. Attach the BVM/ ventilator tubing to the $ETCO_2$ connector and ventilate the patient as required.
12. Secure the iLTS-D with a commercial tube holder.
13. Pre-oxygenate the patient for 3-5 minutes using the BVM.
14. Remove the side stream $ETCO_2$ connector and BVM and place aside.
15. Introduce the pre-lubricated Frova into the iLTS-D. Insert to a depth of 25cm for size 2.5/3 30cm for a size 4/5.
16. Pass the armored ETT over the Frova until the ETT adaptor is in line with the iLTS-D 15mm connector.
17. Inflate the ETT cuff with the appropriate amount of air.
18. Attach the side stream $ETCO_2$ onto the ETT adaptor and note the reading.
19. Attach the BVM/ ventilator tubing to the $ETCO_2$ connector and ventilate the patient as required.
20. Confirm ETT placement with waveform $ETCO_2$.



2.24 Quick trach II Cricothyroidotomy

IV

Indications

1. Cricothyrotomy is indicated in circumstances of actual loss of airway patency and/or airway obstruction, as in: Can't intubate / LTA insertion or can't ventilate using a Face mask/Naso/Oropharyngeal airway LTA/ETT.
2. Several facial or nasal injuries (that do not allow oral or nasal tracheal intubation or LTA insertion.

Contraindications

1. Conscious breathing patients.
2. Patients that you can ventilate via other means.

Adverse Effects

1. Prolonged apnoea: hypoxia.
2. Failed attempt leaving a tracheal opening.
3. Haemorrhage from the cricothyroid vessels.
4. Puncture of the posterior wall of the trachea.
5. Cuff leak, tube, occlusion.
6. Pneumothorax.
7. Hypoventilation: hypoxia, hypercarbia
8. Subcutaneous air due to improper tube or catheter positioning, along with positive ventilation.
9. Bleeding from superficial neck vessels is very common. Use direct pressure after Quicktrach II is in place.
10. Perforations of the back wall of the trachea and the esophagus from excessively deep penetration by Quicktrach II. With stopper in place, this should be an extremely rare complication.

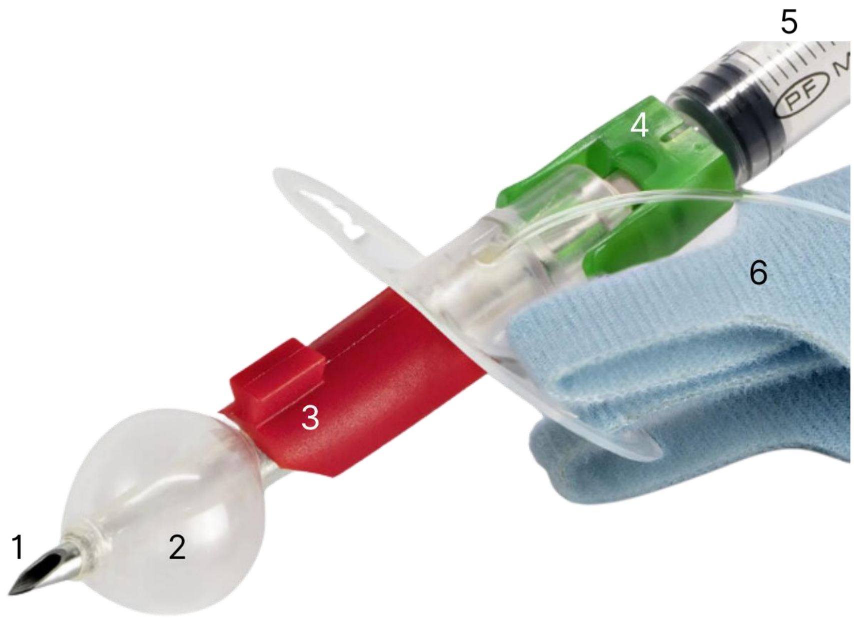
Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms and there is a risk of vomiting.

Procedure

1. Ensure that the patient's BP, ECG, SpO_2, HR, $ETCO_2$ and RR are being monitored.
2. Establish IV access.
3. Place the patient's head and neck in a neutral position or hyperextend if medical case.
4. Disinfect the area using 70% alcohol wipes or Chlorhexidine 2% with alcohol 70%.

5. Stabilize the larynx using the thumb and middle finger of your non dominant hand.
6. Palpate and identify the cricothyroid membrane.
7. Disinfect the area using 70% alcohol wipes or Chlorhexidine 2% with alcohol 70% and index finger, again.
8. Relocate insertion site with index finger to avoid inappropriate insertion.
9. Puncture the cricothyroid membrane with the bevel up.
10. Insert the Quicktrach II (cannula-over- needle attached to a syringe) in the midline at an angle of 90 degree caudally towards the carina (changing angle to 60 degrees) up to the stopper.
11. Aspirate air using the syringe to determine correct position of the cannula as the Quicktrach II is advanced through the cricothyroid membrane.
12. Should aspiration of air be impossible because of an obese neck, then remove the stopper and carefully advance the cannula including the metal needle until air is aspirated.
13. Remove the **RED** stopper from the cannula.
14. Once air is aspirated indicating that the needle is in the trachea, advance the cannula over the needle until the flange rests on the neck.
15. Withdraw the needle from the cannula and dispose of it in sharps container.
16. Inflate the cuff with 5-10 ml of air.
17. Attach the corrugated tube directly onto the Quicktrach II.
18. Attach the BVM to the corrugated tube (This should already have in place a catheter mount, filter and $ETCO_2$ monitor already attached).
19. Direct your assistant to support the tube and ventilate the patient.
20. Secure the cannula with the foam securing device provided in the kit.
21. Observe the chest for equal rise and fall. Auscultate over both axillae to confirm tracheal placement.
22. Reassess patient for any clinical deterioration.



1 **Needle**- Due to the conical needle tip, the opening into the trachea is dilated.

2 **Cannula with cuff**- Enables efficient ventilation with aspiration protection. Inner diameter is adequate for temporary ventilation with standard breathing systems.

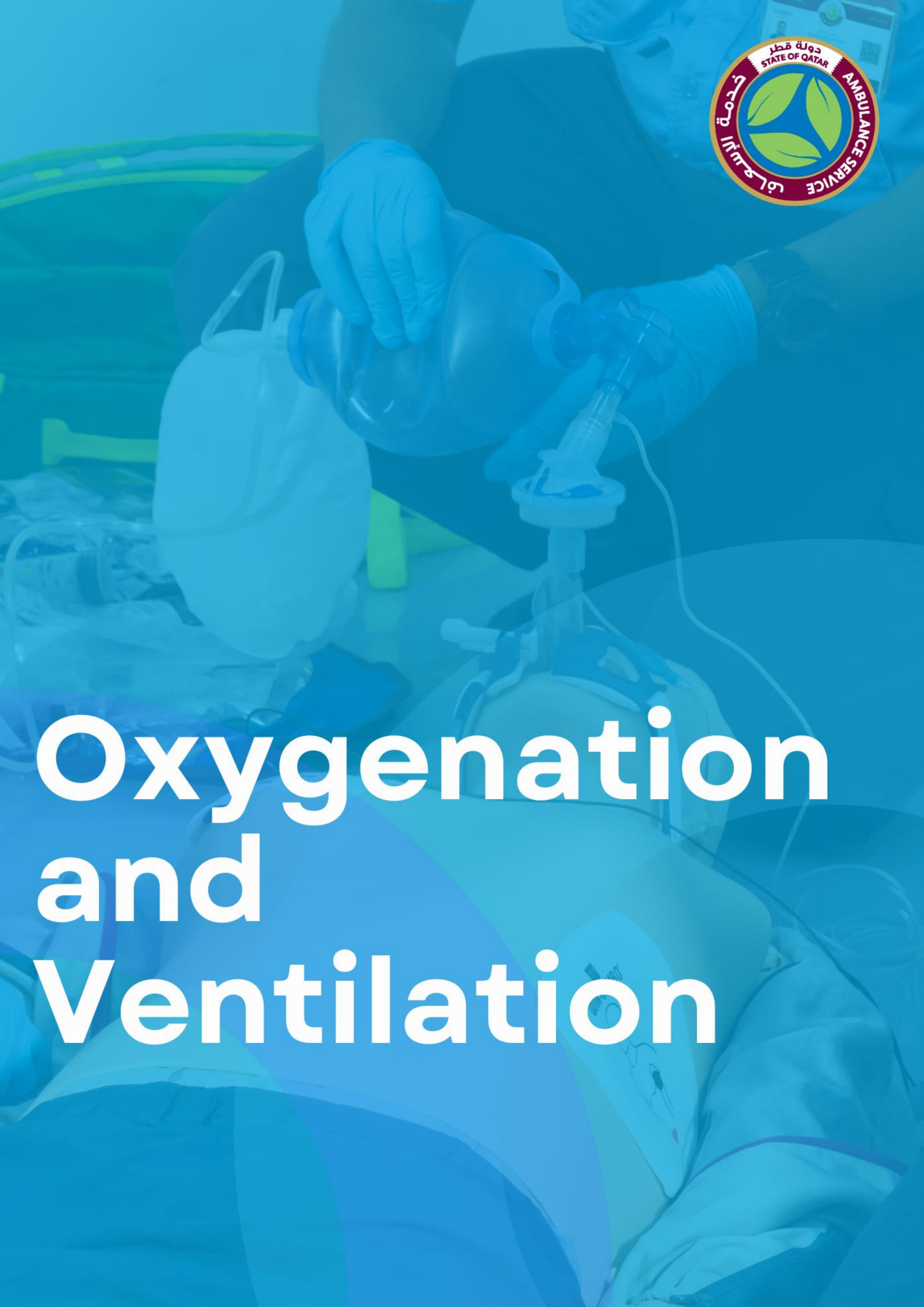
3 **Stopper**- Reduces the risk of posterior tracheal wall perforation while inserting the needle into the trachea.

4 **Safety Clip**- By pushing the cannula forward, the safety clip engages and the needle tip disappears into the cannula to prevent trauma.

5 **Syringe**- For checking the location in the trachea.

6 **Necktape**- For safe fixation.

Quicktrach | VBM Medizintechnik GmbH. (n.d.). https://www.vbm-medical.de/en/products/airway-management/cricothyrotomy/quicktrach/



Oxygenation and Ventilation

3.1 Oxygen therapy: Nasal Cannula

Indications

1. Patient with hypoxia/hypoxemia (SpO_2 less than 94%) who needs low flow oxygen
2. Pre-oxygenation for RSI and DSI.

Contraindications

1. Patients with persistent nasal blockage from vomitus, blood or mucus.
2. Patients with facial injuries that would preclude the use of a cannula.
3. Patient who will not leave the nasal cannula in place.

Adverse Effects

1. Nil in this setting

Standard precautions

Procedure

1. Prepare the oxygen cylinder, pre-pack A6/ A7 and LifePak 15.
2. Ensure that the patient's BP, ECG, SpO_2, HR and RR is being monitored.
3. Ensure that the airway has been opened (if required) using head tilt-chin lift (medical patient) or jaw thrust maneuvers (trauma patient).
4. Ensure that there are no nasal obstructions e.g., vomitus, mucus, blood. Suction and clear as required.
5. Assemble the regulator to the oxygen cylinder, open the cylinder and check for leaks. Check and verbalize tank pressure.
6. Attach the tubing of the nasal cannula to oxygen cylinder.
7. Adjust flow meter to between 2 to 4 LPM. Higher flow rates (up to 15 LPM) may be used during the pre-oxygenation phase of RSI (CCP). This is based on patient tolerance.
8. Apply and adjust the nasal cannula to the patient's face.
9. Reassess the patient and record vital signs after 5 mins. Should the patient's condition not improve consider the use

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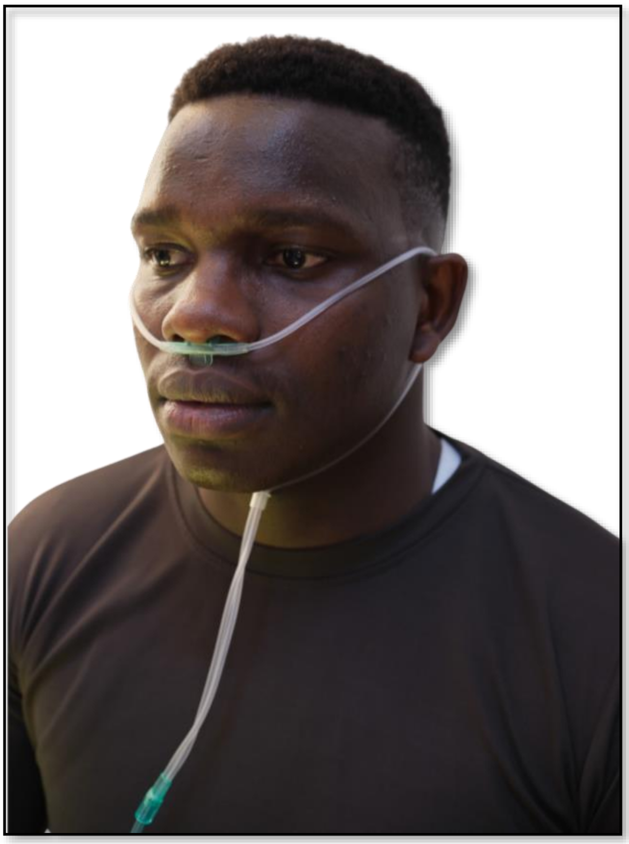
Placement of a nasal cannula on a patient



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3.2 Oxygen therapy: Simple face mask

Indications

1. Patient with Hypoxia/hypoxemia (SpO_2 less than 94%) who needs mild flow oxygen

Contraindications

1. Nil in this setting

Adverse Effects

1. Nil in this setting

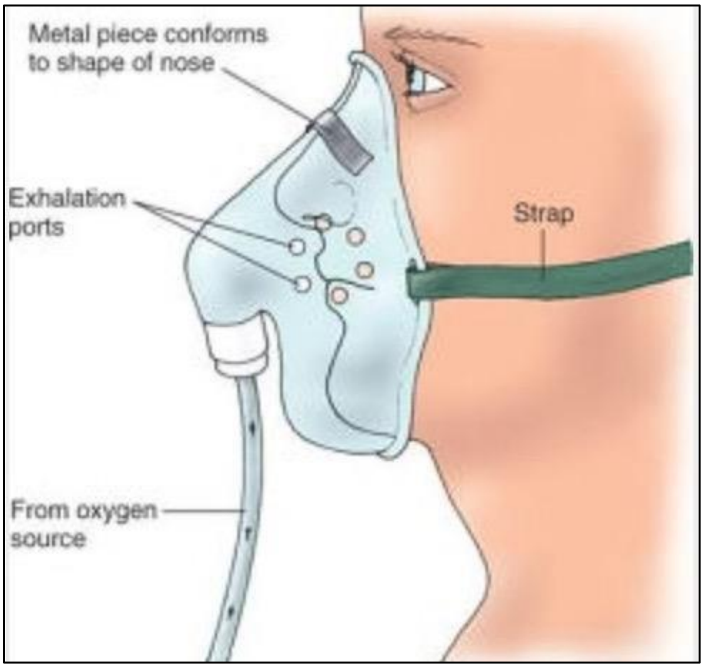
Standard precautions

Procedure

1. Prepare the oxygen cylinder, simple face mask and LifePak 15.
2. Place the patient in their most comfortable position (semi-fowlers or full fowlers).
3. Ensure that the patient's BP, ECG, SpO_2, HR and RR are being monitored.
4. Ensure that the airway has been opened (if required) using head tilt-chin lift (medical patient) or jaw thrust maneuvers (trauma patient).
5. Assemble the regulator to the oxygen cylinder, open the cylinder and check for leaks.
6. Check and verbalise tank pressure.
7. Attach the simple face mask tubing to oxygen cylinder.
8. Adjust flow meter to between 6 to 10 LPM.
9. Apply and adjust the mask to the patient's face.
10. Reassess the patient and record vital signs after 5 mins. Should the patient's condition not improve consider the use of a NRB or BVM.

Simple face mask

Placement of the simple face mask on a patient



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3.3 Oxygen therapy: Non-rebreather mask

Indications

1. Patient with hypoxia/hypoxemia (SpO_2 less than 90%) who needs high flow oxygen.
2. Pre-oxygenation for RSI and DSI.

Contraindications

1. Nil contra-indications in this setting.

Adverse Effects

1. Nil in this setting.

Standard precautions

Procedure

1. Prepare the oxygen cylinder, pre-pack A6/ A7 and LifePak 15.
2. Place the patient in their most comfortable position (semi-fowlers or full fowlers).
3. Ensure that the patient's BP, ECG, SpO_2, HR and RR are being monitored.
4. Ensure that the airway has been opened (if required) using head tilt-chin lift (Medical patient) or jaw thrust maneuvers (Trauma patient).
5. Assemble the regulator to the oxygen cylinder, open the cylinder and check for leaks.
6. Check and verbalise tank pressure.
7. Attach NRB mask tubing to oxygen cylinder and refill reservoir. Ensure that the reservoir fills.
8. Adjust flow meter to between 10-15 LPM.
9. Apply and adjust the mask to the patient's face.
10. Reassess the patient and record vital signs after 5 mins. Should the patient's condition not improve consider the use of a BVM.

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Placement of a non-rebreather mask on the patient

3.4 Nebulization: Mask

II III IV

Indications

1. Difficulty in breathing with the presence of wheezes on chest auscultation and adequate respiratory effort. Patients who commonly present with lower airway obstruction include asthma, COPD, and anaphylactic reactions.

Contraindications

1. Allergy to the respective nebulization medication.

Adverse Effects

1. Adverse effects experienced by the patient are dependent on the nebulization solution being utilized.

Standard precautions

LIFEPAK 1. Wear an N95 mask if the patient exhibits respiratory symptoms.

Procedure

1. Prepare the oxygen cylinder, pre-pack A8 and LIFEPAK15.
2. Ensure that the patient's BP, ECG, SpO_2, HR and RR are being monitored.
3. Place the patient in their most comfortable condition (Semi-fowlers to Full fowlers position preferred).
4. Select correct medication and check its expiry date. Rule out contraindications.
5. Place correct medication into the nebulizer chamber.
6. Assemble nebulizer parts and connect the nebulizer to oxygen supply.
7. Set the oxygen flow rate at 6 LPM and observe for atomization (mist).
8. Apply the mask of the nebulizer to the patient's face.
9. Change to the mouthpiece if the patient does not tolerate the mask.
10. Reassess the patient every 5 minutes.
11. Continue nebulization until the vapour stops.
12. Repeated nebulization may be required with certain categories of patients.
13. Should the patient's condition not improve, AP crew inform NCC and request CCP assistance.

Placement of nebulization mask on the patient

Nebulization mouthpiece assembly

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3.5 Nebulization: Inline

Indications

1. Severe/ life threatening respiratory distress, with wheezing, Stridor, or both with inadequate respiratory effort.
2. Patients who are post-cardiac arrest (ROSC) with asthma or anaphylaxis immediately prior to cardiac arrest.

Contraindications

1. Nil in this setting.

Adverse Effects

1. Adverse events experienced by the patient are dependent on the nebulization solution being utilized.

Standard precautions

1. Wear an N95 mask if the patient exhibits respiratory symptoms.

Procedure

1. Prepare the oxygen cylinder, A8, Airway Roll (Adult Airway Roll/ Paediatric Airway Roll for CCP), Mechanical Ventilator (CCP) and LifePak 15.
2. Ensure that the patient's BP, ECG, SpO_2, $ETCO_2$, HR and RR is being monitored.
3. Place the patient in their most comfortable position. If the patient is unconscious, place the patient supine.
4. Select correct medication and check its expiry date. Rule out contraindications.
5. Select T-piece nebuliser and attach the catheter mount to the corresponding limb of the T-piece.
6. Attach the BVM to other limb of T-piece and attach the catheter mount to a BVM face mask/ LTA/ ETT.
7. Place correct medication into the nebuliser chamber.
8. Assemble nebulizer parts and connect the nebulizer to the lower limb of the T- piece.
9. Set the oxygen flow rate at 6 LPM and observe for atomisation. Increase O_2flowrate to achieve mist.
10. Assist or deliver ventilation and provide in-line nebulisation.
11. When a second oxygen source becomes available (in the ambulance) attach the BVM to oxygen at a flowrate of 10-15 LPM.
12. Continue nebulization until the vapour stops.
13. Repeated nebulization may be required with certain categories of patients.
14. Should the patient's condition not improve, AP crew inform NCC and request CCP

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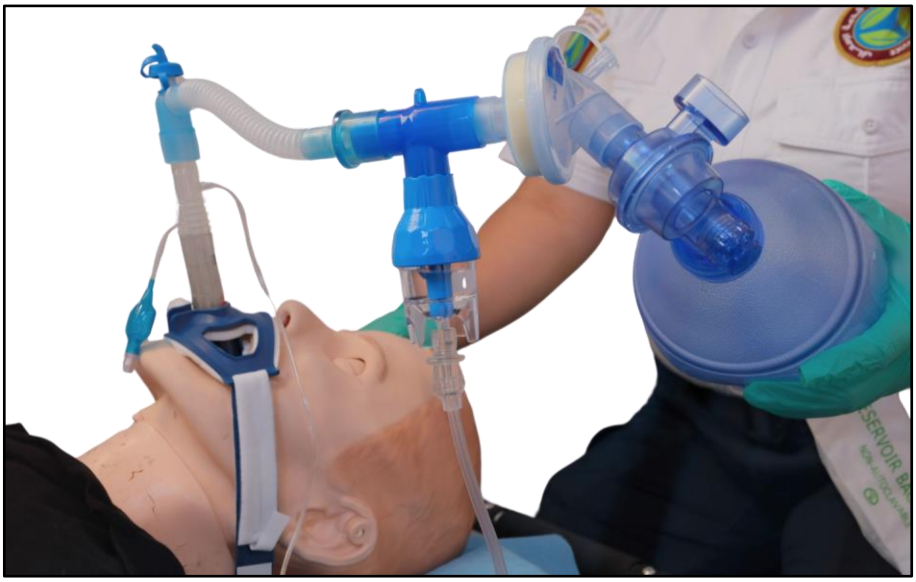
assistance.

Inline nebulization assembly



3.6 Oxygen therapy: Pre-oxygenation for intubation

Indications

1. Pre-oxygenation is indicated for orotracheal intubation using the Frova, Glidescope, Rapid Sequence Intubation and Delayed Sequence Intubation.

Contraindications

1. Nil contra-indications in this setting.

Adverse Effects

1. Nil in this setting.

Standard precautions

1. Wear an N95 mask if the patient exhibits respiratory symptoms.

Procedure

1. Prepare yourself mentally. Identify risks for potential difficult airway using the mnemonic LEON (**L**ook externally, **E**valuate, **O**bstructions and **N**eck mobility).
2. Position the patient appropriately.
   a. If medical patient lying on the floor/ bed, raise the head 10cm and extend in the sniffing position to line the external auditory meatus with the sternal notch. The face will then be parallel to the ceiling.
   b. If medical patient is lying on the ambulance stretcher, elevate the head section of the stretch by 20 degrees. This will line the external auditory meatus with the sternal notch.
   c. If trauma patient and is secured on the scoop stretcher (SMR), elevate the head section of the scoop stretcher by 30 degrees (reverse Trendelenburg position). This will line the external auditory meatus with the sternal notch.
   d. Obese patients should be placed in the **RAMP** position. This will line the external auditory meatus with the sternal notch.
3. Place the nasal $ETCO_2$ and connect to O_2 supply. Start O_2 delivery at 6 LPM (or as much as can be tolerated by the patient) aiming for 15 LPM.
4. Add an NRB at an O2 flowrate of 15 LPM.
5. Allow the patient 3 -5 minutes of tidal volume breathing.
6. If the patient is taking inadequate tidal volume breathes, leave the nasal $ETCO_2$ in place and remove the NRB. Optimize the patient's position and change to the BVM at 15 LPM to achieve an SpO_2 > 95%. Consider using non-invasive CPAP/BIPAP with the mechanical ventilator/ commercial CPAP device for pre- oxygenation. Allow the patient 3 -5 minutes of tidal volume breathing.

PASSIVE APNEIC PERIOD (OXYGENATION AFTER PARALYSIS – BEFORE INTUBATION)

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7. Remove the NRB mask from the patient. Maintain or increase the O_2 flowrate via nasal $ETCO_2$ to LPM.
8. Performance jaw thrust to maintain pharyngeal patency.
9. If the patient is a high risk (required CPAP/BIPAP for pre-oxygenation), then consider leaving the CPAP/BIPAP in place during the apneic period or provide ventilation with the BVM and PEEP valve attached if available.
10. For BIPAP: Expiratory pressure (PEEP) at least 5cmH20, Inspiratory pressure of at least 10cmH2O, Respiratory rate of at least 10bpm and an Fi02 of 100%.
11. Wait for 45-60 seconds for the medication to take effect.

INTUBATION PERIOD (PASSIVE APNEIC OXYGENATION AFTER PARALYSIS DURING INTUBATION)

12. Maintain the nasal $ETCO_2$ in the patient's nares at 15 LPM. Remember **NODESAT** (**N**asal **O**xygen **D**uring **E**fforts **S**ecuring **A** **T**ube).

INTUBATED PERIOD (OXYGENATION AFTER INTUBATION)

13. Connect the mechanical ventilator/ BVM to the ETT as per CPG.
14. Remove the nasal $ETCO_2$.

3.7 BVM Ventilation: One-person-technique

Indications

1. Patient with acute signs of respiratory failure associated with inadequate tidal volume breaths.
2. Pre-oxygenation for RSII or DSI in patients with inadequate tidal volume breathing.
3. Re-oxygenation during RSII
4. Assessing placement of ETT.
5. Place on standby in all patients who are mechanically ventilated.

Contraindications

1. BVM ventilation is contra-indicated in patients with adequate oxygenation and ventilation.
2. Foreign body upper airway obstruction, unless the patient becomes unresponsive.

Adverse Effects

1. Apprehension/ anxiety in the conscious patient.
2. Gastric inflation and passive regurgitation leading to aspiration.
3. Barotrauma
4. Undesirable cardiovascular effects such as hypotension.

Standard precautions

Procedure

1. Ensure that the patient's BP, ECG, SpO_2, HR, $ETCO_2$ (CCP) and RR are being monitored.
2. Prepare Airway Roll (AP), Adult/Paediatric Airway Roll (CCP), suction unit with A5 and oxygen cylinder.
3. If the patient is conscious, reassure them and explain what is happening. Remember that resistance or combativeness may be due to hypoxia.
4. Open oxygen source and ensure pressure is adequate.
5. Connect one end of oxygen tubing to flow meter of O_2 cylinder and the other to the BVM.
6. Set flow meter to 15 LPM (reservoir must fill).
7. Connect the BVM with ETCO2 monitoring and HME filter.
8. Select an appropriate mask size and attach it to BVM.
9. Place an unconscious patient in a supine position or a conscious patient in the semi-fowlers position.
10. Open the airway by head tilt/ chin lift (Medical patient) or jaw thrust (Trauma

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patient).

11. Suction airway, if required.
12. Measure and insert an OPA if no gag reflex. If gag reflex present, consider NPA insertion.
13. The patient attendant should be positioned behind the patient's head.
14. Position the mask on the patient and using the "E-C" technique, seal the mask and maintain an open airway. The technique should bring the jaw into the mask rather than pushing the mask down onto the face.
15. Gently compress the squeeze bag of the BVM to deliver a breath over 1 second, compressing until chest rise is just visible.
16. In the spontaneously breathing patient, the breath should be delivered during the patient's inhalation phase.
17. Release the pressure on the squeeze bag to allow passive exhalation and re-expansion of the bag.
18. Give one breath every 6 seconds (10 per minute) for adults/ 1 breath every 8 seconds if airway is secured. (Follow the indicator light on the LUCAS device if it is in place).

19. During ventilation, check for:
    a. Signs of cyanosis
    b. Adequacy of ventilation by observing chest rise.
    c. Airway pressure based on the effort required to squeeze the bag.
    d. Correct functioning of all valves and tubing.
    e. Continuous supply of oxygen to the resuscitator and inflation of the reservoir bag.

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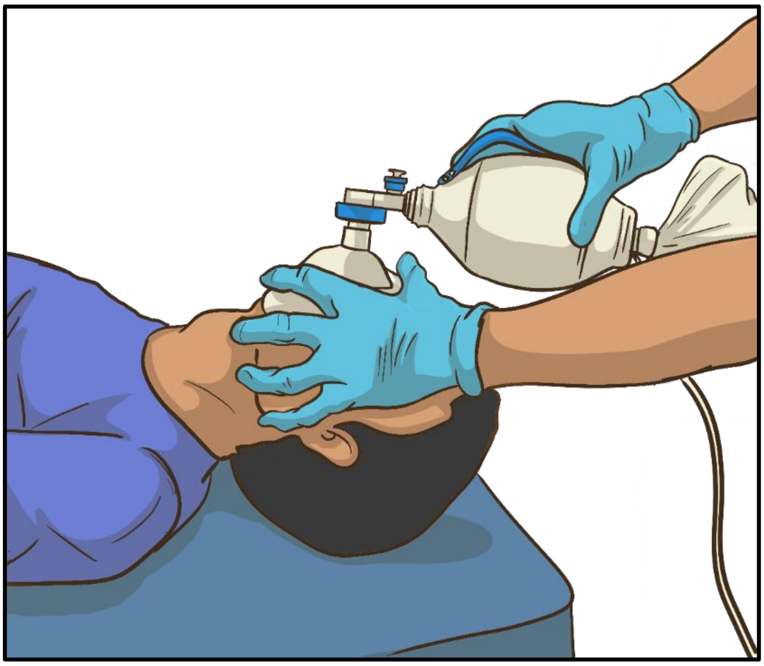
Bag valve mask ventilation using the one-person-technique



3.8 BVM Ventilation: Two-person-technique

Indications

1. Patient with acute signs of respiratory failure associated with inadequate tidal volume breaths.
2. Pre-oxygenation for RSII or DSI in patients with inadequate tidal volume breathing.
3. Re-oxygenation during RSII
4. Assessing placement of ETT.
5. Place on standby in all patients who are mechanically ventilated.

Contraindications

1. BVM ventilation is contra-indicated in patients with adequate oxygenation and ventilation.
2. Foreign body upper airway obstruction, unless the patient becomes unconscious.

Adverse Effects

1. Apprehension/anxiety in the conscious patient.
2. Gastric inflation and passive regurgitation leading to aspiration.
3. Barotrauma
4. Undesirable cardiovascular effects such as hypotension

Standard precautions

Procedure

1. Ensure that the patient's BP, ECG, SpO_2, HR, $ETCO_2$ (CCP) and RR are being monitored.
2. Prepare an Airway Roll (AP), Adult/ Paediatric Airway Roll (CCP), suction unit with A5 and oxygen cylinder.
3. If the patient is conscious, reassure them and explain what is happening. Remember that resistance or combativeness may be due to hypoxia.
4. Open oxygen source and ensure pressure is adequate.
5. Connect one end of oxygen tubing to flow meter of O_2 cylinder and the other to the BVM.
6. Connect the BVM with ETCO2 monitoring and HME filter
7. Set flow meter to 15 LPM.
8. Select an appropriate mask size and attach it to BVM.
9. Place an unconscious patient in a supine position or a conscious patient in the semi-fowlers position.
10. Open airway by head tilt/ chin lift (Medical patient) or jaw thrust (Trauma patient).

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11. Suction airway, if required.
12. Measure and insert an OPA if no gag reflex. If gag reflex present, consider NPA insertion.
13. Patient attendant 1, who is positioned behind the patient's head, uses a two-handed thumbs down technique to seal the mask to the face. The technique should bring the jaw into the mask rather the pushing the mask down onto the face.
14. Patient attendant 2, who is positioned alongside the patient's head, gently compresses the squeeze bag to deliver a breath over 1 second, compressing until chest rise is just visible.
15. In the spontaneously breathing patient, the breath should be delivered during the patient's inhalation phase.
16. Release the pressure on the squeeze bag to allow passive exhalation and re-expansion of the bag.
17. Give one breath every 6 seconds (10 per minute) for adults/ 1 breath every 8 seconds if airway is secured. (Follow the indicator light on the LUCAS device if it is in place).
18. During ventilation, check for:
    - Signs of cyanosis
    - Adequacy of ventilation by observing chest rise.
    - Airway pressure based on the effort required to squeeze the bag.
    - Correct functioning of all valves and tubing.
    - Continuous supply of oxygen to the resuscitator and inflation of the reservoir bag.

Bag valve mask ventilation using the two-person-technique

3.9 BVM Ventilation: Stoma/Tracheostomy

Indications

1. Patient with a stoma/ tracheostomy presenting with acute signs of respiratory failure associated with inadequate tidal volume breaths.

Contraindications

1. BVM ventilation is contra-indicated in patients with adequate oxygenation and ventilation.

Adverse Effects

1. Apprehension/anxiety in the conscious patient.
2. Gastric inflation and passive regurgitation resulting in aspiration.
3. Barotrauma
4. Undesirable cardiovascular effects such as hypotension
5. Subcutaneous emphysema if the tracheostomy tube is displaced

Standard precautions

Procedure

1. Call for CCP assistance.
2. Apply BP, ECG, SpO_2, HR, RR and ETCO2 monitoring.
3. Prepare an Airway Roll (AP), Adult/Paediatric Airway Roll (CCP), suction unit with A5 and oxygen cylinder.
4. If the patient is conscious, reassure them and explain what is happening. Remember that resistance or combativeness may be due to hypoxia.
5. Open oxygen source and ensure pressure is adequate.
6. Connect one end of oxygen tubing to flow meter of O_2 cylinder and the other to the BVM.
7. Connect the BVM with ETCO2 monitoring and HME filter
8. Set flow meter to 15 LPM.
9. Select an appropriate mask size and attach it to BVM.
10. Place patient in a supine position.
11. Suction the stoma/ tracheostomy, if required.
12. Position the paediatric mask (to generate a better seal) on the stoma and close the mouth and nose if required
13. Gently compress the squeeze bag to deliver a breath over 1 second, compressing until chest rise is just visible.
15. In the spontaneously breathing patient, the breath should be delivered during the patient's inhalation phase.

16. Release the pressure on the squeeze bag to allow passive exhalation and re-expansion of the bag.
17. Give one breath every 6 seconds (10 breaths per minute)
18. If unsuccessful in ventilating through the stoma, seal the stoma with a gloved hand and ventilate through the mouth and nose with BVM.
19. If ventilation through tracheostomy tube is required, ensure its patency by inserting a suction catheter.
20. If unsuccessful in ventilating through the tracheostomy tube, deflate the tube cuff and ventilate through the mouth and nose with BVM.
21. if all previous steps unsuccessful remove the tracheostomy tube and start ventilating via the stoma (BVM connected to neonatal mask) or via mouth and nose (and seal the stoma)
22. If the patient has laryngectomy, ventilation should be only via stoma / laryngectomy tube.
15. During ventilation, check for:
    a. Signs of cyanosis
    b. Adequacy of ventilation by observing chest rise.
    c. Airway pressure based on the effort required to squeeze the bag.
    d. Correct functioning of all valves and tubing
    e. Continuous supply of oxygen to the resuscitator and inflation of the reservoir bag.

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3.10 BVM Ventilation: PEEP valve

Indications

1. Patient with acute signs of respiratory failure associated with acute pulmonary oedema and submersion.
2. Pre-oxygenation for RSI.

Contraindications

1. BVM ventilation is contra-indicated in patients with adequate oxygenation and ventilation.
2. Foreign body upper airway obstruction, unless the patient becomes unconscious.

Adverse Effects

1. Apprehension/anxiety in the conscious patient.
2. Gastric inflation and passive regurgitation resulting in aspiration.
3. Barotrauma
4. Undesirable cardiovascular effects such as hypotension

Standard precautions

Procedure

1. Ensure that the patient's BP, ECG, SpO_2, HR, $ETCO_2$ and RR are being monitored.
2. Prepare an Airway Roll (AP), Adult/ Paediatric Airway Roll (CCP), suction unit with A5, PEEP valve and oxygen cylinder.
3. If the patient is conscious, reassure them and explain what is happening. Remember that resistance or combativeness may be due to hypoxia.
4. Open oxygen source and ensure pressure is adequate.
5. Connect one end of oxygen tubing to flow meter of O_2 cylinder and the other to the BVM.
6. Set the O_2 flow meter to 15 LPM.
7. Connect the BVM with ETCO2 monitoring and HME filter.
8. Select an appropriate mask size and attach it to BVM.
9. Place patient in their most comfortable position. For APO place the patient in the semi-fowlers to full fowlers position.
10. Open airway by head tilt/chin lift (Medical patient) or jaw thrust (Trauma patient).
11. Suction airway, if required
12. Measure and insert an OPA if there is no gag reflex.
13. Connect the PEEP valve to the expiration port of the BVM. Set PEEP at 5 cmH_2O. Increase to 10 cmH_2O, as required.

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14. BVM ventilation with the PEEP valve must be performed using the two- person technique.
15. Patient attendant 1, who is positioned behind the patient's head, uses a two-handed thumbs down technique to seal the mask to the face. The technique should bring the jaw into the mask rather the pushing the mask down onto the face.
16. Patient attendant 2, who is positioned alongside the patient's head, gently compresses the squeeze bag to deliver a breath over 1 second, compressing until chest rise is just visible.
17. Gently compress the squeeze bag to deliver a breath over 1 second, compressing until chest rise is just visible.
18. In the spontaneously breathing patient, the breath should be delivered during the patient's inhalation phase.
19. Release the pressure on the squeeze bag to allow passive exhalation and re- expansion of the bag.
20. Give one breath every 6 seconds (10 per minute) for adults/ 1 breath every 6 seconds if airway is secured. (Follow the indicator light on the LUCAS device if it is in place).
21. During ventilation, check for:
    - Signs of cyanosis and adequacy of ventilation by observing chest rise.
    - Airway pressure based on the effort required to squeeze the bag.
    - Correct functioning of all valves and tubing
    - Continuous supply of oxygen to the resuscitator and inflation of the reservoir bag.

Bag valve mask with a PEEP valve connected

3.11 Manual Ventilation with BVM to ETT/LTA

Indications

1. Patient with an ETT/ LTA in place who present with acute signs of respiratory failure associated with inadequate tidal volume breathing.
2. Patient in cardiac arrest who have an ETT/ LTA in place.

Contraindications

1. BVM ventilation is contra-indicated in patients with adequate oxygenation and ventilation.
2. Foreign body upper airway obstruction unless the patient becomes unconscious.

Adverse Effects

1. Gastric inflation if cuff inadequately inflated.
2. Barotrauma
3. Undesirable cardiovascular effects such as hypotension

Standard precautions

1. Wear an N95 mask if the patient exhibits respiratory symptoms.

Procedure

1. Ensure that the patient’s BP, ECG, SpO_2, $ETCO_2$, HR and RR are being monitored.
2. Prepare an Airway Roll (AP), Adult/ Paediatric Airway Roll (CCP), suction unit with A5 and oxygen cylinder.
3. Remember that resistance or combativeness may be due to hypoxia.
4. Ensure that the ETT/ LTA is appropriately positioned and strapped.
5. Open oxygen source and ensure pressure is adequate.
6. Connect one end of oxygen tubing to flow meter of O_2 cylinder and the other to the BVM.
7. Set the O_2 flow meter to 15 LPM.
8. Connect the BVM with ETCO2 monitoring and HME filter to the ETT/ LTA.
9. Gently compress the squeeze bag to deliver a breath over 1 second, compressing until chest rise is just visible.
10. In the spontaneously breathing patient, the breath should be delivered during the patient’s inhalation phase.
11. Release the pressure on the squeeze bag to allow passive exhalation and re- expansion of the bag.
12. Give one breath every 6 seconds (10 per minute) for adults/ 1 breath every 8 seconds if airway is secured. (Follow the indicator light on the LUCAS device if it is in place)
13. During ventilation, check for:

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a. Signs of cyanosis
b. Adequacy of ventilation by observing chest rise.
c. Airway pressure based on the effort required to squeeze the bag.
d. correct functioning of all valves and tubing
e. Continuous supply of oxygen to the resuscitator and inflation of the reservoir bag.

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3.12 Mechanical ventilation using the Oxylog 3000 plus: CMV mode

Indications

1. Any patient with an ETT/ LTA in place who requires ventilation and is not actively receiving chest compressions.
2. Any patient with a cricothyrotomy in place who requires ventilation and is not actively receiving chest compressions.
3. Any patient with a pre-existing advanced airway in place (tracheostomy/ ETT) who requires transport and continuous mechanical ventilation.

Contraindications

1. Any patient with ongoing chest compressions.
2. Failure of the ventilator to pass the device check performed prior to placing on the patient.
3. Any patient who has a pre-existing advanced airway in place (tracheostomy, ETT) who has their own transport ventilator that will accompany the patient with an appropriately trained medical professional who is trained and approved to operate the device

Adverse Effects

1. Hypo/ hyperventilation.
2. Barotrauma resulting in pneumothorax.
3. Diaphragm atrophy.
4. Increased thoracic pressure resulting in reduced cardiac output and hypotension.
5. Oxygen toxicity.

Standard precautions

Procedure

1. Ensure that the patient's BP, ECG, SpO_2, $ETCO_2$, HR and RR are being monitored.
2. Prepare pre-pack Adult/ Paediatric Airway Roll, C3/ C4. Connect the appropriate breathing circuit, ensuring that the flow measurement lines are correctly matched to the nozzles on the ventilator.
3. Ensure sufficient pressure level of O_2.
4. Power device ON and connect the test lung to the **NEW** circuit.
5. Run a device and circuit check, prior to intubation of patient, to ensure proper working of the ventilator. Then disconnect the test lung.
6. Ensure the device is in the VC-CMV mode.

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7. Attach the in-line $ETCO_2$ line to the circuit and connect to the Lifepak 15 prior to attaching circuit to the patient.
8. Attach HME filter to the circuit and under Settings select HME "ON" (found on screen 4/4 under settings).
9. Set the appropriate Tidal Volume, Respiratory Rate, FiO_2 and Pmax using the manual rotary knob based on the patient's ideal body weight and presenting condition as per CPG 3.4.
10. Set the FiO_2 to 100% using the manual rotary knob.
11. Attach the circuit to the patient.
12. Ensure the patient is adequately sedated and paralyzed as necessary.
13. Reassess device to ensure proper patient appropriate settings are being used and adjust as necessary.
14. Continuously assess the patients' air entry looking for signs of barotrauma from mechanical ventilation.
15. Document the mechanical ventilator related settings in the ePCR.
16. Clean and disinfect ventilator machine using HMC-approved disinfectant following manufacturer recommendations

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Recommended Adult Ventilator settings for common conditions

RECOMMENDED ADULT VENTILATOR SETTINGS FOR COMMON CONDITIONS			
SETTINGS	NORMAL LUNGS	ASTHMA/COPD	APO/ARDS
Vent Mode	IPPV	IPPV	IPPV or NIV
Respiratory Rate	12-14 breaths/min*	5-10 breaths/min*	12-16 breaths/min*
Tidal Volume	6-8 ml/kg**	4-6 ml/kg	6-8 ml/kg
Pmax	35-40 mbar	40-45 mbar	40-45 mbar
I:E Ratio	1:2	1:4-1:5	1:1-1:2
PEEP	3-5 mbar	0-5 mbar	Up to 10 mbar
Pressure Support	N/A	N/A	Up to 10 mbar
Trigger	3 L/min	3 L/min	3 L/min
FiO_2	1.0**	1.0**	1.0**

*Respiratory rate should be adjusted according to age for paediatric patients.
** Tidal volume can be increased up to 10 ml/kg if required.
***FiO_2 generally does not need to be adjusted from 1.0 in the acute phase. During interfacility transfers, FiO_2 should be matched to the existing ventilator settings and adjusted as required.

3.13 Mechanical ventilation using the Oxylog 3000 plus: CPAP mode

Indications

1. Medical patients complaining of moderate to severe respiratory distress (as per CPG 1.5) meeting ALL of the following criteria:
   a. Has the ability to maintain an open airway.
   b. Has signs and symptoms of acute pulmonary edema (CPG 4.5).
   c. Has a systolic blood pressure above 100 mmHg.
   d. Is able to fit the CPAP mask.
2. CPAP may be considered in patients with acute hypoxemic respiratory failure due to pneumonia or acute hypercapnic respiratory failure due to COPD as a bridge to hospital to avoid endotracheal intubation.
3. Pre-oxygenation for RSI and DSI.

Contraindications

1. Respiratory or Cardiac Arrest
2. Hemodynamic instability
3. Extreme Agitation – Non-cooperative patient.
4. Acute or imminent airway obstruction (e.g. acute angioedema, inhalation burns)
5. Increased risk of regurgitation and aspiration
6. Presence of a tracheostomy
7. Inability to maintain a mask seal
8. Suspected pneumothorax

Adverse Effects

1. Hypo/ hyperventilation.
2. Barotrauma resulting in pneumothorax.
3. Diaphragm atrophy.
4. Increased thoracic pressure resulting in reduced cardiac output and hypotension.
5. Oxygen toxicity.

Standard precautions

Procedure

1. Ensure that the patient's BP, ECG, SpO2, ETCO2, HR and RR are being monitored.
2. Attach the nasal ETCO2 to the patient, Lifepak 15 and oxygen supply.
3. Administer any indicated oral/sublingual medication prior to placing the CPAP mask on the patient's face.

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4. Prepare suction unit and pre-pack A5.
5. Prepare a Non-invasive Mask (NIV)
6. Ensure sufficient pressure level of O2.
7. Run a device and circuit check, prior to placing the patient on the ventilator to ensure proper working of the ventilator. Connect the appropriate breathing circuit, ensuring that the flow measurement lines are correctly matched to the nozzles on the ventilator.
8. After successful completion of the device check Exit the menu and return to ventilation mode.
9. Select CPAP mode with NIV set to ON.
10. Select an appropriately tight-fitting face mask and connect to breathing circuit. Ensure oxygen is flowing prior to putting mask on patient.
11. Evaluate for excessive leakage and adjust mask if present.
12. Set FiO2 to 1.0 – reduce if needed to a target SpO2 > 94%.
13. Set PEEP to 5mbar and adjust up to 10mbar if patient is comfortable and is needed.
14. Trigger limit set as low as possible (3).
15. Start ventilation and hold the mask on the patient's face. Allow the patient to become comfortable with the mask over their face. Do not rush the placement of the mask over the patient's face as this may increase agitation and decrease success.
16. Add pressure support if the patient requires further support. Start by setting Psupp to 5 mbar. Increase in increments of 5 mbar. Do not exceed 20 mbar of pressure support.
17. If the patient becomes uncomfortable, decrease pressure support by 5 mbar.
18. Increase pressure support to a maximum combined PEEP + pressure support of 20 mbar.
19. Continuously monitor ETCO2.
20. Perform continual reassessment of patient’s condition.
21. Fill in appropriate ePCR documentation requirements.
22. Clean and disinfect ventilator machine using HMC-approved disinfectant following manufacturer recommendations.

3.14 Mechanical ventilation using the Oxylog 3000 plus: BIPAP mode

IV

Indications

1. Medical patients complaining of moderate to severe respiratory distress (as per CPG 1.5) meeting ALL of the following criteria:

a. Has the ability to maintain an open airway.

b. Has signs and symptoms of acute pulmonary edema (CPG 4.5).

c. Has a systolic blood pressure above 100 mmHg.

d. Is able to fit the CPAP mask.

2. BIPAP may be considered in patients with acute hypoxemic respiratory failure due to pneumonia or acute hypercapnic respiratory failure due to COPD as a bridge to hospital to avoid endotracheal intubation.

3. Pre-oxygenation for RSI and DSI.

Contraindications

1. Respiratory or Cardiac Arrest
2. Hemodynamic instability
3. Extreme Agitation – Non-cooperative patient.
4. Acute or imminent airway obstruction (e.g. acute angioedema, inhalation burns)
5. Increased risk of regurgitation and aspiration
6. Presence of a tracheostomy
7. Inability to maintain a mask seal
8. Suspected pneumothorax

Adverse Effects

1.Hypo/ hyperventilation.

2. Barotrauma resulting in pneumothorax.

3. Diaphragm atrophy.

4. Increased thoracic pressure resulting in reduced cardiac output and hypotension.

5. Oxygen toxicity.

Standard precautions

Procedure

1. Ensure that the patient’s BP, ECG, SpO2, ETCO2, HR and RR are being monitored.
2. Attach the nasal ETCO2 to the patient, Lifepak 15 and oxygen supply.
3. Administer any indicated oral/sublingual medication prior to placing the BIPAP mask on the patient’s face.

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4. Prepare suction unit and pre-pack A5.
5. Prepare a Non-invasive Mask (NIV)
6. Ensure sufficient pressure level of O2.
7. Run a device and circuit check, prior to placing the patient on the ventilator to ensure proper working of the ventilator. Connect the appropriate breathing circuit, ensuring that the flow measurement lines are correctly matched to the nozzles on the ventilator.
8. After successful completion of the device check return to the ventilation screen and select PC/BIPAP mode with NIV and HME (filter) set to ON.
9. Select an appropriately tight-fitting face mask and connect to breathing circuit. Ensure oxygen is flowing prior to putting mask on patient.
10. Evaluate for excessive leakage and adjust mask if present.
11. SET IPAP(PinsP): 10-15mbar (Adjust as required. Maximum pressure 25mbar).
12. Set EPAP (PEEP): 5mbar and adjust up to 10mbar if patient is comfortable and is needed.
13. Set FiO2 to 1.0 – reduce if needed to a target SpO2 > 94%.
14. Set Rate: 6-10bpm (set rate low, patient should be breathing spontaneously- this is a backup).
15. Set I:E ratio to 1:2 (adjust as required)
16. Set I-Time: 1.0 seconds (usually a non-issue since rate is low).
17. Set Rise Time: 0.4seconds (only applies to timed breath)
18. Add pressure support if the patient requires further support. Start by setting Psupp between 3- 5mbar. Increase in increments of 2mbar. Do not exceed 20mbar of pressure support.
19. Trigger limit set as low as possible (3).
20. Start ventilation and hold the mask on the patient's face. Allow the patient to become comfortable with the mask over their face. Do not rush the placement of the mask over the patient's face as this may increase agitation and decrease success.
21. If the patient becomes uncomfortable, decrease pressure support by 5 mbar.
22. Increase pressure support to a maximum combined PEEP + pressure support of 20 mbar.
23. Continuously monitor ETCO2.
24. Perform continual reassessment of patient’s condition.
25. Fill in appropriate ePCR documentation requirements.
26. Clean and disinfect ventilator machine using HMC-approved disinfectant following manufacturer recommendations.

3.15 Mechanical ventilation using the Oxylog 1000

Indications

1. Any patient with an ETT or LTA in place who requires ventilation and is not actively receiving chest compressions.
2. Any patient with a cricothyrotomy in place who requires ventilation and is not actively receiving chest compressions.
3. Any patient with a pre-existing advanced airway in place (tracheostomy/ ETT) who requires transport and continuous mechanical ventilation.

Contraindications

1. Any patient with ongoing chest compressions.
2. Failure of the ventilator to pass the device check.
3. Any patient with a (tracheostomy, ETT) who has their own transport ventilator that will accompany the patient with an appropriately trained medical professional who is trained and approved to operate the device.
4. Patients under 30 kgs.

Adverse Effects

1. Hypo/ hyperventilation.
2. Barotrauma resulting in pneumothorax.
3. Diaphragm atrophy.
4. Increased thoracic pressure resulting in reduced CO and hypotension.
5. Oxygen toxicity

Standard precautions

Procedure

1. Ensure that the patient's BP, ECG, SpO2, ETCO2, HR and RR are being monitored.
2. Open the O2 supply, ensure the O_2 bottle has sufficient pressure level and check for leaks in the high-pressure system.
3. Connect the reusable/ disposable low-pressure circuit to the Oxylog 1000. Connect a PEEP valve (if required) to the exhalation port of the valve set.
4. Attach the in-line ETCO2 to the circuit and connect to the Lifepak 15.
5. Attach HME filter to the circuit (if available).
6. Set the appropriate Minute Volume, Frequency, and Pmax based on the patient's ideal body weight and presenting condition, as per CPG 3.4
7. Select "Air MIX" or "NO AIR MIX" based on the patients' requirements.
8. Power the device to I (ON).
9. Attach the circuit to the patient.

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10. Confirm adequate patient ventilation (ETCO2, SpO2 and airway pressures).
11. Reassess device to ensure proper patient appropriate settings are being used and adjust as necessary.
12. Complete the ePCR with the appropriate mechanical ventilator details.
13. Clean and disinfect ventilator machine using HMC-approved disinfectant following manufacturer recommendations.

RECOMMENDED ADULT VENTILATOR SETTINGS FOR COMMON CONDITIONS			
SETTINGS	**NORMAL LUNGS**	**ASTHMA/COPD**	**PULMONARY EDEMA/ ARDS**
Frequency	8-12 breaths/min	4-10 breaths/min	12-16 breaths/min
Tidal Volume	6-8ml/kg	4-6ml/kg	6-8ml/kg
Pmax	35-40mbar	40-45mbar	40-45mbar
PEEP	5mbar	0-5mbar	Up to 10mbar
Air Mix or No Air Mic	As required	As required	As required

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Troubleshooting the Oxylog 1000

Fault	Cause	Remedy
Ventilator does not build up any airway pressure. "Psupply" indicator red	Gas capacity of O_2 cylinder is empty.	Immediately connect the ventilator to a new O_2 supply.
	Supply pressure at ventilator inlet too low; no central gas supply or O_2 cylinder empty.	Establish sufficient supply pressure.
Ventilator remains on "Inspiration"	Supply pressure at ventilator inlet too low.	Establish sufficient supply pressure.
	Oxylog 1000 faulty.	Arrange with Production to change Oxylog 1000
Patient cannot exhale, or can only exhale	Ventilation hose kinked.	Route the ventilation hose without kinking it.
	Red non-return valve in the diaphragm is faulty or "creased".	Open the ventilation valve and assemble correctly.
"Paw" indicator red (HIGH) Audible alarm is sounded The minute volume is not fully applied	Stenosis in the airways.	Free the airways.
	Ventilation hose kinked.	Route the ventilation hose without
	Lung compliance reduced.	Set the "Pmax" higher.
	Patient is "fighting the ventilator"	Change the ventilation pattern or sedate the patient.
"Paw" indicator red (LOW) Audible alarm is sounded	Disconnection/ leakage at the patient connection, ventilation valve or ventilation hose.	Ensure that all connections are leakproof.
	Diaphragm of the ventilation valve incorrectly fitted or damaged.	Mount the diaphragm correctly or replace diaphragm if faulty.
	Leaky cuff.	Inflate the cuff and check for leaks.

3.16 Mechanical ventilation using the Hamilton T1

IV

Indications

1. Any patient with an ETT/ LTA in place who requires ventilation and is not actively receiving chest compressions.
2. Any patient with a cricothyrotomy in place who requires ventilation and is not actively receiving chest compressions.
3. Any patient with a pre-existing advanced airway in place (tracheostomy/ ETT) who requires transport and continuous mechanical ventilation.

Contraindications

1. Any patient with ongoing chest compressions.
2. Failure of the ventilator to pass the device check performed prior to placing on the patient.
3. Any patient who has a pre-existing advanced airway in place (tracheostomy, ETT) who has their own transport ventilator that will accompany the patient with an appropriately trained medical professional who is trained and approved to operate the device

Adverse Effects

1. Hypo/ hyperventilation.
2. Barotrauma resulting in pneumothorax.
3. Diaphragm atrophy.
4. Increased thoracic pressure resulting in reduced cardiac output and hypotension.
5. Oxygen toxicity.

Standard precautions

Procedure

1. Ensure that the patient's BP, ECG, SpO_2, $ETCO_2$, HR and RR are being monitored.
2. **To install the expiratory valve set**
   a. Remove the safety cover.
   b. Ensure the membrane is properly aligned with the expiratory valve housing, and the metal plate faces up.
   c. Position the expiratory valve set in the expiratory port and twist the locking ring clockwise until it locks into place.
3. Attach the circuit to the inspiratory and expiratory port and ensure the flow sensors are correctly attached to the sensor connections and in the correct place on the breathing circuit (in front of the ETco2/airway adapter)
4. Attached the Coaxial inspiratory/expiratory limb to the circuit.

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5. Connect the mainstream CO2 sensor to the airway adapter.
6. Connect the sensor/adapter to the breathing circuit.
7. Connect the HMEF to the circuit (in front of the Flow sensor)
8. If a nebulizer is required, it should be positioned between the Coaxial breathing circuit and the flow sensor.
9. Attach the Spo2 pulse oximeter
10. Connect the ventilator to AC power and oxygen supply.
11. Assemble and connect the patient breathing circuit. Press (Power/Standby)
12. The ventilator runs a self-test and, when complete, displays the Standby window. (Use the ventilator only if it passes all tests).
13. Enable the sensory monitoring by touching **System**>**Sensors**> **On/Off.**
14. Select the O2 sensor, CO2 and/or SpO2 sensor checkboxes as required and close the window.
15. Activate the Patient group and specify patient date by Touching **Neonatal, Adult/Ped,** or **Last patient** (uses the last-specified settings)
16. If Adult/Ped is selected, set the patient sex and height. The device calculates the ideal body weight (IBW).
17. Touch **Preop check** to perform the preoperational check.
18. To perform the Leak Test:
    a. Perform these steps disconnected from the patient. (Prompts are provided in the System > Tests & Calib window)
    Step one (Do either of the following:)
    b. Touch **System** > **Tests & Calib**. In the Standby window, touch **Preop check**. Touch **Leak test** and when prompted, block the patient end of the breathing circuit. Hold until instructed to stop on the display.
    c. Pass or fail and date/time of the completed test are displayed.
    Step two
    d. Touch **Flow sensor** to calibrate the flow sensor. Calibration starts automatically.
    e. When prompted, attach the calibration adapter to the flow sensor and flip them both 180° so the adapter is directly connected to the limb. Calibration starts automatically. When prompted, flip the flow sensor/adapter 180° again, so the flow sensor is directly connected to the limb and remove the calibration adapter.
    f. Pass or fail and date/time of completed test are displayed.
    Step three
    g. If an X is displayed next to O2 sensor, touch the O2 sensor button to calibrate the sensor. If the O2 sensor calibration needed alarm is generated, repeat the calibration.
    Step four
    h. During ventilator startup, the ventilator performs a self-check that also verifies proper alarm function, including generation of an audible alarm sound.
    You are not required to perform additional alarm tests.

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i. If desired, you can test any adjustable alarm by manually changing the set limit such that the ventilator exceeds or fails to reach this limit, thereby generating the associated alarm.

19. To change the mode, do either of the following: Touch the mode name at the top left of the display or touch **Modes** at the top right of the display. The Modes window will open OR touch the desired ventilation mode and **confirm**. The control window will open for review and adjust settings in the Basic and More windows. Touch **Confirm**. The mode and settings become active.
20. To adjust ventilation setting:
    a. Touch **Controls**. The Controls > Basic window opens.
    b. Adjust control settings as needed.
    c. Touch **More** to access additional controls and make changes as needed.
    d. If displayed, touch **Confirm** (C). If not, changes are applied immediately.
21. To start ventilating the patient:
    a. Touch **Start ventilation** or press ⏻ to start ventilating the patient.
22. To stop ventilation and enter Standby:
    a. Press ⏻ and in the confirmation window, touch Activate standby.
23. Adjust the Alarms if needed
24. Attach the Ventilator to the patient
25. Attach the circuit to the patient.
26. Ensure the patient is adequately sedated and paralyzed as necessary.
27. Reassess device to ensure proper patient appropriate settings are being used and adjust as necessary.
28. Continuously assess the patients' air entry looking for signs of barotrauma from mechanical ventilation.
29. Document the mechanical ventilator related settings in the ePCR
30. Clean and disinfect ventilator machine using HMC-approved disinfectant following manufacturer recommendations

3.17 CPAP: Flow-safe II

Indications

1. Medical patients complaining of moderate to severe respiratory distress (as per CPG 1.5) meeting ALL of the following criteria.
   a. Has the ability to maintain an open airway.
   b. Has signs and symptoms of acute pulmonary edema (CPG 4.5).
   c. Has a systolic blood pressure above 100 mmHg.
   d. Is able to fit the CPAP mask.
2. CPAP may be considered in patients with acute hypoxemic respiratory failure due to pneumonia or acute hypercapnic respiratory failure due to COPD as a bridge to hospital to avoid endotracheal intubation.
3. Pre-oxygenation for RSI and DSI.

Contraindications

1. Respiratory or Cardiac Arrest
2. Hemodynamic instability
3. Extreme Agitation – Non-cooperative patient.
4. Acute or imminent airway obstruction (e.g. acute angioedema, inhalation burns)
5. Increased risk of regurgitation and aspiration
6. Presence of a tracheostomy
7. Inability to maintain a mask seal
8. Suspected pneumothorax

Adverse Effects

1. Hypo/ hyperventilation.
2. Barotrauma resulting in pneumothorax.
3. Diaphragm atrophy.
4. Increased thoracic pressure resulting in reduced cardiac output and hypotension.
5. Oxygen toxicity.

Standard precautions

Procedure

1. Identify the need to perform CPAP using the Flow-safe II.
2. Standard safety precaution.
3. Get consent from the patient/ family, if applicable. Explain the procedure to the

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patient/ family.

4. Ensure that the patient's BP, ECG, SpO_2, $ETCO_2$, HR and RR are being monitored.
5. Attach the nasal $ETCO_2$ to the patient, Lifepak 15 and oxygen supply.
6. Administer sublingual GTN (ONLY if indicated) prior to placing the CPAP Flow-safe II mask on the patient's face.
7. Prepare the Flow-safe II. Attach the mask to the manometer/valve assembly.
8. Ensure sufficient pressure level of O_2.
9. Connect the O_2 tubing to the oxygen source.
10. Turn O_2 flowrate to 8-9 LPM. This will give you a starting PEEP of 5 cm H_2O.
11. Confirm PEEP setting on the Flow-safe II gauge, once applied to patient. 1
12. Place the mask (only Adult Large size available) over the patients face and encourage them to breathe normally. Ensure a tight fit of the mask.
13. Once the patient is settled and comfortable with the mask, then apply the head harness.
14. Evaluate for excessive leakage and adjust mask and head harness, if needed.
15. Continuously monitor the SpO_2 and target $SpO_2 > 94\%$
16. Adjust the O_2 flowrate until the desired PEEP is achieved based on the patient's condition. Oxygen flowrate of 10-12 LPM delivers 7.5 cm H_2O PEEP.
17. If the patient's condition does not improve, adjust the O_2 flowrate further to 13-14 LPM to achieve a maximum PEEP of 10 cm H_2O.
18. If the patient becomes uncomfortable, decrease the flowrate to a lower setting.
19. Continuously monitor $ETCO_2$
20. Perform continual reassessment of patient's condition.
21. Complete the appropriate ePCR documentation requirement

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Circulatory Management

4.1 Intravenous cannulation: Extremities

Indications

1. Fluid administration
2. Access for IV medication administration

Contraindications

1. Whenever possible avoid sites of burn, infection or localised cellulitis.

Adverse Effects

1. Air embolus
2. Arterial puncture
3. Cannula shear or breakage
4. Medication/ fluid extravasation
5. Haematoma or haemorrhage from the site.
6. Infection or phlebitis
7. Irritation to the vein wall
8. Nerve damage
9. Vasovagal syncope

Standard precautions

Procedure

1. Ensure that the patient’s BP, ECG, SpO2, HR and RR are being monitored (Time permitting).
2. Prepare appropriate pack (A10M/ A10T- Adult/ A11-Pediatric, A12/ A13), check the expiry date.
3. Prepare and assemble fluid and appropriate administration set (macro drip/ micro drip).
4. Prepare administration set by filling drip chamber reaching 1/3 to 1/2 and flush tubing. Ensure all air is removed from the administration set.
5. Select the appropriate cannulation site and apply the tourniquet correctly.
6. Allow the veins time to fill up by ensuring the arm is below the level of the heart. Then select and palpate a suitable vein.
7. Disinfect the area using 70% alcohol wipes (approximately 3 inches radius).
8. Select and prepare a correct size IV cannula.
9. Stabilise the vein and insert the needle with the bevel pointing up, at a 30-degree angle to the skin.
10. Check primary flashback, lower the angle and advance further and check.
11. Hold the grip plate in place and advance the plastic catheter fully.

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12. Stabilize the wings and occlude the vein proximal from the insertion site (apply gauze if needed).
13. Release the tourniquet.
14. Withdraw the needle and discard directly into sharps container.
15. Connect the IV tubing to catheter.
16. Open the roller flow clamp for a brief period of time to ensure patency then regulate to the required rate.
17. Check for signs of infiltration on the insertion site.
18. Secure catheter with Tegaderm. Use additional tape if necessary.
19. Adjust flow rate as appropriate.
20. Inspect the cannulation site regularly for infiltration

4.2 Intravenous cannulation: External jugular vein

Indications

1. Any adult patient requiring immediate vascular access for treatment of a life-threatening condition where attempts at other sites have either been unsuccessful or are contraindicated

Contraindications

1. Paediatric patient (< 14 years old).
2. IV or IO can be established at another site.
3. Unable to visualize and palpate the external jugular vein.
4. Access through visibly dirty, infected or burned tissue should be avoided.

Adverse Effects

1. Air embolus
2. Arterial puncture
3. Cannula shear or breakage
4. Medication/ fluid extravasation
5. Haematoma or haemorrhage from the site.
6. Infection or phlebitis
7. Irritation to the vein wall
8. Nerve damage
9. Vasovagal syncope
10. Pneumothorax
11. Surgical emphysema from tracheal puncture.

Standard precautions

Procedure

1. Ensure that the patient’s BP, ECG, SpO2, HR and RR are being monitored.
2. Prepare appropriate pack (A10M/ A10T, C6, A12/ A13), check the expiry date.
3. Prepare and assemble fluid and appropriate administration set (macro drip).
4. Prepare administration set by filling drip chamber reaching 1/3 to 1/2 and flush tubing. Ensure all air is removed from the administration set.
5. Select the appropriate cannulation site. Place the patient supine or in Trendelenburg position if possible and turn patient’s head towards contralateral side (for medical patients only).
6. Identify the external jugular vein.
7. Allow the veins time to fill up.

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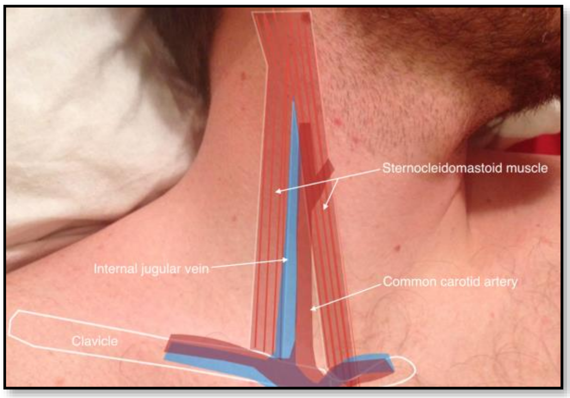
8. Disinfect the area using 70% alcohol wipes (approximately 3 inches radius)
9. Select and prepare a correct size IV cannula.
10. Lightly place a finger of your non- dominate hand just above the clavicle to produce a tourniquet effect.
11. Use the thumb of that same hand to pull above the puncture site to stabilize the vein.
12. Insert the catheter correctly with the bevel pointing up.
13. Puncture the vein as far above the clavicle as is possible and cannulate the vein in a shallow and superficial manner.
14. Observe flashback and advance slightly further.
15. Hold the grip plate in place and advance the plastic catheter only.
16. Stabilize the wings and occlude the vein proximal to the end of the catheter.
17. Release tourniquet effect.
18. Remove needle and place directly into the sharp's container.
19. Connect IV tubing to catheter.
20. Run IV for a brief period to ensure patency and look for signs of infiltration.
21. Apply Tegaderm and tape catheter as needed.
22. Adjust flow rate as appropriate.



(*Venous Access*, n.d.)

4.3 Intra-osseous cannulation in adults: Proximal Humerus

IV

Indications

1. Adult patient requiring immediate vascular access for treatment of a life-threatening condition where peripheral vascular access appears difficult or impossible.
2. Adult patient requiring immediate vascular access for treatment of a life-threatening condition where 2 or more attempts at peripheral vascular access by an experienced clinician have failed.
3. Preferred IO site for adult patients in cardiac arrest.

Contraindications

1. Fracture of the targeted bone.
2. Previous significant orthopaedic procedures at the insertion site
3. IO in the target bone within the last 48 hours.
4. Infection at the area of insertion.
5. Excessive tissue or absence of adequate anatomical landmarks.

Adverse Effects

1. Pain
2. Fat micro emboli
3. Air embolus
4. Fracture
5. Arterial puncture
6. Medication/ fluid extravasation
7. Haematoma or haemorrhage from the site.
8. Osteomyelitis
9. Nerve damage
10. Vasovagal syncope
11. Compartment syndrome

Standard precautions

Procedure

1. Ensure that the patient's BP, ECG, SpO2, ETCO2, HR and RR are being monitored.
2. Prepare appropriate pre-pack (A10M/ A10T, A12/ A13, C7), check the expiry date.
3. Prepare and assemble fluid and appropriate administration set (macro drip).
4. Prepare administration set by filling drip chamber reaching 1/3 to 1/2 and flush tubing. Ensure all air is removed from the administration set.
5. Prepare the following and inspect package for sterility, review expiration date:

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a. EZ-IO Power Driver and C7 Pre-pack.
b. Appropriate EZ-IO Needle Set (45mm Yellow needle for all patients > 40kg utilizing the humeral head site).
c. EZ-Connect Extension Set
d. Unlock Clamp
e. Prime set and purge air
f. EZ Stabilizer dressing
g. Luer lock syringe with sterile normal saline flush (5-10 mL for adults).
h. Sharps container.

6. Assess for potential pain response prior to insertion (if patient conscious).
7. Expose the selected humeral head area.
8. Place the patient's hand over the abdomen, elbow adducted and humerus internally rotated.
9. Locate the correct anatomical site for proximal humerus IO insertion:
   a. Place palm on the patient's shoulder anteriorly to identify the "ball" under the palm as the target area.
   b. Place the ulnar aspect of one hand vertically over the axilla and the ulnar aspect of the other hand along the midline of the upper arm laterally.
   c. Place the thumbs together over the arm to identify the vertical line of insertion on the proximal humerus.
   d. Palpate deeply up the humerus to the surgical neck then move 1-2cm proximal to the most prominent aspect of the greater tubercle.
10. Disinfect the area using 70% alcohol wipes or 2% Chlorhexidine with 70% alcohol
11. Stabilize the insertion site using a clean "no touch" technique once site has been cleaned.
12. Aim the needle set at a 45º angle to the anterior plane and posteriomedially.
13. Push the needle tip through the skin at the selected location and advance until the needle rests on the bone.
14. Ensure that at least one black line (5mm) is visible on the needle shaft. If a black line is not visible, a longer needle or a different site must be selected.
15. While applying gentle pressure, depress the drill trigger. Continue drilling until a "pop" or "give" is felt, indicating entry into the medullary space.
16. Continue to advance the needle until the hub rests against the skin and then release the drill trigger. (This recommendation is for adults only at the humeral head site).
17. Remove driver and stylet:
   a. Hold the hub in place and pull the driver straight off; continue to hold the hub while twisting the stylet off the hub with counterclockwise rotations.
   b. Note catheter feels firmly seated in bone (1st Confirmation).

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c. Dispose of all sharps and biohazard materials using standard practices.

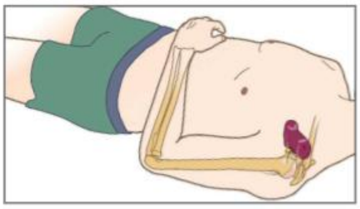
18. Place the EZ-Stabilizer Dressing over the hub – do not remove tabs at this step.
19. Attach a primed EZ-Connect Extension Set to the catheter hub, firmly securing by twisting clockwise.
20. Pull the tabs off the dressing to expose the adhesive, apply to skin.
21. Aspirate for blood/ bone marrow (2nd Confirmation). Note inability to withdraw/ aspirate blood from the catheter does not mean the insertion was unsuccessful.
22. Flush the IO catheter with 5-10 mL of normal saline.
23. Verify placement/ patency prior to all infusion.
24. If high flow is required for resuscitation, wrap a manual BP cuff around the IV fluid bag and inflate to 300mmHg. Maintain a target infusion pressure of 300mmHg.
25. Assess for any signs of extravasation/ complications.
26. Document date and time on wristband and place on the patient.
27. Stabilize arm using swath technique and monitor site and limb for extravasation or other complications during transport

EZ-IO® Proximal Humerus Identification & Insertion Technique – Adult

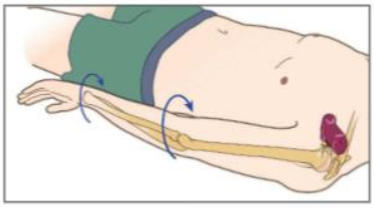
Proximal Humerus

Arm Positioning

Using either method below, adduct elbow, rotate humerus internally.



Place the patient's hand over the abdomen with arm tight to the body.



Place the arm tight against the body, rotate the hand so the palm is facing outward, thumb pointing down.

Landmarking

Place your palm on the patient's shoulder anteriorly.

- The area that feels like a "ball" under your palm is the general target area
- You should be able to feel this ball, even on obese patients, by pushing deeply

Place the ulnar aspect of one hand vertically over the axilla. Place the ulnar aspect of the opposite hand along the midline of the upper arm laterally.

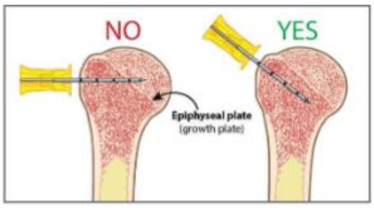
Place your thumbs together over the arm.

- This identifies the vertical line of insertion on the proximal humerus

Palpate deeply as you climb up the humerus to the surgical neck.

- It will feel like a golf ball on a tee – the spot where the "ball" meets the "tee" is the surgical neck

The insertion site is on the most prominent aspect of the greater tubercle, 1 to 2 cm above the surgical neck.



Point the needle tip at a 45-degree angle to the anterior plane and posteromedial.

Proximal Humerus | *LA* | *Teleflex*. (n.d.). https://www.teleflex.com/la/en/product-areas/emergency-medicine/intraosseous-access/arrow-ez-io-system/proximal-humerus/index.html

4.4 Intra-osseous cannulation in adults: Proximal Tibia

Indications

1. Adult patient requiring immediate vascular access for treatment of a life-threatening condition where peripheral vascular access appears difficult or impossible.
2. Adult patient requiring immediate vascular access for treatment of a life-threatening condition where 2 or more attempts at peripheral vascular access by an experienced clinician have failed.

Contraindications

1. Fracture of the targeted bone.
2. Previous, significant orthopaedic procedures at the insertion site
3. IO in the target bone within the last 48 hours.
4. Infection at the area of insertion.
5. Excessive tissue or absence of adequate anatomical landmarks.

Adverse Effects

1. Pain
2. Fat micro emboli
3. Air embolus
4. Fracture
5. Arterial puncture
6. Medication/ fluid extravasation
7. Haematoma or haemorrhage from the site.
8. Osteomyelitis
9. Nerve damage
10. Vasovagal syncope
11. Compartment syndrome

Standard precautions

Procedure

1. Ensure that the patient’s BP, ECG, SpO2, HR and RR are being monitored.
2. Prepare appropriate pre-pack (A10M/ A10T, A12/ A13, C7), check the expiry date.
3. Prepare and assemble fluid and appropriate administration set (macro drip).
4. Prepare administration set by filling drip chamber reaching 1/ 3 to 1/ 2 and flush tubing. Ensure all air is removed from the administration set.
5. Prepare the following and inspect package for sterility, review expiration date:
   a. EZ-IO Power Driver and C7 Pre-pack.

   b. Appropriate EZ-IO Needle Set (45mm Yellow needle for Humeral head insertion site/Proximal Tibia for overweight or muscular patients. 25mm Blue needle for all patients>40kg and 15mm Pink needle for pediatrics weight range 3-39kg).
   c. EZ-Connect Extension Set
   d. Unlock Clamp
   e. Prime set and purge air
   f. EZ Stabilizer dressing
   g. Luer lock syringe with sterile normal saline flush (5-10 mL for adults).
   h. Sharps container.
6. Assess for potential pain response prior to insertion.
7. Extend the selected leg. Expose the chosen site. Insertion site is approximately 2cm medial to the tibial tuberosity, or approximately 3cm (two finger widths) below the patella and approximately 2cm medial, along the flat aspect of the tibia.
8. Disinfect the area using 70% alcohol wipes or 2% Chlorhexidine with 70% alcohol.
9. Stabilize the insertion site using a clean “no touch” technique once site has been cleaned.
10. Stabilize the extremity.
11. Aim the needle set at a 90-degree angle to centre of the bone.
12. Push the needle set tip through the skin until the tip rests against the bone.
13. The 5mm mark must be visible above the skin for confirmation of adequate needle set length.
14. Gently drill, advancing the needle set approximately 1-2cm after entry into the medullary space or until the needle set hub is close to the skin.
15. While applying gentle pressure, depress the drill trigger. Continue drilling until a “pop” or “give” is felt, indicating entry into the medullary space.
16. Hold the hub in place and pull the driver straight off.
17. Continue to hold the hub while twisting the stylet off the hub with counterclockwise rotations. The catheter should feel firmly seated in the bone (1st confirmation of placement).
18. Remove driver and stylet:
   a. Hold the hub in place and pull the driver straight off; continues to hold the hub while twisting the stylet off the hub with counterclockwise rotations.
   b. Note catheter feels firmly seated in bone (1st Confirmation).
   c. Dispose of all sharps and biohazard materials using standard practices.
19. Place the EZ-Stabilizer Dressing over the hub – does not remove tabs at this step.
20. Attach a primed EZ-Connect Extension Set to the catheter hub, firmly securing by twisting clockwise.
21. Pull the tabs off the dressing to expose the adhesive, apply to skin.

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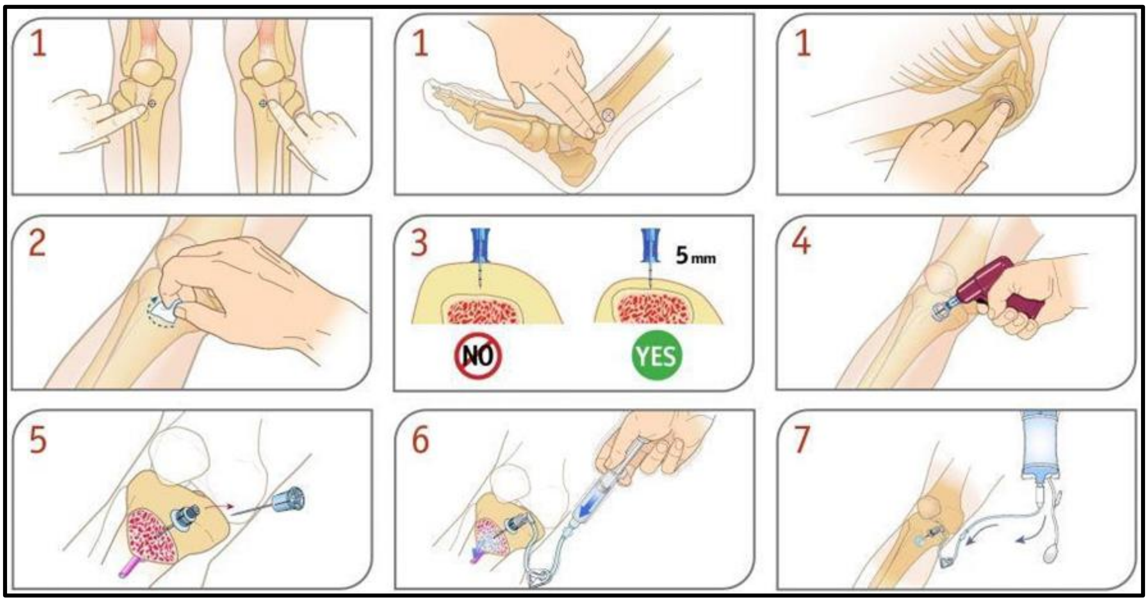
22. Aspirate for blood/bone marrow (2nd Confirmation). Note inability to withdraw/aspirate blood from the catheter does not mean the insertion was unsuccessful.
23. Flush the IO catheter with 5-10 mL of normal saline.
24. Verify placement/patency prior to all infusion.
25. If high flow is required for resuscitation, wrap a manual BP cuff around the IV fluid bag and inflate to 300mmHg. Maintain a target infusion pressure of 300mmHg.
26. Assess for any signs of extravasation/complications.
27. Documents date and time on wristband and places on the patient.
28. Monitor site and limb for extravasation or other complications during transport.



Shade. (2021, May 7). *Intraosseous (IO) access*. Manual of Medicine. https://manualofmedicine.com/topics/emergency-acute-medicine/intraosseous-access/

4.5 Intra-osseous cannulation in paediatrics: Proximal Tibia

Indications

1. Paediatric patient requiring immediate vascular access for life-threatening treatment where peripheral access is difficult, impossible, or has failed after two or more attempts by an experienced clinician.

Contraindications

1. Fracture of the targeted bone.
2. Previous, significant orthopaedic procedures at the insertion site
3. IO in the target bone within the last 48 hours.
4. Infection at the area of insertion.
5. Excessive tissue or absence of adequate anatomical landmarks.

Adverse Effects

1. Pain
2. Fat micro emboli
3. Air embolus
4. Fracture
5. Arterial puncture
6. Medication/ fluid extravasation
7. Haematoma or haemorrhage from the site.
8. Osteomyelitis
9. Nerve damage
10. Vasovagal syncope
11. Compartment syndrome

Standard precautions

Procedure

1. Ensure that the patient's BP, ECG, SpO_2, HR and RR are being monitored.
2. Prepare appropriate pre-pack (A10M/ A10T, A12/ A13, C7), check the expiry date.
3. Prepare and assemble fluid and appropriate administration set (macro drip).
4. Prepare administration set by filling drip chamber reaching 1/ 3 to 1/ 2 and flush tubing. Ensure all air is removed from the administration set.
5. Prepare the following and inspect package for sterility, review expiration date:
    a. EZ-IO Power Driver and C7 Pre-pack.
    b. Appropriate EZ-IO Needle Set (45mm Yellow needle for Humeral head insertion site/Proximal Tibia for overweight or muscular patients. 25mm Blue needle for all

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patients>40kg and 15mm Pink needle for pediatrics weight range 3-39kg).
  c. EZ-Connect Extension Set
  d. Unlock Clamp
  e. Prime set and purge air
  f. EZ Stabilizer dressing
  g. Luer lock syringe with sterile normal saline flush (5-10 mL for adults).
  h. Sharps container.
6. Assess for potential pain response prior to insertion.
7. Extend the selected leg. Insertion site is approximately 1cm medial to the tibial tuberosity, or just below the patella (approximately 1cm or one finger width) and slightly medial (approximately 1cm or one finger width), along the flat aspect of the tibia. Pinch the tibia between your fingers to identify the center of the medial and lateral borders.
8. Disinfect the area using 70% alcohol wipes or 2% Chlorhexidine with 70% alcohol.
9. Stabilize the insertion site using a clean "no touch" technique once site has been cleaned.
10. Remove the needle set cap.
11. Stabilize the extremity.
12. Aim the needle set at a 90-degree angle to the center of the bone.
13. Push the needle set tip through the skin until the tip rests against the bone.
14. Push the needle tip through the skin at the selected location and advance until the needle rests on the bone.
15. Ensure that at least one black line (5mm) is visible on the needle shaft. If a black line is not visible, a longer needle or a different site must be selected.
16. While applying gentle pressure, depress the drill trigger. Continue drilling until a "pop" or "give" is felt, indicating entry into the medullary space.
17. Remove driver and stylet:
    a. Hold the hub in place and pull the driver straight off; continues to hold the hub while twisting the stylet off the hub with counterclockwise rotations.
    b. Note catheter feels firmly seated in bone (1st Confirmation).
    c. Dispose of all sharps and biohazard materials using standard practices.
18. Place the EZ-Stabilizer Dressing over the hub – does not remove tabs at this step.
19. Attach a primed EZ-Connect Extension Set to the catheter hub, firmly securing by twisting clockwise.
20. Pull the tabs off the dressing to expose the adhesive, applies to skin
21. Aspirate for blood/ bone marrow (2nd Confirmation). Note inability to withdraw/aspirate blood from the catheter does not mean the insertion was unsuccessful.
22. Flush the IO catheter with 2-5 mL of normal saline.

23. Verify placement/patency prior to all infusion. If high flow is required for resuscitation, wrap a manual BP cuff around the IV fluid bag and inflate to 300mmHg. Maintain a target infusion pressure of 300mmHg.
24. Assess for any signs of extravasation/complications.
25. Documents date and time on wristband and places on the patient.
26. Monitor site and limb for extravasation or other complications during transport

4.6 Intra-osseous cannulation in paediatrics: Distal Femur

Indications

1. Paediatric patient requiring immediate vascular access for life-threatening treatment where peripheral access is difficult, impossible, or has failed after two or more attempts by an experienced clinician.

Contraindications

1. Fracture of the targeted bone.
2. Previous, significant orthopaedic procedures at the insertion site
3. IO in the target bone within the last 48 hours.
4. Infection at the area of insertion.
5. Excessive tissue or absence of adequate anatomical landmarks.

Adverse Effects

1. Pain
2. Fat micro emboli
3. Air embolus
4. Fracture
5. Arterial puncture
6. Medication/ fluid extravasation
7. Haematoma or haemorrhage from the site.
8. Osteomyelitis
9. Nerve damage
10. Vasovagal syncope
11. Compartment syndrome

Standard precautions

Procedure

1. Ensure that the patient’s BP, ECG, SpO2, HR and RR are being monitored.
2. Prepare appropriate pre-pack (A10M/ A10T, A12/ A13, C7), check the expiry date.
3. Prepare and assemble fluid and appropriate administration set (macro drip).
4. Prepare administration set by filling drip chamber reaching 1/ 3 to 1/ 2 and flush tubing. Ensure all air is removed from the administration set.
5. Prepare the following and inspect package for sterility, review expiration date:
   a. EZ-IO Power Driver and C7 Pre-pack.
   b. Appropriate EZ-IO Needle Set (45mm Yellow needle for Humeral head insertion

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site/Proximal Tibia for overweight or muscular patients. 25mm Blue needle for all patients>40kg and 15mm Pink needle for pediatrics weight range 3-39kg)
   c. EZ-Connect Extension Set
   d. Unlock Clamp
   e. Prime set and purge air
   f. EZ Stabilizer dressing
   g. Luer lock syringe with sterile normal saline flush (5-10 mL for adults).
   h. Sharps container.
6. Secure the selected leg outstretched to ensure the knee does not bend. Identify the patella by palpation.
7. The insertion site is just proximal to the patella (maximum 1cm) and approximately 1-2 cm medial to midline.
8. Disinfect the area using 70% alcohol wipes or 2% Chlorhexidine with 70% alcohol.
9. Stabilize the insertion site using a clean “no touch” technique once site has been cleaned.
10. Remove the EZ-IO® Needle Set cap.
11. Aim the needle set toward the centre of the bone at a 90-degree angle.
12. Push the needle tip through the skin at the selected location and advance until the needle rests on the bone.
13. Ensure that at least one black line (5mm) is visible on the needle shaft. If a black line is not visible, a longer needle or a different site must be selected.
14. While applying gentle pressure, depress the drill trigger. Continue drilling until a “pop” or “give” is felt, indicating entry into the medullary space.
15. Remove driver and stylet:
    a. Hold the hub in place and pull the driver straight off; continue to hold the hub while twisting the stylet off the hub with counterclockwise rotations.
    b. Note catheter feels firmly seated in bone (1st Confirmation).
    c. Dispose of all sharps and biohazard materials using standard practices.
16. Place the EZ-Stabilizer Dressing over the hub – does not remove tabs at this step.
17. Attach a primed EZ-Connect Extension Set to the catheter hub, firmly securing by twisting clockwise.
18. Pull the tabs off the dressing to expose the adhesive, applies to skin
19. Aspirate for blood/ bone marrow (2nd Confirmation). Note inability to withdraw/aspirate blood from the catheter does not mean the insertion was unsuccessful.
20. Flush the IO catheter with 2-5 mL of normal saline.
21. Verify placement/patency prior to all infusion.
22. If high flow is required for resuscitation, wrap a manual BP cuff around the IV fluid bag

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and inflate to 300mmHg. Maintain a target infusion pressure of 300mmHg.

23. Assess for any signs of extravasation/ complications.
24. Documents date and time on wristband and places on the patient.
25. Monitor site and limb for extravasation or other complications during transport

4.7 Buretrol set for fluid administration to pediatrics

Indications

1. Controlled IV medication or fluid volume administration to patients < 14 years of age (if available).

Contraindications

1. There are no contraindications in this setting

Adverse Effects

1. Over transfusion

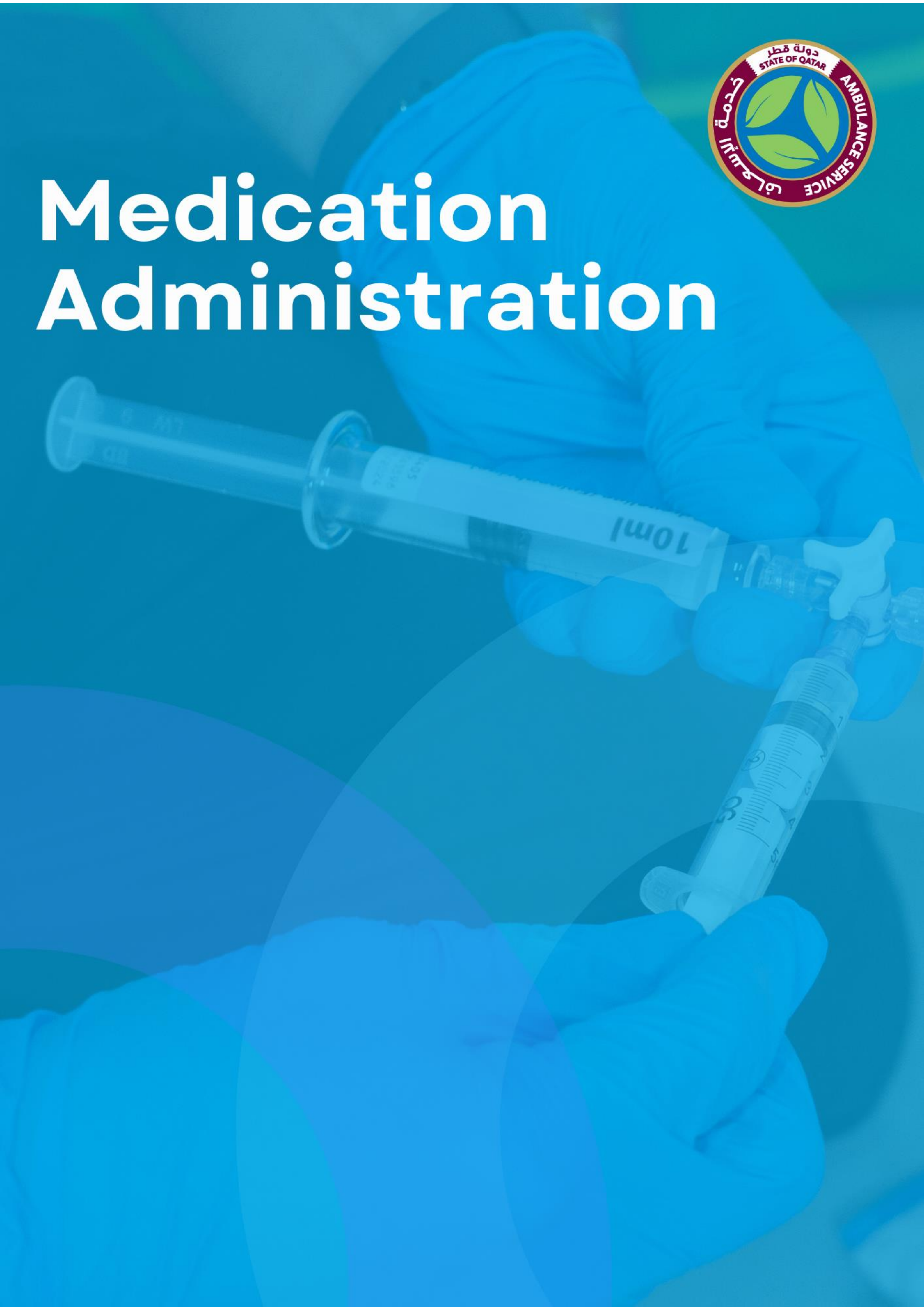
Standard precautions

Procedure

1. Ensure that the patient's BP, ECG, SpO2, HR and RR are being monitored.
2. Prepare appropriate pre-pack (A11, A12/ A13, C12), check the expiry date.
3. Prepare and assemble fluid and paediatric IV set (C12).
4. Close the clamp fully.
5. Push the spike of the Buretrol into the IV port solution container.
6. Open the burette air filter and upper clamp to fill the burette up to 30mls.
7. Squeeze the drip chamber gently to initiate the diaphragm.
8. Open the control clamp to expel any air from the line.
9. When the level reaches 20mls, close the control clamp to refill the burette up to 100mls. Then close the upper clamp.
10. Start fluid administration, establish the infusion rate by using the control clamp. Re-adjust after 3 minutes.
11. The diaphragm will close automatically as the solution reaches the zero level on the burette.

Buretrol IV set

Medication Administration



5.1 Medication preparation and labelling

I II III IV

Indications

1. As indicated by each specific medication.

Contraindications

1. As indicated by each specific medication. Paediatric patients
2. OB patients
3. Non-emergent scheduled transfers (PTS), unless the patient is observed to be unwell, develops a new complaint during transport or if the patient/ family requests.

Adverse Effects

1. As indicated by each specific medication.

Standard precautions

Procedure

1. Ensure that the patient’s BP, ECG, SpO_2, HR and RR are being monitored.
2. Prepare pre-packs C6/ A16 (if needed).
3. Safely open and lay out the CCP/ AP medication bag.
4. Select the medication to be administered.
5. Check the five “R” (Right patient, Right medication, Right dose, Right route, Right time).
6. Inspect the medication for correct name, expiry date, clarity, concentration, and integrity.
7. Double check the medication with your partner/ other QCHP licenced practitioner (where possible) prior to preparation. Get your partner to read the medication name, dose and or dilution out loud to you to verify the correct medication.
8. For IV/ IO medication, label the syringes correctly annotating the concentration.
9. Prepare inhaled medication as per procedure 3.4, 3.5 and 5.8.
10. Prepare intranasal medication as per procedure 5.4.
11. Prepare buccal medication as per procedure 5.5.
12. Prepare sublingual medication as per procedure 5.6.
13. Prepare oral medication as per procedure 5.7.
14. Double check the medication with your partner (where possible) prior to administration (second check). Get your partner to read the medication name, dose and or dilution out loud to you to verify the correct medication.

5.2 Intravenous/Intraosseous medication administration

Indications

1. Medication directed for IV/IO administration, as per CPGs.

Contraindications

1. Absence of IV/ IO access.
2. Contraindication based on the specific medication required to be administered.

Adverse Effects

1. Air embolus
2. Arterial puncture
3. Cannula shear or breakage
4. Medication/ fluid infiltration.
5. Haematoma or haemorrhage from the site.
6. Infection or phlebitis
7. Irritation to the vein wall
8. Nerve damage
9. Vasovagal syncope
10. Specific adverse effects will be based on the medication being administered.

Standard precautions

Procedure

1. Ensure that the patient's BP, ECG, SpO2, HR and RR are being monitored.
2. Assess the IV/ IO for patency or any signs of infiltration on the insertion site.
3. Prepare the medication to be administered as per Procedure 5.1. Prepare pre-pack C6/ A16.
4. Close the slide clamp/ roller of the IV tubing.
5. Open the injection port/ cleanse the Y injection port.
6. Administer the medication (depends on what medication needs to be administered via IV/ IO).
7. Flush with 10 mls Normal Saline/ open the roller or the slide clamp.
8. Regulate IV fluids accordingly.
9. Ensure proper disposal into the sharp container.
10. Reassess the patient's vital signs. Regularly assess the patient for side effects of the medication administered and complete an OVA for all side effects.

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5.3 Intramuscular medication administration

Indications

1. Administration of medications via the IM route (e.g. Epinephrine and Ketorolac).

Contraindications

1. If there is any evidence of infection or trauma at the injection site, then choose an alternate IM site.
2. Contraindication based on the specific medication require to be administered.

Adverse Effects

1. Pain
2. Bleeding

Standard precautions

Procedure

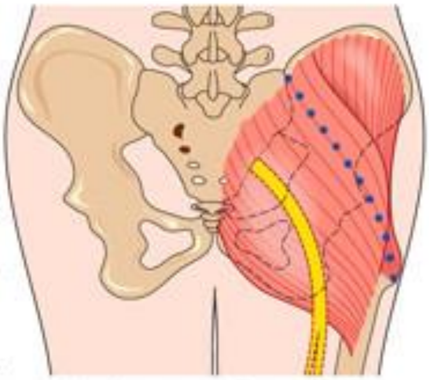
1. Ensure that the patient’s BP, ECG, SpO2, HR and RR are being monitored.
2. Prepare the medication to be administered as per Procedure 5.1. Prepare pre-pack C6/ A16.
3. Change the needle and dispel any air from the syringe.
4. Expose the area for IM injection with discretion (maintain patient privacy).
5. Cleanse the injection site with alcohol.
6. Aspirate slightly by withdrawing the plunger of the syringe to ensure no blood return.
9. If no blood is withdrawn, then inject the medication and withdraw needle and syringe.
10. Apply digital pressure over the injection site and place a band-aid over the injection site

Mid-deltoid site

The mid-deltoid site is easily accessible but due to the size of the muscle the area should not be used repetitively and only small volumes should be injected. The maximum volume should be 1ml (Rodger and King 2000). The denser part of the deltoid must be used. It is useful to visualise a triangle whereby the horizontal line is located 2.5-5cm below the acromial process and the mid-point of the lateral aspect of the arm in line with the axilla forms the apex. The injection is given about 2.5cm down from the acromial process, avoiding the radial and brachial nerves (Workman 1999, Rodger and King 2000).

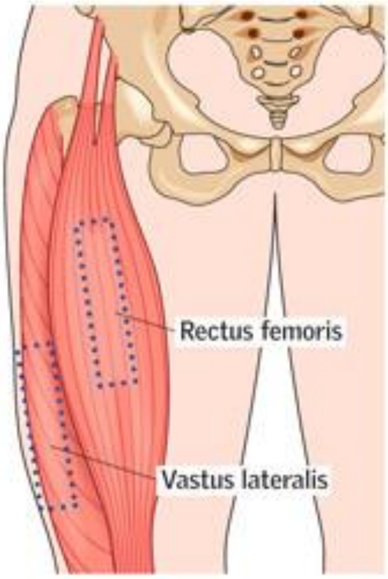
Dorsogluteal site

This area is used for deep IM and Z-track injections. Up to 4ml can be injected into this muscle (Workman 1999, Rodger and King 2000). Commonly referred to as the outer upper quadrant, it is located by using imaginary lines to divide the buttocks into four quarters. To identify the gluteus maximus, picture a line that extends from the iliac spine to the greater trochanter of the femur. Draw a vertical line from the midpoint of the first line to identify the upper aspect of the upper outer quadrant. This location avoids the superior gluteal artery and sciatic nerve (Workman 1999, Small 2004).



Rectus femoris site

This site is used for deep IM and Z-track injections. Between 1-5ml can be injected, although for infants this would be 1-3 ml. The rectus femoris is a large and well-defined muscle and is the anterior muscle of the quadriceps. It is located halfway between the superior iliac crest and the patella (Workman 1999).

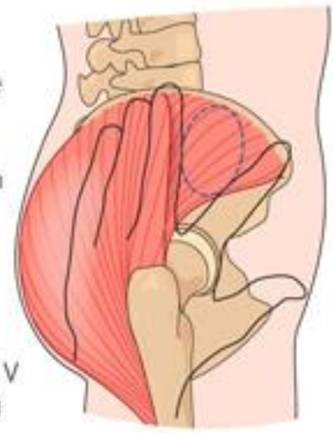


Vastus lateralis site

The vastus lateralis site: used for deep IM and Z-track injections. Up to 5ml can be administered (Rodger and King 2000). The muscle forms part of the quadriceps femoris group of muscles and is located on the outer side of the femur. It is found by measuring a hand's breath from the greater trochanter and the knee joint, which identifies the middle third of the quadriceps muscle (Workman 1999). There are no major blood vessels or structures which could cause an injury in this area (Rodger and King 2000).

Ventrogluteal site

This site is used for deep IM and Z-track injections. This site is located by placing the palm of the nurse's hand on the patient's opposite greater trochanter (for example, the nurse's right palm on the patient's left hip), then extending the index finger to the anterior superior iliac spine to make a 'V'. The injection is then given into the gluteus medius muscle, which is the centre of the V (Workman 1999, Rodger and King 2000).



5.4 Intranasal medication administration

Indications

1. To deliver fast acting medications where IM and IV/ IO access is inappropriate and/or difficult (e.g. rapid analgesia in children).

Contraindications

1. Based on the specific medication being administered.
2. Suspected nasal fractures.
3. Blood obstructing the nasal passage.
4. Mucus obstructing the nasal passage.
5. Medication not approved to be administered via atomiser

Adverse Effects

1. Based on the specific medication being administered.
2. Underdose is possible if not administered properly.
3. Mild, short-lasting discomfort (typically burning) from the drug itself.

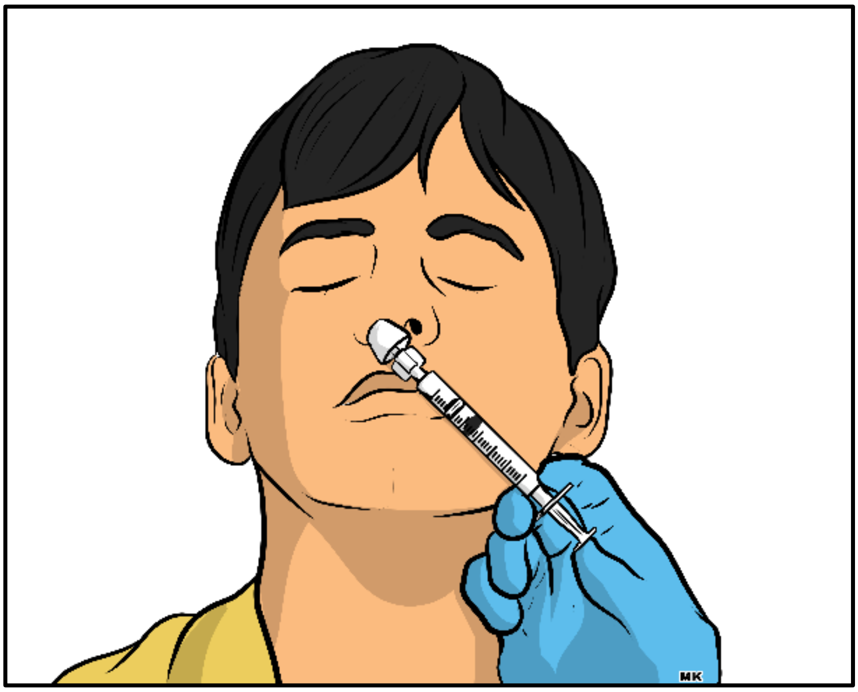
Standard precautions

Procedure

1. Ensure that the patient's BP, ECG, SpO2, HR and RR are being monitored.
2. Prepare the medication to be administered as per procedure 5.1. Prepare pre-pack C6.
3. Calculate and draw up the medication dosage required in a 3 mL syringe. Attach a three-way tap. Attach the atomizer to the 1ml syringe and draw out required half dose. The dose must be split into two 1 mL syringes (Half dose in each syringe).
4. The maximum volume per nostril is 0.5 ml (paediatrics) and 1ml (adults).
5. Control and stabilise the patient's head with one hand. Apply the atomiser gently but firmly into the nostril.
6. Aim slightly upward and towards the ear on the same side as the nostril.
7. Briskly compress the syringe plunger to deliver the medication into the nostril.
8. Remove the device, connect the second syringe and repeat the procedure above.
9. Move the device over to the opposite nostril and briskly administer the remaining half of the medication into the second nostril.
10. Reassess the patient's vital signs. Regularly assess the patient for adverse effects of the medication administered

Intranasal medication preparation with atomizer

Intranasal medication administration with atomizer



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5.5 Buccal medication administration

Indications
1. To deliver fast acting medications where IM and IV/ IO access is inappropriate and/ or difficult (e.g. oral glucose administration)
Contraindications
1. Oral glucose is contraindicated in patients who do not have airway control and are at risk of aspiration.
Adverse Effects
1. Aspiration 2. Tissue necrosis
Standard precautions

Procedure
1. Ensure that the patient's BP, ECG, SpO2, HR and RR are being monitored. 2. Prepare the medication to be administered as per Procedure 5.1. 3. Inspect the mouth for any airway obstruction. 4. Apply the medication e.g. oral glucose on the practitioner's gloved hand. 5. Lift the lips with the other hand and apply the medication e.g. oral glucose on the buccal membrane. 6. Be cautious not to insert your fingers between the patient's teeth for fear of a bite injury. 7. Reassess the patient's vital signs. Regularly assess the patient for side effects of the medication administered

Buccal medication administration

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5.6 Sublingual medication administration

Indications

1. To deliver medication sublingually e.g. Glyceryl Trinitrate (GTN).

Contraindications

1. Contraindications are related to the specific medication being administrated.

Adverse Effects

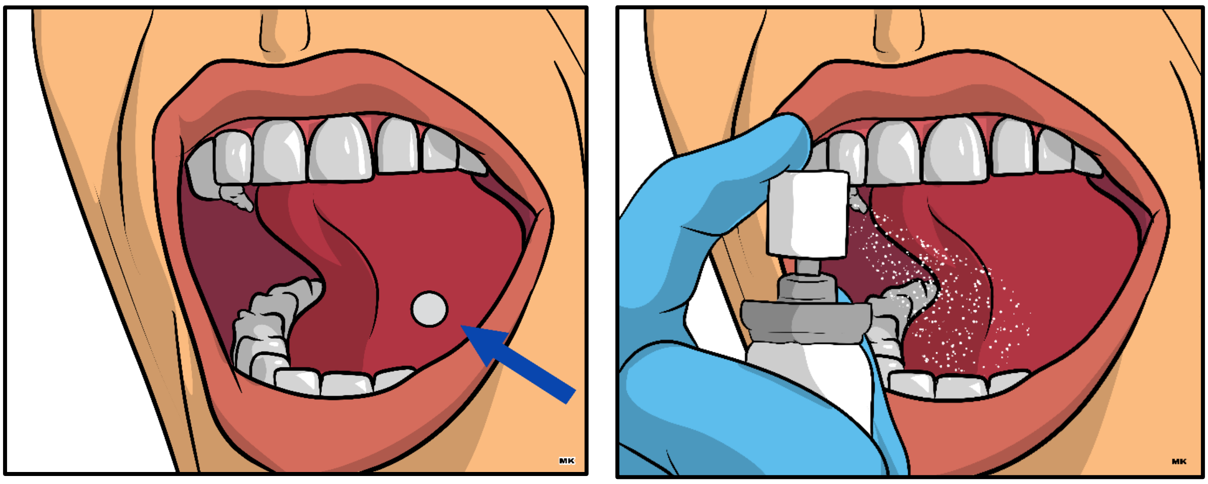
1. Are related to the specific medication being administered.

Standard precautions

Procedure

1. Ensure that the patient's BP, ECG, SpO2, HR and RR are being monitored.
2. Place the patient in the semi-fowlers position.
3. Prepare the medication to be administered as per Procedure 5.1.
4. Inspect the mouth for any airway obstruction.
5. Instruct the patient to open their mouth, raise their tongue and close it after the medication is administered. Example, 1 spray of GTN.
6. Inform patient to report any possible adverse effects.
7. Reassess patient after 5 minutes and repeat medication administration e.g. up to 3 doses of GTN, if needed.
8. Reassess the patient's vital signs. Regularly assess the patient for adverse effects of the medication administered

Sublingual medication administration



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5.7 Oral medication administration

Indications

1. Oral medication administration (e.g. Aspirin, Clopidogrel, Ibuprofen and Paracetamol).

Contraindications

1. Impaired conscious state or swallowing ability.
2. Are related to the specific medication being administered.

Adverse Effects

1. Aspiration and airway compromise.
2. Are related to the specific medication being administered.

Standard precautions

Procedure

1. Ensure that the patient’s BP, ECG, SpO2, HR and RR are being monitored.
2. Place the patient in the full-fowlers or semi-fowlers position.
3. Prepare the medication to be administered as per Procedure 5.1.
4. Inspect the mouth for any airway obstruction.
5. Place the medication in the patients mouth with a gloved hand or give the medication to the patient to place in their mouth.
6. Provide an adequate volume of water with tablets.
7. Monitor patient until all medications are swallowed. If tablets are dissolvable/ dispersible, ensure the whole volume of solution is taken.
8. Reassess the patient’s vital signs. Regularly assess the patient for side effects of the medication administered

5.8 Inhaled medication administration: Methoxyflurane

Indications

1. Mild to moderate pain in patients who are co-operative and able to understand and comply with instructions of self- administration.
2. Labour and obstetric pain and cardiac chest pain

Contraindications

1. Unconscious patients and patients who cannot self-administer the Penthrox®.
2. Exceeding 6 ml in 24 hours
3. Known hypersensitivity
4. History of Malignant hyperthermia

Precautions

1. Pre-existing renal disease/renal impairment.

Adverse Effects

1. Nausea and vomiting
2. Drowsiness
3. Coughing
4. A decrease in BP
5. Bradycardia
6. Mild amnesia (memory loss)
7. Exceeding the maximum dose of 6 mls in 24 hours may cause renal toxicity.

Standard precautions

Procedure

1. Ensure that the patient’s BP, ECG, SpO2, HR and RR are being monitored.
2. Prepare the medication as per Procedure 5.1
3. Attach the charcoal filter to the Penthrox® inhaler.
4. Tilt the Penthrox® inhaler and slowly pour the entire 3ml of the Penthrox® solution into the base of the inhaler while rotating it.
5. Shake lightly and ensure that the liquid is absorbed by the gauze pad located in the inhaler.
6. Explain the process of self- administration to the patient and place the wrist-loop around patient's wrist.
7. Ask the patient to inhale deeply and exhale through the mouthpiece at least 6 times. Once pain relief is achieved, ask the patient to breathe normally.
8. Monitor the patient for pain relief and/ or drowsiness (hands dropping, closing eyes).
9. Keep the empty ampoule to return to HMCAS Stores for replacement.

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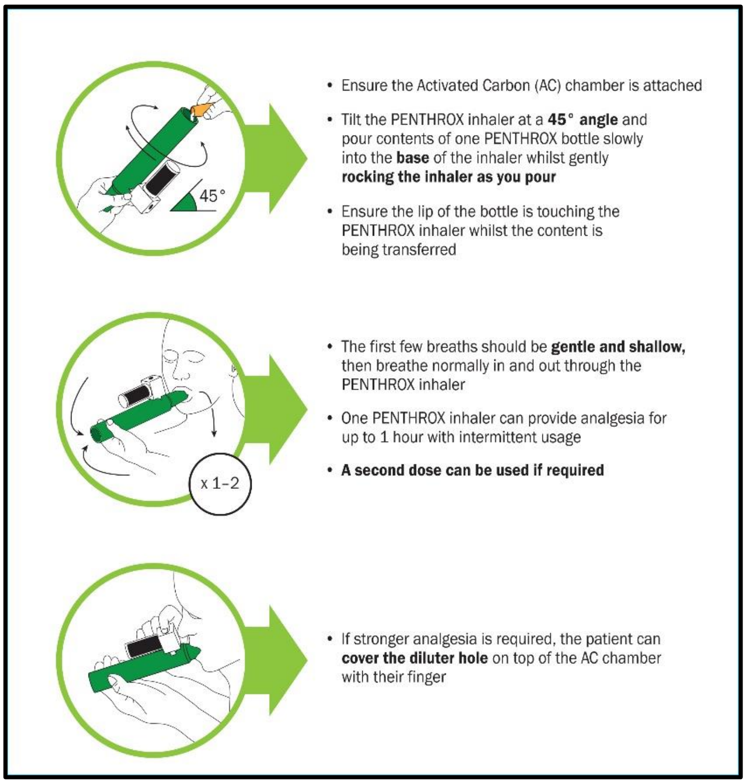
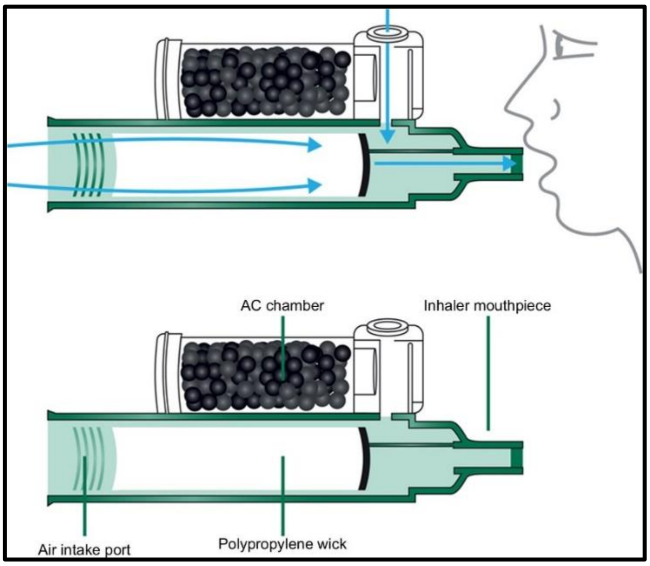
10. Record ALL changes in pain scores.
11. Reassess the patient's vital signs. Regularly assess the patient for adverse effects of the medication administered

Methoxyflurane preparation and administration





(International Paramedic College Australia, 2024)

5.9 Use of syringe drivers

Indications

1. Medication infusion required via syringe pump (Example GTN, Adrenaline, etc.)

Contraindications

1. Nil in this setting.

Adverse Effects

1. Fluid/ Medication infiltration
2. Compartment syndrome
3. Low battery
4. Over transfusion

Standard precautions

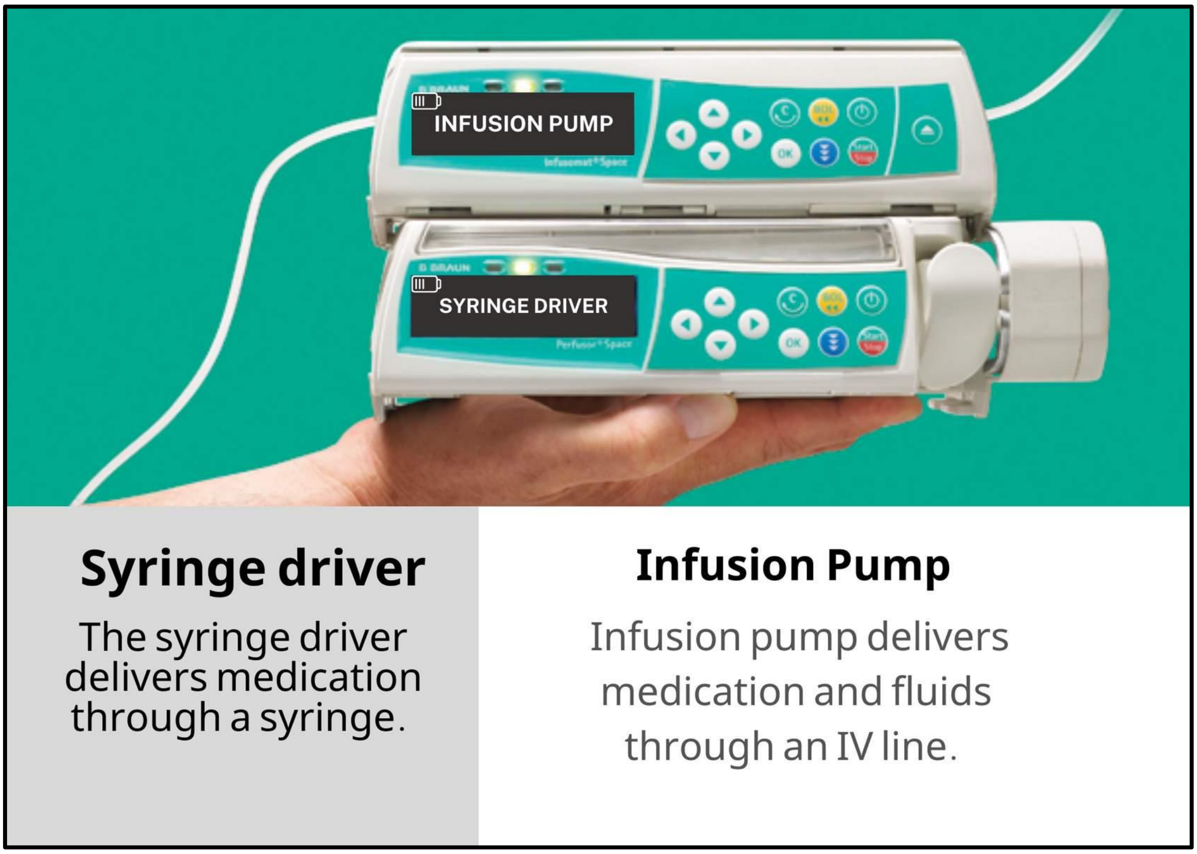
Procedure

1. Ensure that the patient's BP, ECG, SpO2, HR and RR are being monitored.
2. Prepare appropriate pre-pack (A10M/ A10T - adult/ A11-paediatric, A12/ A13, C8), check the expiry date.
3. Prepare the appropriate medication and solution in a 20 or 50ml syringe and label it appropriately, as per Procedure 5.1.
4. Attach the administration set to the syringe and prime the line manually.
5. Switch ON the Braun syringe pump and wait for the drive arm to fully open until it comes to a stop.
6. Open the front screen downwards. Pull and rotate the syringe holder clockwise.
7. Insert the prepared syringe and ensure correct placement in the syringe pump. Lock the syringe in place by rotating the syringe holder anticlockwise and close the front screen.
8. Scroll, press on the arrows to select the appropriate syringe from the displayed SYRINGE SELECTION menu and wait for syringe drive to configure settings.
9. Scroll using the down arrow to select VTBI (Volume to be infused) from the displayed main menu and enter the appropriate volume (20ml or 50ml).
10. Scroll down arrow to select Rate and enter the appropriate time as per guidelines. Use special functions as required.
11. Review and confirm all indicators for the infusion from the final menu displayed and push on the START button for the infusion to commence.
12. Confirm that the infusion is running.
13. Reassess the patient during the infusion administration and adjust infusion as

required.

14. Stop the infusion by pressing on the STOP button.
15. Clean and disinfect the infusion pumps using HMC-approved disinfectant according to the manufacturer's recommendations.

Syringe driver and Infusion pump



SYRINGE DRIVER BASIC SCREEN COMMANDS

STANDBY MODE

In the case of extended interruption, the user has the option to maintain the set values.

- Press to stop the infusion. Then press for less than 3 sec.
- Confirm that the pump is supposed to switch into standby by pressing .
- The pump is now in Standby.

While the pump is in the standby mode, it's display shows the drug and the remaining time for this mode. Change of remaining time by pressing . Exit standby by pressing .

As long as a disposable is inserted in the pump will use standby also in case is pressed for at least or more than 3 sec.

Press and hold for 3 sec to turn pump off. A white bar stretches from left to right and counts down the 3 sec.
As long there is a syringe inserted the pump will not turn off but will use stand-by.

2/2

SYRINGE DRIVER BASIC SCREEN COMMANDS

END OF INFUSION

- Press to stop the infusion. The green LED disappears. Disconnect the pump from the patient.
- Open the syringe holder. Answer the question whether a syringe change should be performed with . The drive moves backwards into the starting position.
- Open pump cover. Remove the syringe, move the syringe holder into an upright position and close the front door.
- Press for 3 sec. to switch the pump off. The drive moves into parking position.

Note: The settings will be permanently saved by the switched off device.

Note: Pump cannot be powered off with syringe inserted.

1/2

5.10 Use of infusion pumps

Indications

1. Medication infusion required via infusion pump (if available).

Contraindications

1. Nil in this setting.

Adverse Effects

1. Fluid/ Medication infiltration
2. Compartment syndrome
3. Low battery
4. Over transfusion

Standard precautions

Procedure

1. Ensure that the patient's BP, ECG, SpO2, HR and RR are being monitored.
2. Prepare appropriate pre-pack (A10M/ A10T-adult/A11-pediatric, A12/ A13, C8), check the expiry date.
3. Prepare the appropriate drug and solution in an appropriate size IV bag and label it appropriately, as per Procedure 5.1.
4. Insert administration set into IV bag and manually prime the tubing.
5. Switch ON the Braun syringe pump and wait for the drive arm to fully open until it comes to a stop.
6. Open the front screen by pressing the correct button.
7. Insert the prepared IV tubing correctly in the pump and close the mechanical door.
8. Follow prompts on the infusion pump screen.
9. Scroll using the down arrow to select VTBI (Volume to be infused) from the displayed main menu and enter the appropriate volume (100ml, 250ml or 500ml).
10. Scroll down arrow to select Rate and enter the appropriate rate as per AS guidelines.
11. Review and confirm all indicators for the infusion from the final menu displayed and push on the START button for the infusion to commence.
12. Confirm that the infusion is running.
13. Reassess the patient during the infusion administration and adjust infusion as required.
14. Stop the infusion by pressing on the STOP button.
15. Clean and disinfect the infusion pumps using HMC-approved disinfectant according to the manufacturer's recommendations.

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INFUSION PUMP BASIC SCREEN COMMANDS

IV LINE CHANGE

NOTE: Always close roller clamp and disconnect the patient before changing IV line to avoid inadvertent free flow.

1. Press Start/Stop to stop the infusion.
2. Press and respond to prompt to clamp IV line.
3. Open the pump door by pressing . Once door finishes opening automatically, pull down on door to fully open. Expect some resistance.
4. Press down the green opening lever completely until it locks in place and yellow triangle light is flashing. Remove IV line from right to left.
5. Insert new IV line following steps in "Open Door and Load IV Line" section above.

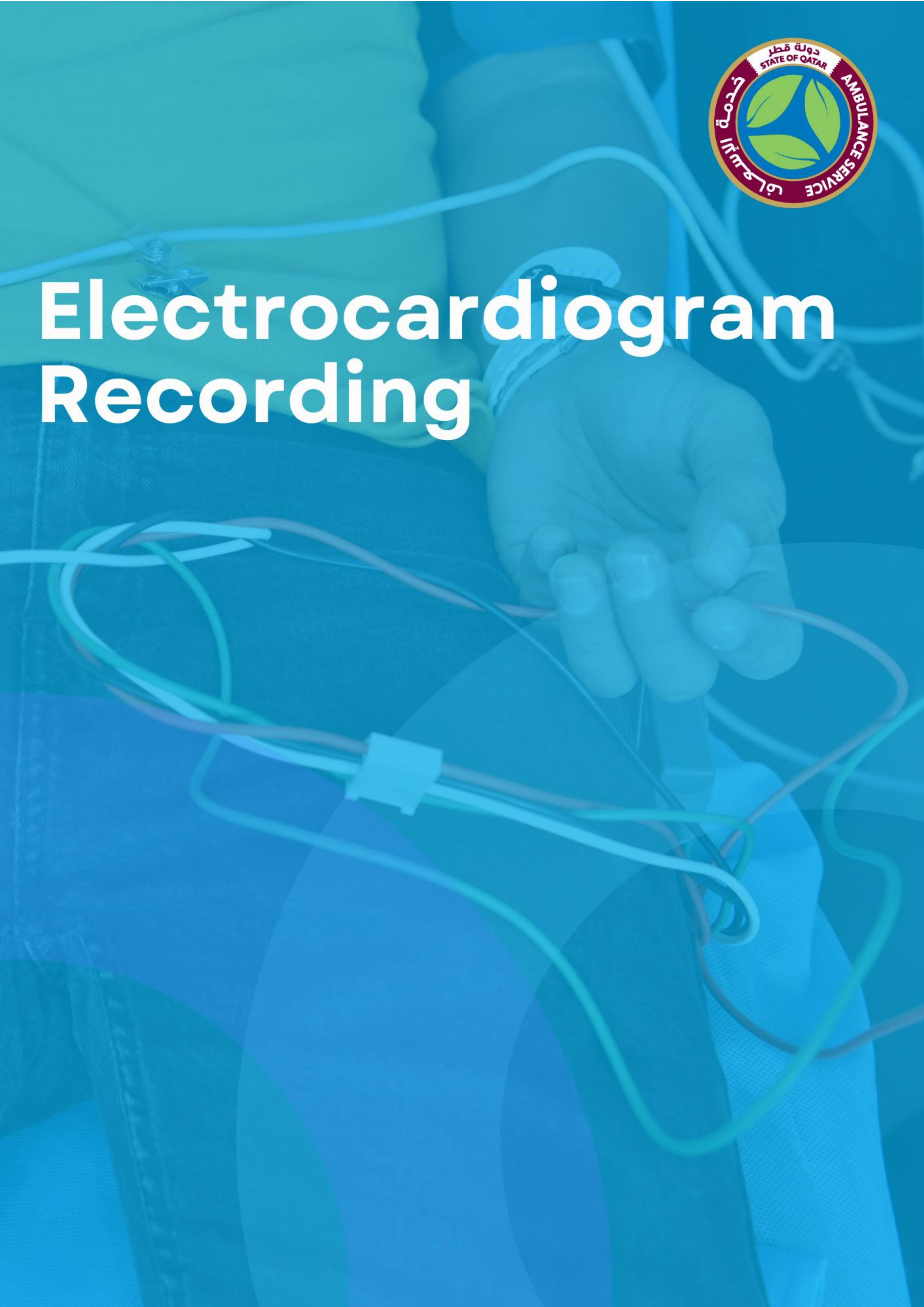
END INFUSION AND POWER OFF

1. Press Start/Stop to stop the infusion.
2. Follow steps 2 through 4 to remove IV line.
3. Close the pump door.
4. Press for 3 seconds to power off the pump.

NOTE: Pump cannot be powered off with IV line inserted.

STANDBY MODE

1. Press Start/Stop to stop the infusion. Leave the IV line inserted in the pump.
2. Press and hold for 3 seconds.
3. Exit Standby by pressing C.



Electrocardiogram Recording

6.1 Cardiac Monitoring: 4-Lead ECG

I II III IV

Indications

1. All unconscious patients or collapsed patients (or those who have recently been unconscious).
2. Patients complaining of chest pain or dyspnea.
3. Patients with suspected poisoning.
4. Patients who are poorly perfused/ shocked or hypoxic, or with abnormal vital signs.
5. When a CCP/ CTL requests the patient to be monitored during transport to hospital.

Patients in cardiac arrest are also to be monitored but this is done through the defibrillation pads in anticipation of a shockable rhythm.

Contraindications

1. Nil in this setting.

Adverse Effects

1. Nil in this setting.

Standard precautions

Procedure

1. Ensure that the patient's BP, SpO_2, and RR are being monitored.
2. Prepare LifePak 15 (4 limb lead cables attach electrodes to leads before placement on patient).
3. Make the patient comfortable.
4. Expose the distal parts of the patient's extremities (with discretion).
5. Ensure the patient is not in contact with any conductive material (remove jewellery and handover to patient/ family).
6. Attach limb leads with electrode in the correct location.
7. Continuously re-check that all cables are connected and confirm 4 lead attachments are connected.
8. Advise the patient to remain still during monitoring, if possible.
9. The leads primarily used in continuous cardiac monitoring are those referred to as 'limb leads', identified as follows:
   a. RA = right arm
   b. LA = left arm
   c. RL = right leg
   d. LL = left leg
10. If any abnormalities are noted in the 4 lead ECG, undertake a 12 lead ECG.

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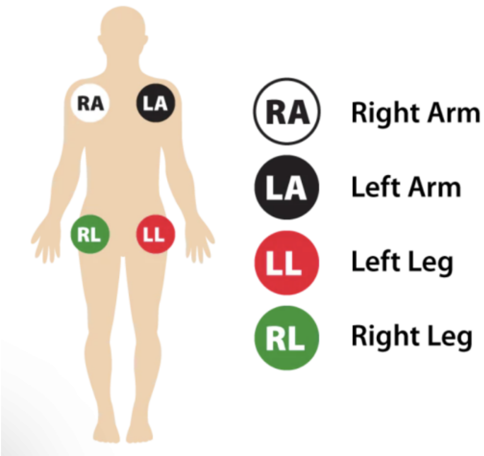
Limb (extremity) Electrodes and Placement



» RA (Right Arm) - Anywhere between the right shoulder and right elbow
» RL (Right Leg) - Anywhere below the right torso and above the right ankle
» LA(Left Arm) - Anywhere between the left shoulder and the left elbow
» LL (Left Leg) - Anywhere below the left torso and above the left ankle

6.2 Cardiac Monitoring: 6-Lead ECG acquisition

Indications

1. A patient ≥ 30 years of age with chest pains/ palpitations.
2. A patient ≥ 50 years of age with dyspnea/ decreased level of consciousness/ upper extremity pain/ syncope/ weakness.
3. A patient ≥ 60 years of age with a history of diabetes mellitus/ female patient.
4. A patient ≥ 70 years of age with abdominal pain/ nausea and vomiting.

Contraindications

1.Nil in this setting.

Adverse Effects

1. Nil in this setting.

Standard precautions

Procedure

1. Ensure that the patient's BP, SpO_2, HR and RR are being monitored.
2. Prepare LifePak 15 (4 limb lead cables attach electrodes to leads before placement on patient).
3. Make the patient comfortable and position the patient in the supine position.
4. Expose the distal parts of the patient's extremities (with discretion).
5. Ensure the patient is not in contact with any conductive material (remove jewelry).
6. Attach limb leads with electrode in the correct location.
7. Advise the patient to remain still, if possible.
8. Select 12-lead button once on the LifePak 15 and enter patient's age and gender.
9. Change the radio channel to Clinical 1 and contact CCE if indicated.
10. Send 6-Lead ECG with call sign and HBE number of LifePak 15 if indicated.
11. Perform ongoing exam while waiting for CCE confirmation.
12. Confirm the transmission by ECG printout or CCE.

6.3 Cardiac Monitoring: 12-Lead ECG acquisition

Indications

1. A patient ≥ 30 years of age with chest pains/ palpitations.
2. A patient ≥ 50 years of age with dyspnea/ decreased level of consciousness/ upper extremity pain/ syncope/ weakness.
3. A patient ≥ 60 years of age with a history of diabetes mellitus/ female patient.
4. A patient ≥ 70 years of age with abdominal pain/ nausea and vomiting.

Contraindications

1. Nil in this setting.

Adverse Effects

1. Nil in this setting.

Standard precautions

Procedure

1. Ensure that the patient's BP, SpO_2, HR and RR are being monitored.
2. If chest pain, treat with Aspirin and GTN prior to acquiring 12-lead ECG.
3. Prepare LifePak 15 (4 limb lead cables + 6 chest lead cables, attach electrodes to leads before placement on patient).
4. Make the patient comfortable and position the patient in the supine position (if possible, if APO record ECG with patient in full-fowlers position).
5. Expose the distal parts of the patient's extremities and chest (with discretion).
6. Shave and dry the chest, if necessary.
7. Ensure the patient is not in contact with any conductive material (remove jewellery and handover to patient/ family).
8. Attach limb leads with electrode in the correct location.
9. Attach precordial leads in the correct location:
   a) V1 on fourth intercostal space, right of sternum.
   b) V2 on fourth intercostal space, left of sternum.
   c) V4 on fifth intercostal space, mid- clavicular line.
   d) V3 in between V2 and V4.
   e) V5 on anterior axillary line, fifth intercostal space.
   f) V6 on mid-axillary line, fifth intercostal space
10. Re-check that all cables are connected and confirm 12-lead attachments is connected.
11. Advise the patient to remain still, if possible.

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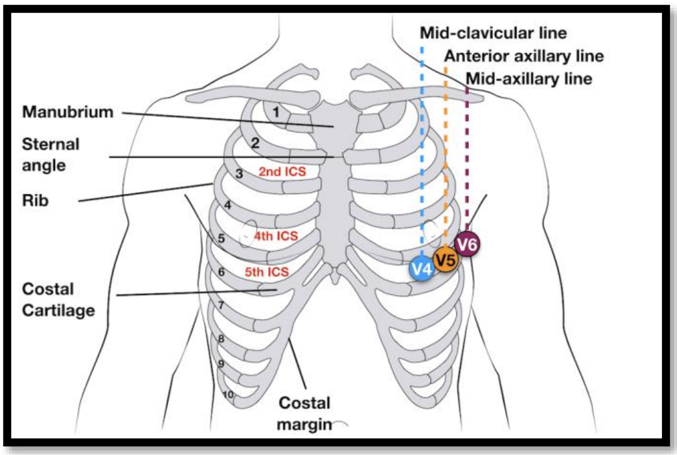
12. Select 12-lead button once on the LifePak 15 and enter patient's age and gender.
13. Inform the NCC to change radio channel to Clinical 1 and contact CCE for ECG interpretation if meets the criteria.
14. Ensure transmission setting is programmed to NCC and not ePCR prior to transmission.
15. Perform ongoing exam while waiting for CCE confirmation.
16. Administer analgesia as required.
17. Confirm the transmission by ECG printout or CCE.
18. If STEMI confirmed, remove the chest leads and attach the defibrillation pads.
19. If STEMI identified, then update CTL with patients' information to notify Heart Hospital.
20. If STEMI identified, minimise time on scene. Load the patient and transport as soon as possible. Meet Charlie en-route.
21. Endorse the patient with the next level caregiver.

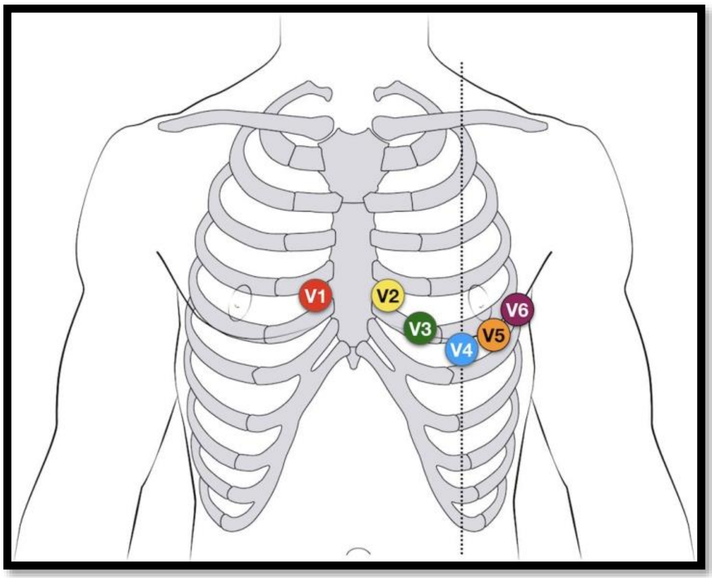
15 Lead-ECG acquisition

22. If ST-segment depression in V1, V2 and V3 then record a V7, V8 and V9 looking for the presence of a posterior STEMI.
23. If Inferior STEMI then record a V4r, V5r and V6r looking for Right ventricular.
24. Clean and disinfect the infusion pumps using HMC-approved disinfectant according to the manufacturer's recommendations.

12-Lead ECG precordial lead placement

- V1: 4th intercostal space (ICS), RIGHT margin of the sternum
- V2: 4th ICS along the LEFT margin of the sternum
- V4: 5th ICS, mid-clavicular line
- V3: midway between V2 and V4
- V5: 5th ICS, anterior axillary line (same level as V4)
- V6: 5th ICS, mid-axillary line (same level as V4)

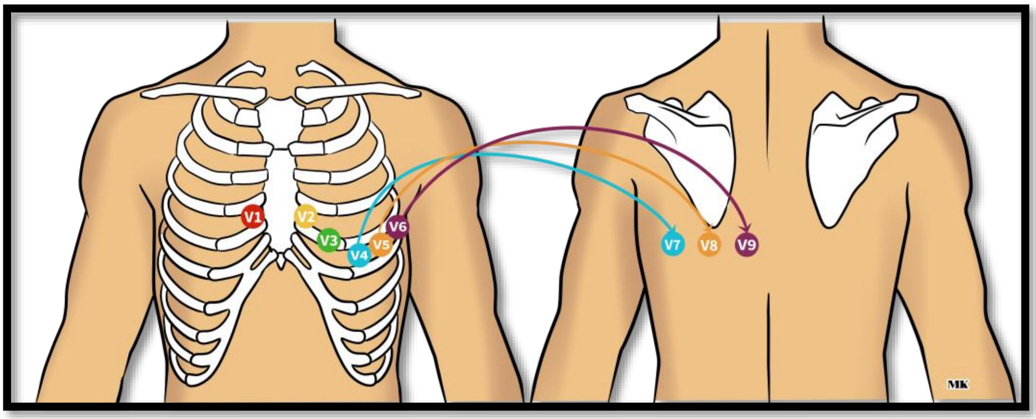




Cadogan, M. (2022, January 30). *ECG Lead positioning*. Life in the Fast Lane • LITFL. https://litfl.com/ecg-lead-positioning/

15 Lead-ECG electrodes placement if inferior STEMI

- ✓ Locate V4R position: 5th intercostal space, right side mid-clavicular line
- ✓ Move **V1** lead to the **V4R** position
- ✓ Locate **V8** position: in line with anterior 5th intercostal space,
- ✓ mid scapular line
- ✓ Move **V5** lead to **V8** position
- ✓ Locate **V9** position: in line with **V8** position,
- ✓ left paraspinal border
- ✓ Move **V6** lead to **V9** position



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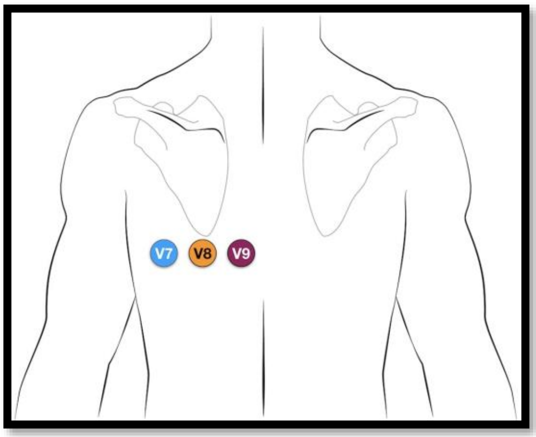
15 Lead-ECG electrodes placement if posterior STEMI

Leads V7-9 are placed on the posterior chest wall in the following positions:

- **V7** – Left posterior axillary line, in the same horizontal plane as V6.
- **V8** – Tip of the left scapula, in the same horizontal plane as V6.
- **V9** – Left paraspinal region, in the same horizontal plane as V6.



Cadogan, M. (2022, January 30). *ECG Lead positioning*. Life in the Fast Lane • LITFL. https://litfl.com/ecg-lead-positioning/

6.4 Cardiac Monitoring: 12-Lead ECG telemetry

Indications

1. A patient ≥ 30 years of age with chest pains/ palpitations.
2. A patient ≥ 50 years of age with dyspnea/ decreased level of consciousness/ upper extremity pain/ syncope/ weakness.
3. A patient ≥ 60 years of age with a history of diabetes mellitus/ female patient.

A patient ≥ 70 years of age with abdominal pain/ nausea and vomiting.

Contraindications

1. Nil in this setting.

Adverse Effects

1. Nil in this setting.

Standard precautions

Procedure

1. Ensure that the patient's BP, SpO_2, HR and RR are being monitored.
2. If chest pain, treat with Aspirin and GTN prior to acquiring 12-lead ECG.
3. Prepare LifePak 15 (4 limb lead cables + 6 chest lead cables, attach electrodes to leads before placement on patient).
4. Check to ensure that the Lifepak 15 is fitted with a modem device.
5. Make the patient comfortable and position the patient in the supine position (if possible), if APO record ECG with patient in full-fowlers position).
6. Expose the distal parts of the patient's extremities and chest (with discretion).
7. Shave and dry the chest, if necessary.
8. Ensure the patient is not in contact with any conductive material (remove jewellery and handover to patient/ family).
9. Attach limb leads with electrode in the correct location.
10. Attach precordial leads in the correct location:
    - V1 on fourth intercostal space, right of sternum.
    - V2 on fourth intercostal space, left of sternum
    - V4 on fifth intercostal space, mid- clavicular line.
    - V3 in between V2 and V4.
    - V5 on anterior axillary line, fifth intercostal space.
    - V6 on mid-axillary line, fifth intercostal space
11. Re-check that all cables are connected and confirm 12-lead attachments is connected.

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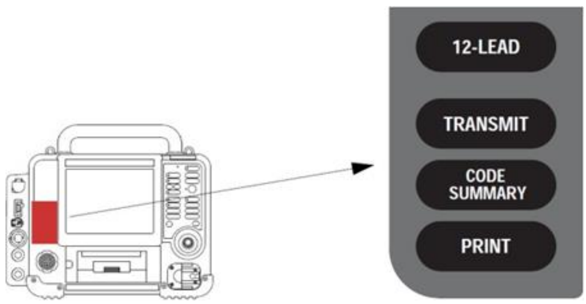
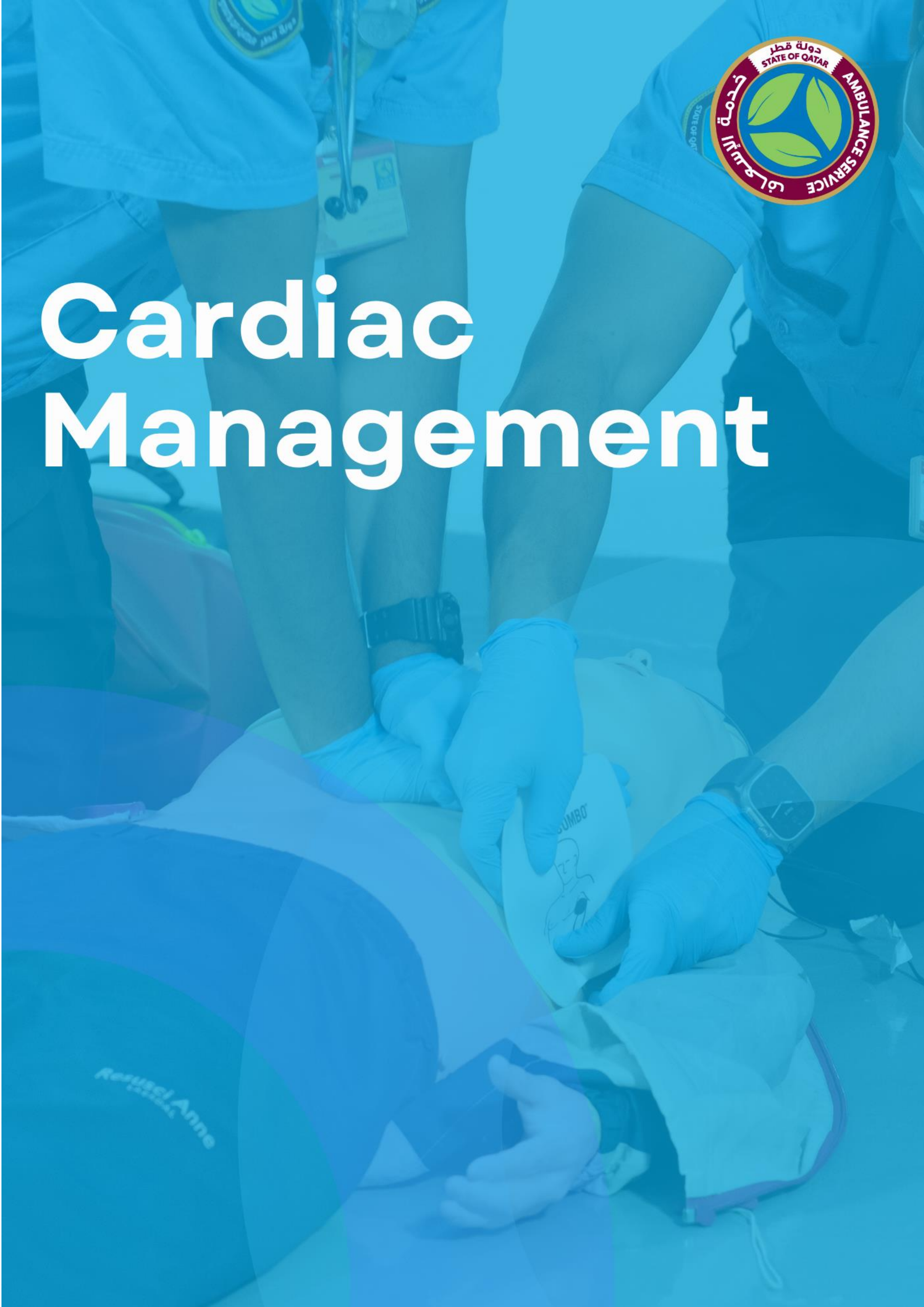
12. Advise the patient to remain still, if possible.
13. Select 12-lead button once on the LifePak 15 and enter patient's age and gender, ensuring transmission to NCC not to ePCR.
14. Contact the CCD via clinical 1 channel for ECG interpretation if meets the criteria
15. Perform ongoing exam while waiting for CCE confirmation.
16. Confirm the transmission by ECG printout or CCE.
17. If STEMI confirmed, remove the chest leads and attach the defibrillation pads.
18. If STEMI identified, then update CCE with patients' information to notify Heart Hospital.
19. If STEMI identified, minimise time on scene. Load the patient and transport as soon as possible. Meet Charlie en-route.

    Clean and disinfect the infusion pumps using HMC-approved disinfectant according to the manufacturer's recommendations.

LIFEPAK 15 12-lead ECG acquisition/transmission control area



Cardiac Management



7.1 Declaration of death: withholding resuscitation

Indications

1. Decapitation
2. Incineration
3. Decomposition
4. Rigor mortis
5. Dependent lividity
6. Evisceration of major organs
7. Foetal maceration
8. Putrefaction
9. Cranial/cerebral destruction
10. Hemicorporectomy or similar significant injury
11. Pre-term infant <20 weeks gestation
12. A valid “Do Not Resuscitate” (DNR) Order

Contraindications

1. Nil contra-indications based on the indications

Adverse Effects

1.Nil contra-indications based on the indications

Standard precautions

Procedure

1. Assess the patient for the absence of signs of life.
2. Confirm the patient is unresponsive (AVPU)
3. Confirm no breathing.
4. Confirm no peripheral and central pulses
5. Record a 4-lead ECG. The 4- lead ECG rhythms must show asystole in ALL views.
6. Pupils must be fixed and dilated.
7. If in doubt about withholding resuscitation, initiate resuscitation while seeking consultation and advice from the CCE
8. Update NCC accordingly.

Coordinate the movement of the body of the deceased with family, Delta and Police (If required).

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7.2 Declaration of death: Withdrawal of resuscitation efforts

Indications

1. Resuscitation efforts on adult patients may be terminated if the criteria below are met:

- No breathing and no pulses
- Asystole or broad PEA (agonal rhythm) < 40 bpm for ≥20 minutes
- Reversible causes have been managed
- The patient's family or significant other gives verbal consent to stop CPR (if present on scene)
- Where no CCP is on scene, APs must contact the CC: Emergency to confirm termination of resuscitation.

Contraindications

1. Nil in this setting.

Adverse Effects

1. Nil in this setting

Standard precautions

Procedure

1. Update NCC accordingly and request backup of Delta, CCP and Charlie (as required).
    - No breathing and no pulses
    - Asystole or broad PEA (agonal rhythm) < 40 bpm for ≥20 minutes
    - Reversible causes have been managed
    - The patient's family or significant other gives verbal consent to stop CPR (if present on scene)
    - Where no CCP is on scene, APs must contact the CC: Emergency to confirm termination of resuscitation.
2. NOTE: Points (e) and (f) must be confirmed by at least 2 HMCAS staff.
3. Update NCC accordingly and note the time of death

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7.3 Chest compression: Manual (Adult)

Indications

1. Adult medical cardiac arrest.
2. Adult trauma cardiac arrest.

Contraindications

1. Nil in this setting

Adverse Effects

1. Nil in this setting

Standard precautions

Procedure

1. Begin manual chest compressions if the patient is UNRESPONSIVE, no breathing or no normal breathing (agonal breathing) for medical cardiac arrest patients. And UNRESPONSIVE, no breathing or no normal breathing (agonal breathing), and no carotid pulse for trauma cardiac arrest.
2. Place the patient on a hard surface.
3. If at any time during MANUAL compressions, the victim speaks out, moves, or moans, compressions should stop immediately.
4. If the patient is on the floor, kneel next to the patient's chest.
5. Place the palm of your hand on the lower half of the sternum.
6. With straight arms compress the chest to a depth of 5 cm in an adult patient at a rate of between 100 to 120 per minute.
7. Allow for full recoil of the chest and ensure an equal compression and release ratio.
8. Request feedback on compressions from your partner.
9. Update NCC of cardiac arrest and request backup.
10. Treat the patient accordingly based on the ECG rhythm as per CPG 2.1 and 2.2.

7.4 Chest compression: Mechanical (LUCAS device 2/3)

Indications

1. Adult (above 14 years old) medical cardiac arrest. Adult (above 14 years old) trauma cardiac arrest during transportation.

Contraindications

1. If it is not possible to position LUCAS 2/3 safely or correctly on the patient's chest.
2. Too small patient: If LUCAS 2/3 alerts with 3 fast signals when lowering the Suction Cup, and you cannot enter the PAUSE mode or ACTIVE mode.
3. If the patient is too large: if you cannot lock the Upper Part of LUCAS 2/3 to the Back Plate without compressing the patient's chest.

Adverse Effects

1. Bruising and soreness of the chest.
2. Fractured ribs

Standard precautions

Procedure

1. Your partner should position the LUCAS 2/3 bag with its top nearest to you. Put their left hand on the black strap on the left-hand side and pull the red handle so that the bag unfolds.
2. They should turn the device on by pushing ON/OFF on the User Control Panel for 1 second to power up LUCAS 2/3 in the bag and start the self-test. The green LED adjacent to the ADJUST key illuminates when LUCAS is ready for use
3. They must then remove the LUCAS 2/3 Back Plate from the Carrying Bag.
4. When applying the LUCAS 2/3 device to the patient, stop manual chest compressions. Make sure that you support the patient's head. Carefully put the LUCAS 2/3 Back Plate under the patient, immediately below the arm pits. Use one of these procedures.
5. slightly raise the patient's upper body,
6. Lie the patient onto the backplate.
7. Get your partner to hold the handles on the support legs to remove the LUCAS 2/3 Upper Part from the bag. Pull the release rings once to make sure that the claw locks are open. Then let go of the release rings.
8. Do not interrupt chest compressions for longer than 10 seconds.
9. Attach the support leg that is nearest to you to the Back Plate. Attach the other support leg to the Back Plate, so that the two support legs lock against the Back Plate. Listen for click. Pull up once to make sure that the parts are correctly attached.
10. Push STEP 1 button, then the suction cup down with two fingers until the pressure pad

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touches the patient's chest and confirm correct placement.

11. Push STEP 2, the PAUSE to lock the Start Position of the suction cup - then remove your fingers from the suction cup.
12. Push STEP 3 ACTIVE: Continuous For intubated patient and 30:2 for not intubated (LTA and BVM).
13. Confirm appropriate placement of the LUCAS 2/3 device by standing at the patient's feet.
14. Reposition the LUCAS 2/3 device if required.
15. Place LUCAS 2/3 support strap behind patient's lower neck and then secure to LUCAS 2/3 device – tighten to ensure LUCAS 2/3 does not move.
16. Strap the patient's arms to the LUCAS 2/3 device after IV insertion and or before transportation.
17. Pause the LUCAS 2/3 device to analyse the rhythm after every 2 minutes.
18. Remember to regularly assess the LUCAS 2/3 position as the device may move out of position.
19. Clean and disinfect the LUCAS 2/3 device using HMC-approved disinfectant according to the manufacturer's recommendations.

LUCAS 2

LUCAS 3

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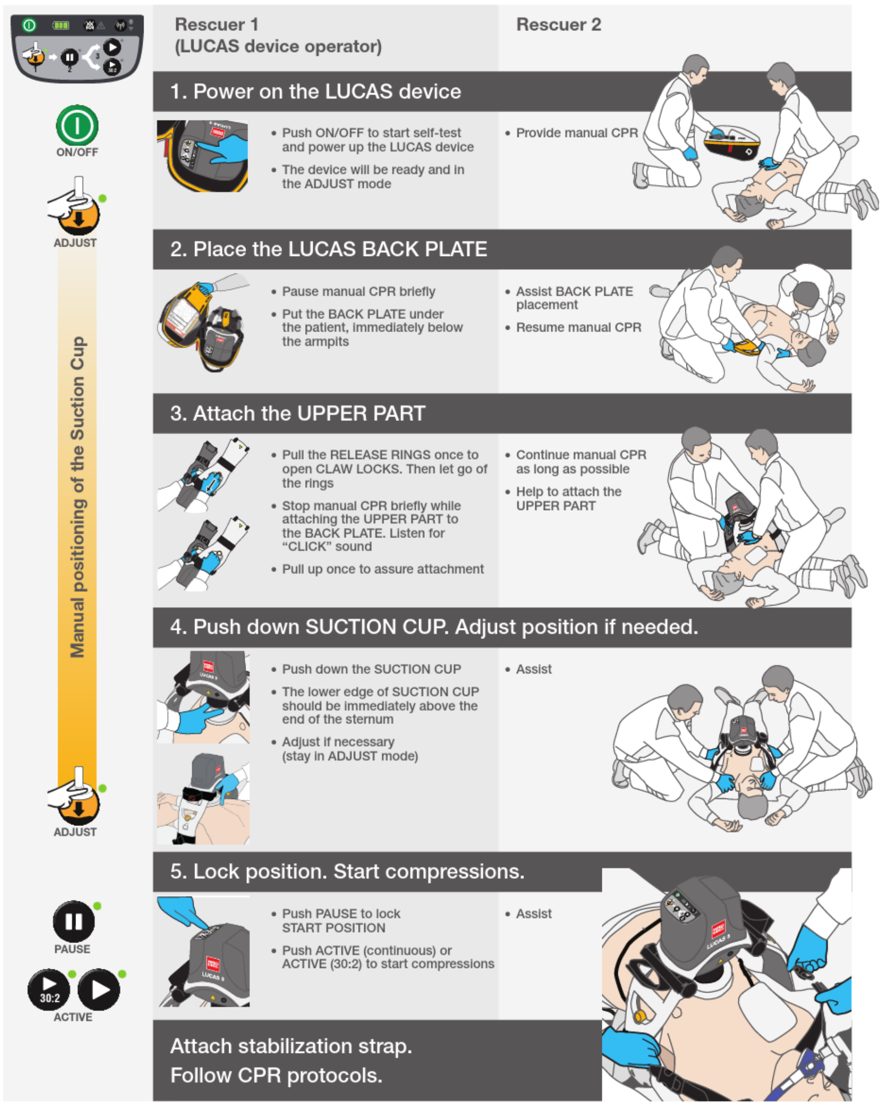

PHYSIO CONTROL

LUCAS® 3

CHEST COMPRESSION SYSTEM

QUICK REFERENCE GUIDE

	Rescuer 1 (LUCAS device operator)	Rescuer 2
1. Power on the LUCAS device (ON/OFF)	• Push ON/OFF to start self-test and power up the LUCAS device • The device will be ready and in the ADJUST mode	• Provide manual CPR
2. Place the LUCAS BACK PLATE (ADJUST)	• Pause manual CPR briefly • Put the BACK PLATE under the patient, immediately below the armpits	• Assist BACK PLATE placement • Resume manual CPR
3. Attach the UPPER PART	• Pull the RELEASE RINGS once to open CLAW LOCKS. Then let go of the rings • Stop manual CPR briefly while attaching the UPPER PART to the BACK PLATE. Listen for "CLICK" sound • Pull up once to assure attachment	• Continue manual CPR as long as possible • Help to attach the UPPER PART
4. Push down SUCTION CUP. Adjust position if needed. (ADJUST)	• Push down the SUCTION CUP • The lower edge of SUCTION CUP should be immediately above the end of the sternum • Adjust if necessary (stay in ADJUST mode)	• Assist
5. Lock position. Start compressions. (PAUSE, ACTIVE, 30:2)	• Push PAUSE to lock START POSITION • Push ACTIVE (continuous) or ACTIVE (30:2) to start compressions	• Assist

Manual positioning of the Suction Cup

Attach stabilization strap.
Follow CPR protocols.

The LUCAS 3 device is for use as an adjunct to manual CPR when effective manual CPR is not possible (e.g., transport, extended CPR, fatigue, insufficient personnel). Refer to the Instructions For Use for complete directions for use, indications, contraindications, warnings, precautions and potential adverse events.

www.physio-control.com/LUCAS

Physio-Control is now part of Stryker.

GDR 3328215_B |USA| Rx Only

stryker

7.5 Cardiac arrest: Adult medical resuscitation

I II III IV

Indications

1. Unconscious medical adult patient with absence of normal breathing or presence of agonal breathing on visual inspection of the chest wall for 5-10 seconds.

Contraindications

1. Undeniable death).

Adverse Effects

1. CPR induced consciousness
2. Rib fractures.
3. Bruising and chest discomfort.

Standard precautions

1. A face shield and gown are required when treating patients with a travel history from high-risk countries. (emerging and reemerging infectious diseases)

Procedure

1. Determine if the patient is unconscious by using the “shout and shake" principle. Immediately tap or shake the patient on their shoulders and ask loudly, “ARE YOU OK?”
2. Open and assess the airway using the head tilt and chin lift technique, as required.
3. Expose the patient’s chest so you can correctly determine if there is normal equal rise and fall of the chest. (Care should be taken to maintain patient privacy – particularly in females).
4. To assess breathing, kneel at the patient’s side and look at their exposed chest for at least 5 seconds but not more than 10 seconds. Normal breathing is equal rise and fall of the chest. Do not try and determine the rate at this stage. If there is no rise and fall of the chest, or short, sudden intakes or gasps of air usually associated with a snore or grunting noise – this patient is considered as to have no normal breathing.
5. Begin manual chest compressions if normal breathing absent or agonal breathing present as per Procedure 7.3. If at any time during MANUAL compressions, the victim speaks out, moves, or moans, compressions should be stopped immediately.
6. Perform high quality manual chest compressions immediately.
7. Place the palm of your hand on the lower half of the sternum. Compression depth should be 5 cm in an adult patient at a rate of between 100 to 120 per minute.
8. Allow for full recoil of the chest and ensure an equal compression and release ratio.
9. Confirm case of cardiac arrest with NCC and request for Delta and Charlie help.
10. Request your partner to apply defibrillation pads and to prepare the LUCAS 2/3 device.
11. Request for feedback on compressions from your partner.

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12. Once the defibrillation pads are attached, stop chest compressions momentarily and analyse the ECG rhythm.
13. If Pulseless Ventricular Tachycardia or Ventricular Fibrillation provide a shock
14. Resume chest compressions while the Defib is charging for200 j
15. If Asystole or Pulseless Electrical Activity continue with manual chest compressions
16. Insert/delegate correct LTA size, confirm placement by EtCO2 wave form then perform 30 compressions to 2 ventilations (30:2 ratio)
17. Perform rhythm checks every 2 min in less than 10 seconds.
18. Recognize the need to check pulse (except for VF or Asystole)
19. Ensure that any interruption in chest compressions will be less than 10 seconds.
20. Insert or delegate partner to insert IV.
21. Administer 1 mg Adrenaline 1:10000 every 4 minutes.
22. Attach LUCAS after 6 minutes and place it in correct position in less than 10 seconds set the Lucas device to 30:2 mode.
23. Consider Vector Change Defibrillation for refractory ventricular fibrillation.
24. Administer Amiodarone 300mg IV/IO for VF and VT after the 3rd defibrillation. Second dose of Amiodarone 150mg IV/IO after 4th defibrillation.
25. Check for expiry date of medications and verify with partner, keep a record of the time and what medications were given.
26. Treat the possible reversible causes.
27. Stop CPR if ROSC is achieved. Positive signs of ROSC include a palpable central pulse, a persistent rise in ETCO2 to the > 40 mm Hg.
28. Post ROSC care:10 mins on scene, SpO2 > 90%, ETCO2 35-45 mmHg, Systolic blood pressure above 90mmHg, RBS (4-6.7mmol/L),12 Lead ECG treat H's & T's.
29. Consider the need to load and go using of Combi-carrier/scoop stretcher/carry sheet to package the patient

Vector change (VC) defibrillation

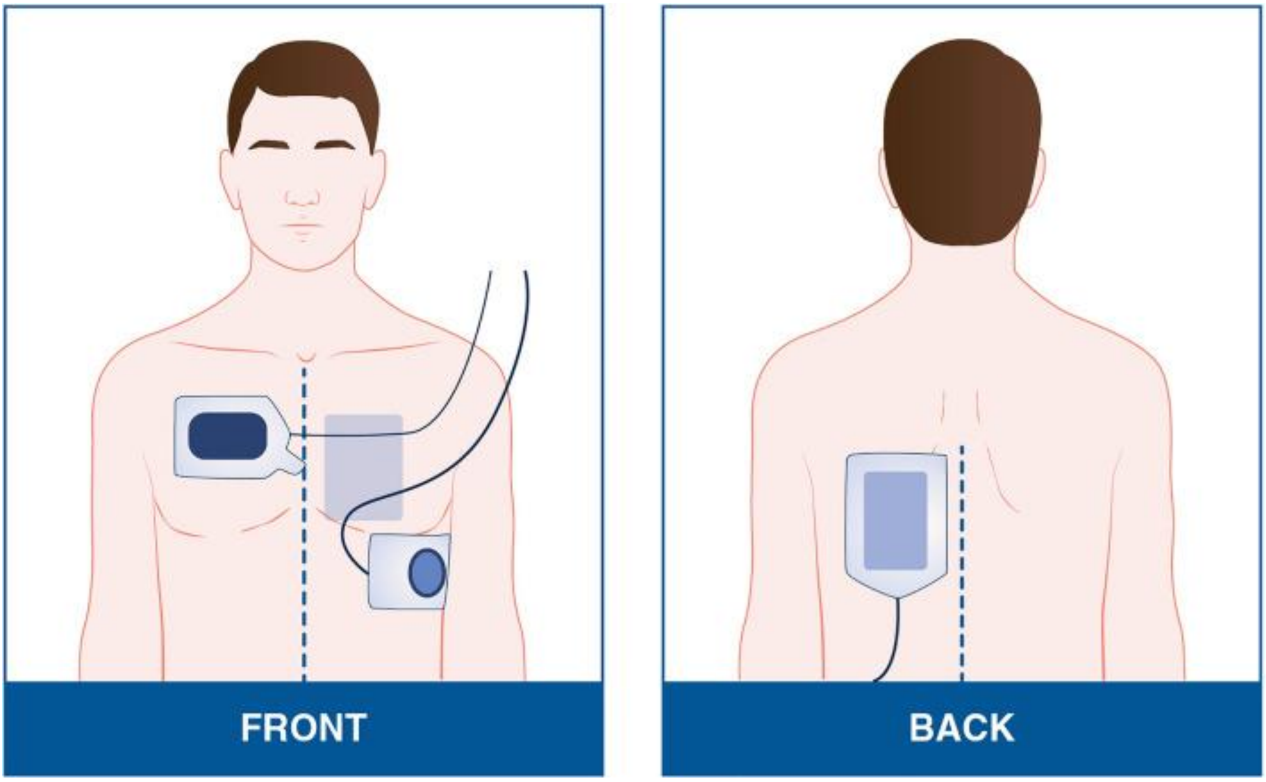
Vector change: Is a defibrillation strategy to improve outcomes in patient with refractory VF. to be performed if a patient shows a shockable rhythm. Pad changes should occur during the placement of the backplate or prior to applying the Lucas device.

Refractory VF definition: After 2 minutes of compressions and the patient receiving the maximum shock of 360J, the patient remains in ventricular fibrillation (VF).

Pads placement for VC:

Pad 1 – left anterior chest between midline of chest and left nipple.

Pad 2 – left posterior back, left of spine, just below the scapula.



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7.6 Cardiac arrest: Adult trauma resuscitation

Indications

1. Unconscious adult trauma patient with the absence of breathing on visual inspection of the chest wall for 5-10 seconds and the absence of a carotid pulse.
2. Unconscious adult trauma patient with the presence of agonal breathing on visual inspection of the chest wall for 5-10 seconds and the absence of a carotid pulse.

Contraindications

1. Nil in the setting of adult traumatic cardiac arrest (except in situations of undeniable death).

Adverse Effects

1. Rib fractures.
2. Bruising and chest discomfort.

Standard precautions

1. A face shield and gown are required if the patient has catastrophic bleeding.

Procedure

1. Stop all detected major external bleeding.
2. Consider Cervical spine manual immobilisation
3. Determine if the patient is unconscious by using the “shout and shake" principle. Immediately tap or shake the patient on their shoulders and ask loudly, “ARE YOU OK?” If the patient is NOT unconscious, they are likely to answer, move, or moan. If the patient remains unresponsive that the person is considered to be unconscious
4. Assess for airway patency and open the airway using the jaw thrust technique, as required.
5. If the patient is unconscious, expose the patient’s chest so you can correctly determine if there is normal equal rise and fall of the chest. (Care should be taken to maintain patient privacy – particularly in females).
6. To assess breathing, kneel at the patient’s side and look at their exposed chest for at least 10 seconds. Normal breathing is equal rise and fall of the chest. Do not try and determine the rate at this stage. If there is no rise and fall of the chest, or short, sudden intakes or gasps of air usually associated with a snore or grunting noise – this patient is considered to have NO normal breathing.
7. Check the carotid pulse for 10 seconds simultaneously while checking for breathing.
8. Begin manual compressions if normal breathing is absent or agonal breathing present and absent carotid pulse.
9. Perform high quality manual chest compressions immediately.

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10. Second practitioner to release the c-spine, Inform NCC and request Delta and Charlie help, apply defib pads and give feedback on chest compressions
11. Bilateral chest decompression based on privileging (unless isolated head trauma).
12. The following must happen simultaneously for a patient with suspected polytrauma, when possible:
    1. Stop major external bleeding.
    2. Bilateral chest decompression based on privileging / Bilateral Finger Thoracostomies (CCP ONLY), unless isolated head trauma.
    3. Apply the pelvic binder.
    4. Establish IV/ IO (CCP only) access and administer up to 1 L of Normal Saline.
    5. Establish a patent airway when practical.
13. If it's not possible to perform the skills/ procedures listed above simultaneously with manual chest compressions, then stop catastrophic bleeding and perform chest decompression/Thoracostomy before initiating chest compressions.
14. Once the defibrillation pads are applied, stop manual chest compressions momentarily and analyse the ECG rhythm.
15. Provide shock for shockable rhythms and resume compressions for non-shockable rhythms
16. Resume CPR and monitor for 2 minutes.
17. Insert ILTS-D and confirm its position with EtCO2 and attach the BVM to ILTS-D (Passive oxygenation).
18. Start positioning Pelvic Binder
19. Stop chest compression to help with the pelvic binder application.
20. Every 2 min, analyze the rhythm and manage accordingly.
21. Insert IV line and give fluid (500 to 1000 ml) then first dose of Epinephrine.
22. Start chest compression / ventilation 30:2
23. Stop CPR if ROSC is achieved. Positive signs of ROSC include a palpable central pulse, a persistent rise in ETCO2 to the > 40 mm Hg.
24. Post ROSC care:10 mins on scene, SpO2 > 90%, ETCO2 35-45 mmHg, Systolic blood pressure above 90mmHg, RBS (4-6.7mmol/L),12 Lead ECG treat H's & T's.
25. Should the patient meet transportation criteria to hospital whilst in cardiac arrest, then apply the LUCAS 2/3 device and activate. Place the patient on the CombicarrierII with SMR.
26. Communicate with the Command Desk to update the Trauma Resuscitation Unit.
27. Minimize time on scene and load the patient. Rendezvous point with Charlie en- route.

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7.7 Cardiac arrest: Pediatric medical resuscitation

Indications

1. Unconscious paediatric patient with the absence of normal breathing on visual inspection of the chest wall for 5-10 seconds and the absence of a carotid/brachial pulse.
2. Unconscious patient with the presence of agonal breathing on visual inspection of the chest wall for 5-10 seconds and the absence of a carotid/ brachial pulse

Contraindications

1. Undeniable death.

Adverse Effects

1. Nil in this setting

Standard precautions

Procedure

1. Ask the family members for the patient's weight. Alternatively estimate the weight using the formula: [0 – 12 Months: (Age x 0.5) + 4] [1 – 5 Years: (Age x 2) + 8] [6 – 14 Years: (Age x 3) + 7].
2. Determine if the patient is unconscious by using the "shout and shake" principle. Immediately tap or shake the patient on their shoulders and ask loudly, "ARE YOU OK?" If the patient is NOT unconscious, they are likely to answer, move, or moan. If the patient remains unresponsive that person is UNCONSCIOUS. Determine LOC (< 5 SECONDS)
3. Assess for airway patency and open the airway using the head tilt and chin lift technique, apply pads under the shoulder as required.
4. If the patient is unconscious, expose the patient's chest so you can correctly determine if there is normal equal rise and fall of the chest. (Care should be taken to maintain patient privacy – particularly in females).
5. To assess breathing, kneel at the patient's side and look at their exposed chest for no more than 10 seconds. Simultaneously palpate for a central pulse. Normal breathing is equal rise and fall of the chest. Do not try and determine the rate at this stage. If there is NO rise and fall of the chest, or short, sudden intakes or gasps of air usually associated with a snore or grunting noise – this patient is considered to have NO normal breathing.
6. Check the carotid pulse (brachial < 2 years old) 10 seconds simultaneously while checking breathing.
7. Confirm cardiac arrest in less than 15 seconds.
8. Start chest compressions, instruct partner to call for help, prepare BVM, and provide

real time feedback on chest compressions.

9. Provide 5 rescue breaths.
10. Perform chest compressions on the lower half of the sternum at a compression/ ventilation ratio of 30:2 (1 rescuer) at a rate of 100 to 120 compressions/ minute. Depth of compression of 4cm in infants and 5cm in children. Allow for full chest recoil, Equal compression and release ratio. Request feedback on chest compression quality.
11. Place a pad beneath the shoulders to open the airway.
12. Change person providing compressions every 2 minutes.
13. Delegate defibrillation pad placement and provide immediate defibrillation at 4J/ Kg if required (VF/ pulseless VT). Any subsequent shocks thereafter at > 4J/ Kg (maximum 10J/Kg or adult dose).
14. Direct resumption of CPR for 2-minute cycles (30:2) two or more person CPR, change the compressor after every 2 minutes.
15. Prepare LTA with ECO2 monitoring, once inserted continue 15:2 compression ventilation ratio.
16. IF ET is inserted, continuous compressions + ventilations 1 breath every 6 – 8 seconds...
17. If non-shockable rhythm (Asystole/ PEA), continue with 2 minutes cycles of CPR before rhythm checks.
18. Obtain more information about arrest (SAMPLE history and possible reversible causes)
19. Insert IV line and give fluid bolus (10-20 ml/kg x 3)
20. After 2 minutes analyse the rhythm(<10sec) (ensure patient is clear), Repeat defibrillation as needed (2nd shock 4J/kg)
21. Direct resumption of CPR starting with chest compressions for 2 mins compression / ventilation 30:2
22. Administer 0.01mg/kg (1:10 000) Adrenaline IV followed by saline flush or raise the arm and repeat every other cycle or 4mins if no ROSC; check RBS.
23. After 2 minutes analyse the rhythm (<10 sec) (ensure patient is clear)- Repeat defibrillation as needed (4J/kg for the 3rd shock)
24. Administer 1st dose Amiodarone 5mg/kg (if CCP on scene) after 3rd shock IV. If it exceeds 300 mg, use adult dose (300 mg after 3rd shock then 150 mg after 4th shock)
25. After 2 minutes analyse the rhythm(<10sec)
26. Person doing compressions clears everybody and then defibrillates patient safely. Administer Amiodarone 2nd dose 5mg/kg after 4th shock, flush line then administers 2nd adrenaline dose
27. Exclude / treat reversible causes (H’s & T’s)
28. After 2 minutes analyse the rhythm(<10sec) (ensure patient is clear)
29. Person doing compressions clears everybody and then defibrillates patient safely.

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continue 15:2 compression ventilation ratio for 2 mins.
30. Consider amiodarone 3rd/final dose 5mg/kg after 5th shock.
31. Stop CPR if ROSC is achieved. Positive signs of ROSC include a palpable central pulse, a persistent rise in ETCO2 to the > 40 mm Hg.
32. Post Cardiac Arrest Care: Confirmation of ROSC, sinus rhythm, pulse present HR > 60; increase in ETCO2.
33. Ventilate as needed, remove LT if Patient is breathing/ gag reflex and no CCP on scene – maintain SpO2 > 94 %, ETC02 (35 -45)
34. Treat hypotension- SBP < [Age x 2 + 70=SBP] - 10 to 20 ml/kg
35. 2.5ml/kg 10% dextrose if needed (RBS<4mmol) max of 50ml
36. Consider transport to hospital using Pedi Pac or CombiCarrierII as indicated

The WAAFELSS: This mnemonic may be used to guide care while managing a paediatric cardiac arrest:

W	Weight: 0-12 months = (age x 0.5) + 4, 1-5 years = (age x 2) + 8, 6-14 years = (age x 3) + 7
A	Adrenaline: 0.01mg/kg (0.1ml/kg 1:10,000)
A	Amiodarone: 5mg/kg
F	Fluid: 10-20 ml/kg x 3
E	Energy: 4J/kg & CCPs can escalate to max 10J/kg
L	LT-size: size 1 = 5-12 kg, size 2 = 12-25 kg. ET-size: uncuffed (age÷4) + 4 and cuffed (age÷4) + 3.5. Depth: (age ÷ 2) + 12
S	Sugar: 2.5ml/kg of 10% Dextrose
S	Systolic BP: (age x 2) + 70

7.8 Cardiac arrest: Pediatric trauma resuscitation

I II III IV

Indications

1. Unconscious paediatric trauma patient with the absence of breathing on visual inspection of the chest wall for 5-10 seconds and the absence of a carotid pulse.
2. Unconscious paediatric trauma patient with the presence of agonal breathing on visual inspection of the chest wall for 5-10 seconds and the absence of a carotid pulse.

Contraindications

1. Undeniable death.

Adverse Effects

1. Nil in this setting

Standard precautions

1. A face shield and gown are required if the patient has catastrophic bleeding.

Procedure

1. Stop all detected major external bleeding.
2. Consider Cervical spine manual immobilisation.
3. Determine if the patient is unconscious by using the “shout and shake" principle. Immediately tap or shake the patient on their shoulders and ask loudly, “ARE YOU OK?” If the patient is NOT unconscious, they are likely to answer, move, or moan. If the patient remains unresponsive that the person is considered to be unconscious
4. Assess for airway patency and open the airway using the jaw thrust technique, as required.
5. If the patient is unconscious, expose the patient’s chest so you can correctly determine if there is normal equal rise and fall of the chest. (Care should be taken to maintain patient privacy – particularly in females).
6. To assess breathing, kneel at the patient’s side and look at their exposed chest for at least 10 seconds. Normal breathing is equal rise and fall of the chest. Do not try and determine the rate at this stage. If there is no rise and fall of the chest, or short, sudden intakes or gasps of air usually associated with a snore or grunting noise – this patient is considered to have NO normal breathing.
7. Check the carotid pulse for 10 seconds simultaneously while checking for breathing.
8. Begin manual compressions if normal breathing is absent or agonal breathing present and absent carotid pulse.
9. Perform high quality manual chest compressions immediately.
10. Second practitioner to release the c-spine, Inform NCC and request Delta and Charlie

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help, apply defib pads and give feedback on chest compressions.

11. Bilateral chest decompression based on privileging (unless isolated head trauma).
12. The following must happen simultaneously for a patient with suspected polytrauma, when possible:
13. Stop major external bleeding.
14. Bilateral chest decompression based on privileging (unless isolated head trauma).
15. Apply the pelvic binder.
16. Establish IV/ IO (CCP only) access and administer 20ml/kg max 1 L of Normal Saline.
17. Establish a patent airway when practical.
18. If it's not possible to perform the skills/ procedures, listed above simultaneously with manual chest compressions, then stop catastrophic bleeding and perform chest decompression before initiating chest compressions.
19. Once the defibrillation pads are applied, stop manual chest compressions momentarily and analyse the ECG rhythm.
20. Provide shock for shockable rhythms and resume compressions for non-shockable rhythms.
21. Resume CPR and monitor for 2 minutes.
22. Insert ILTS-D and confirm its position with $EtCO_2$ and attach the BVM to ILTS-D (Passive oxygenation).
23. Start positioning Pelvic Binder
14. Stop chest compression to help with the pelvic binder application.
15. Every 2 min, analyse the rhythm and manage accordingly.
16. Insert IV line and give fluid based on patient's weight (20 ml/Kg max 1 L)
17. Administer 0.01mg/kg (1:10 000) Adrenaline IV followed by saline flush or raise the arm and repeat every other cycle or 4mins if no ROSC; check RBS.
18. Start chest compression / ventilation 30:2
19. Stop CPR if ROSC is achieved. Positive signs of ROSC include a palpable central pulse, a persistent rise in $ETCO_2$ to the > 40 mm Hg.
20. Post ROSC care: SpO_2 > 90%, $ETCO_2$ 35-45 mmHg, Systolic blood pressure above 90mmHg, RBS (4-6.7mmol/L),12 Lead ECG treat H's & T's.
21. Should the patient meet transportation criteria to hospital whilst in cardiac arrest, place the patient on the CombicarrierII or PediPac with SMR.
22. Communicate with the Command Desk to update the Trauma Resuscitation Unit.
23. Minimize time on scene and load the patient. Rendezvous point with Charlie en- route

7.9 Neonatal resuscitation

Indications

1. Newborn/neonate patient with no signs of life: Absent or poor tone and no normal breathing /absent breathing
2. Newborn/neonate patient with Signs of inadequate perfusion: Absent or poor tone, Unable to establish normal breathing and Heart rate < 100 beats/minute

Contraindications

1. Undeniable death.

Adverse Effects

1. Nil in this setting

Standard precautions

Procedure

1. Initial Stabilization (30 Seconds) Suction, Dry, Warm, Stimulate:
   - Ensure the neonate is kept warm.
   - Clear secretions, if necessary.
   - Dry and stimulate the neonate thoroughly.
2. Apnea or Gasping? (30 Seconds):
   - Provide Positive Pressure Ventilations (PPV) without PEEP or supplemental oxygen at a rate of 60 inflations per minute
   - Attach SpO_2 sensor to right hand
   - Attach ECG / Pads
3. Assess Heart Rate and Respond Appropriately (30 Seconds):
   a. **Heart Rate between 60 and 100/min:**
   - Continue PPV at 60/min
   - Assess SpO_2 using the targeted levels (refer to SpO_2 table)
   - If target SpO2 is not being achieved, apply MR SOPA method:

     M: Mask adjustment
     R: Reposition airway
     S: Suction
     O: Open mouth
     P: Pressure increase (PEEP valve set to 5cm H2O)
     A: Alternative airway (LT Size 0 or ETT)
   - Consider oxygen at 5 L/min if SpO2 is not being achieved after applying MR SOPA method.
   - Evaluate other potential causes (e.g. hypothermia, hypoglycaemia, toxins from

maternal history).

b. Heart Rate is < 60 bpm:

- Insert advanced airway (LT Size 0 or ETT)
- Initiate 3:1 CPR (3 compressions to 1 ventilation)
- Continue BVM ventilation with PEEP valve (5cm H2O) with an oxygen flow rate of 5 L/min. Increase oxygen flow by 1 L/min increments, if necessary.
- If these interventions do not improve heart rate to more than 60/min, establish IO access and administer 0.01 mg/kg Epinephrine every 4 minutes.
- Consider hypovolemia if heart rate does not respond despite ventilations, compressions and epinephrine. Administer NaCl at 10 mL/kg.

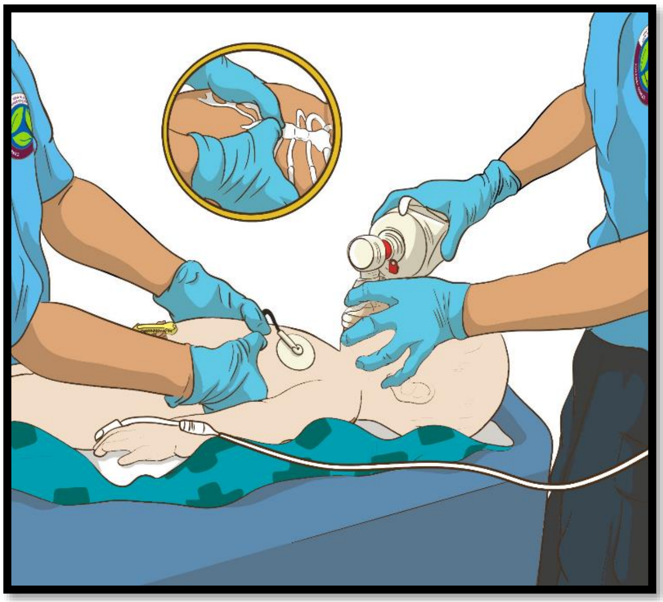
4. Continuous Re-Evaluation:

Continuously monitor and reassess the neonate's tone, breathing and heart rate. Incorporate assessment of Hs and Ts, including hypoglycaemia. Ensure effective ventilation using MR SOPA method.

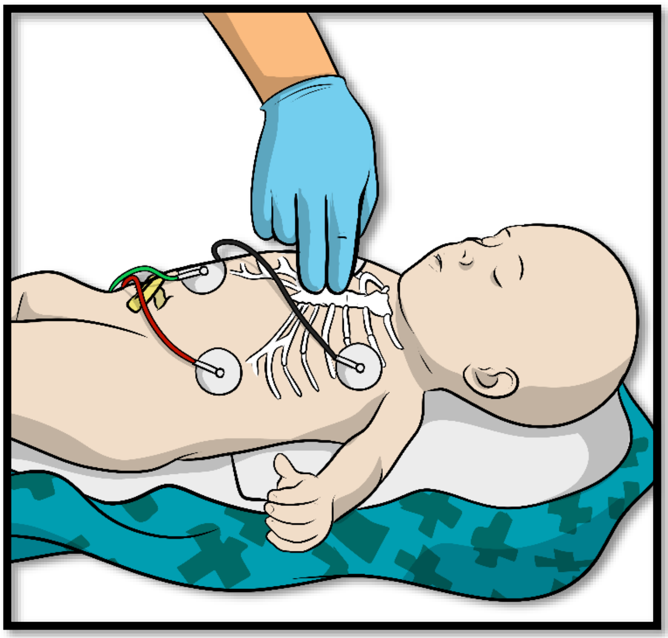
Targeted Preductal SpO_2 After Birth	
1 min	**60%–65%**
2 min	**65%–70%**
3 min	**70%–75%**
4 min	**75%–80%**
5 min	**80%–85%**
10 min	**85%–95%**

Neonatal chest compression technique

a. **Two-thumb technique**



b. **Two fingers technique**



7.10 Cardiac arrest: Obstetric patient resuscitation

Indications

1. Unconscious pregnant patient with the absence of normal breathing on visual inspection of the chest wall for 5-10 seconds.
2. Unconscious pregnant patient with the presence of agonal breathing on visual inspection of the chest wall for 5-10 seconds.

Contraindications

1. Undeniable death.

Adverse Effects

1. Nil in this setting

Standard precautions

Procedure

1. Determine if the patient is unconscious by using the “shout and shake" principle. Immediately tap or shake the patient on their shoulders and ask loudly, “ARE YOU OK?” If the patient is NOT unconscious, they are likely to answer, move, or moan. If the patient remains unresponsive that person is considered to be UNCONSCIOUS.
2. If the patient is unconscious, expose the patient’s chest so you can correctly determine if there is normal equal rise and fall of the chest. (Care should be taken to maintain patient privacy – particularly in females).
3. To assess breathing, kneel at the patient’s side and look at their exposed chest for at least 5 seconds but not more than 10 seconds. Normal breathing is equal rise and fall of the chest. Do not try and determine the rate at this stage. If there is NO rise and fall of the chest, or short, sudden intakes or gasps of air usually associated with a snore or grunting noise – this patient is considered to have NO normal breathing.
4. Begin manual compressions if normal breathing absent or agonal breathing present. Perform manual compressions in a slightly higher (1 cm) position than normal on the sternum.
5. If at any time during MANUAL compressions, the victim speaks out, moves, or moans, compressions should be stopped immediately.
6. Place the palm of your hand on the lower half of the sternum. Compression depth should be 5 cm at a rate of between 100-120/ minute.
7. Allow for full recoil of the chest and ensure an equal compression and release ratio.
8. Inform NCC of cardiac arrest and request CCP and Delta help.
9. Perform left lateral uterine displacement.
10. Request your partner for defibrillation pads placement and LUCAS 2/3 preparation.

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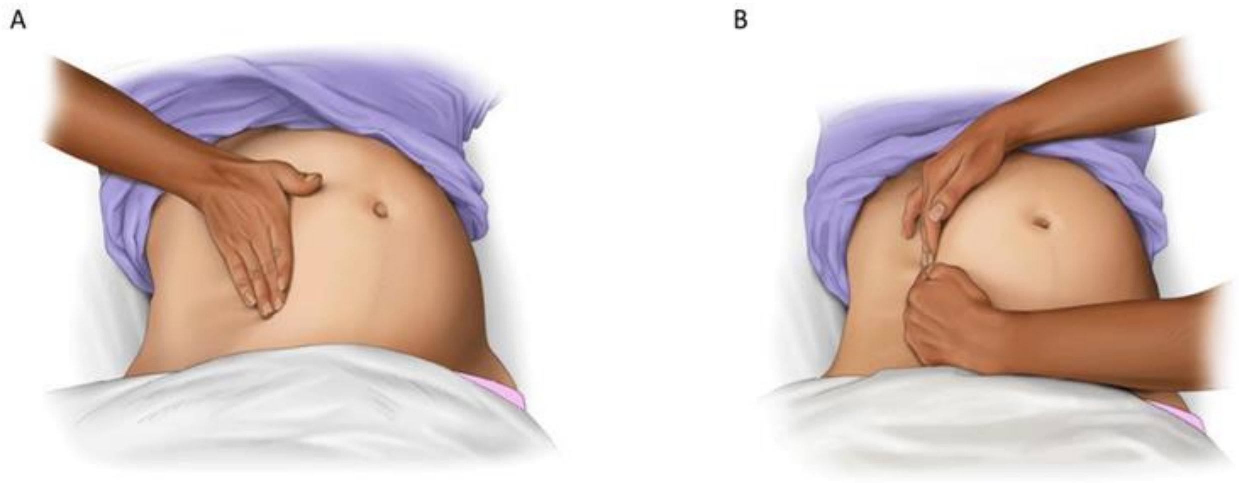
11. Request for feedback on compressions from your partner.
12. After defibrillation pads are attached, stop chest compressions momentarily and analyse the ECG rhythm.
13. Analyse the ECG rhythm:
    a. Pulseless Ventricular Tachycardia or Ventricular Fibrillation
    b. Asystole or Pulseless Electrical Activity.
14. Treat the patient accordingly based on the ECG rhythm and the algorithm for adult medical cardiac arrest.
15. If non-shockable rhythm (Asystole/ PEA), apply the LUCAS 2/ 3 after the 3rd cycle and activate, as per Procedure 7.4.
16. Continue with 2 minutes cycles of CPR before rhythm checks.
17. Epinephrine/Adrenaline is administered every 4 minutes during resuscitation 1mg IV/ IO.
18. Amiodarone (if available) at 300mg IV/IO is administered for VF and pulseless VT after the 3rd defibrillation. Second dose of Amiodarone 150mg IV/IO after 4th defibrillation.
19. Consider a fluid bolus administration of Sodium Chloride.
20. Consider Magnesium toxicity as a cause of maternal cardiac arrest in patients who have been given Magnesium. Stop any Magnesium infusions and administer Calcium Chloride IV.
21. Continue with resuscitation to the nearest hospital after pre-notification with CCE.
22. Stop CPR if ROSC achieved. Positive signs of ROSC include a palpable central pulse, a persistent rise in ETCO2 to the normal range (> 40 mm Hg).
23. Minimize time on scene and load the patient. Rendezvous with Charlie en- route.

Left Uterine Displacement technique

Left uterine displacement (LUD) is a recommended maneuver during obstetric patient resuscitation to minimize the gravid uterus compression on the abdominal aorta and inferior vena cava.

Left uterine displacement (LUD) should be performed with one hand (A) or two hands(B) by pushing or pulling the abdomen to the patient's left side.



A, Manual LUD, performed with one-handed technique. **B,** Two-handed technique during resuscitation.

7.11 Defibrillation with the LIFEPAK15: AED

Indications

1. Unconscious medical or trauma patient (more than 8 years old) with the absence of normal breathing on visual inspection of the chest wall for 5-10 seconds.
2. Unconscious patient (more than 8 years old) with the presence of agonal breathing on visual inspection of the chest wall for 5-10 seconds.
3. The AED advice "SHOCK ADVISED" By voice and screen message

Contraindications

1. Patient less than 8 years old
2. LIFEPAK 15 machine not calibrated for AED mode

Adverse Effects

1.Burns to the chest

Standard precautions

Procedure

1. Determine if the patient is unconscious by using the "shout and shake" principle. Immediately tap or shake the patient on their shoulders and ask loudly, "ARE YOU OK?" If the patient is NOT unconscious, they are likely to answer, move, or moan. If the patient remains unresponsive that person is considered to be UNCONSCIOUS.
2. If the patient is unconscious, expose the patient's chest so you can correctly determine if there is normal equal rise and fall of the chest. (Care should be taken to maintain patient privacy – particularly in females).
3. To assess breathing, kneel at the patient's side and look at their exposed chest for at least 5 seconds but not more than 10 seconds. Normal breathing is equal rise and fall of the chest. Do not try and determine the rate at this stage. If there is NO rise and fall of the chest, or short, sudden intakes or gasps of air usually associated with a snore or grunting noise – this patient is considered to have NO normal breathing.
4. Begin manual compressions if normal breathing absent or agonal breathing present. If at any time during MANUAL compressions, the victim speaks out, moves, or moans, compressions should be stopped immediately.
5. Place the palm of your hand on the lower half of the sternum. Compression depth should be 5 cm in an adult patient at a rate of between 100 to 120 per minute.
6. Allow for full recoil of the chest and ensure an equal compression and release ratio.
7. Request for feedback on compressions from your partner.
8. Inform NCC of cardiac arrest and request CCP and Delta help.
9. Request your partner to attach the defibrillation pads in the anterior-lateral position.

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10. Turn on the LIFEPAK 15 and select the AED mode.
11. Prepare and place the LUCAS 2/3 after the 3rd cycle/ 6 minutes of CPR in a Medical adult patient.
12. After the AED pads are applied, stop chest compressions momentarily and analyse the ECG rhythm.
13. Push the ANALYZE button. The "PUSH ANALYZE" prompts occur when the patient is properly connected to the AED.
14. Do not move the AED during analysis. Moving the AED during analysis may affect the ECG signal resulting in an inappropriate "SHOCK" or "NO SHOCK ADVISED" decision. Do not touch the patient or the AED during analysis.
15. The "ANALYZING NOW—STAND CLEAR"
16. prompts occur. Analysis of the patient's ECG takes approximately 6 to 9 seconds and then advises either "SHOCK ADVISED" or "NO SHOCK ADVISED".
17. Continue to follow the screen messages and voice prompts provided by the AED.
18. If the AED detects a shockable rhythm, the "SHOCK ADVISED" prompts occur. Charging to the joule setting for Shock begins. A charging bar appears and a ramping tone sound.
19. When charging is complete, the available energy is displayed. The "STAND CLEAR, PUSH SHOCK BUTTON!"
20. message occurs, followed by a "Shock ready" tone. Clear everyone away from touching the patient, bed, or any equipment that is connected to the patient. Press (shock) to deliver energy to the patient.
21. When the (shock) button is pressed, the "ENERGY DELIVERED" message occurs indicating that the energy transfer was completed.
22. If you do not press the shock button within 60 seconds, or the SPEED DIAL is pressed to cancel charging, the defibrillator disarms, and the DISARMING message appears.
23. After a shock is delivered, the "START CPR" prompts occur. A countdown timer (min: sec format) continues for the duration specified in the CPR TIME 1 setup option.
24. The CPR metronome automatically provides audible compression "tocks" and ventilation prompts or tones only during CPR intervals.
25. When the CPR countdown time ends, the "PUSH ANALYZE" prompts occur. These prompts repeat every 20 seconds until you press ANALYZE.
26. If the AED detects a non-shockable rhythm, the "NO SHOCK ADVISED" prompts occur. The defibrillator does not charge, and no shock can be delivered.
27. After "NO SHOCK ADVISED", the "START.
28. CPR" prompts occur. A countdown timer (min: sec format) continues for the duration specified in the CPR TIME 2 setup option.
29. Subsequent analysis for "SHOCK ADVISED" and "NO SHOCK ADVISED"

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30. sequences are the same as described above. The energy level for Shock 2, 3, and greater depends on the ENERGY PROTOCOL setup and the analysis decision. When a "NO SHOCK ADVISED" decision follows a shock, the energy level does not increase for the next shock. When a "SHOCK ADVISED" decision follows a shock, the energy level increases for the next shock.

7.12 Defibrillation with the LIFEPAK 1000: AED

Indications

1. Unconscious medical or trauma patient (more than 8 years old) with the absence of normal breathing on visual inspection of the chest wall for 5-10 seconds.
2. Unconscious medical or trauma patient (more than 8 years old) with the presence of agonal breathing on visual inspection of the chest wall for 5-10 seconds.
3. The AED advice "SHOCK ADVISED" By voice and screen message

Contraindications

1. Patient less than 8 years old

Adverse Effects

1. Burns to the chest

Standard precautions

Procedure

1. Determine if the patient is unconscious by using the "shout and shake" principle. Immediately tap or shake the patient on their shoulders and ask loudly, "ARE YOU OK?" If the patient is NOT unconscious, they are likely to answer, move, or moan. If the patient remains unresponsive that person is considered to be UNCONSCIOUS.
2. If the patient is unconscious, expose the patient's chest so you can correctly determine if there is normal equal rise and fall of the chest. (Care should be taken to maintain patient privacy – particularly in females).
3. To assess breathing, kneel at the patient's side and look at their exposed chest for at least 5 seconds but not more than 10 seconds. Normal breathing is equal rise and fall of the chest. Do not try and determine the rate at this stage. If there is NO rise and fall of the chest, or short, sudden intakes or gasps of air usually associated with a snore or grunting noise – this patient is considered to have NO normal breathing
4. Begin manual compressions if normal breathing absent or agonal breathing present. If trauma cardiac arrest, ensure absent pulse. If at any time during MANUAL compressions, the victim speaks out, moves, or moans, compressions should be stopped immediately.
5. Place the palm of your hand on the lower half of the sternum. Compression depth should be 5 cm in an adult patient at a rate of between 100 to 120 per minute.
6. Allow for full recoil of the chest and ensure an equal compression and release ratio.
7. Request for feedback on compressions from your partner.
8. Inform NCC of cardiac arrest and request CCP and Delta help.

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9. Request your partner to attach the defibrillation pads in the anterior-lateral position of the LIFEPAK 1000.
10. Turn on the LIFEPAK 1000. The voice prompt will advise you to connect electrodes, if not done already.
11. Prepare and place the LUCAS 2/3 after the 3rd cycle/ 6 minutes of CPR in a Medical adult patient, if available. If LUCAS device not available, then continue with manual chest compressions.
12. After the AED pads are applied, stop chest compressions momentarily. The AED will analyse the ECG rhythm. Voice prompts will advise you to 'STAND CLEAR, ANALYZING NOW. STAND CLEAR"
13. Do not move the AED during analysis. Moving the AED during analysis may affect the ECG signal resulting in an inappropriate "SHOCK" or "NO SHOCK ADVISED" decision. Do not touch the patient or the AED during analysis.
14. The "ANALYZING NOW—STAND CLEAR"
15. prompts occur. analysis of the patient's ECG takes approximately 6 to 9 seconds and then advises either "SHOCK ADVISED" or "NO SHOCK ADVISED".
16. Continue to follow the screen messages and voice prompts provided by the AED.
17. If the AED detects a shockable rhythm, the "SHOCK ADVISED" prompts occur. Charging to the joule setting for Shock begins. A charging bar appears and a ramping tone sound.
18. When charging is complete, the available energy is displayed. The "STAND CLEAR, PUSH SHOCK BUTTON!"
19. message occurs, followed by a "Shock ready" tone. Clear everyone away from touching the patient, bed, or any equipment that is connected to the patient. Press (shock) to deliver energy to the patient.
20. When the (shock) button is pressed, the "ENERGY DELIVERED" message occurs indicating that the energy transfer was completed. Voice prompt will then advise you to "START CPR"
21. After a shock is delivered, the "START CPR" prompts occur. A countdown timer (min: sec format) continues for the duration specified in the CPR TIME 1 setup option.
22. The CPR metronome automatically provides audible compression "tocks" and ventilation prompts or tones only during CPR intervals.
23. When the CPR countdown time ends, 'STAND CLEAR, ANALYZING NOW.
24. STAND CLEAR". These prompts repeat every 20 seconds until you press ANALYZE.
25. If the AED detects a non-shockable rhythm, the "NO SHOCK ADVISED" prompts occur. The defibrillator does not charge, and no shock can be delivered.
26. After "NO SHOCK ADVISED", the "START.
27. CPR" prompts occur. A countdown timer (min: sec format) continues for the duration

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specified in the CPR TIME 2 setup option.

28. Subsequent analysis for “SHOCK ADVISED” and “NO SHOCK ADVISED”
29. sequences are the same as described above. The energy level for Shock 2, 3, and greater depends on the ENERGY PROTOCOL setup and the analysis decision. When a “NO SHOCK ADVISED” decision follows a shock, the energy level does not increase for the next shock. When a “SHOCK ADVISED” decision follows a shock, the energy level increases for the next shock

7.13 Defibrillation with the LIFEPAK 15: Manual

Indications

1. Ventricular fibrillation (VF).
2. Pulseless ventricular tachycardia (pVT).

Contraindications

1. PEA
2. Asystole

Adverse Effects

1. Burns at the pad location

Standard precautions

Procedure

1. If a second provider is present, high-quality CPR should be performed while the preparation of defibrillation is being performed.
2. Press ON on the LIFEPAK 15.
3. If the defibrillation pads are not already attached, identify the most appropriate site for the pads (Anterior-Lateral for Adults or Anterior-Posterior for Paediatrics) and prepare the skin. (Remove excessive hair. Dry and clean skin, if necessary.) Ensure good contact of the pad to the patient's skin.
4. Once the pads are applied, press LEAD to select PADDLES view on the LIFEPAK 15. Pause compressions to confirm that a shockable rhythm is present. After confirming the rhythm, continue high-quality chest compressions.
5. Confirm the desired energy is selected. The LIFEPAK 15 is programmed to automatically begin at 200J for the initial shock and then escalate for the second (300J) and third (360J) shocks. If the desired energy is not selected, press "ENERGY SELECT" or rotate the SPEED DIAL to the desired energy.
6. Press CHARGE. High-quality chest compressions should be performed while the defibrillator is charging regardless of the number of providers on scene.
7. Before delivering the shock, ensure safety of the scene. Remove the oxygen supply from the patient. Stop CPR and ensure that no person is in contact with the patient or equipment attached to the patient.
8. Press SHOCK.

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7.14 Defibrillation with the LIFEPAK 1000: Manual

Indications

1. Ventricular fibrillation (VF).
2. Pulseless ventricular tachycardia (pVT).

Contraindications

1. PEA
2. Asystole

Adverse Effects

1. Pain
2. Burns to the chest

Procedure

1. Begin manual chest compressions if normal breathing is absent or agonal breathing present. If at any time during MANUAL compressions, the victim speaks out, moves, or moans, compressions should be stopped immediately.
2. Delegate to your partner to apply the defibrillation pads. Remove all clothing from the patient’s chest. Remove excessive chest hair. Clean and dry the skin, if necessary.
3. Place the defibrillator pad lateral to the patient's left nipple in the midaxillary line, with the centre of the electrode in the midaxillary line, if possible. Place the other pad on the patient’s upper right torso, lateral to the sternum and below the clavicle.
4. Step 1: Turn the LIFEPAK 1000 ON.
5. Press the MENU button and select “Yes” when “Enter Manual Mode?” message displays.
6. Identify a shockable rhythm (VF/ pVT).
7. Step 2: Activate the charge button. Once the charge option is activated the LifePak 1000 will automatically charge to 200J for the initial shock. The LifePak 1000 is pre-programmed to escalate the energy levels. The second charge will be at 300J and the third at 360J. All subsequent energy levels will be at 360J.
8. Ensure safety of the scene before delivering a shock. Remove the oxygen supply from the patient to reduce a fire hazard. Ensure NO PERSON is in contact with the patient before a shock is delivered. If a drip is being held up, please place the drip on the floor.
9. Step 3: Deliver the shock by depressing the SHOCK Icon on the LifePak 1000 and holding until the shock is delivered.
10. Following the shock, treat the patient accordingly based on the ECG rhythm and the algorithm.

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7.15 Defibrillation with the Propaq MD: Manual

Indications
1. Ventricular fibrillation (VF). 2. Pulseless ventricular tachycardia (pVT).
Contraindications
1. PEA 2. Asystole
Adverse Effects
1. Pain 2. Burns to the chest
Procedure
1. Begin manual chest compressions if normal breathing is absent or agonal breathing present. If at any time during MANUAL compressions, the victim speaks out, moves, or moans, compressions should be stopped immediately. 2. Delegate to your partner to apply the defibrillation pads. Remove all clothing from the patient's chest. Remove excessive chest hair. Clean and dry the skin, if necessary. 3. Place the defibrillator pad lateral to the patient's left nipple in the midaxillary line, with the centre of the electrode in the midaxillary line, if possible. Place the other pad on the patient's upper right torso, lateral to the sternum and below the clavicle. 4. Turn the Propaq MD ON. 5. Identify a shockable rhythm (VF/ pVT). a. Step 1: Select energy level 120 J b. Step 2: Charge Defibrillator c. Step 3: Deliver Shock 6. NOTE: PROPAQ MD is NOT calibrated to escalate shocks 7. Ensure safety of the scene before delivering a shock. Remove the oxygen supply from the patient to reduce a fire hazard. Ensure NO PERSON is in contact with the patient before a shock is delivered. If a drip is being held up, please place the drip on the floor. 8. The second SHOCK will be at 150 Joules and the third SHOCK at 200 Joules. All subsequent SHOCKS will be at 200 Joules. NB Manually select energy level prior to each shock. 9. Following the shock, treat the patient accordingly based on the ECG rhythm and the algorithm.

7.16 3-Stack Defibrillation with Propaq MD: Manual

IV

Indications

1. Witnessed Cardiac Arrest of an Adult patient presenting with a shockable rhythm:
    a. Ventricular fibrillation (VF)
    b. Pulseless ventricular tachycardia (pVT)

Contraindications

1. Unwitnessed Cardiac Arrest
2. PEA
3. Asystole

Adverse Effects

1. Pain
2. Burns to the chest

Procedure

1. If a second provider is present, high-quality CPR should be performed while the preparation of defibrillation is being performed.
2. Press ON on the LIFEPAK 15.
3. If the defibrillation pads are not already attached, identify the Anterior-Lateral site for the pads and prepare the skin. (Remove excessive hair. Dry and clean skin, if necessary.) Ensure good contact of the pad to the patient’s skin.
4. Once the pads are applied, press LEAD to select PADDLES view on the LIFEPAK 15. Pause compressions to confirm that a shockable rhythm is present. After confirming the rhythm, continue high-quality chest compressions.
5. Confirm the desired energy of 200J is selected. The LIFEPAK 15 is programmed to automatically begin at 200J for the initial shock and then escalate for the second (300J) and third (360J) shocks. If the desired energy is not selected, press “ENERGY SELECT” or rotate the SPEED DIAL to the desired energy.
6. Press CHARGE. High-quality chest compressions should be performed while the defibrillator is charging regardless of the number of providers on scene.
7. Before delivering the shock, ensure safety of the scene. Remove the oxygen supply from the patient. Stop CPR and ensure that no person is in contact with the patient or equipment attached to the patient.
8. Press SHOCK.
9. Following the first shock, immediately analyze the rhythm. If a shockable rhythm is identified (VF/ pVT), then immediately press CHARGE. The second charge should be at 300J. High-quality chest compressions should be performed while the defibrillator is charging. If the rhythm is not shockable, reassess the patient.
10. Ensure safety of the scene and press SHOCK.

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11. Following the second shock, immediately analyze the rhythm. If a shockable rhythm is identified (VF/ pVT), then immediately press CHARGE. The third charge should be at 360J. High-quality chest compressions should be performed while the defibrillator is charging. If the rhythm is not shockable, reassess the patient.
12. Ensure safety of the scene and press SHOCK.
13. Following the third shock, immediately analyze the rhythm. If a shockable rhythm is identified (VF/ pVT), then immediately begin chest compressions and proceed with management of cardiac arrest according to CPG 2.1. If the rhythm is not shockable, reassess the patient.

7.17 Post Resuscitation (ROSC) management

I II III IV

Indications

1. Patients who present with Return of Spontaneous Circulation (ROSC) post cardiac arrest.

Contraindications

1. Nil in this setting

Adverse Effects

1. Medication specific adverse effects

Standard precautions

Procedure

1. Stop CPR if ROSC is achieved and release the Lucas device suction cup pressure. Positive signs of ROSC include a palpable central pulse, a persistent rise in ETCO2 to the > 40 mm Hg.
2. APs to call for CCP assistance or consult CCE.
3. Assess patient’s level of consciousness (AVPU), airway, breathing and circulation (LOC-ABC)
4. Continue monitoring the patient’s vital signs RR, ECG, HR, BP, SpO2 and ETCO2.
5. Continue to manage identified reversible causes (4 Hs and 4 Ts)
6. Consider inserting an advanced airway if not already done so during the cardiac arrest if the
   patient’s airway is compromised or to maximise oxygenation and ventilation.
7. Consider replacing the LTA with an ETT (age dependent) if not done so during the cardiac arrest.
8. Ensure adequate tidal volumes based on the patient’s ideal body weight. Do not hyperventilate the patient.
9. Do not try to normalise EtCO2 prehospitally as this requires correction over time in-hospital. Adopt an appropriate ventilation strategy based on the patient’s lung pathology and the cause of the cardiac arrest.
10. Aim to maintain SpO2 > 90%.
11. Consider performing a 12-lead ECG on all ROSC patients following a medical cardiac arrest,
    particularly if the cause of arrest is suspected to be cardiac related. Patients with a pre-arrest
    12-lead ECG indicating a STEMI should be transported to an appropriate facility for PCI.

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12. BP and MAP should be monitored regularly (minimum of every 5 minutes)
13. Maintain SBP > 90 mmHg/MAP >65 mmHg or MAP of 70-80 for isolated TBI in adults and according to age appropriate SBP range in paediatrics using (age x 2) + 70. Avoid spikes in BP, particularly in trauma. The target MAP is to achieve adequate cerebral perfusion.
14. For adults administer fluids as required using 250ml boluses to a maximum of 1-2L depending on the cause of arrest. Limit fluids in the presence of major haemorrhage to a maximum of 1L. For paediatrics administer a fluid bolus of 10-20ml/kg repeated once if required.
15. Consider vasopressor/inotrope to maintain BP and MAP. Phenylephrine and Noradrenaline may be considered for adults only. Adrenaline infusions for adults are not for routine use and are only to be considered for specific indications (see formulary). Adrenaline infusion may be considered for paediatrics requiring vasopressor/inotrope support.
16. Provide sedation/analgesia and paralysis as required.
    Correct hypoglycaemia (RBS < 4.0 mmol/L) if present. For adults administer 50ml bolus of Dextrose 10% and reassess after 10 minutes. Repeat if required. Avoid increasing RBS to > 6.7 mmol/L. For newborns, administer 2ml/kg (max 50ml) and for infants/children, administer 2.5ml/kg (max 50ml) of Dextrose 10% and reassess after 10 minutes. Repeat as required to normalise RBS.
17. Monitor the patient's temperature and avoid hypo/hyperthermia.
18. Avoid rapid movement of the medical cardiac arrest patient for up to 10 minutes following ROSC. Rapid movement may result in re-arrest. Transport should then be initiated if not already loaded in the ambulance.
19. Post ROSC for trauma cardiac arrest, do not wait on scene for rhythm stabilisation. Transport the patient immediately.
20. Provide hospital prenotification when transporting. Transport to the most appropriate Facility according to the cause of arrest.

7.18 Standard Valsalva manoeuvre

Indications

1. Patient experiencing symptomatic tachycardia with moderate end organ dysfunction (dizziness, mild chest pain or SOB without gross pulmonary oedema)
2. Atrial origin tachycardia (haemodynamically stable SVT)

Contraindications

1. Bradycardia
2. Heart blocks.
3. Atrial fibrillation

Adverse Effects

1. Syncope
2. Hypotension
3. Bradycardia

Standard precautions

Procedure

1. APs call for CCP assistance and/ or consult CCE.
2. Connect the patient to the LifePak 15 and monitor vital signs RR, ECG, HR, BP, SpO_2.
3. Prepare pre-pack A16/ C6 for a 10 ml syringe (without the hypodermic needle).
4. Calm and reassure the patient.
5. Place the patient in the supine position.
6. Rule out any history of coronary artery disease or reduction in blood volume.
7. Ask the patient to blow into the 10 ml syringe (pressure of 40 mmHg) trying to move the plunger (the 10 mL syringe is useful in generating the recommended standard of 40 mmHg intra-thoracic pressure).
8. Advice the patient to maintain the 40 mmHg intra-thoracic pressure for 15 seconds.
9. Be prepared that the patient may feel dizzy or faint during the procedure.
10. Maintain haemodynamic monitoring of the patient.
11. Establish IV access with NaCl 0.9%.
12. Consider 12 lead ECG after treating the life-threatening situation.
13. AP crew to load the patient and rendezvous with the CCP en-route

7.19 Modified Valsalva manoeuvre

Indications

1. Patient experiencing symptomatic tachycardia with moderate end organ dysfunction (dizziness, mild chest pain or SOB without gross pulmonary oedema)
2. Atrial origin tachycardia (haemodynamically stable SVT)

Contraindications

1. Bradycardia
2. Heart blocks
3. Atrial fibrillation

Adverse Effects

1. Syncope
2. Hypotension
3. Bradycardia

Standard precautions

Procedure

1. APs call for CCP assistance and/ or consult CTL.
2. Connect the patient to the LifePak 15 and monitor vital signs RR, ECG, HR, BP, SpO2.
3. Prepare a manual BP cuff gauge and Oxygen tubing. Attach a 10-15cm piece of Oxygen tubing to the pressure gauge.
4. Calm and reassure the patient.
5. Place the patient in the semi-/ full- fowlers position.
6. Rule out any history of coronary artery disease or reduction in blood volume.
7. Instruct the patient to forcefully exhale (strain) into the oxygen tubing attached to the manometer to generate a pressure of 40mmHg for 15 seconds.
8. Placed the patient into a supine position and raise the legs to 45 degrees for 15 seconds.
9. Return the patient to the semi fowlers position for a further 45 seconds before cardiac rhythm reassessment.
10. Be prepared that the patient may feel dizzy or faint during the procedure.
11. Maintain haemodynamic monitoring of the patient.
12. Establish IV access with NaCl 0.9%.
13. Consider 12 lead ECG after treating the life-threatening situation.
14. Clean and disinfect the manual BP apparatus using HMC-approved disinfectant according to the manufacturer's recommendations.

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7.20 Synchronized cardioversion using the LIFEPAK 15

IV

Indications

1. Unstable tachyarrhythmias (SVT, AFib, AF and VT with a pulse)

Contraindications

1. Tachyarrhythmia from Digitalis toxicity
2. Non-sustained tachyarrhythmia
3. Rhythm that is not a tachyarrhythmia

Adverse Effects

1. Pain
2. Burns at the pad locations
3. Hypoventilation (if sedation is used)
4. Hypotension
5. Arrhythmias (VF/VT or Asystole)

Standard precautions

Procedure

1. Attach the 4-lead ECG. Perform a 12-lead ECG, if time permits.
2. Provide appropriate level of procedural sedation (CPG 11.3), if time permits.
3. Select Lead II or lead with greatest QRS complex amplitude.
4. Press SYNC and confirm that the SYNC MODE message is present at the at the bottom of the screen.
5. Observe the ECG rhythm and confirm that a triangle sense marker appears near the middle of each QRS complex. If the markers do not appear or are showing in the wrong location, increase the ECG size or select a different lead.
6. Connect the defibrillation pads to the cable and confirm cable connection to the defibrillator.
7. Identify the Anterior-Posterior sites and prepare the patient's skin. (Remove excessive hair. Dry and clean skin, if necessary.) If unable to apply pads to Anterior-Posterior sites, Anterior-Lateral sites can be utilized.
8. Apply defibrillation pads to the patient. Ensure good contact of the pad to the patient's skin.
9. Press ENERGY SELECT or rotate the SPEED DIAL to the desired energy.
10. Press CHARGE.
11. Before delivering the shock, ensure safety of the scene. Remove the oxygen supply from the patient. Ensure that no person is in contact with the patient or equipment attached to the patient.
12. When the LIFEPAK 15 is fully charged, confirm the ECG rhythm,
13. Once the charge is reached, confirm ECG rhythm, triangle sense markers, and correct energy. To cancel the charge, press the SPEED DIAL.

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14. Press and hold the SHOCK button until the ENERGY DELIVERED message appears on the screen.
15. Reassess the patient and the ECG. If necessary, repeat this procedure from Step 4.

7.21 Synchronized cardioversion using the Propaq MD

Indications

1. Unstable tachycardias (SVT, AF and VT with a pulse) with signs of imminent death (severe hypotension, decreasing GCS, and/or severe pulmonary oedema).

Contraindications

1. Digitalis toxicity
2. Sinus tachycardia
3. Ventricular fibrillation
4. Bradycardia
5. Heart blocks

Adverse Effects

1. Pain
2. Burns to the chest
3. Hypoventilation from the sedation
4. Hypotension
5. Arrhythmias (Bradycardia, asystole)

Standard precautions

Procedure

1. Connect the patient to the Propaq MD and monitor vital signs RR, HR, BP, SpO2 and 4-lead ECG.
2. Calm and reassure the patient.
3. Place the patient in the supine position.
4. Establish IV access with NaCl 0.9%.
5. If time and patient condition permits, obtain 12-lead ECG and direct your crew for equipment to be prepared in anticipation of possible cardiac arrest.
6. Select Lead II or lead with greatest QRS complex amplitude.
7. Sedate the patient and administer pain relief if necessarily indicated, as per CPG.
8. Press the SYNC button and verify that the (S) display marker appears near the middle of each QRS complex. A Sync indicator appears at the top of the display screen, and a green light next to the indicator flashes with each Sync marker. The unit automatically exits Sync mode after each shock.
9. Connect the defibrillation pads to the cable and confirm cable connection to the defibrillator.
10. Prepare the patient's skin and apply defibrillation pads to the patient in the anterior-lateral position.
11. Press ENERGY SELECT arrows up or down to select the desired energy:
    a. SVT 100J
    b. AF 200J

   c. Wide Complex (VT with pulse) 200J
   d. Paeds 0.5-1 J/ Kg
12. Press CHARGE.
13. Ensure all personnel stand clear of the patient, bed, and any equipment connected to the patient.
14. Once the charge is reached, reconfirm ECG rhythm, sense markers, and correct Joules. Press and hold the “shock” button until the ENERGY DELIVERED message appears on the screen.
15. To disarm (cancel charge) reduce energy level.
16. Reassess the patient and confirm ECG rhythm.
17. Repeat steps 14-17, if necessary.
18. In addition to successful cardioversion, drug therapy as per CPG should be considered.
19. In the event that the Propaq MD cannot synchronize (documented in cases of wide complex rapid irregular rhythms with Propaq MD e.g. accelerated, aberrantly conducted Atrial Fibrillation) and the patient is near death, perform unsynchronized defibrillation immediately.
20. Transport the patient to the appropriate hospital with CTL pre-notification.

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7.22 Transcutaneous pacing using the LIFEPAK 15

Indications

1. Immediately life-threatening bradycardia.

Contraindications

1. None in the presence of a life- threatening bradycardia with a pulse.
2. Active Cardiac Arrest

Adverse Effects

1. Pain
2. Burns to the chest
3. Hypoventilation from the sedation

Standard precautions

Procedure

1. Attach the 4-lead ECG. Perform a 12-lead ECG, if time permits.
2. Provide appropriate level of procedural sedation (CPG 11.3), if time permits. This step may necessary later in the process to accommodate the painful procedure.
3. Select Lead II or lead with greatest QRS complex amplitude.
4. Identify the Anterior-Posterior sites and prepare the patient's skin. (Remove excessive hair. Dry and clean skin, if necessary.) If unable to apply pads to Anterior-Posterior sites, Anterior-Lateral sites can be utilized.
5. Apply defibrillation pads to the patient. Ensure good contact of the pad to the patient's skin.
6. Press PACER.

NOTE: A heart rate will not be displayed after you have engaged the PACER. Dashes (- - -) will be displayed in place of the heart rate on the home screen.

7. Observe the ECG rhythm and confirm that a triangle sense marker appears near the middle of each QRS complex. If the markers do not appear or are showing in the wrong location, increase the ECG size or select a different lead.
8. Press RATE (increments of 10 PPM) or rotate the SPEED DIAL (increments of 5 PPM) to select the desired pacing rate. The default pacing rate on the LIFEPAK 15 is 70 PPM.
9. Press CURRENT (increments of 10 PPM) or rotate the SPEED DIAL (increments of 5 PPM) to increase current until electrical capture occurs. Electrical capture is indicated by a pace marker (spike) followed by a **wide** QRS complex.
10. Palpate the patient's pulse to assess for mechanical capture. If the pulse rate does not correlate to the PPM, continue increasing the CURRENT by 5mA increments until mechanical capture is accomplished.
11. Reassess and monitor the patient's perfusion status (level of consciousness, pulse, blood pressure, etc.). If pacing fails to improve perfusion, optimize cardiac output with

appropriate amounts of fluid boluses, vasopressor administration, as well as considering an increase in PPM.

12. To stop pacing, reduce current to zero or press PACER.

7.23 Transcutaneous pacing using the Propaq MD

Indications

1. Immediately life-threatening bradycardia.

Contraindications

1. None in the presence of a life- threatening bradycardia with a pulse.
2. Any patient with signs of cardiac arrest (no signs of life, absent or agonal breathing) regardless of ECG rhythm.

Adverse Effects

1. Pain
2. Burns to the chest
3. Hypoventilation from the sedation

Standard precautions

Procedure

1. Connect the patient to the Propaq MD and monitor vital signs RR, HR, BP, SpO2 and 4-lead ECG. (KEEP 4-LEAD CONNECTED)
2. Calm and reassure the patient.
3. Place the patient in the supine position.
4. Establish IV access with NaCl 0.9%.
5. If time and patient condition permits, obtain 12-lead ECG and direct your crew for equipment to be prepared in anticipation of possible cardiac arrest.
6. Select Lead II or lead with greatest QRS complex amplitude.
7. Sedate the patient and administer pain relief if necessary.
8. Apply the defibrillation pads using either the anterior-lateral or anterior-posterior position after preparing the patient's skin.
9. Press PACER to bring up the pacer menu. Confirm Pacer Mode and Pacer Rate (70bpm)
10. Press the up/Clockwise and down/counterclockwise arrow to highlight **Start Pacer**.
11. Press the select button to begin pacing.
12. Press the up/Clockwise and down/counterclockwise arrow to highlight the Pacer Output. Press Select.
13. Press the up/Clockwise and down/counterclockwise arrow to adjust the pacer output until capture occurs, then decrease the output to the lowest level that still maintains capture
14. Assess the patient's BP and perform a perfusion status assessment.
15. If there is no improvement in BP with poor signs of perfusion, increase the rate to a maximum of 90/bpm.
16. After pacing is initiated and the BP has stabilized, provide further sedation/analgesia as per CPG if the patient is experiencing discomfort.

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17. If pacing fails to improve the BP and signs of perfusion, the initiation of a vasopressor should be considered.
18. Perform continuous monitoring during transport with vital signs assessed every 5 minutes.
19. Press the Display/Home button to exit the Pacer menu.
20. Clean and disinfect the Propaq MD using HMC-approved disinfectant according to the manufacturer's recommendations

Wound Care

8.1 External Haemorrhage Control: Direct Pressure

Indications

1. Major external venous/arterial bleeding.

Contraindications

1. Nil in the setting of a catastrophic haemorrhage.

Adverse Effects

1. Nil in the setting of a catastrophic haemorrhage.

Standard precautions

1. A face shield and gown are required if the patient has catastrophic bleeding.

Procedure

1. Elevate the wound above the heart to reduce blood flow to the wound, if possible.
2. Prepare pre-pack A14 and A20, as required.
3. Inspect the wound for fractured bone fragments. If fractured bone fragments are present, do not apply direct pressure with a gloved hand.
4. Apply firm pressure with gauze or a gloved hand if fractured bone fragments absent. Alternatively, you can use the pressure points to occlude major arteries of the limbs such as brachial or femoral arteries.
5. When the bleeding slows or stops completely place dressing over wound with the dressing extending 1 inch beyond edges of wound.
6. Apply bandage firmly and securely but not tight enough to restrict blood supply.
7. Continuously assess PMS distal to the injury. If practical, assess PMS prior and post dressing application.
8. Make several anchoring wraps, overlapping each wrap to secure in place.
9. Continue overlapping wrap circumferentially. Ensure bandage holds dressing securely and controls bleeding.
10. Securely tie or fasten bandage in place so it will not move.
11. Monitor the wound for any further development in bleeding.
12. Administer analgesia, as required.
13. Ensure that the patient's BP, ECG, SpO2, HR and RR are being monitored.
14. Establish IV access en-route to hospital and administer fluids, as required.
15. If the patient meets the Trauma Bypass Criteria, then consult with CCE to pre- inform TRU.
16. Meet Charlie en-route, if required.

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8.2 External Haemorrhage Control: Chito gauze

I II III IV

Indications

1. Haemostatic dressing for the external, temporary control of catastrophic haemorrhage.

Contraindications

1. Nil in the setting of a catastrophic haemorrhage.

Adverse Effects

1. Nil in the setting of a catastrophic haemorrhage.

Standard precautions

1. A face shield and gown are required if the patient has catastrophic bleeding.

Procedure

1. Prepare pre-packs A14 and A20.
2. Direct your partner to compress the pressure point of the affected area, if practical.
3. Open the foil pouch and remove the z- folded chito gauze.
4. Apply chito gauze onto and in contact with the source of bleeding.
5. Continue packing the dressing completely into the wound.
6. For larger wounds, more than one dressing pack of chito gauze may be required.
7. Note the time of application.
8. After packing the wound with chito gauze, apply pressure to the wound for 2-3 minutes.
9. Continuously assess PMS distal to the injury. If practical, assess PMS prior and post dressing application.
10. Ensure that the patient's BP, ECG, SpO2, HR, RR are being monitored.
11. Establish IV/ IO (CCP only) access and administer Normal Saline in 250 ml increments, as required.
12. Administer analgesia, as required.
13. Monitor the wound for any further development in bleeding.
14. If the patient meets the Trauma Bypass Criteria, then consult with CCE to pre- inform TRU.
15. Meet Charlie en-route, if required.
16. Ensure endorsement to the receiving facility that the dressing may stick to the wound if left on beyond 48 hours.

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8.3 External Haemorrhage Control: Blast bandage

Indications

1. Dressing for the external, temporary control of severe bleeding.

Contraindications

1. Nil in the setting of a catastrophic haemorrhage.

Adverse Effects

1. Nil in the setting of a catastrophic haemorrhage.

Standard precautions

1. A face shield and gown are required if the patient has catastrophic bleeding.

Procedure

1. Prepare pre-packs A14 and A20, as required.
2. Direct your partner to compress the pressure point of the affected area.
3. Apply direct pressure until bleeding is controlled for about 2-3 minutes.
4. Remove the occlusive layer of the BLAST bandage from the pouch.
5. If extremity injuries, then wrap the wound pad onto the extremity with the plastic side of the wound pad on the outside.
6. Continuously assess PMS distal to the injury. If practical, assess PMS prior and post dressing application.
7. In other instances, cover the injury with the occlusive layer.
8. Secure wound pad with elastic bandage and Velcro on a figure-of-eight manner.
9. Note the time of application.
10. Ensure that the patient's BP, ECG, SpO2, HR and RR are being monitored.
11. Establish IV/ IO (CCP only) access and administer Normal Saline in 250 ml increments, as required.
12. Administer analgesia, as required.
13. Monitor the wound for any further development in bleeding.
14. If the patient meets the Trauma Bypass Criteria, then consult with CTL to pre- inform TRU.
15. Meet Charlie en-route, if required.
16. Complete the ePCR appropriately.
17. Ensure endorsement to the receiving facility that the dressing may stick to the wound if left on beyond 48 hours.

8.4 Application of tourniquet

I II III IV

Indications

1. Extreme life-threatening limb haemorrhage, limb amputation or mangled limb with multiple bleeding points.

Contraindications

1. Nil in the setting of a catastrophic haemorrhage.

Adverse Effects

1. Compartment syndrome.
2. Pain or discomfort (may require analgesia).
3. Reperfusion injury when released.
4. Embolism.
5. Permanent nerve damage, muscle injury, vascular injury, and/or skin necrosis.
6. Ischaemia.
7. Fractures – use on tibia and fibula or radius or ulna, as the device has the potential to cause fracture

Standard precautions

1. A face shield and gown are required if the patient has catastrophic bleeding.

Procedure

1. Immediately apply/ instruct your partner to apply direct pressure. Be cautious of fractured bone fragments.
2. Prepare pre-packs A14 and A20, as required.
3. Expose the extremity proximal to the injury. Apply direct pressure as described in 8.1.
4. Continuously assess PMS distal to the injury. If practical, assess PMS prior and post dressing application.
5. Slide the wounded extremity through the loop of the CAT or route the band around the limb and pass the tip through the buckle.
6. Position the CAT directly over exposed skin, 5-10 cm proximal to the injury.
7. Pull the band tight and pass the tip through the outside slit of the buckle.
8. Pull the band tight and securely fasten the band back on itself with the Velcro fastener.
9. Twist the rod until bright red bleeding has stopped.
10. Place the rod inside the triangular ring, tighten safety screw and mark the time and location of application.
11. Dress wound with Blast bandage, if required.

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12. Do not apply the tourniquet over joints.
13. Do not loosen or remove an applied tourniquet.
14. Wrap amputated body part in Transparent plastic bag covered with bed sheet.
15. Transport amputated body part with the patient.
16. Administer analgesia, as required.
17. Transport the patient to HGH TRU after endorsing the patient with CCE.
18. Meet Charlie en-route, if required.

8.5 Impaled object management

Indications

1. Patients presenting with impaled objects.

Contraindications

1. Nil in the setting of impaled object.

Adverse Effects

1. Nil in the setting of impaled object.

Standard precautions

1. A face shield and gown are required if the patient has catastrophic bleeding.

Procedure

1. Rule out any other injury near the impaled object.
2. Prepare pre-packs A14 and non-sterile gauze.
3. Expose the area of the impaled object for inspection without moving the object.
4. Continuously assess PMS distal to the injury. If practical, assess PMS prior and post dressing application.
5. Check the object orientation and check for exit wounds.
6. Make several layers of bulky dressing or roll bandage from the two opposite sides of the object.
7. If possible, secure the impaled object with tape or crepe bandage.
8. If an extremity is affected, splint the extremity to avoid movement.
9. If the impaled object is interfering with lifesaving interventions (CPR, airway management etc.), ask for assistance and consult CCD.
10. Establish IV/ IO (CCP only) access and administer Normal Saline in 250 ml increments, as required.
11. Administer analgesia, as required.
12. Transport the patient to hospital or TRU with pre-notification via CCE.
13. Meet Charlie en-route, if required

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8.6 Application of dressings: Arm and leg

Indications

1. External bleeding to the arm and legs.

Contraindications

1. Nil in the setting of external bleeding to the arm and legs.

Adverse Effects

1. Applying the bandage too tightly may impact on the patients PMS.

Standard precautions

1. A face shield and gown are required if the patient has catastrophic bleeding.

Procedure

1. Prepare pre-pack A14 and A20, as required.
2. Expose the entire wound.
3. Handle dressings following aseptic techniques. Use sterile water as necessary for cleaning and irrigating the wound.
4. Place the dressing over wound. If sterile dressing is not available use gloved hand or other available material. Be cautious of fractured bone fragments.
5. Ensure dressing extends 3 cm beyond the edges of wound.
6. Apply direct pressure as needed to stop the bleeding.
7. Continuously assess PMS distal to the injury. If practical, assess PMS prior and post dressing application.
8. Apply bandage firmly and securely but not tight enough to restrict blood supply.
9. Make several anchoring wraps, overlapping each wrap to secure in place.
10. Continue overlapping wrap circumferentially.
11. Ensure bandage holds dressing securely and controls bleeding.
12. Securely tie or fasten bandage in place so it will not move. Tuck or tape loose ends.
13. Apply sling and swathe as needed.
14. Reassess PMS and bandage to ensure bleeding control.
15. Establish IV access and administer Normal Saline in 250 ml increments, as required.
16. Administer analgesia, as required.
17. Transport the patient to hospital or TRU with pre-notification via CCE.
18. Meet Charlie en-route, if required

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8.7 Application of dressings: Elbow and knee

Indications

1. External bleeding to the elbow and knee.

Contraindications

1. Nil in the setting of external bleeding to the elbow and knee.

Adverse Effects

1. Applying the bandage too tightly may impact on the patients PMS.

Standard precautions

1. A face shield and gown are required if the patient has catastrophic bleeding.

Procedure

1. Prepare pre-pack A14 and A20, as required.
2. Continuously assess PMS distal to the injury. If practical, assess PMS prior and post dressing application.
3. Expose the entire wound.
4. Handle dressings following aseptic techniques. Use sterile water or saline as necessary for cleaning and irrigating the wound.
5. Make several anchoring wraps starting either above or below the wound, overlapping each wrap to secure in place.
6. Apply a sterile dressing to wounds on elbow or knee.
7. Proceed diagonally across the dressing.
8. Circle below the joint and diagonally back across the dressing.
9. Repeat until dressing and area are sufficiently covered.
10. Splint the extremity to avoid movement.
11. Reassess PMS and bandage to ensure bleeding control.
12. Establish IV/ IO (CCP only) access and administer Normal Saline in 250 ml increments, as required.
13. Administer analgesia, as required.
14. Transport the patient to hospital or TRU with pre-notification via CCE.
15. Meet Charlie en-route, if required.

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8.8 Application of dressings: Neck, shoulder and hip

Indications

1. External bleeding to the neck, shoulder and hip.

Contraindications

1. Nil in the setting of external bleeding to the neck, shoulder and knee.

Adverse Effects

1. Applying the bandage too tightly may impact on the patients PMS.

Standard precautions

1. A face shield and gown are required if the patient has catastrophic bleeding.

Procedure

1. Prepare pre-pack A14 and A20, as required.
2. Expose the entire wound.
3. Handle dressings following aseptic techniques. Use sterile water or saline as necessary for cleaning and irrigating the wound.
4. Make several anchoring wraps around arm or leg opposite injury site, overlapping to secure bandage.
5. Apply a sterile dressing to wounds on neck, shoulder or hip.
6. Proceed diagonally across the dressing.
7. Circle shoulder and arm or thigh and hip overlapping each time.
8. Be cautious not to apply the bandage too tightly around the neck as it may compromise airway and breathing.
9. Assess patient comfort of the bandage around the neck regularly. Assess for swelling under the dressing. Any sudden drop in the patient's LOC, increase in HR, and drop in BP or hypoventilation may indicate swelling under the neck bandage.
10. Continuously assess PMS distal to the injury. If practical, assess PMS prior and post dressing application.
11. Repeat until dressing and area are sufficiently covered.
12. Ensure bandage allows the joint to move without becoming loose.
13. Establish IV/ IO (CCP only) access and administer Normal Saline in 250 ml increments, as required.
14. Administer analgesia, as required.
15. Transport the patient to hospital or TRU with pre-notification via CCE.
16. Meet Charlie en-route, if required.

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8.9 Application of dressings: Ears, eyes and head

Indications

1. External bleeding to the ear, eyes and head.

Contraindications

1. Nil in the setting of external bleeding to the ear, eyes and head.

Adverse Effects

1. Dry eye, corneal hypoxia and corneal edema.
2. Blood supply restriction
3. Bradycardia: occulo-cardiac reflex when direct pressure is applied.

Standard precautions

1. A face shield and gown are required if the patient has catastrophic bleeding.

Procedure

1. Prepare pre-pack A14, as required.
2. Expose the entire wound.
3. Handle dressings following aseptic techniques. Use sterile water or saline as necessary for cleaning and irrigating the wound.
4. Place sterile padding behind the ears.
5. Apply bandage firmly and securely enough to control bleeding but not tight enough to restrict blood supply.
6. Secure dressing in place by wrapping with roller bandage around the head.
7. Ensure bandage does not occlude airway
8. Reassess bandage to ensure bleeding control.
9. Establish IV/ IO (CCP only) access and administer Normal Saline in 250 ml increments, as required.
10. Administer analgesia, as required.
11. Transport the patient to hospital or TRU with pre-notification via CCD.
12. Meet Charlie en-route, if required

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8.10 Application of dressings: Top of head

Indications

1. External bleeding to the top of head.

Contraindications

1. Nil in the setting of external bleeding to the top of head.

Adverse Effects

1. Blood supply restriction

Standard precautions

1. A face shield and gown are required if the patient has catastrophic bleeding.

Procedure

1. Prepare pre-pack A14, as required.
2. Expose the entire wound.
3. Handle dressings following aseptic techniques. Use sterile water or saline as necessary for cleaning and irrigating the wound.
4. Place padding behind ears.
5. Using roller bandage, start at top of head and make loop over each ear hanging to just above shoulder.
6. Bring end back to top centre of head making sure that all dressings are covered.
7. Twist and take roller bandage to the back of the head below the occiput.
8. Wrap securely 2-3 times from below occiput to forehead being sure not to cover patient's eyes.
9. Pull loops down securely and tuck into wrapping.
10. Continue wrapping securely from forehead to occiput.
11. Apply bandage firmly and securely but not tight enough to restrict blood supply.
12. Establish IV/ IO (CCP only) access and administer Normal Saline in 250 ml increments, as required.
13. Administer analgesia, as required.
14. Transport the patient to hospital or TRU with pre-notification via CCD.
15. Meet Charlie en-route, if required.

8.11 Application of dressings: Burns

Indications

1. Burns injuries to the body.

Contraindications

1. Nil in the setting of burns injuries to the body.

Adverse Effects

1. Blood supply restriction

Standard precautions

Procedure

1. Ensure that the patient's BP, ECG, SpO2, HR and RR are being monitored (Time permitting).
2. Establish IV access.
3. Prepare pre-pack A15 and A14, as required.
4. Expose all burn wounds.
5. Handle dressings in an aseptic manner.
6. Select the appropriate size Burnshield, according to the extent of burn injuries.
7. Open and remove the burns dressing.
8. If face burns, cut to accommodate the face.
9. Apply the Burnshield and secure it with an appropriate bandage.
10. Note the percentage of body surface area burnt.
11. For partial and full thickness burns in adults, administer 250 ml boluses up to 1000 ml of Normal Saline over the first hour following the burns injury.
12. Administer analgesia, as required.
13. Transport the patient to HGH TRU after endorsing the patient with CCD.
14. Meet Charlie en-route, if required.

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8.12 Eye irrigation

Indications

1. Foreign body in eye (Dust, chemicals, etc.).
2. Exposure to acidic or alkaline chemicals
3. Exposure to bodily fluid

Contraindications

1. Eye related trauma.

Adverse Effects

1. Eye irritation and redness
2. Eye pain

Standard precautions

Procedure

1. Ensure that the patient’s BP, ECG, SpO2, HR and RR are being monitored.
2. Prepare pre-pack A16/ A14 and Sterile Water.
3. Draw up 10mls of Sterile Water in a 10 ml syringe. Remove the hypodermic needle from the syringe and dispose the needle in the sharp’s container.
4. Place the patient supine with the head turned laterally. The affected eye should be positioned closest to the bed.
5. Place a kidney dish under the patient’s affected eye.
6. Using one hand, open the eye by moving the eyelids apart. Grasp the syringe in the other hand and slowly irrigate the affected eye. Irrigate the affected eye from side to side. The irrigated water will collect in the kidney dish.
7. Repeat the process until eye discomfort stops.
8. Repeat the process for the other eye, if affected.
9. Ask the patient to close the affected eye/s and apply an Eye Pad.
10. Consult with CTL, if required. Meet Charlie en-route, if required
11. Administer analgesia, as required.
12. Transport the patient to the appropriate facility.

8.13 Bowel Evisceration Management

Indications

1. EVISCERATION: An abdominal evisceration occurs when organs protrude from a penetrating wound.
2. It can be a small loop of the intestine leaking out of a stab wound or an entire open abdomen with many exposed organs.
3. The intestines, liver, spleen, and stomach are commonly involved in eviscerations.

Contraindications

1. Nil in this setting.

Adverse Effects

1. Torsion of the blood supply to the eviscerated organs

Standard precautions

Procedure

1. Expose the wound and control any associated visible bleeding,
2. Inspect the wound for bowel leakage, protruding viscera, and other associated injuries (exit wounds)
3. Place the patient in the supine position with flexed knees, if no suspected spinal injury, allow the patient to be in the most comfortable position
4. Do not force eviscerated content, or actively bleeding viscera back into abdomen.
5. Stabilize any associated impaled object.
6. Do not touch exposed organs with bare hands.
7. Cover exposed bowel with sterile saline-soaked dressing.
8. Cover the dressed eviscerated organs with transparent plastic from blast bandage.
9. Provide analgesia and manage associated signs of haemorrhagic shock as needed

8.14 Pressure Bandage

Indications

1. Limb Bites and stings by the following creatures:
    a. All venomous snakes, including sea snakes.
    b. Funnel Web spider.
    c. Blue-ringed octopus.
    d. Cone Shell

2. The pressure bandage retards the flow of lymph by which venoms gain access to the circulation. It has also been shown that there may be inactivation of certain venoms and venom components when the injected venom remains trapped in the tissues by the pressure bandage.

Contraindications

1. Pressure bandage is not to be used on spider or insect bites (except the Funnel web spider).
2. Other spider bites including redback.
3. jellyfish stings.
4. fish stings including stonefish bites.
5. stings by scorpions, centipedes or beetle.
6. If the bite is not on the limb.

Adverse Effects

1. Neurovascular compromise (such as a capillary refill time greater than three seconds on the affected limb) and/or
2. Pain that is disproportionate to the presenting injury,
3. Compartment syndrome

Standard precautions

Procedure

1. Keep the patient and the limb completely at rest.
2. Assess pulse, motor and sensory (PMS) in the affected limb.
3. Apply a broad elasticated pressure bandage (10-15cm wide) over the bite site as soon as possible.
4. Start at the distal end of the affected limb (i.e. fingers or toes of the bitten limb) and extend upwards, covering as much of the limb as possible.
5. Apply enough tension to the bandage so the bandage should be firm and tight, and you should be unable to easily slide a finger between the bandage and the skin.
6. Assess pulse, motor and sensory (PMS) in the affected limb.
7. Apply a splint to immobilize the limb.

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8. Monitor the patient limb for excessive swelling.
9. If there is neurovascular compromise after applying the pressure dressing, contact CCE for take down the dressing.
10. Provide analgesia as needed.

Spinal Motion Restriction

9.1 Manual in-line stabilization: MILS

Indications

1. Stabilization of the head and neck in a patient with suspected cervical spine injury.
    - Extricating or moving a patient.
    - Performing a log roll if indicated
    - Transferring a patient to and from a stretcher

Contraindications

1. Nil contraindication in this setting.

Adverse Effects

1. Patient pain and discomfort.
2. May cause difficult laryngoscopy

Standard precautions

Procedure

1. MILS may be performed from behind the patient, side of the patient, in front of the patient and on supine patients.
2. From behind
    - Hands are placed over the patient's ears.
    - Thumbs are placed behind the patient's head.
    - Little fingers are placed just under the angle of the jaw.
    - Remaining fingers are spread across the flat planes of the patient's head.
3. From the side
    - An arm is placed across the patient's shoulder and the back of the head is cupped in the hand.
    - The thumb and first finger of the other hand are placed either side of the patient's face.
4. From the front
    - Both hands are placed either side of the patient's head.
    - Little fingers are placed at the rear of the head.
    - Thumbs are placed in the notch between the upper teeth and maxilla on each cheek.
    - Remaining fingers are spread across the flat planes of the patient's head.
5. Supine patient
    - The paramedic is positioned above the supine patient's head.
    - Hands are placed either side of the patient's head covering the ears with the palms of the hand.

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- Fingers are spread to stabilise the head with the tips pointing toward the patient's feet.
- The fourth and fifth digits should slightly wrap around to the rear of the patient's head.
- After MILS is applied the second patient attendant measures and applies a cervical collar.
- Appropriate SMR is then applied, as required.
- If the patient meets the Trauma Bypass Criteria, then consult with CCD to inform TRU.
- Meet Charlie en-route, if required.

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9.2 Application of cervical collar

Indications

1. Suspicion of a cervical spine or spinal cord injury based on the mechanism of injury.
2. Patient meets one or more of NEXUS criteria

Contraindications

1. Nil in the setting of suspected cervical spinal injury.

Adverse Effects

1. Discomfort
2. Anxiety

Standard precautions

Procedure

1. Align the head to neutral or 'eyes forward' position if there are no contra-indications.
2. Bring your fingers and thumb together. Place your outstretched hand on the base of the patient's shoulder, Pinky finger side down.
3. With your eye, draw an imaginary line sticking straight forward from the bottom of the patient's chin. Note which finger matches the level of that imaginary line.
4. Count the number of your fingers starting from the Pinky finger.
5. Direct your partner to maintain manual in-line stabilization of the head until the cervical collar is applied.
6. Measure the number of finger breadths between the patient's shoulder and chin.
7. Flatten the cervical collar out so that the outside of the collar is facing up.
8. Align the number of fingers measured with the plastic edge of the collar at the bottom and the assembly port (hole) at the top.
9. Ensure the top of your fingers are at the same level as the assembly port to determine the appropriate size.
10. Configure the cervical collar so that black post is clipped into the assembly port.
11. Slide the longer end of the collar between the patient's occiput and shoulders.
12. Position the chin support and sternal support and ensure midline alignment.
13. Secure the ends of the collar using the Velcro straps, ensuring maximum Velcro attachment and a snug fit.
14. Provide SMR for transportation, as required.
15. If the patient meets the Trauma Bypass Criteria, consult with CCD to pre- inform TRU.
16. Meet Charlie en-route, if required

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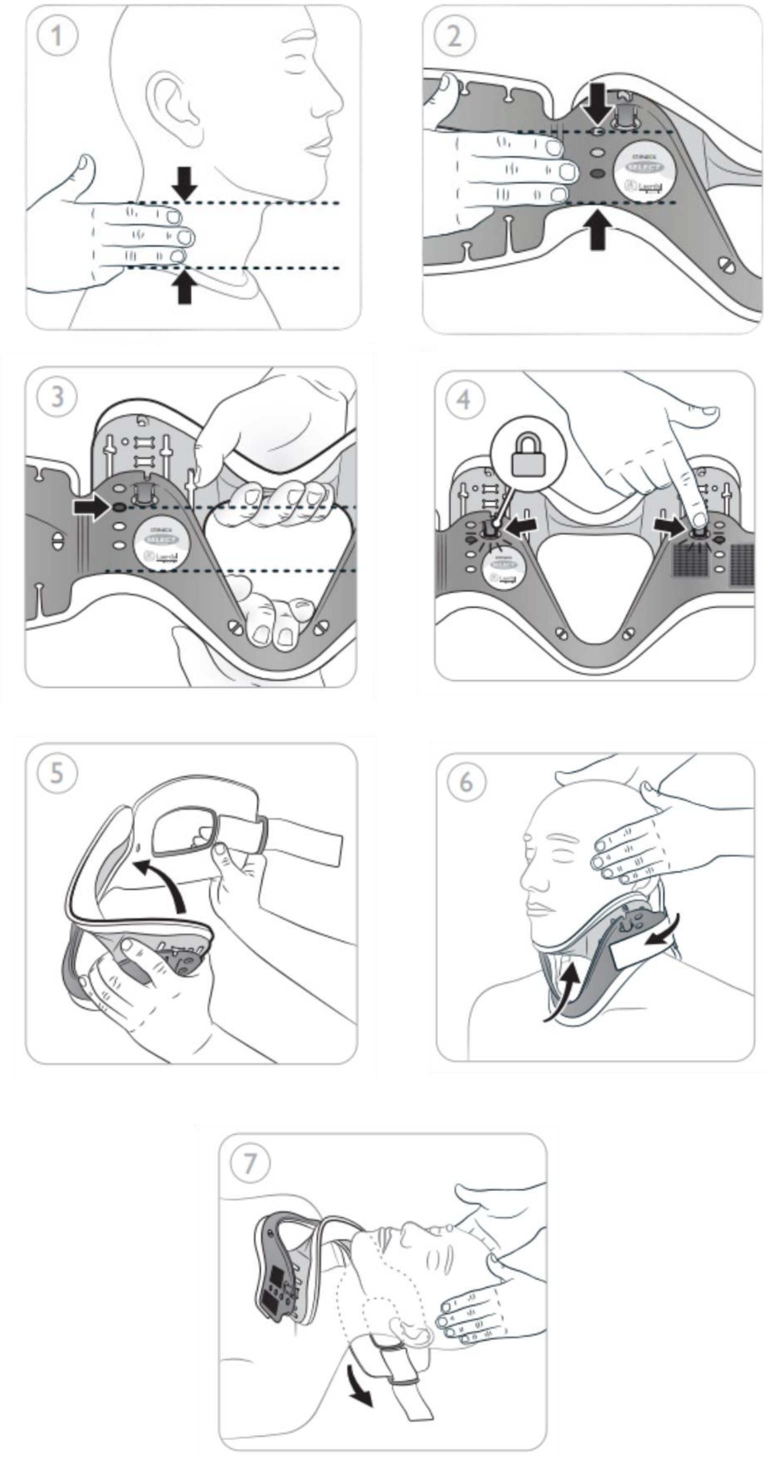
Application of adjustable cervical collar



Laerdal Medical. (n.d.). *Stifneck Select Collar*. https://laerdal.com/item/98001033/

9.3 Combi-Carrier II application

Indications

1. Patient with specific injuries which complicate logrolling including Femur fractures, Unstable pelvis, Multiple trauma.
2. For spinal motion restriction
3. Patient meets one or more of NEXUS criteria

Contraindications

1. Nil contraindication in this setting.

Adverse Effects

1. Patient pain and discomfort.
2. Misalignment of the spine.
3. Pressure points.

Standard precautions

Procedure

1. Manual stabilization of patient C-spine
2. Prepare the Combi-Carrier and straps (SPIDER-HARNESS) as indicated.
3. Check PMS prior and post application of the equipment.
4. Ensure enough helpers.
5. Measure and apply an appropriately sized cervical collar, if indicated.
6. Place the Combi-Carrier next to the patient so that the top of the head grooves is lined up with the top of the patient’s head.
7. Separate Combi-Carrier and place one half on each side of the patient
8. Gently slide the Combi-Carrier under the patient by slightly lifting the shoulder and the hips of the patient.
9. Apply the head blocks and spider harness (Chest-pelvis-shoulder-head-legs), as required. Recheck PMS.
10. Check the patient back prior to safely moving the patient to the ambulance stretcher

9.4 Scoop stretcher application

Indications

1. Patient with specific injuries which complicate logrolling including Femur fractures, Unstable pelvis, Multiple trauma.
2. For spinal motion restriction.
3. Patient meets one or more of NEXUS criteria

Contraindications

1. Nil contraindication in this setting.

Adverse Effects

1. Patient pain and discomfort.
2. Misalignment of the spine.
3. Pressure points.

Standard precautions

Procedure

1. Direct assistant to maintain manual in- line stabilisation of the head, at all times.
2. Preparation of the scoop stretcher and straps (SPIDER-HARNESS) as indicated.
3. Continuously assess PMS distal to the injury. If practical, assess PMS prior and post application.
4. Measure and apply an appropriately sized cervical collar, if indicated.
5. Place scoop beside patient and adjust scoop stretcher to appropriate length.
6. Separate the scoop stretcher and place one half on each side of the patient.
7. Gently slide the wings of the scoop stretcher under the patient being careful not to pinch any part of the body.
8. Apply the headblocks and spider harness, as required.
9. Safely move the patient to the ambulance stretcher.
10. Administer analgesia, as required.
11. If the patient meets the Trauma Bypass Criteria, then consult with CCE to pre- inform TRU.

9.5 Application of a spider straps/harness

Indications

1. Immobilization of a patient onto a scoop stretcher for spinal motion restriction.
2. Immobilization of a patient onto a LBB who will be carried across a distance.
3. Movement of a patient on a scoop.
4. Patient meets one or more of NEXUS criteria

Contraindications

1. Nil contraindication in this setting.

Adverse Effects

1. Airway compromise to a patient secured supine that cannot manage his or her own airway.

Standard precautions

Procedure

1. The spider straps/ harness should be applied to all patients that are placed onto a scoop stretcher or LBB and are carried across a distance.
2. In a suspected spinal injury, one person should maintain manual in-line stabilization of the head until the head blocks and cervical collar are applied.
3. Place the spider harness on top of the patient and lay out the straps.
4. Position the top end of the spider straps level with the sternal notch of the patient.
5. Attach the left shoulder strap by placing the strap over the left clavicle and feed through the hand hold under the patient's left shoulder and Velcro into place.
6. Now attach the right shoulder strap similarly.
7. These first two straps will prevent upward sliding of the patient’s body when the scoop is tilted head down.
8. Attach the chest straps by placing the straps under the patient’s armpits and feed through the hand hold and Velcro into place.
9. Position the pelvic straps across the patient’s pelvis.
10. Attach the pelvic straps by placing the straps over the pelvic bone and feed through the pelvic hand hold and Velcro into place.
11. Place the femur strap across the patient’s femur, and feed through the femur hand holds and Velcro into place.
12. Place the lower leg straps across the patient’s lower legs, feed through the lower leg hand holds and secure.
13. Continuously assess PMS distal to the injury, prior and post spider straps application.

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9.6 Extrication using the long backboard (LBB): Supine

Indications

1. Extrication of an adult patient with suspected spinal injury onto an ambulance stretcher.
2. A critical patient lying on the ground located in a site in whom it is difficult to perform appropriate life-saving care due to patient's location or position.
3. If the scene is unsafe and the crew's safety is in jeopardy.
4. Patient meets one or more of NEXUS criteria

Contraindications

1. Nil contraindication in this setting.

Adverse Effects

1. Patient pain and discomfort.
2. Misalignment of the spine.
3. Pressure points

Standard precautions

Procedure

1. Direct assistant to maintain manual in- line stabilisation of the head.
2. Assess PMS in each extremity.
3. Measure and apply the appropriately sized cervical collar.
4. Prepare the spider straps/ harness.
5. Place the LBB on the appropriate side of the patient.
6. If needed, log roll the patient onto his/ her side without compromising the integrity of the spine.
7. Slide the LBB under the patient and log
8. Roll the patient back onto the LBB. Continuously assess PMS distal to the injury, prior and post spider straps application.
9. Remember to examine the back whilst the patient is log rolled.
10. Adjust the patient into the appropriate position, if required.
11. Secure the patient to the LBB, by placing the spider straps over the patient. Only apply the spider straps if the patient is carried across a distance.
12. Secure the patient's head to the board.
13. Reassess distal pulse, motor function and sensation (PMS) in extremities.
14. Remove the patient from the LBB by either sliding the patient onto the ambulance stretcher or scooping the patient and performing SMR, as required.
15. Do not transport the patient on an LBB. The patient must be scooped off the LBB and

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placed on the ambulance stretcher prior to transportation to hospital.

16. Administer analgesia, as required.
17. If the patient meets the Trauma Bypass Criteria, then consult with CCE to pre- inform TRU.
18. Meet Charlie en-route, if required

9.7 Extrication using the long backboard (LBB): Vehicle

Indications

1. A critical patient located in a vehicle or any other site in whom it is difficult to perform appropriate life-saving care due to patient's location or position.
2. If the scene is unsafe and the crew's safety is in jeopardy.
3. Patient meets one or more of NEXUS criteria

Contraindications

1. Nil contraindication in this setting.

Adverse Effects

1. Patient pain and discomfort.
2. Misalignment of the spine.
3. Pressure points

Standard precautions

Procedure

1. Direct assistant to maintain manual in- line stabilisation of the head.
2. Prepare the LBB, c-collar, spider straps/ harness and stretcher.
3. Ensure additional rescuers are available.
4. First rescuer steps into the vehicle behind the patient and carefully manoeuvres the cervical spine into neutral position and provides manual in- line stabilisation.
5. The airway is then quickly assessed and opened by using a jaw thrust if necessary.
6. Second rescuer approaches the patient from the side, quickly examines the neck, applies the cervical collar and supports the torso.
7. Third rescuer places a LBB near the door and moves to the opposite side of the car to free patient's legs.
8. The patient is rotated in several, short, coordinated moves until the patient's back is in the open doorway with feet on the passenger seat.
9. Assess the patient's back quickly.
10. A fourth rescuer must step in and support the patient's head and neck, as the first rescuer cannot do so throughout this process.
11. The end of LBB is placed on the seat, under the patient's buttocks.
12. The neck and back of the patient are moved as one unit.
13. All rescuers slide the patient into proper position on the LBB in short, coordinated moves, as directed by the rescuer at the patient's head. Apply the spider straps.
14. Move the patient quickly to the ambulance for further evaluation, treatment and removal of LBB. Then treat patient as per respective CPG.

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15. Administer analgesia, as required.
16. Meet Charlie en-route, if required

9.8 Extrication using Kendrick Extrication Device: KED

Indications

1. Stable sitting patient with suspected traumatic spinal injury.
2. To facilitate safe extrication from a confined space.
3. Patient meets one or more of NEXUS criteria

Contraindications

1. When the patient is actual time critical and the application of the KED will delay transport to a trauma centre, or appropriate hospital.

Adverse Effects

1. Chest straps that are too tight tend to interfere with respiratory effort.
2. Groin straps need to be firmly secured to minimise jacket and neck movement during extrication.
3. Incorrect head padding can lead to C- spine hyperextension or hyperflexion.
4. Immobilising the head without properly securing the torso section may cause C- spine movement

Standard precautions

Procedure

1. Direct assistant to maintain manual in- line stabilisation of the head.
2. Assess PMS in each extremity.
3. Measure and apply the appropriately sized cervical collar.
4. While supporting the patient's spine, carefully lean the patient forward or roll the seat back down (if in a car).
5. Perform a rapid physical exam of the back.
6. Slide the KED behind the patient while asking the patient to hold the KED under his/ her arm pits, if practical.
7. Ensure the leg straps are correctly pulled out, then move the patient backwards.
8. Secure the torso straps without compromising the patient's breathing using the middle and lower straps.
9. Slide in seesaw movement the leg straps without compromising the genitalia.
10. Pad the void between the patient head and the KED, if required.
11. Secure the head strap.
12. Secure straps sequentially; Middle Chest Strap, Bottom Chest Strap, Leg, Head and then Top Chest Strap.
13. Secure the top torso, strap (without compromising the patient's breathing) and readjust all torso straps.

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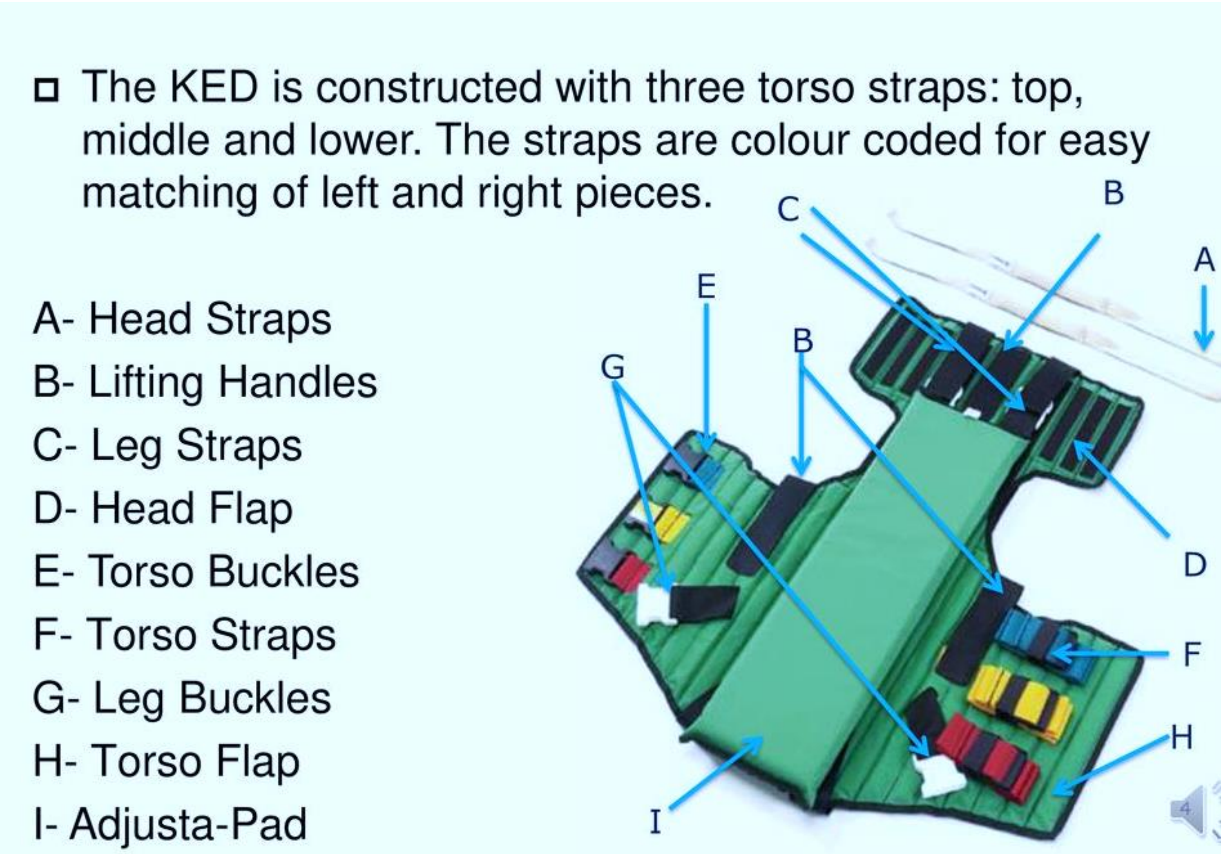
14. Move the patient to the LBB and stretcher.
15. Release the legs strap and then remove the LBB.
16. Remove the KED (time permitting).
17. Perform reassessment including PMS.
18. Administer analgesia, as required.
19. Meet Charlie en-route, if required

Kendrick Extrication Device Components



9.9 Motorcycle helmet removal

Indications

1. Difficulty in assessing or managing patient's airway and breathing due to helmet.
2. Difficulty in performing proper SMR due to helmet.
3. Patient meets one or more of NEXUS criteria

Contraindications

1. Nil contraindication in this setting.

Adverse Effects

1. Possible exacerbation of cervical spine injury

Standard precautions

Procedure

1. Evaluate bilateral PMS in all extremities.
2. Approach the patient from the front and instruct them not to move.
3. Direct the second patient attendant to maintain manual in-line stabilisation at the top of the patient's head, holding the helmet stable, with palms on the sides of the helmet and fingers on the jaw.
4. Remove the patient's eyeglasses if present.
5. The first patient attendant must loosen or cut the chin straps.
6. The first patient attendant places one hand on the mandible at the angle of the jaw and the other hand posterior at the occipital region.
7. The second patient attendant holding the helmet apart gently slips the helmet halfway off the patient's head and then stops.
8. The first patient attendant maintaining stabilisation of the neck repositions and slides the posterior hand superiorly to prevent the head from falling back after complete helmet removal.
9. Remove the helmet completely.
10. Assess the airway.
11. Proceed with SMR, as indicated.
12. Administer analgesia, as required.
13. If the patient meets the Trauma Bypass Criteria, then endorse the patient with CCE to pre-inform TRU.
14. Meet Charlie en-route, if required

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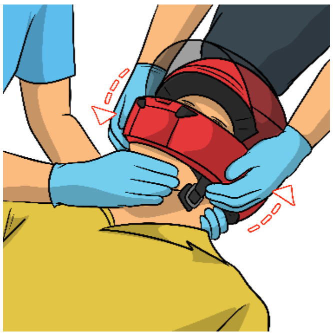
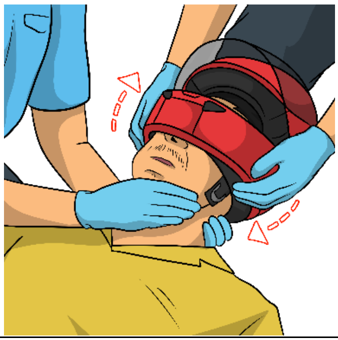
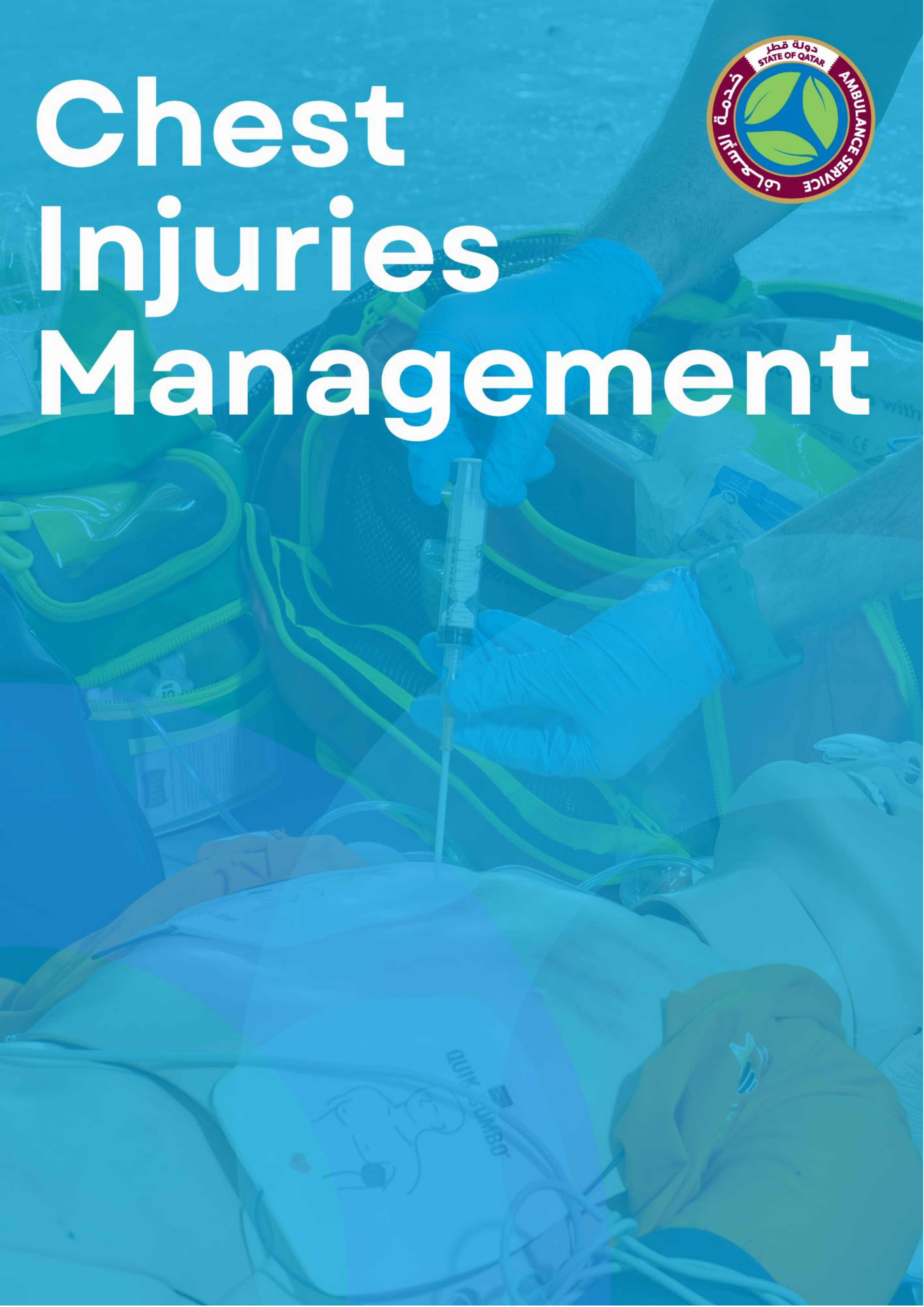
Motorcycle helmet removal procedure

1. One paramedic provides inline immobilisation by placing hands on each side of the helmet with fingers on the victim’s mandible.	2. The second AP cuts or loosens the strap at the D-rings
	
3.The second AP places one hand on the mandible and with the other hand, applies pressure from the occipital region (this manoeuvre transfers inline immobilisation responsibility to the second AP.).	4.The first AP at the top removes the helmet. Helmet must be expanded laterally to clear the ears. The second AP maintains inline immobilisation from below to prevent unnecessary neck motion.
	
5. After the helmet has been removed, AP1 at the top will maintain the inline immobilisation of the head.	6. Inline immobilisation is maintained from above until a combicarrier is in place and a cervical collar is applied.
	

Chest Injuries Management



10.1 Anterior chest wall decompression using the ARS needle

Indications

1. Suspected tension Pneumothorax with respiratory and/or haemodynamic compromise.
2. The patient may present with the following signs and symptoms.
3. Respiratory: Chest pain, dyspnea, tachypnoea, surgical emphysema, diminished breath sounds on the affected side, tracheal deviation, and cyanosis.
4. Cardiovascular: Tachycardia, decreased level of consciousness, hypotension, JVD (may not be present with hypotension).
5. Traumatic cardiac arrest with possible chest trauma.

Contraindications

1. Nil in the setting of a tension pneumothorax

Adverse Effects

1. Improper diagnosis and insertion of the ARS needle may lead to the creation of a simple or tension pneumothorax.
2. Incorrect placement of the ARS needle may result in life-threatening injury to the heart, great vessels, or damage to the lung.
3. Bilateral pleural decompression in the spontaneously breathing patient may result in significant respiratory compromise.
4. Airway compromise to a patient secured supine that cannot manage his or her own airway

Standard precautions

Procedure

1. Ensure that the patient’s BP, ECG, SpO2, Pulse and respiratory rate is being monitored (Time permitting).
2. Establish IV access (Time permitting).
3. Identify the affected side of the chest (left/ right). Locate the second intercostal space in the midclavicular line of the affected side.
4. Disinfect the area using 70% alcohol wipes or 2% Chlorhexidine with 70% alcohol.
5. Remove the RED CAP from the 14-gauge ARS needle with a twisting motion.
6. Remove the ARS needle from the casing and attach a 10 mL syringe.
7. Insert the ARS needle at a 90˚ angle to the chest wall along the superior border of the third rib on the affected side whilst drawing on the syringe, until a sudden loss of resistance is felt.
8. Stabilise the stylet and advance the cannula fully into the pleural cavity until the hub

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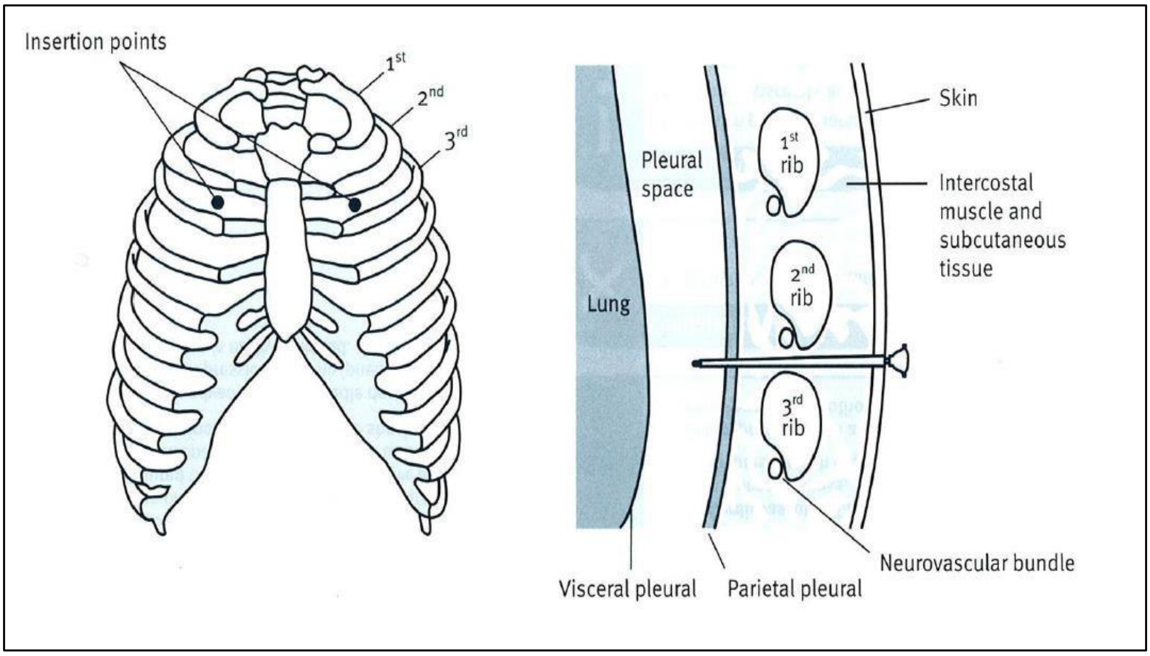
is flushed with the skin. Remove the stylet and dispose of it in a sharp container.
9. Secure the catheter in place with tape.
10. There is no need to attach a flutter valve or 3-way stop cock to the cannula.
11. Re-evaluate breath sounds and haemodynamic status continuously

Decompression of a tension pneumothorax using an ARS needle



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10.2 Lateral chest wall decompression using the ARS needle

Indications

1. Suspected tension Pneumothorax with respiratory and/or haemodynamic compromise.
2. The patient may present with the following signs and symptoms.
3. Respiratory: Chest pain, dyspnea, tachypnoea, surgical emphysema, diminished breath sounds on the affected side, tracheal deviation, and cyanosis.
4. Cardiovascular: Tachycardia, decreased level of consciousness, hypotension, JVD (may not be present with hypotension).
5. Traumatic cardiac arrest with possible chest trauma.

Contraindications

1. Nil in the setting of a tension pneumothorax

Adverse Effects

1. Improper diagnosis and insertion of the ARS needle may lead to the creation of a simple or tension pneumothorax.
2. Incorrect placement of the ARS needle may result in life-threatening injury to the heart, great vessels, or damage to the lung.
3. Bilateral pleural decompression in the spontaneously breathing patient may result in significant respiratory compromise.
4. Airway compromise to a patient secured supine that cannot manage his or her own airway

Standard precautions

Procedure

1. Ensure that the patient's BP, ECG, SpO2, Pulse and respiratory rate is being monitored (Time permitting).
2. Establish IV access (Time permitting).
3. Identify the affected side of the chest (left/ right). Locate the second intercostal space in the midclavicular line of the affected side.
4. Disinfect the area using 70% alcohol wipes or 2% Chlorhexidine with 70% alcohol.
5. Remove the RED CAP from the 14-gauge ARS needle with a twisting motion.
6. Remove the ARS needle from the casing and attach a 10 mL syringe.
7. Insert the ARS needle at a 90˚ angle to the chest wall along the superior border of the 5th or 6th rib on the affected side whilst drawing on the syringe, until a sudden loss of resistance is felt.
8. Stabilise the stylet and advance the cannula fully into the pleural cavity until the hub is flushed with the skin. Remove the stylet and dispose of it in a sharp container.

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9. Secure the catheter in place with tape.
10. There is no need to attach a flutter valve or 3-way stop cock to the cannula.
11. Re-evaluate breath sounds and haemodynamic status continuously.

10.3 Finger Thoracostomy

IV

Indications

1. Traumatic Cardiac Arrest (>14 years old) with suspected chest trauma.
2. Bilateral finger thoracostomies should be performed by a multi-member clinical team (CCP & CCA).

Contraindications

1. Nil in the setting of a tension pneumothorax
2. Obvious non-survivable injury in the traumatic cardiac arrest.

Adverse Effects

1. Improper diagnosis and finger insertion may lead to the creation of a simple or tension pneumothorax.
2. Incorrect placement of the ARS needle may result in life-threatening injury to the heart, great vessels, or damage to the lung.
3. Bilateral pleural decompression in the spontaneously breathing patient may result in significant respiratory compromise. Airway compromise to a patient secured supine that cannot manage his or her own airway.

Standard precautions

Procedure

1. Ensure that the patient's BP, ECG, SpO2, Pulse and respiratory rate is being monitored (Time permitting).
2. Establish IV access (Time permitting).
3. Identify the affected side of the chest (left/ right).
4. Abduct the arm(s) to >90˚ if possible and locate the 'triangle of safety' identified by:
   a. Lateral border of the Pectoralis major (anteriorly).
   b. Anterior border of the Latissimus dorsi (posteriorly).
   c. The axilla (apex).
5. Prepare the incision site with antiseptic solution.
6. Identify appropriate incision site (4th intercostal space, anterior to the mid axillary line) and ensure you are within the 'triangle of safety'.
7. Using a disposable scalpel, make a 30-40mm (3-4cm) incision into the subcutaneous fat. Dispose of the scalpel in the sharp's container once bilateral thoracostomies have been performed.
8. With a 4cm forceps grip, supported by the non-dominant hand, firmly push the forceps through the intercostal muscles and pleura on top of the rib while simultaneously opening and closing the forceps to create an opening. A tract capable of having a finger inserted should be achieved.

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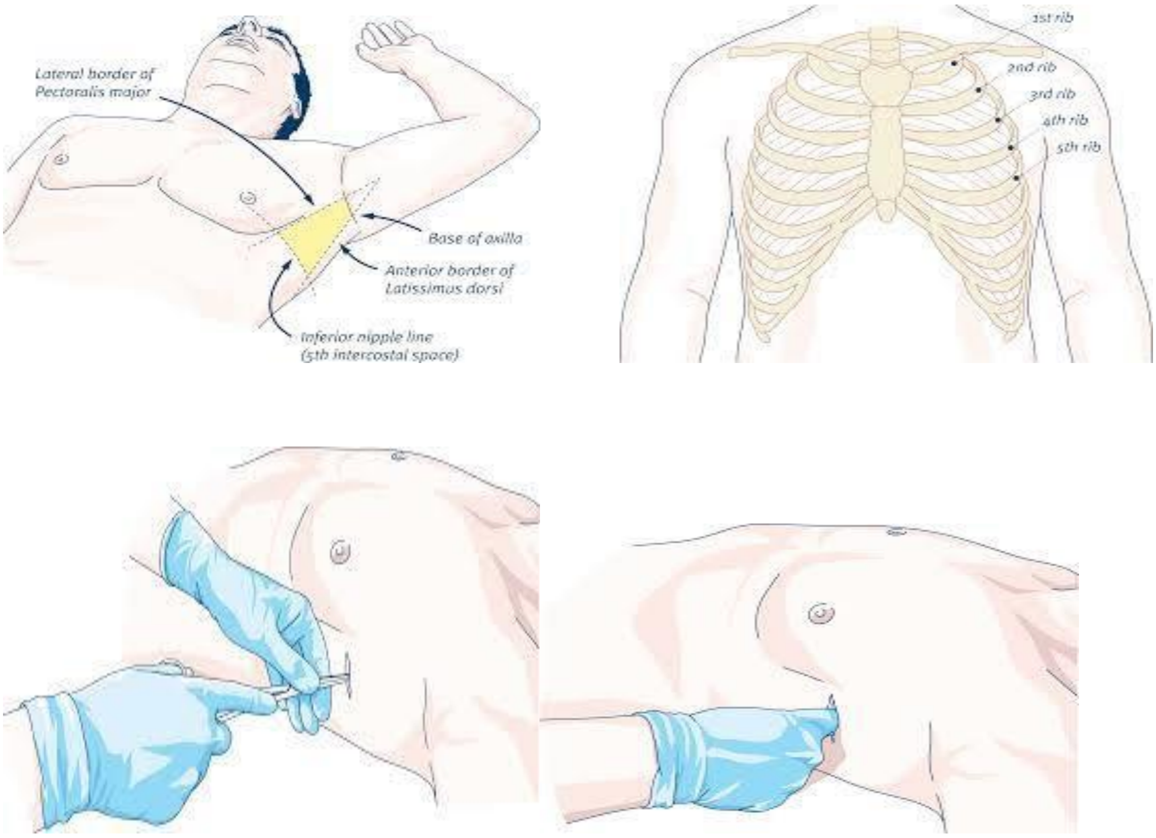
9. With the forceps in position, insert a finger into the pleural space to ensure an opening has been achieved.
10. Remove forceps.
11. Perform a finger sweep to assess for the release of air and/or blood and lung inflation or deflation.
12. Re-evaluate breath sounds and haemodynamic status continuously.

Additional information

1. Clinicians should be double gloved with sterile gloves on the outside.
2. If bilateral finger thoracostomy is anticipated (e.g. traumatic cardiac arrest), then the side with the likely pathology should be decompressed first.
3. Frequently check for re-development of a tension pneumothorax.
4. Finger thoracostomy should happen simultaneously to other cardiac arrest interventions if possible (CPR, LTA etc.)

Finger Thoracostomy



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10.4 Sucking chest wound seal

Indications

1. Penetrating trauma to the chest wall (gunshot wound, stab wound, impaled object that was removed, etc.)
2. Signs of air movement through the wound (bubbles at wound site, spray of blood on exhalation, sound of "sucking" or "gurgling" from the wound with breathing).
3. Any open wound on the thorax caused by penetrating trauma that the practitioner is concerned may be a sucking chest wound should be treated as such.

Contraindications

1. Nil contraindication in this setting.

Adverse Effects

1. A simple pneumothorax may tension.

Standard precautions

Procedure

1. Ensure that the patient's BP, ECG, SpO2, HR and RR is being monitored (time permitting).
2. Prepare an A9 pre-pack.
3. Identify the affected side of the chest.
4. Seal the open chest wound as quickly as possible. If need be, use your gloved hand. Do not delay sealing the wound.
5. Administer high flow oxygen to the patient.
6. Apply the chest seal ensuring that the chest wound is adequately sealed.
7. For multiple sucking chest wounds (example: entrance and exit wound from a gunshot) and no additional chest seal devices being available, use an improvised occlusive dressing for the additional wounds. Use the chest seal for the largest wound and occlusive dressings for smaller wounds.
8. Re-evaluate breath sounds and haemodynamic status continuously.
9. Place the patient in their most comfortable position (if possible, on affected side). Remember to SMR the patient if indicated.
10. Administer analgesia as required.
11. Transport the patient to the appropriate hospital based on Trauma Bypass

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Sucking chest wound seal

A

B

C

D

E

F

10.5 Flail chest stabilization

Indications

1. Blunt chest trauma associated with:
    - Paradoxical chest wall movement.
    - Decreased lung sounds on the injured side.
    - Shortness of breath and chest pain in the conscious patient.

Contraindications

1. Nil contraindication in this setting.

Adverse Effects

1. Nil

Standard precautions

Procedure

1. Initiate high flow oxygen administration.
2. AP crew to request CCP assistance early.
3. Prepare A9 pre-pack and ARS needle.
4. Administer analgesia as required.
5. Identify the affected side.
6. Ask the patient or one of your colleagues to perform manual stabilization of the flail chest segment.
7. Apply a 500 ml vacoliter (A12/ A13) over the flail chest. Then tape the 500ml bag from side to side to stabilize the flail segment on the chest wall.
8. Keep the patient in the position of comfort.
9. Consider the use of BVM assisted ventilation in the patient with a large flail segment and inadequate tidal volume breathes who continues to exhibit respiratory distress and low SpO_2 after stabilization.
10. Re-evaluate breath sounds and haemodynamic status continuously.
11. AP crew to load the patient and rendezvous with the CCP en-route to hospital. Consider the Trauma Bypass Criteria.

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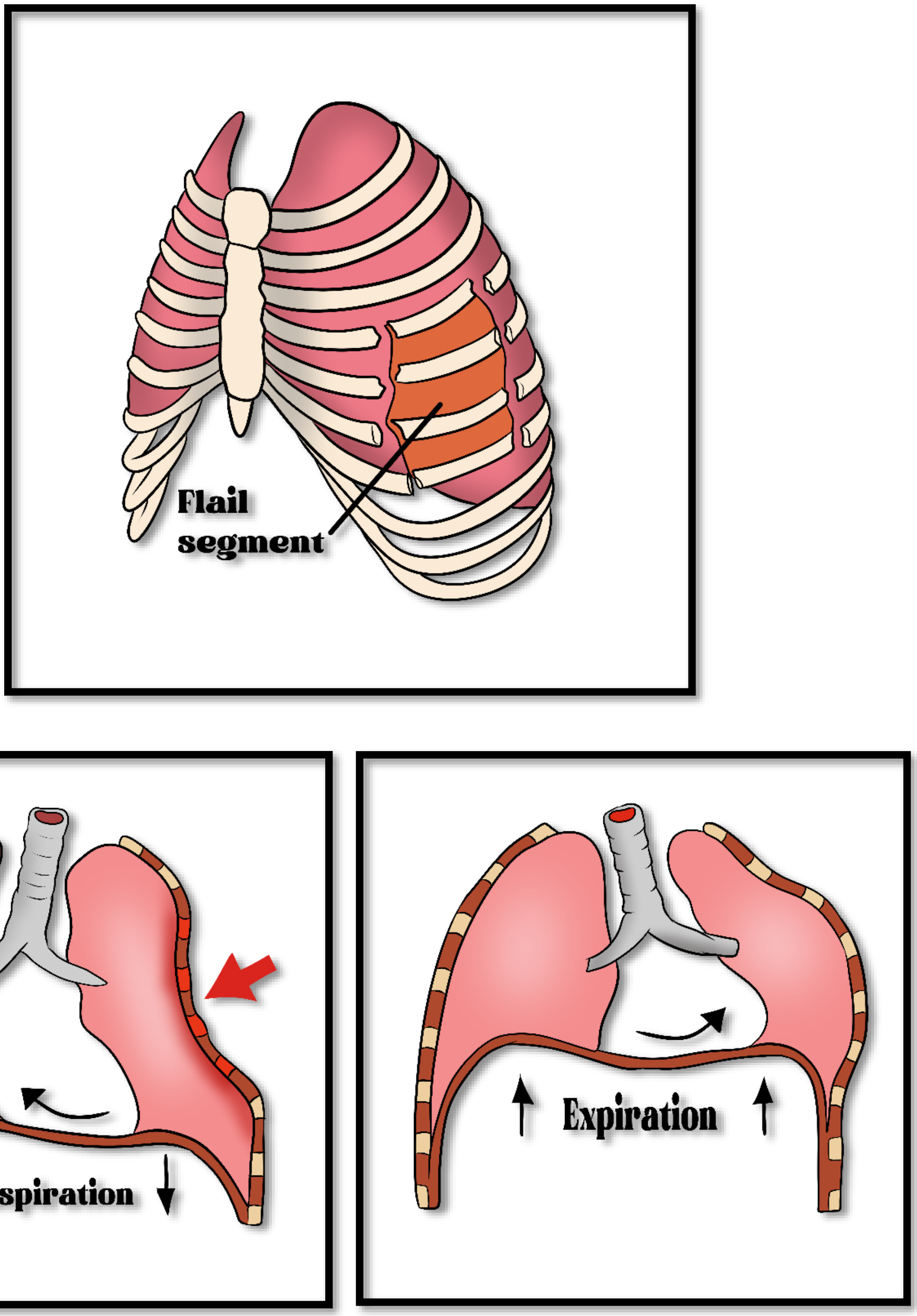
Identifying a flail chest



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Splinting

11.1 Pelvic immobilization with Pelvic Sling

Indications

1. Pelvic pain with associated injury.
2. Unexplained hypotension in a polytrauma patient without other identifiable causes.
3. Traumatic cardiac arrest (DO NOT APPLY FOR ISOLATED HEAD INJURY).

Contraindications

1. Suspected isolated neck of femur fracture.
2. Suspected traumatic hip dislocation.

Adverse Effects

1. May cause pelvic area discomfort and bruising.

Standard precautions

Procedure

1. Assess the patient's lower limbs for PMS, (Not for traumatic cardiac arrest).
2. Consider the patient for manual in-line stabilization of the head.
3. Measure the appropriate length and attach the black strap velcro to the sling surface.
4. Remove objects from patient's pockets.
5. Locate the greater trochanter on either side of the hips, which is at the same level as the pubis symphysis.
6. Gently apply the sling under the patient's knees. With your hands under the buttocks gently lift the patient 1 cm of the ground and slide the sling into position. The sling should be centred over the greater trochanter.
7. Insert the black strap through the buckle.
8. Instruct the partner to hold the orange/ yellow strap firmly, while pulling the black strap in the opposite direction until first "click" sound is heard.
9. Maintain tension while securing the strap to the pelvic sling surface (second click).
10. Reassess PMS continuously.
11. Apply the CombiCarrier II/scoop stretcher to the patient without unnecessary movement of the pelvis. Then apply SMR if required.
12. Move the patient to the ambulance stretcher.
13. Administer further analgesia, as required.
14. Endorse the patient to the CCE as the patient meets Trauma Bypass Criteria, to pre-inform TRU.
15. Meet Charlie en-route, if required.

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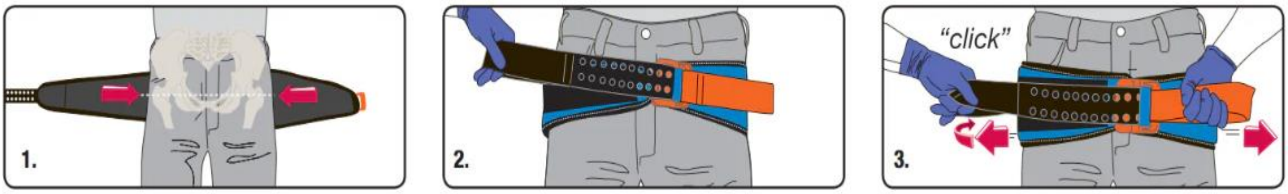
Application of Pelvic Sling

Applies in 3 Easy Steps

no trimming, no cutting, no guessing



1. Remove objects from patient's pocket or pelvic area. Place SAM Pelvic Sling II black side up beneath patient at level of trochanters (hips).

2. Place **BLACK STRAP** through buckle and pull completely through.

3. Hold **ORANGE STRAP** and pull **BLACK STRAP** in opposite direction until you hear and feel the buckle click. Maintain tension and immediately press **BLACK STRAP** onto surface of SAM Pelvic Sling II to secure. You may hear a second click as the sling secures.

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11.2 Application of traction splint

Indications

1. Suspected isolated, closed midshaft femur fracture.

Contraindications

1. Suspected isolated neck of femur fracture.
2. Suspected traumatic hip dislocation.

Adverse Effects

1. Patient pain and discomfort.

Standard precautions

Procedure

1. Assess the patient’s limbs for PMS.
2. Achieve appropriate analgesia before starting the procedure.
3. Expose the affected limb and remove footwear. Assess the limb for wounds.
4. Prepare the traction splint for application (undo the Velcro straps).
5. Attach the ankle strap to the affected limb, then reassess PMS.
6. Delegate your partner to perform manual traction and stabilization to the injured limb above and below the injury site.
7. Measure the length of the traction splint and position the straps appropriately using the unaffected limb. Ensure to extend the splint 10 to 15cm to facilitate the application of traction.
8. Move the traction splint to the affected leg. Position the device under the limb until the Ischial pad rests against patient’s Ischial area.
9. Secure the proximal Ischial/ groin strap.
10. Raise the heel stand. Recheck the traction level and adjust as necessary.
11. Secure the loop to the ankle hitch.
12. Apply mechanical traction until mechanical traction is equal to manual traction.
13. Recheck the ischial strap.
14. Do not overstretch the leg but use enough traction to maintain limb alignment.
15. Apply the leg straps. One should be above the ankle strap, one strap below the knee, one strap above the knee, and one strap at the top of the thigh below the ischial strap. Do not fasten a strap directly over the injury site.
16. Reassess PMS.
17. Scoop the patient and transport the patient on the stretcher. The traction splint should be strapped to the Combicarrier/scoop stretcher during transportation.
18. Administer further analgesia, as required.

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19. Endorse the patient with CCE to pre-notify TRU. Meet Charlie en- route, if required.

11.3 Application of vacuum splints

Indications
1. Immobilization of upper and lower extremities with joints or bones injuries 2. Control of pain involving joint and long bones injuries.
Contraindications
1. There are no absolute contraindications to the use of splints in the emergency setting or in the field to stabilize for transport. For use as a temporary immobilizing device until definitive treatment can be performed.
Adverse Effects
1. Splints applied too tight may cause the patient discomfort and weakening or loss of distal pulses. Compartment syndrome. 2. Pain or discomfort (may require analgesia). 3. Reperfusion injury when released. 4. Embolism. 5. Permanent nerve damage, muscle injury, vascular injury, and/or skin necrosis.
Standard precautions

Procedure
1. Stop all external bleeding to the limb. 2. Select appropriate vacuum splint size. 3. Expose the patient's extremity and assess for PMS. 4. Achieve appropriate analgesia before starting the procedure. 5. Check for any associated injuries to the extremity. 6. Direct second rescuer to apply manual stabilization of the joints adjacent to the injured bone. 7. Whilst the second rescuer is providing manual stabilization throughout the splinting procedure and the first rescuer applies the splint. 8. Lay the splint flat with the valve upward and smooth the split to a uniform thickness. 9. Using HMCAS-approved immobilization techniques, place the splint (valve on the outside) around the injury. Make sure you have access to the valve. 10. Apply the splint with the opening positioned to allow access to check the patient's injury and/or the pulse in the patient's extremity. 11. Keep the hand (upper extremities) or the foot (lower extremities) in the position of function. 12. Secure the splint in place using the hook-and-loop adjustment straps and fastening strip.

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13. Connect the vacuum pump to the valve.
14. Begin evacuating the air from the splint by repeatedly squeezing the pump handle; tighten the valve knob if needed.
15. Continue the procedure until the splint is as rigid as desired.
16. Disconnect the vacuum pump from the splint and tighten the adjustment straps if necessary.
17. Provide reassurance throughout the procedure.
18. Reassess PMS after splinting.
19. Administer further analgesia, as required.
20. If the patient meets the Trauma Bypass Criteria, then consult with CCE to inform TRU.
21. Meet Charlie en-route, if required.

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11.4 Application of splints for joint injuries

Indications

1. Joint injuries. Temporary immobilization of sprains, fractures, and dislocations.
2. Control of pain involving joint injuries.
3. Prevention of further soft tissue or neurovascular injuries.

Contraindications

1. There are no absolute contraindications to the use of splints in the emergency setting or in the field to stabilize for transport.

Adverse Effects

1. Patient discomfort and weakening or loss of distal pulses. Compartment syndrome.
2. Pain or discomfort (may require analgesia).
3. Reperfusion injury when released.
4. Embolism.
5. Permanent nerve damage, muscle injury, vascular injury, and/or skin necrosis.

Standard precautions

Procedure

1. Stop all external bleeding to the limb.
2. Select appropriate splinting material.
3. Assess the patient's limbs for PMS.
4. Achieve appropriate analgesia before starting the procedure.
5. Check for any direct associated injuries to the extremity.
6. Direct second patient attendant to apply manual stabilization of the bones adjacent to the injured joint (one bone above and one bone below).
7. Maintain the joint in the position it is found.
8. Provide reassurance throughout the procedure.
9. Position splint in place under the injured joint. Splint above and below of the injured joint. Pad the voids if necessary.
10. Secure splint in place with the most appropriate fasteners.
11. Splint feet or hands on position of function.
12. Immobilize the injured joint by securing it to the body (Sling and Swathe for upper extremities and other leg for lower extremities).
13. Reassess PMS after splinting.
14. Administer further analgesia, as required.
15. If the patient meets the Trauma Bypass Criteria, then consult with CTL to inform TRU.
16. Meet Charlie en-route, if required

11.5 Application of splints for long bone injuries

Indications

1. Long bone injuries.
2. Temporary immobilization of sprains, fractures, and dislocations.
3. Control of pain involving joint injuries.
4. Prevention of further soft tissue or neurovascular injuries.

Contraindications

1. There are no absolute contraindications to the use of splints in the emergency setting or in the field to stabilize for transport.

Adverse Effects

1. Patient discomfort and weakening or loss of distal pulses. Compartment syndrome.
2. Pain or discomfort (may require analgesia).
3. Reperfusion injury when released.
4. Embolism.
5. Permanent nerve damage, muscle injury, vascular injury, and/or skin necrosis.

Standard precautions

Procedure

1. Stop all external bleeding to the limb.
2. Select appropriate splinting material.
3. Assess the patient's limbs for PMS.
4. Achieve appropriate analgesia before starting the procedure.
5. Check for any direct associated injuries to the extremity.
6. Direct second patient attendant to apply manual stabilization of the joints adjacent to the injured bone (one joint above and one joint below).
7. Maintain the joint in the position it is found.
8. Provide reassurance throughout the procedure.
9. Position the splints in place surrounding the injured bone including the joint above and the joint below. Pad as necessary.
10. Secure splint in place with the most appropriate fasteners.
11. Splint feet or hands on position of function.
12. Immobilize the injured joint by securing it to the body (Sling and Swathe for upper extremities and other leg for lower extremities).
13. Reassess PMS after splinting.
14. Administer further analgesia, as required.
15. If the patient meets the Trauma Bypass Criteria, then consult with CTL to inform TRU.
16. Meet Charlie en-route, if required.

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11.6 Application of splints: Sling and swathe

Indications

1. Immobilization of upper extremity injuries
2. Shoulder dislocation
3. Humerus fractures after splinting
4. Acromioclavicular joint separation
5. Forearm or elbow injuries after splinting (a sling without the addition of the swathe is adequate)

Contraindications

1. There are no absolute contraindications to the use of splints in this setting

Adverse Effects

1. Patient discomfort
2. Pain

Standard precautions

Procedure

1. Ensure that the patient’s BP, ECG, SpO2, HR and RR is being monitored.
2. Prepare pre-pack A14.
3. Continuously assess PMS distal to the injury. If practical, assess PMS prior and post dressing application.
4. Administer analgesia, as required.
5. Direct your partner to apply manual stabilization of the bones above and below the injury site.
6. Position triangular bandage under the injured arm with the acute angle of the bandage over the uninjured shoulder and the right angle of the bandage at the elbow of the injured arm.
7. Bring the lower portion of the bandage up over the injured arm and over the shoulder on the same side.
8. Draw up the ends of the bandage until the hand is several inches above the elbow. Only If you do not suspect a spinal injury
9. Tie the two ends of the sling together.
10. Immobilize the shoulder girdle and upper extremity by tying a swathe around the chest and injured arm with another bandage.
11. Transport the patient on the stretcher in their most comfortable position

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Sling and swathe with two triangular bandages

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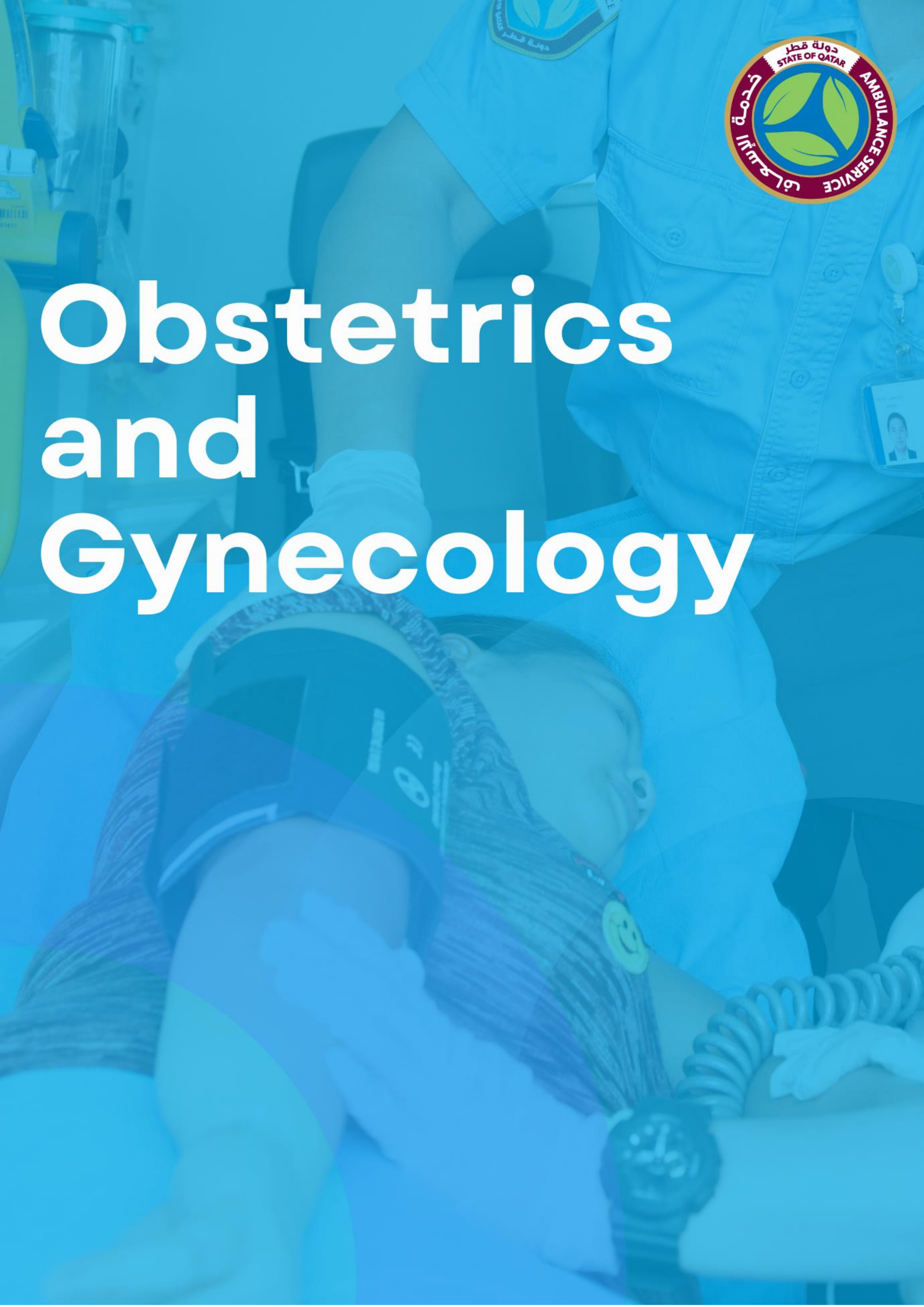


Obstetrics and Gynecology

12.1 Obstetric patient assessment

I II III IV

Indications

1. All pregnant patients

Contraindications

1. Nil in this setting

Adverse Effects

1. Nil in this setting

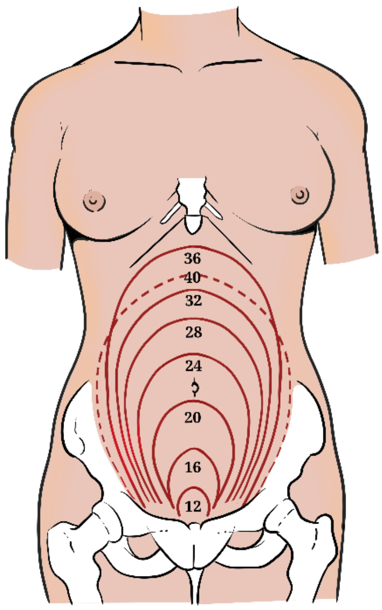
Standard precautions

Procedure

1. Initial approach as per Procedure 1.15: Adult medical patient assessment and Procedure 1.16: Adult trauma patient assessment.
2. Gather a full maternal history (past medical history, past obstetric history, current pregnancy history, etc.).
3. Ensure that the patient's BP, ECG, SpO2, HR and RR are being monitored.
4. Prepare pre-pack A17.
5. Expose the patient's abdomen, with discretion. Inspect for abnormal masses, scars from previous surgery and asymmetry.
6. Palpate the abdomen assessing the height of fundus and the presence of any abdominal tenderness utilizing Leopold's Maneuver.
7. First perform the fundal grip. While facing the woman, palpate the woman's upper abdomen with both hands. The foetal head is hard, round, and moves independently of the trunk while the buttocks feel softer, are symmetric, and the shoulders and limbs have small bony processes; unlike the head, they move with the trunk.
8. Second perform the umbilical grip. Palpate the abdomen with gentle but also deep pressure using the palm of the hands. First the right hand remains steady on one side of the abdomen while the left hand explores the right side of the woman's uterus. This is then repeated using the opposite side and hands. The foetal back will feel firm and smooth while foetal extremities (arms, legs, etc.) should feel like small irregularities and protrusions.
9. Third perform the 1st pelvic grip. Grasp the lower portion of the abdomen just above the pubic symphysis with the thumb and fingers of the right hand. Palpate for the foetus head.
10. Lastly perform the 2nd pelvic grip. The fingers of both hands are moved gently down the sides of the uterus toward the pubis palpating for the foetus forehead.
11. If imminent delivery is not expected, place the patient in the left lateral position on

the stretcher for transportation.
12. Transport the patient to the nearest maternal facility. Consider the Obstetric/ Gynae bypass criteria

Fundus height landmarks

12 weeks gestation	Height of symphysis pubis	
20 weeks gestation	Height of the umbilicus	
36 weeks gestation	Height of xiphoid process	
37-40 weeks gestation	Fundus regresses to between 32-36cm	

12.2 Normal Vaginal Delivery

Indications

1. Full term pregnancy with criteria of imminent delivery, cephalic presentation

Contraindications

1. Nil in this setting

Adverse Effects

1. Retained placenta
2. Ruptured uterus
3. Perineal lacerations
4. Post-partum haemorrhage
5. Neonatal resuscitation

Standard precautions

Procedure

1. Initial approach as per Procedure 1.15: Adult medical patient assessment.
2. Gather a full maternal history (past medical history, past obstetric history, current pregnancy history, etc.).
3. Ensure that the patient's BP, ECG, SpO2, HR and RR are being monitored.
4. Prepare pre-pack A17, A6, A7 and Airway roll.
5. Expose the patient's abdomen and pelvic area, with discretion.
6. Position the mother. Elevate the head and the hips by putting pillows (if available) behind then place plastic lined underpad under the buttocks. Flex the hips and knees spread apart and the feet flat on the stretcher.
7. Create a sterile field around the vaginal opening.
8. Avoid pain management at late stage of labour. Must be considered after delivery and cutting of the umbilical cord.
9. Deliver the head of the foetus, providing gentle pressure on the perineum and over the bony part of the skull to slow the delivery.
10. Look for umbilical cord around the neck and remove it if present.
11. Support the head of the foetus as it
12. If required suction the foetus's mouth first, then the nose (squeeze bulb syringe before entering nose and mouth).
13. Guide the head of the foetus downwards to deliver the first shoulder.
14. Guide the head of the foetus upward to deliver the other shoulder.
15. Deliver the rest of the foetus and grasp the feet.
16. Place newborn at the level of the mother's vagina.

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17. Dry the newborn with baby wrap.
18. Warm with thermal blanket and stimulate the baby.
19. Position the airway and suction if needed.
20. Evaluate the newborn's RR, HR, Tone and colour.
21. Calculate the APGAR score at one minute. If low, request CCP assistance.
22. After pulsations in the umbilical cord cease, clamp the cord four-finger widths (approx. 10-15 cm) away from the baby and the second clamp 5-10 cm further away from the first clamp. Cut the cord between the 2 clamps.
23. After delivery report to NCC, the baby's gender, time of delivery and APGAR.
24. Calculate the APGAR score at five minutes. If low, request CCP assistance.
25. Prepare for delivery of the placenta (Encourage the woman to push/ bear down).
26. After delivery of the placenta, place it in a yellow plastic bag and transport it with mother and baby.
27. If placental delivery is delayed, initiate transport and attempt to deliver the placenta en-route to hospital.
28. Use fundal massage and breast-feeding by the newborn to assist delivery of the placenta.
29. Apply sanitary pads for bleeding.
30. Apply oxygen and treat the mother with IV Normal Saline for shock, if present.
31. Continuously monitor the vital signs of the mother and baby.
32. Transport mother and baby with pre- notification of receiving facility.
33. Meet Charlie en-route, if required.

12.3 Prolapsed cord management

Indications

1. Umbilical cord is presenting through the vaginal opening, with or without the presenting part of the fetus.

Contraindications

1. Nil in this setting

Adverse Effects

1. Spasm of the umbilical cord from excessive handling.

Standard precautions

Procedure

1. Initial approach as per Procedure 1.15: Adult medical patient assessment.
2. Gather a full maternal history (past medical history, past obstetric history, current pregnancy history, etc.).
3. Ensure that the patient’s BP, ECG, SpO2, HR and RR are being monitored.
4. Prepare pre-pack A17, A18 and Airway roll.
5. Position the mother by elevating her buttocks, using the exaggerated sims position.
6. Administer high flow oxygen to the mother, if required.
7. Administer analgesia, as required.
8. Check the cord for any pulsations; NOTE: minimal handling.
9. If cord is still pulsating and no pressure is applied to it from the presenting part, place wet sterile gauze on the cord.
10. If the cord has no pulsation and the presenting part is applying pressure on it, apply pressure to the presenting part to ease the pressure and allow circulation to the foetus.
11. Do not remove your hand until in the delivery suite/ operating theatre.
12. If delivery is imminent, then immediately assist delivery and prepare for newborn resuscitation.
13. Meet Charlie en-route, if required. Evaluate the newborn’s RR, HR, Tone and colour.

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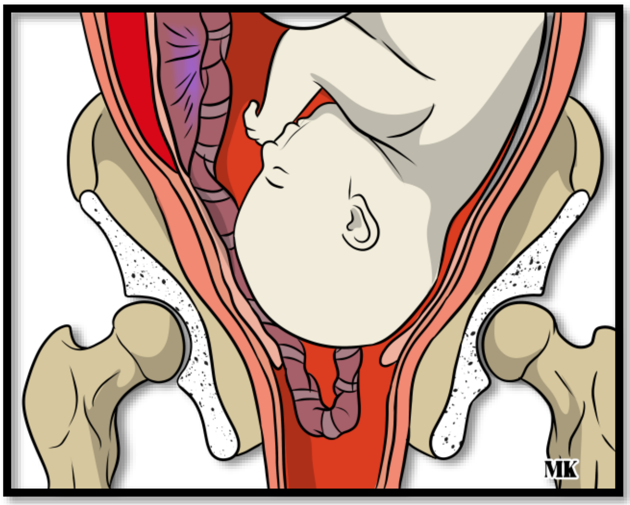
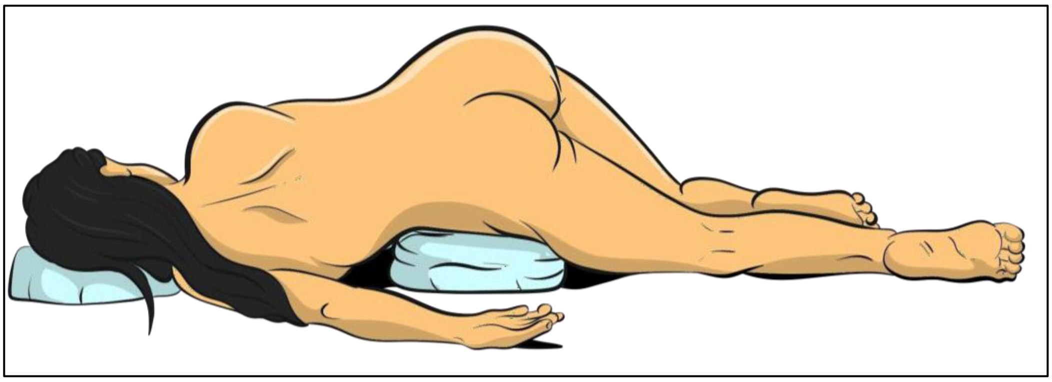
Cord Prolapse



Exaggerated sims position



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12.4 Breech presentation management

I II III IV

Indications

1. Umbilical cord is presenting through the vaginal opening, with or without the presenting part of the fetus.

Contraindications

1. Nil in this setting

Adverse Effects

1. Spasm of the umbilical cord from excessive handling.

Standard precautions

Procedure

1. Initial approach as per Procedure 1.15: Adult medical patient assessment.
2. Gather a full maternal history (past medical history, past obstetric history, current pregnancy history, etc.).
3. Ensure that the patient's BP, ECG, SpO2, HR and RR are being monitored.
4. Prepare pre-pack A17, A18, A6, A7 and Airway roll.
5. Expose the patient's abdomen and pelvic area, with discretion.
6. Position the mother. Elevate the head and the hips by putting pillows (if available) behind then place plastic lined underpad under the buttocks. Flexes the hips and knees spread apart and the feet flat on the stretcher.
7. Create a sterile field around the vaginal opening.
8. Support the perineum by applying pads.
9. Allow the foetus to be delivered spontaneously, supporting the legs, buttocks and torso by wrapping with sterile sheet. NEVER PULL the feet!
10. Once the umbilicus is past the perineum, gentle downwards and outwards traction of the foetus by holding the bony parts of the pelvis until the scapula are visible.
11. Once the scapula is visible, rotate the infant 90 degrees and gently sweep the anterior arm out of the vagina.
12. Rotate the foetus 180 degrees in the reverse direction and sweep the other arm out of the vagina.
13. Then rotate the foetus 90 degrees to the original position.
14. Instruct your partner to apply suprapubic pressure.
15. Maintain the foetal head in flexed position during delivery by gently lifting the foetal maxillary prominences using index and middle finger.
16. During delivery of the head, avoid overextending the neck and foetal body by ensuring it is held in neutral position.

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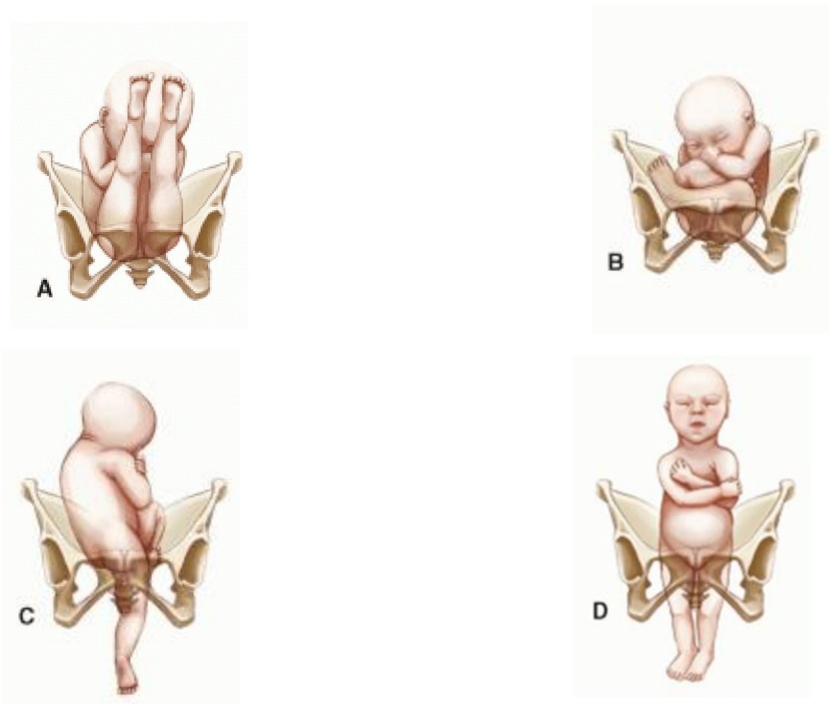
17. Once the foetus is delivered, perform newborn care and resuscitation, if required.
18. Instruct partner to report APGAR score of newborn and status of the mother to CTL.
19. Dry and warm the newborn with baby wrap. Stimulate the baby.
20. Position the airway and suction if needed.
21. Evaluate the baby's RR, HR, Tone and colour.
22. Prepare for the delivery of the placenta.
23. After delivery of the placenta, place it in yellow plastic bag and transport it with mother and baby.
24. Administer analgesia, as required.
25. Use fundal massage, breast-feeding and pads for bleeding. Apply oxygen and treat the mother for shock, if present.
26. Transport mother and baby with pre- notification of receiving facility.
27. Meet Charlie en-route, if required.

Classification of Breech presentations

A: Frank Breech **B: Complete breech** **C: Footling breech** **D: Double Footling breech**



12.5 External fundal massage

Indications

1. Post-partum vaginal bleeding

Contraindications

1. Nil in this setting

Adverse Effects

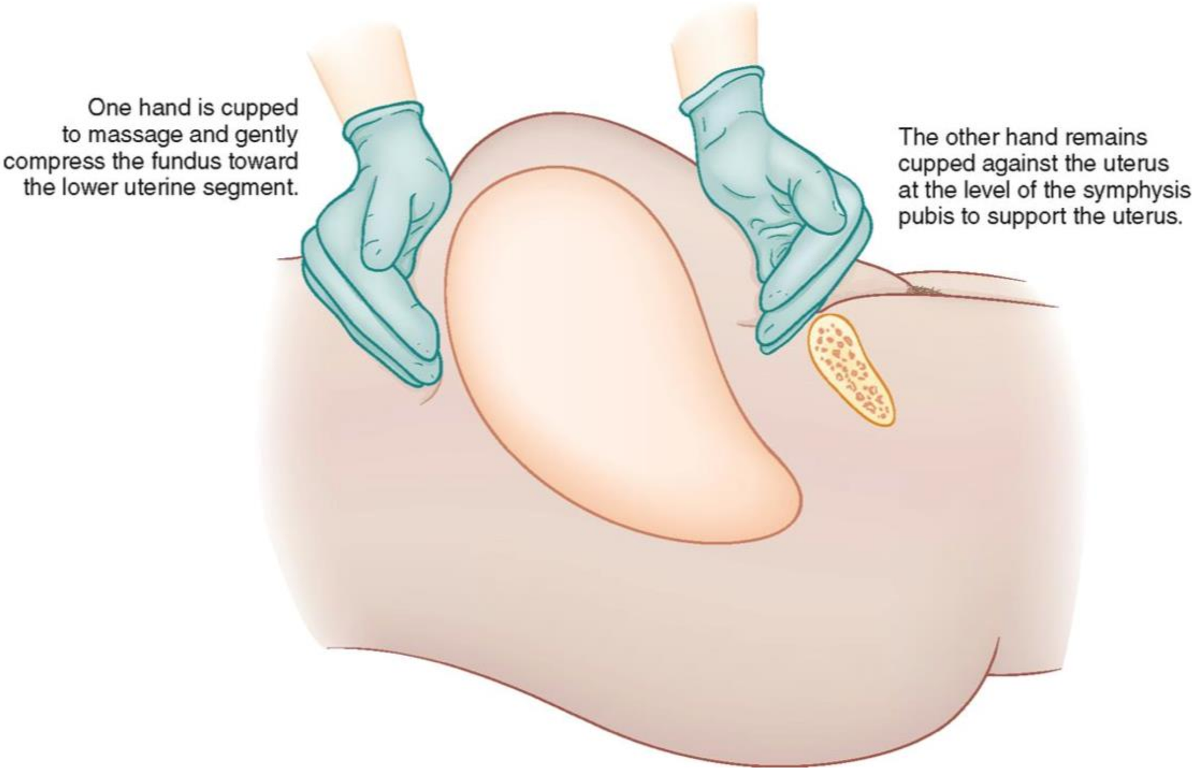
1. Pain
2. Discomfort
3. Trauma to the uterus

Standard precautions

Procedure

1. Initial approach as per Procedure 1.15: Adult medical patient assessment.
2. Gather a full maternal history (past medical history, past obstetric history, current pregnancy history, etc.).
3. Ensure that the patient’s BP, ECG, SpO2, HR and RR are being monitored.
4. Prepare pre-pack A17, A18, A6, A7 and Airway roll.
5. Expose the patient’s abdomen and pelvic area, with discretion.
6. Position the mother. Elevate the head and the hips by putting pillows (if available) behind then place plastic lined underpad under the buttocks. Flexes the hips and knees spread apart and the feet flat on the stretcher.
7. After delivering the baby and the neonatal assessment reveals a stable baby, get the baby to breastfeed.
8. Place one hand over the pubic bone and firmly massage the uterine fundus to stimulate uterine contraction.
9. Continue the procedure until post- partum bleeding stops.
10. Transport the patient P1 to the nearest maternal facility.
11. Meet Charlie en-route, if required. CCP to consider TXA administration

External fundal massage



Paediatrics

13.1 Paediatric patient assessment

Indications

1. All paediatric patients

Contraindications

1. Nil in the setting.

Adverse Effects

1. Patient discomfort

Precautions

1. Expose the patient with discretion
2. Involve next of kin (family) in assessment process

Standard precautions

1. Wear an N95 mask if the patient exhibits respiratory symptoms.
2. A face shield and gown are required when treating patients with a travel history from high-risk countries. (emerging and reemerging infectious diseases)

Procedure

1. Keep the family/ parents/ next-of-kin close to the paediatric patient.
2. Assess the 3 essential components of paediatric assessment. The paediatric Assessment Triangle, Secondary Survey and Ongoing assessment.
3. Complete the Paediatric Assessment Triangle in a sequential assessment concentrating on the patient's appearance, work of breathing and circulation.
4. For appearance complete TICLS assessing the paediatrics tone, interactivity, Consolability, look/ gaze and speech/ cry.
5. When assessing the work of breathing:
   a. Look for abnormal positioning: Head bobbing, tripod position and/or sniffing position.
   b. Listen for abnormal audible breath sounds (snoring, stridor, wheezing, grunting and hoarse voice).
   c. Look for visual signs on increase work of breathing (retractions, tachypnoea, tracheal tugging, abdominal breathing and nasal flaring).
6. Assess the circulation by assessing the pulse rate and volume, skin colour and mucous membrane (pallor, mottling, cyanosis and petechiae), skin turgor and capillary refill.
7. Complete a comprehensive secondary survey and an appropriate history taking
8. Continuously reassess the paediatric determining the effectiveness of the emergency interventions.
9. Assess the relevant parts of the various systems based on the patient's chief complaint.

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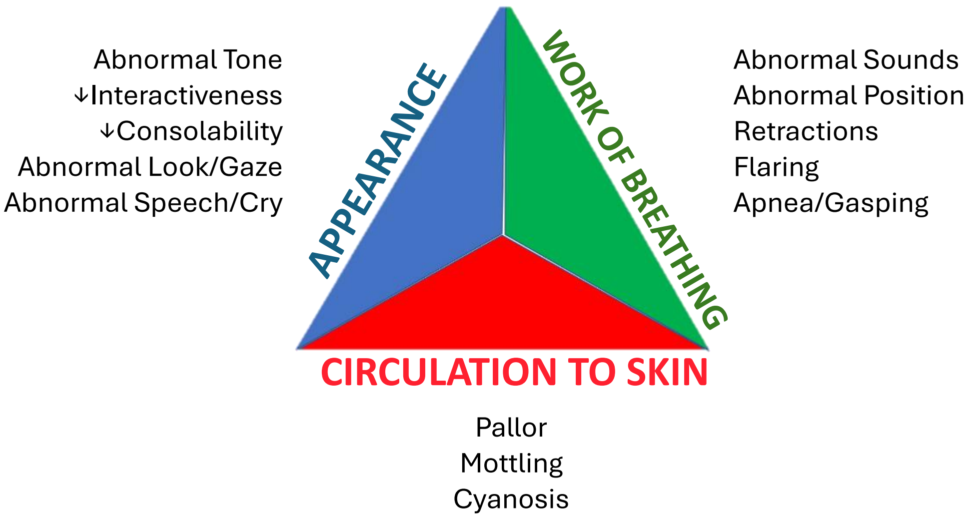
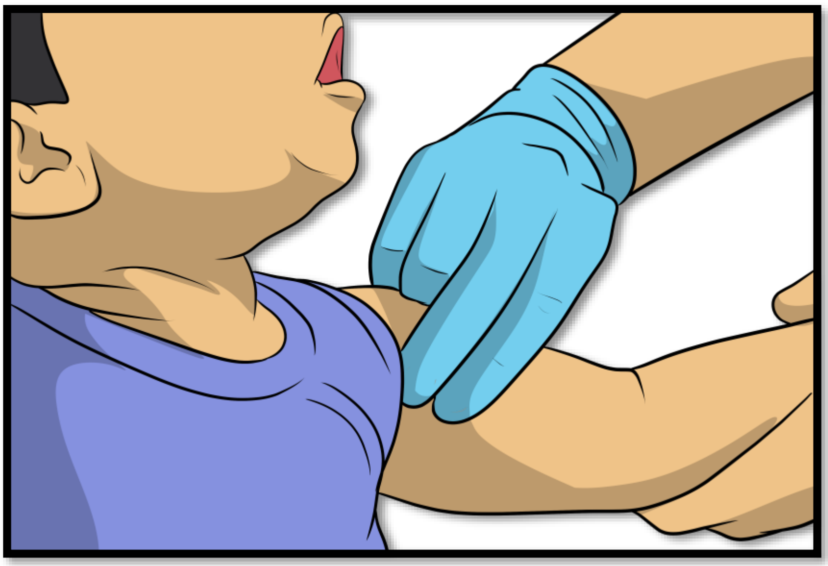
10. Manage the patient according to the appropriate CPG.
11. Reassess the patient's condition and interventions.
12. Ensure notification of the receiving facility as indicated (through CTL).
13. Maintain good communication with the family/ parents/ next of kin.
14. Transport the patient to the appropriate facility.

Pediatric assessment Triangle (PAT)



Neonatal brachial pulse check site



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ESTIMATE WEIGHT CALCULATIONS:

0-12 months: (age x 0.5) + 4

1-5 years: (age x 2) + 8

6-14 years: (age x 3) + 7

Circulating blood volume for paediatrics: 80-85

NORMAL VITAL SIGNS FOR PAEDIATRICS							
Age Group	Weight	RR (breaths/min)	HR (beats/min)	SBP (mmHg)	DBP (mmHg)	Temp (°C)	RBS (mmol/L)
Preterm	2kg	40-70	110-170	55-75	35-45	36.5	2.6-6
Term – 2 months	3 – 4kg	35-55	110-160	65-85	45-55	36.5	2.6-6
3 – 7 months	5 – 7kg	35-45	110-160	70-90	50-65	36.5	4-6
8 – 12 months	8 – 11kg	22-38	90-160	80-100	55-65	36.5	4-6
1 – 3 years	12 – 15kg	22-30	80-150	90-105	55-70	36.5	4-6
4 – 5 years	16 – 21kg	20-24	70-120	95-110	60-75	36.5	4-6
6 – 7 years	22 – 28kg	20-24	70-120	95-110	60-75	36.5	4-6
8 – 9 years	29 – 34kg	16-22	60-110	100-120	60-75	36.5	4-6
10 – 11 years	35 – 40kg	16-22	60-110	100-120	60-75	36.5	4-6
12 – 13 years	41 – 45kg	12-20	60-100	110-135	65-85	36.5	4-6
≥ 14 years	46 – 50kg	12-20	60-100	110-135	65-85	36.5	4-6

13.2 TICLS: Paediatric patient assessment

Indications

1. All paediatric patients

Contraindications

1. Nil in this setting.

Adverse Effects

1. Patient discomfort

Standard precautions

Procedure

1. Keep the family/ parents/ next-of-kin close to the paediatric patient.
2. Assess the 3 essential components of paediatric assessment viz. The paediatric Assessment Triangle, Secondary Survey and Ongoing assessment.
3. Complete the Paediatric Assessment Triangle in a sequential assessment concentrating on the patient's appearance, work of breathing and circulation.
4. For appearance complete TICLS assessing the paediatrics tone, interactivity, Consolability, look/ gaze and speech/ cry.
5. T - Tone: Is the child active, listless or lethargic? Is the child moving spontaneously, sitting or standing as appropriate for age or supporting the weight of their head?
6. I - Interactivity: Does the child interact with you or their parents or caregiver? Are you able to gain the child's interest or engage them in play or activity? Does the child react to sounds?
7. C - Consolability: Can the child be comforted by their parents or caregiver? Is the child behaving to external stimuli as you would predict?
8. L - Look/gaze: Are the child and infant able to fix their gaze on his parents, caregiver or the clinician?
9. S - Speech/cry: Are the child using speech appropriate for age, do the child or infant have a strong and vigorous or weak cry?
10. Complete a comprehensive secondary survey and an appropriate history taking.
11. Assess the relevant parts of these various systems based on the patient's chief complaint.
12. Manage the patient according to the appropriate CPG.
13. Reassess the patient's condition and interventions.
14. Ensure notification of the receiving facility as indicated (through CCE).
15. Maintain good communication with the family/ parents/ next of kin.
16. Transport the patient to the appropriate facility.

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13.3 Modified Westley croup score assessment

Indications

1. Paediatric patients with suspected croup

Contraindications

1. There are no contra-indications to the use of the Modified Westley Croup score during the assessment of croup.

Adverse Effects

1. Patient discomfort from the assessment.
2. Drug specific adverse effects (Epinephrine/ Adrenaline).

Standard precautions

1. N95 mass and face shields are required if a patient exhibits respiratory symptoms, such as coughing

Procedure

1. Keep the family/ parents/ next-of-kin close to the paediatric patient.
2. Assess the 3 essential components of paediatric assessment viz. The paediatric Assessment Triangle, Secondary Survey and Ongoing assessment.
3. Complete the Paediatric Assessment Triangle in a sequential assessment concentrating on the patient's appearance, work of breathing and circulation.
4. For the Modified Westley Croup Score, assess for the presence of chest wall retractions, stridor, cyanosis, the level of consciousness and air entry.
5. Each of the five parameters are then scored based on the table below.
6. The total possible Modified Westley Croup Score is: 0 – 17.
7. For mild croup (score ≤ 2) administer Dexamethasone 0.6mg/kg orally OR Budesonide 2mg via Nebuliser (CCP ONLY).
8. For moderate croup (score 3 – 7) administer Dexamethasone 0.6mg/kg orally OR Budesonide 2mg via Nebuliser. Consider Epinephrine/ Adrenaline 0.5 mg / kg (maximum 5 mg) (CCP ONLY) .
9. For severe croup (≥ 8) administer Dexamethasone 0.6mg/kg orally OR Budesonide 2mg via Nebuliser. Nebulize with Epinephrine/ Adrenaline 0.5 mg / kg (maximum 5 mg) (CCP ONLY).
10. Complete a comprehensive secondary survey and an appropriate history taking.
11. Reassess the patient's condition and interventions.
12. Ensure notification of the receiving facility as indicated (through CCE).
13. Maintain good communication with the family/ parents/ next of kin.
14. Transport the patient to the appropriate facility.

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Modified Westley croup score

	0 POINTS	1 POINT	2 POINTS	3 POINTS	4 POINTS	5 POINTS
Inspiratory stridor	None	When agitated	At rest			
Intercostal recession	None	Mild	Moderate	Severe		
Air entry	Normal	Mildly decreased	Severely decreased			
Cyanosis	None				With agitation/ activity	At rest
LOC	Awake					Altered
Total possible score = 0-17. Mild croup = ≤2; Moderate croup = 3-7; Severe croup = ≥8						

13.4 Paediatric in-line Nebulization

Indications
1. Paediatric patient requiring assisted ventilation with in-line Nebulization: ○ Acute Bronchoconstriction (wheezes) – Salbutamol. ○ Upper Airway obstruction (Stridor) – Epinephrine.
Contraindications
1. Drug specific contra-indications
Adverse Effects
1. Patient discomfort from the assessment. 2. Drug specific adverse effects from the Salbutamol or Epinephrine/ Adrenaline
Standard precautions

1. N95 mass and face shields are required if a patient exhibits respiratory symptoms, such as coughing
Procedure
1. Place the patient in their most comfortable condition. If the patient is unconscious, place the patient supine. 2. Complete the paediatric assessment as per Procedure 10.1. 3. Prepare the oxygen cylinder, pre-packs A8, Airway Roll, Lifepak 15 and monitor the patient. 4. Select the correct medication and check its expiry date. Rule out allergy to medication. Also check for contraindications. 5. Inform CCE and request CCP assistance. 6. Select T-piece nebuliser and attach the catheter mount to the corresponding limb of the T-piece. 7. Attach the BVM to other limb of T-piece and attach the catheter mount to a BVM face mask/ LTA/ ETT. 8. Prepare and place correct medication into the nebuliser chamber. 9. Assemble nebulizer parts and connect the nebulizer to the lower limb of the T- piece. 10. Set the oxygen flow rate at 6 litres per minute and observe for atomisation. 11. Assist or deliver ventilation and provide in-line nebulisation. 12. When a second oxygen source becomes available (in the ambulance) attach the BVM to oxygen. 13. Continue nebulization until the vapour stops. 14. Repeated nebulization may be required with certain categories of patients. 15. Reassess the patient's condition and interventions.

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16. Maintain good communication with the family/ parents/ next of kin.
17. Transport the patient to Sidra Hospital/ Closest appropriate facility.
18. Meet Charlie en-route

13.5 Quick Track: Child Cricothyroidotomy

Indications

1. Cricothyrotomy is indicated in circumstances of actual loss of airway patency and/or airway obstruction, as in: Can't intubate / LTA insertion or can't ventilate using a Face mask/ Naso/ Oro- pharyngeal airway LTA /ETT.
2. Severe facial or nasal injuries (that do not allow oral or nasal tracheal intubation or LTA insertion).

Contraindications

1. Conscious breathing patients.
2. Patients that you can ventilate via other means.

Adverse Effects

1. Prolonged apnoea: hypoxia.
2. Failed attempt leaving a tracheal opening.
3. Haemorrhage from the cricothyroid vessels
4. Cuff leak, tube, occlusion.
5. Pneumothorax, hypoventilation: hypoxia, hypercarbia
6. Subcutaneous air due to improper tube or catheter positioning, along with positive ventilation.
7. Bleeding from superficial neck vessels is very common. Use direct pressure after Quicktrach is in place.
8. Perforations of the back wall of the trachea and the oesophagus from excessively deep penetration by the Quicktrach.

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms and there is a risk of vomiting

Procedure

1. Ensure that the patient’s BP, ECG, SpO2, HR and RR are being monitored.
2. Place the patient’s head and neck in neutral position or hyperextend if medical case.
3. Sterilize the puncture area.
4. Stabilize the larynx using the thumb and middle finger of your non dominant hand.
5. Palpate and identify the cricothyroid membrane.
6. Sterilize the area and index finger, again.
7. Relocate insertion site with index finger to avoid inappropriate insertion.
8. Puncture the cricothyroid membrane with the bevel up.
9. Insert the Quicktrach Child (cannula- over-needle attached to a syringe) at a 90-degree angle caudally (towards the carina) initially prior to changing to 60 degrees during the insertion process up to the red stopper.

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10. Aspirate air using the syringe to determine correct position of the cannula as the Quicktrach Child is advanced through the cricothyroid membrane.
11. Should aspiration of air be impossible because of an obese neck, then remove the stopper and carefully advance the cannula including the metal needle until air is aspirated.
12. Remove the stopper from the cannula.
13. Once air is aspirated indicating that the needle is in the trachea, advance the cannula over the needle until the flange rests on the neck.
14. Withdraw the needle from the cannula and dispose in Sharps container.
15. Attach the BVM to Quicktrach (prepare in advance a catheter mount, filter and ETCO2).
16. Direct your assistant to support the tube and ventilate the patient.
17. Secure the cannula with the foam securing device provided in the kit.
18. Observe the chest for equal rise and fall. Auscultate over both axillae to confirm tracheal placement.
19. Reassess patient for any clinical deterioration.

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13.6 Paediatric orotracheal intubation using Rapid Sequence Induction

Indications

1. Paediatric patient patients ≥8 years old in both the medical and trauma setting.
2. Failure of airway maintenance or protection.
3. Hypercapnic or Hypoxemic ventilatory failure.
4. Expected clinical course of the patient will require intubation.

Contraindications

1. Known hypersensitivity to any medication used in RSI.
2. Patient in whom cricothyrotomy would be impossible.
3. Patient in whom intubation would be impossible (e.g. complete upper airway obstruction)
4. Paediatric patients under the age of 8 years.
5. Procedure should not be performed prior to evaluating for, and attempting to treat, any reversible causes e.g. hypoglycaemia, seizure, etc.

Adverse Effects

1. Failure to intubate
2. Trauma (dental, vocal cords etc.)
3. Failure to oxygenate and ventilate
4. Oesophageal intubation
5. Aspiration
6. Bronchospasm
7. Hypotension
8. Raised ICP

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms and there is a risk of vomiting

Procedure

1. Prepare yourself mentally. Identify risks for potential difficult airway using the mnemonic LEON (**L**ook externally, **E**valuate, **O**bstructions and **N**eck mobility).
2. Position the patient appropriately.

a. If a medical patient is lying on the floor/ bed, raise the head 10 cm and extend in the sniffing position to line the external auditory meatus with the sternal notch. The face will then be parallel to the ceiling.

b. If a medical patient is lying on the ambulance stretcher, elevate the head section of the stretch by 20 degrees. This will line the external auditory meatus with the sternal notch.

c. If trauma patient and is secured on the scoop stretcher (SMR), elevate the head section of the scoop stretcher by 30 degrees (reverse Trendelenburg position). This will line the external auditory meatus with the sternal notch.

d. Obese patients should be placed in the **RAMP** position. This will line the external auditory meatus with the sternal notch.

3. Pre-oxygenate the patient appropriately (refer to Procedure 3.6).
4. Place the nasal $ETCO_2$ and connect to O_2 supply. Start O_2 delivery at 6 LMP (or as much as can be tolerated by the patient) aiming for 15 LPM.
5. Prepare your team by giving them clear and concise commands.
6. Establish 2 IV access and infuse Normal Saline, as required. (Up to 1 liter)
7. Ensure that the patient's BP, ECG, SpO_2, HR, $ETCO_2$ and RR are being monitored. Attach the defibrillation pads to the patient.
8. Change BP time interval on the Lifepak 15 to 2/ 3 minutes. Vitals signs should be monitored pre-RSI, per-RSI and post-RSI.
9. Prepare for standard intubation using the Adult Airway Roll, C6, RSI medication administration pack and medication.
10. Prepare an appropriately sized OPA and oxygen with adult BVM.
11. Tested suction unit with Yankeur suction catheter.
12. Prepare laryngoscope with appropriate blade and size. Prepare the spare laryngoscope handle as backup.
13. Select ETT with 3 size (1 correct size, 1 size big and 1 size small) options and lubricated Frova.
14. Prefill a 10 ml syringe with the desired amount of air to inflate the ETT cuff.
15. Prepare back up equipment, LTA for failed intubation (Safeguard) and Quick Trach II for failed ventilation (Rescue).
16. Prepare the Glidescope/ McGrath video laryngoscope, if available.
17. Prepare and place the tube holder in position under the patient's neck.
18. Preload an appropriately sized, lubricated ETT onto the Adult Frova device (14.0 French).
19. Optimize haemodynamics
20. Calculate Shock Index
21. Prepare pharmacology: Fentanyl, Ketamine, Rocuronium, Push dose pressor (Phenylephrine/Adrenaline) and Adrenaline IM dose for anaphylaxis.
22. Prepare for mechanical ventilation.
23. Proceed with RSI Checklist.
24. Final briefing to team on roles and responsibilities during procedure.
25. Paralyze and induce the patient:
26. Administer:

a. **Fentanyl 1mcg/kg** based on hemodynamic status (Fentanyl to be diluted to 10ml).

Administer Fentanyl 5 minutes prior to induction and paralysis. Administer Fentanyl slowly for over 3 minutes.

b. **Ketamine 1-2mg/kg** based on hemodynamic status.

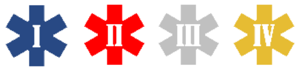
c. **Rocuronium 100-150mg (1.5mg/kg)** for adults and **1.5mg/kg** for paediatrics.

27. During the apneic phase, jaw thrust is performed to maintain pharyngeal patency and increase apneic O_2 to 15 LPM.
28. Wait 45-60 seconds. Then check for jaw flaccidity before attempting laryngoscopy.
29. Change $ETCO_2$ probe on Lifepak 15 from nasal (Smart CapnoLine) to the Filter-Line Set during the apneic phase.
30. Adjust step 30-37 based on device and technique use for laryngoscopy. Place the patient's head in desired position (Align the laryngeal, pharyngeal and oral airway axes) and place the McGrath in the right side of the patient's mouth with your left hand.
31. Move the device to the central position of the patient's mouth while sweeping the tongue to the left.
32. Advance the tip of the McGrath blade into the vallecula.
33. Visualize the epiglottis on the screen. Then lift the anatomy forward and upwards to expose the glottic aperture. When the device is in the optimal position, the glottis should be viewed in the central upper section of the screen.
34. Introduce the tip of the Frova beyond the epiglottis and advance it in a straight line toward the glottis.
35. Advance the Frova into the trachea. If resistance is encountered, do not force the Frova; gently rotate and advance.
36. Maintain position of the Frova; advance the preloaded ETT into the trachea to the appropriate distance.
37. Maintain the position of the ETT, remove the Frova and laryngoscope.
38. Inflate ETT cuff inlet port using a 10ml syringe with air to an appropriate amount.
39. Place BVM on ETT with ETCO2 monitoring, ventilate while listening over epigastrium, then left lung and then right lung to ensure proper placement.
40. Confirm ETT placement with ETCO2 waveform on monitor.
41. If ETT cannot be confirmed through ETCO2 monitoring or confirmation of lung sounds, the ETT should be removed, and an alternate airway device should be utilized (Follow the Failed Intubation Drill).
42. After ETT is confirmed to be in the trachea, secure the ETT with standard commercial tube holder.
43. Ensure that the ETT cuff is adequately inflated.
44. Place the patient on the mechanical ventilator.
45. Continue to ventilate patient while monitoring vital signs.

46. Plan for post intubation hypotension and desaturation.
47. Follow the post intubation checklist.
48. Post intubation management, maintain sedation and paralysis.
    d. **Fentanyl 0.5 mcg/kg**, repeated every 15-20 minutes, as necessary OR Infusion of 1-10mcg/kg/hr.
    e. **Ketamine 0.25 - 0.5mg/ kg** every 15-20 minutes, as necessary OR infusion of 0.01-0.05mg/kg/min.
    f. **Rocuronium 0.25mg/kg** after 30-45 minutes of initial dose.

Fill in appropriate ePCR documentation requirements.

13.7 Paediatric spinal motion restriction: Pedi-Pac



Indications

1. Paediatric patient with suspected spinal injury. Maximum paediatric weight of 41 kg and maximum height of 120 cm.

Contraindications

1. Nil in this setting

Adverse Effects

1. Patient discomfort

Standard precautions

Procedure

1. Complete the paediatric assessment as per Procedure 13.1.
2. Adjust the straps and the head support.
3. Consider pain management during the intervention.
4. Instruct your partner to immobilise the patient's head and neck.
5. Check PMS in all extremities.
6. Apply appropriately sized c-collar and place the Pedi-Pac on the appropriate side.
7. Transfer the patient to the device using the approved method while maintaining manual in-line stabilisation.
8. Avoid excessive movement for suspected spinal injury and internal bleeding.
9. Use the CombiCarrierII as alternative if applicable.
10. Pad any voids as needed.
11. Fasten the restraints to limit movement starting by securing the torso.
12. Reassess PMS in four extremities.
13. Transfer the patient and secure the device to the stretcher using the cot- attachment loops.
14. Request CCP assistance early, if required.
15. Reassess the patient's condition and interventions.
16. Maintain good communication with the family/ parents/ next of kin.
17. Transport the patient to the appropriate facility

13.8 Paediatric/Infant car seat immobilization

Indications

1. Conscious paediatric/ infant patient seated in a car seat with a high index of suspicion for spinal injury

Contraindications

1. Nil in this setting

Adverse Effects

1. Patient discomfort

Standard precautions

Procedure

1. Complete the paediatric assessment as per Procedure 13.1.
2. Consider pain management during the intervention.
3. Instruct your partner to immobilise the patient's head and neck.
4. Check PMS in four extremities.
5. Apply appropriately sized c-collar.
6. Immobilise the infant's torso into the seat with towel rolls on either side of body.
7. Secure the cloth and child with tape.
8. Immobilise the infant's head with towel rolls on either side of the head or one large towel in a horseshoe shape over the infant's head.
9. Secure the towel/s with tape, starting at one side of the seat, crossing the forehead and attaching the tape on either side of the seat.
10. Reassess PMS in each extremity after immobilisation.
11. Remove the safety seat from the car
12. Keep the safety belt of the safety seat.
13. Request CCP assistance early, if required.
14. Manage the patient according to the appropriate CPG.
15. Reassess the patient's condition and interventions.
16. Maintain good communication with the family/ parents/ next of kin.
17. Transport the patient to the appropriate facility.

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Lifting and Moving

14.1 Ferno Mondial Stretcher operations

Indications

1. Raising and lowering of the Ferno Mondial Stretcher
2. Removing and replacing the stretcher top from the transporter of the Ferno Mondial ST stretcher
3. Locking the wheels

Contraindications

1. Nil in this setting

Adverse Effects

2. Lock wheels Patient discomfort

Standard precautions

Procedure

1. Ensure that the patient is secured on the stretcher with the patient straps.
2. One practitioner must position at the top end of the stretcher and the second practitioner at the foot end of the stretcher.
3. Activate the stretcher wheel brakes on the rear wheels.
4. To lower the stretcher, pull the EZ Pull Release Handle (Red Lever) at the rear of the stretcher.
5. Assume the weight of the stretcher by lifting slightly before lowering the stretcher.
6. Hold the EZ Pull Release Handle until the desired height is reached then release the EZ Pull Release Handle to lock the position of the stretcher.
7. To raise the stretcher, do not pull the EZ Pull Release Handle.
8. One practitioner must position at the top end of the stretcher and the second practitioner at the foot end of the stretcher.
9. Activate the stretcher wheel brakes on the rear wheels.
10. Grasp the transporter lift bars or the lift handles on the stretcher top end and lower end.
11. Lift the stretcher to the desired height. The legs will ratchet and lock into position automatically.
12. Unlock the wheel brakes and transport the patient safely.

Removing and replacing the stretcher top from the transporter of the Ferno Mondial ST stretcher

1. To remove the stretcher top from the transporter, one practitioner must be positioned at the top end of the stretcher and the second practitioner at the foot end of the stretcher.
2. Activate the stretcher wheel brakes on the rear wheels.
3. Push the yellow release lever forward and push/ pull the stretcher top towards the rear end of the stretcher.

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4. When the stretcher top is disengaged from the latch mechanism, the practitioners can then remove the stretcher top from the transporter.
5. To attach the stretcher top onto the transporter, one practitioner must be positioned at the top end of the stretcher and the second practitioner at the foot end of the stretcher.
6. Activate the stretcher wheel brakes on the rear wheels.
7. Lift the stretcher top and position it onto the transporter slightly towards the rear legs of the stretcher glides resting on the stretcher rails.
8. Then push the stretcher top towards the front legs until it engages the slam latch mechanism, and you will hear an audible click.
9. The stretcher top can be placed on the transporter facing any direction.
10. Confirm that the stretcher top is locked in place before moving the stretcher.

Other operations of the Ferno Mondial ST stretcher

Lock wheels

1. One practitioner must be positioned at the top end of the stretcher and the second practitioner at the foot end of the stretcher.
2. Both practitioners grasp the stretcher ends.
3. The wheel lock lever of the stretcher is situated on the rear wheels.
4. **To lock** the wheels and activate the wheel lock, the practitioner situated at the foot end of the stretcher presses down on the wheel lock lever with their foot.
5. This process is repeated with the second rear wheel. Lock both rear wheels to prevent the stretcher from moving.
6. **To unlock** the wheel, press the wheel lock lever in the opposite direction. The wheels will be free to move.
7. Prior to disengaging the wheel locks ensure that the stretcher is being supported to prevent fall.

Adjust backrest

1. **To raise** the backrest, press down on the red release levers situated towards the top end of the stretcher. Lift up to raise the backrest to the desired height.
2. **To lower** the backrest, press down on the red release levers and push down on the frame. Stop at the desired height.

Other operations of the Ferno Mondial ST stretcher

Swivel lock

1. One practitioner must be positioned at the top end of the stretcher and the second practitioner at the foot end of the stretcher.
2. Both practitioners grasp the stretcher ends.
3. **To manoeuvre** the stretcher, pull the blue front wheel swivel lock release lever to

allow the front wheels to swivel in any direction. The blue front wheel swivel lock release lever is situated on the front end of the stretcher.

4. When the lever is released, and the stretcher rolls forward the wheels will lock into place.

Additional lift assistance

1. If additional lift assistance is required for the stretcher, pull out on the red FW tab to engage the side lift handles. These additional lift handles are situated on either side of the stretcher.
2. Push the tab in to hide the side release handles.

Raising and lowering stretcher side rails

1. To lower the stretcher side rails, pull out on the red nob and rotate the stretcher rail downwards.

2. To raise the stretcher side rails, lift on the side rail until the red knob locks into place.

3. Sanitize the stretcher, including the handles and belts, after each patient contact, using HMC-approved disinfectant wipes

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14.2 Lifting a stretcher using body mechanics

Indications

1. Lifting a patient secured on a stretcher

Contraindications

1. Nil in this setting

Adverse Effects

1. Injury too self or others lifting the stretcher.
2. Stretcher failure.
3. Patient fall.

Standard precautions

Procedure

1. Know the physical limitations of the crew.
2. Secure the patient with straps on the stretcher.
3. Stand facing the stretcher with feet shoulder width apart.
4. Squat down to the stretcher, bending your knees.
5. Keep the back tight with the abdominal muscles locking the back in a normal slight inward curve.
6. Use the power grip to get the maximum force from your hands.
7. Ensure the palm and fingers come into complete contact with the object and all fingers should bend at the same angles.
8. Place the hands at least +25 cm apart.
9. Keep your feet flat and distribute the weight to the balls of the feet or just behind them.
10. Stand up, making sure that the back is locked, and the upper body comes up before the hips.
11. When lowering the stretcher, reverse the steps.

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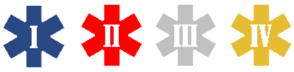
14.3 Stretcher to bed transfer



Indications

1. All patients transported on a stretcher (Excluding patients with SMR and suspected spinal cord injury).

Contraindications

1. Patients with suspected spinal cord injury.
2. Patients who have SMR.

Adverse Effects

1. Patient discomfort
2. Patient fall

Standard precautions

Procedure

1. Apply the brakes on the bed.
2. Place the stretcher next to the bed.
3. Lower/ raise the stretcher to an appropriate level of the bed. Ensure that the stretcher brakes are engaged.
4. Ensure that patient monitoring of RR, HR, ECG, SpO2, and BP is being undertaken (If required).
5. Get the patient to cross their arms over their chest.
6. Lower the bed and stretcher rails, when ready.
7. If the patient is stable, they can be asked to move across onto the bed with minimal assistance. The practitioners will be positioned at the top and foot ends of the stretcher providing support to the stretcher.
8. If the patient is unstable, the practitioner at the top end of the patient will grasp the patient under the patient's arms and the practitioner at the foot end will grasp the patient's feet. In one swift motion the patient will be lifted and moved over onto the bed.
9. Once the patient is completely on the bed, raise the bed rails, if available.
10. Sanitize the stretcher, including the handles and belts, after each patient contact, using HMC-approved disinfectant wipes.

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14.4 Floor to stretcher transfer

Indications

1. Patients requiring to be moved from the floor onto a stretcher.

Contraindications

1. Nil in this setting

Adverse Effects

1. Patient fall
2. Patient agitation.

Standard precautions

Procedure

1. Direct assistant to maintain manual in- line stabilisation (MILS) of the head (based on the patients MOI). Always maintain MILS in trauma related case.
2. Measure and apply an appropriately sized cervical collar, if indicated. Check PMS.
3. Place scoop beside patient and adjust scoop stretcher to appropriate length.
4. Separate scoop stretcher and place one half on each side of the patient.
5. Gently slide the wings of the scoop stretcher under the patient being careful not the pinch any part of the body.
6. Secure the patient appropriately.
7. Lower the stretcher to an appropriate level. Ensure that the stretcher brakes are engaged.
8. Safely lift he scoop stretcher and place onto the ambulance stretcher while maintaining MILS.
9. Ensure full immobilisation of the patient to the ambulance stretcher as required.
10. Continuously reassesses PMS.
11. If SMR not required, remove the scoop stretcher and transport the patient on the ambulance stretcher. Secure the patient straps.
12. Lift the stretcher to the appropriate height and release the brakes.
13. Wheel the stretcher to the ambulance.

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14.5 Chair to stretcher transfer

I II III IV

Indications

1. A conscious patient on a chair/ wheelchair requiring to be placed onto a stretcher for transportation.

Contraindications

1. Suspected spinal injury.
2. Unconscious patients.

Adverse Effects

1. Patient fall
2. Patient discomfort.

Standard precautions

Procedure

1. Activate the wheelchair brakes.
2. Lower the stretcher to an appropriate level. Ensure that the stretcher brakes are engaged.
3. Place the wheelchair next to the stretcher.
4. Explain the procedure to the patient.
5. Stand in front of the seated patient. Place your feet shoulder width apart. Place your hands on either side of the patient’s hips.
6. Encourage the patient to place their hands on your shoulder.
7. In one swift motion, support the patients hips as they stand up.
8. Rotate the patient from the standing position to sit onto the stretcher.
9. Place the patient in the semi-fowlers position and apply the patient straps.
10. Lift the stretcher to the appropriate height and release the brakes.
12. Wheel the stretcher to the ambulance.

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14.6 Patient walking

Indications

1. Any walking patient on scene who needs to be placed onto a stretcher for transportation

Contraindications

1. Patients requiring SMR.
2. Unconscious patients.

Adverse Effects

1. Patient fall
2. Patient discomfort.

Standard precautions

Procedure

1. Remove the stretcher from the ambulance and place the stretcher on secure ground. Lower the stretcher to an appropriate height. Ensure that the stretcher brakes are engaged.
2. Practitioners position themselves on either side of the standing patient (Left and right of the patient).
3. The patient's arms are then placed on the practitioners' shoulders.
4. Supporting the patient, the patient is safely moved towards the stretcher.
5. The side rails of the stretcher are lowered.
6. The patient is safely supported to sit onto the stretcher.
7. The patient's head is safely placed onto the top end of the stretcher whilst the patient's legs are lifted onto the lower end on the stretcher.
8. Cover the patient with a blue sheet to maintain privacy. Secure the patient straps.
9. Raise the side rails of the stretcher.
10. Lift the stretcher, release the brakes and safely transport the patient to the ambulance.
11. Monitor the patients RR, HR, BP, ECG, SpO_2 (As required).

14.7 Use of the Stair chair

Indications

1. Moving patients downstairs or narrow hallways.
2. Patients with specific indication for the stair chair including:
    - Fully conscious
    - No signs of spinal injury
    - Patient can sit on the stair chair
    - Presence of stairs
    - Narrow hallways where stretchers cannot fit.

Contraindications

1. Nil contra-indications in the setting.

Adverse Effects

1. An overtightened chest strap may cause chest discomfort and bruising.

Standard precautions

Procedure

1. Unfold the stair chair and ensure that it is properly locked.
2. Safely place the patient onto the stair chair. Secure the patient on the chair with the straps provided.
3. Front and rear operators: Move to positions front and rear of the chair.
4. Assisting operator: When going downstairs, stand behind the front operator. When going up stairs, stand behind the rear operator.
5. Front operator: Grasp and extend carrying handles, position the handles to fit comfortably. Always face the patient.
6. Rear operator: Grasp the rear carrying handles. Tilt chair back, slowly, until weight is balanced on transport wheels.
7. Front and rear operators: Position Chair near first step.
8. Front and rear operators, in a coordinated movement, lift the chair and carry it up or down stairs.
9. The patient should always be facing down.

12\. Once at the bottom of the stairs, place the patient on the ambulance stretcher.

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14.8 Use of Xtract 2 Carry Sheet

Indications

1. Patient needs rapid extraction and carrying across uneven terrain from a remote location that is inaccessible by ambulance or another vehicle.
2. Patients requiring extraction and carrying from confined spaces e.g. stadiums.

Contraindications

1. Nil contra-indications in the setting.

Adverse Effects

1. Patient pain and discomfort.
2. Misalignment of the spine.
3. Pressure points.
4. Airway compromise .

Standard precautions

Procedure

1. Roll out the Xtract 2 Carry Sheet next to the patient. Ensure the red starps on either side of the Xtract 2 are in line with the patients shoulders to allow for appropriate positioning on the device.
2. Perform a log roll technique and safely load the patient onto the Xtract 2. Ensure the victim is loaded centrally on the Xtract 2.
3. Once the patient is loaded onto the Xtract 2, use the red drawstring at the feet to adjust the stretcher to the correct length.
4. Additional security can be achieved by lacing the elastic straps in holes provided on both sides of the Xtract 2 and secure with the Karabina.
5. Use the red drawstring at the head to adjust the stretcher.
6. Before moving, recheck that the patient is well positioned and secured in the Xtract 2 Carry Sheet.
7. Utilize as many rescuers as are available to carry/move the patient to a safe location

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14.9 Use of Carry Sheet

Indications

1. Lifting a medical patient from the floor onto a bed/ ambulance stretcher.
2. Transferring a medical patient from the bed onto an ambulance stretcher.
3. Transferring a patient from ambulance stretcher onto hospital bed.
4. Carrying medical patients’ downstairs.
5. Carrying of patients out of or through confined spaces

Contraindications

1. Trauma patients with suspected spinal injuries

Adverse Effects

1. Patient pain and discomfort.
2. Patient fall

Standard precautions

Procedure

1. Ensure Minimum 4 staff members required to effectively utilize the carry sheet.
2. Fold the carry sheet.
3. 2 staff members log roll the patient.
4. Staff member 3 and 4 places the folded carry sheet behind the patient.
5. The patient is then log rolled back onto the carry sheet.
6. Staff member 3 and 4 then log roll the patient in the opposite direction.
7. Staff 1 and 2 pull the carry sheet straps.
8. The patient is then returned to the supine position onto the carry sheet.
9. Carry sheet straps are then applied. After applying a blue sheet on the patient
10. The 4 staff then lift the patient holding onto 2 straps each on the 4 corners of the carry sheet.
11. The patient is then safely transferred onto a bed/ ambulance stretcher

14.10 Use of Stokes Basket (Basket Stretcher)

Indications

1. Patient needs rapid extraction and carrying across uneven terrain from a remote location that is inaccessible by ambulance or another vehicle.
2. Patients requiring extraction and carrying from confined spaces e.g. stadiums and Field of play.

Contraindications

1. Nil contra-indications in this setting.

Adverse Effects

1. Patient pain and discomfort.
2. Misalignment of the spine.
3. Pressure points.
4. Airway compromise.

Standard precautions

Procedure

1. Place the vinyl-covered pad in the stretcher and press the pad against the stretcher to join the hook-and-loop strips.
2. Line the basket with one or more blankets for additional patient warmth or protection, if needed.
3. Transfer the patient to the basket stretcher using the appropriate HMCAS guidelines (patient specific needs).
4. Secure the patient in the stretcher with the supplied straps and secure the head using disposable head blockers/head rest.
5. Attach the foot plate.
6. Slide the single hook onto the stretcher shell at the handhold opening closest to the patient's feet.
7. Attach the other end of the strap to the handhold opening on the opposite side of the stretcher.
8. Center the foot plate on the strap, parallel with the end of the stretcher the foot plate should press firmly against the bottom of the patient's feet
9. Carrying the stretcher require a minimum of two trained operators positioned at opposite ends of the stretcher, facing each other.

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Miscellaneous

15.1 Nasogastric tube insertion

IV

Indications

1. Relieve stomach distention.
2. Remove fluid from the stomach.
3. Aeromedical evacuation.

Contraindications

1. Conscious patients
2. Patients with base of skull fractures
3. Severe midface trauma
4. Recent nasal surgery

Adverse Effects

1. Patient discomfort
2. Nausea and vomiting.

Standard precautions

1. N95 masks and face shields are required if a patient exhibits respiratory symptoms and there is a risk of vomiting

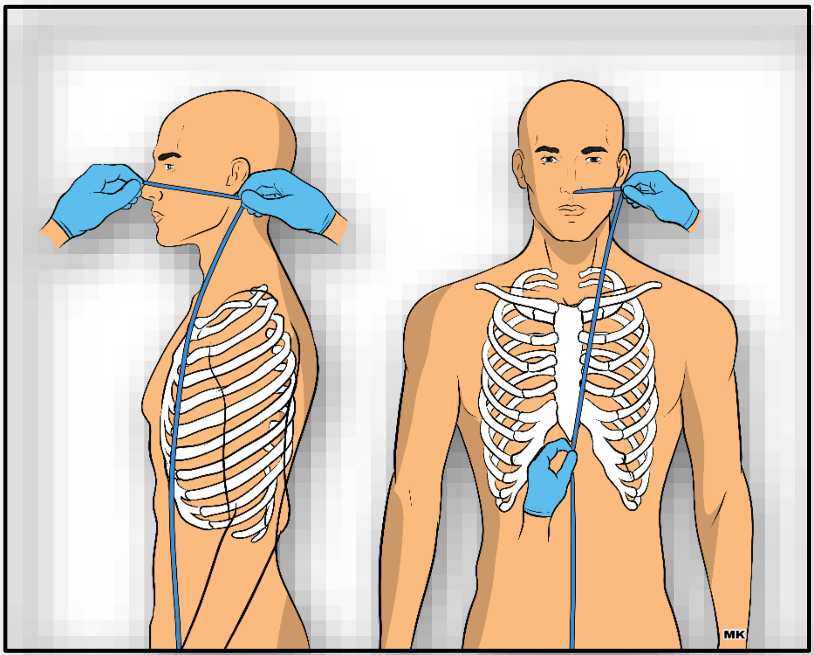
Procedure

1. Identify the need to insert a nasogastric tube (NGT).
2. Standard safety precautions.
3. Get consent from the patient/ family if possible. Explain the procedure to the patient.
4. Prepare pre-pack C10.
5. Measure the length of the NGT (Fr10 for paediatrics and Fr16 for adults) to be inserted by placing the tip of the tube at the patient's nose and running the tube past the angle of the jaw to the xiphoid process. Place a piece of tape at this point to mark the length.
6. Lubricate the first 20 cm of the tube with the water-based lubricant.
7. Inspect the patient's nostrils to exclude anatomical obstructions.
8. Gently insert the tube in an anterior-to- posterior direction (at right angles to the patient's face, and not in a cephalad direction).
9. A gentle twisting action will assist the insertion of the tube.
10. Inspect the patient's mouth to ensure that the tube is not curling up.
11. Confirm the position of the tube by auscultating for air entering the stomach as a result of 20 ml of air injected via the tube or drawing stomach content with the 50 ml syringe.
12. After confirming that the NGT is correctly positioned, connect the drainage bag to

the end of the NGT.

13. Strap the NGT in place.
14. Place the drainage bag below the patient to allow for gravitational emptying of the stomach.

Measuring of nasogastric tubes for insertion



15.2 Clinical Handover: IMIST-AMBO

Indications

1. Handover occurs every time the care for a patient is transferred from one healthcare provider to another, either within the Ambulance Service or to another healthcare facility.
2. Every patient must be formally handed over and received through verbal and/or written communication.
3. Ambulance Paramedic requesting assistance or advice from a Critical Care Paramedic or reporting to CCD.

Contraindications

1. Nil contra-indications in this setting.

Adverse Effects

1. Interruption of the clinical treatment being administered to the patient

Standard precautions

Procedure

1. Review handover details before arrival at hospital
2. Clearly identify self as the treating clinician
3. Introduce the patient's family / friends / colleagues to the receiving HMCAS Clinician
4. Use the IMIST-AMBO mnemonic to guide verbal handover.
5. Deliver the IMIST report uninterrupted over 20 to 30 seconds while the patient remains on the ambulance stretcher.
6. **I:** Identify Age, Gender and Time of Injury or Illness
7. **M:** Describe the Mechanism of Injury or the Medical Complaint
8. **I:** List the Injuries sustained or list the illness or information related to medical complaint.
9. **S:** List the Signs and Symptoms (Look at monitor for latest vital signs)
10. **T:** List the Treatment given and response thereto
11. Safely move patient to hospital bed after IMIST report
12. Continue with AMBO report after patient has been.
13. moved to hospital bed.
14. **A:** List any Allergies the patient has.
15. **M:** List any Medications the patient is taking.
16. **B:** Summarize the patient's Background, i.e. Past Medical History
17. **O:** Describe any Other issues, e.g. social situation, organ donation.
18. Ensure efficient transfer of the patient to alternate monitoring and treatment devices, minimizing interruption of the clinical treatment being administered to the patient.
19. The HMCAS treating clinician should briefly re-evaluate the patient's clinical

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presentation during the handover.

20. If the patient's clinical presentation or parameters differ significantly from the initial handover then this could indicate a pre- morbid state and should be highlighted to the receiving doctor / nurse and documented accordingly in the "Treatment" section of the PCR.
21. Once the receiving HCF doctor or nurse has indicated that they are satisfied with the handover, ensure that they sign the PCR to acknowledge receipt and responsibility for the patient.
22. Remove HMCAS equipment that is not needed and notify the NCC of unit availability accordingly

Pre-arrival IMIST-AMBO checklist

خـدمـة الإسعـاف
Ambulance Service

IMIST-AMBO

Clinical Handover Protocol

I Identify **Age, Gender and Time of Injury or Illness**	**Approximate Age:** _______ **years** ☐ Adult ☐ Pediatric **Nationality:** _______________ **Gender:** ☐ Male ☐ Female **Time of injury/illness onset:** ____h______ ***OR*** _________minutes ago
M Describe the **Mechanism of Injury or the Medical Complaint**	Mechanism of injury/Medical Complaint
I List the **Injuries** sustained or List the illness or information related to medical complaint	**Injuries or Illnesses** (e.g. tension pneumothorax OR severe asthma)
S List the **Signs** and **Symptoms** (Look at monitor for latest vital signs)	**First set of Vital Signs:**
T List the **Treatment** given and response thereto	**Treatment:**

BP =	HR=	GCS: E = V= M= (/15) A V P U	
SpO_2 =	RR =	$EtCO_2$ =	ECG:
Symptoms :			
Signs:			

Treatment:	☐ O_2 ☐ETT	☐ LTA Tube Size____	☐ Crich.	☐ IPPV
☐ CPAP	☐Chest seal	☐ Needle decomp.	☐ Nebulizer	
☐ Defib	☐ Cardiovert	☐ Pacing	☐ Fluids ______ml	
☐CPR	☐ Suction	☐ Analgesia		

Other treatment/response:

Medications and doses:

Allergies	
Medications	
Background	
Other	

Pre-arrival quick checklist. To be read out during handover.